

# SuiteScript 2.1 Complete API Reference & Script Samples Manual

Authoritative technical reference manual detailing all 55 SuiteScript 2.1 modules, including core business logic, query, workbook, UI widgets, machine learning, and security APIs, enriched with official runnable script samples for each module directly from Oracle NetSuite documentation.

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**55**

TOTAL MODULES

**2,065+**

METHODS & MEMBERS

**160**

OFFICIAL CODE SAMPLES

**250+**

OBJECT CLASSES

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| 37 | <a href="#">N/render Module</a>                          | 4 sample(s)  | Server scripts                                                         | Advanced PDF/HTML Templates<br>Custom PDF Layouts                     |
| 38 | <a href="#">N/runtime Module</a>                         | 2 sample(s)  | Client and server scripts                                              | Â                                                                     |

| #  | Module Name & Link                                   | Code Samples | Supported Script Types    | Permissions Required                                       |
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| 39 | <a href="#">N/scriptTypes/restlet Module</a>         | 1 sample(s)  | RESTlet scripts           | -                                                          |
| 40 | <a href="#">N/search Module</a>                      | 9 sample(s)  | Client and server scripts | Perform Search Persist Search<br>Publish Search            |
| 41 | <a href="#">N/sftp Module</a>                        | 3 sample(s)  | Server scripts            | Key Access (needed when public key authentication is used) |
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# 1. N/action Module

Load the N/action module to execute business logic to update the state of a record. Action APIs emulate NetSuite user interface buttons.

```
define(['N/action'], (action) => { ... })
```

✓ 3 Code Sample(s)

Scripts: Client and server scripts

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/action'], (action) => { // Access methods and properties on action });
```

Use the N/action module to execute business logic to update the state of records in view mode. To execute business logic on records in edit mode, use the record macro APIs, which are included in the N/record Module module. See Record Object Members and Macro Object Members. Action and Macro APIs are the programmatic equivalent to clicking a button in the UI. To learn more, see Overview of Record Action and Macro APIs.

The changes that you make to records with N/action module APIs are persisted in the database immediately. For example, consider the timebill record. After you click the Approve button in the UI, the timebill and its entries are saved in an approved state, and this change is immediately updated in the database.

Governance for action module APIs varies for actions and record types. See the action help for governance information specific to actions and record types.

A limited number of individual actions for specific record types are supported. For details, see Supported Record Actions.

For supported script types, see individual member topics listed below.

## N/action Module Members

Return Type / Value Type

Supported Script Types

Client and server scripts

Encapsulates a NetSuite record action.

Plain JavaScript Object

Client and server scripts

| MEMBER TYPE | NAME                                   | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                             |
|-------------|----------------------------------------|--------------------------|---------------------------|---------------------------------------------------------------------------------------------------------|
| OBJECT      | <b>action.Action</b>                   | Object                   | Client and server scripts | Encapsulates a NetSuite record action.                                                                  |
| OBJECT      | <b>Plain JavaScript Object</b>         | Object                   | Client and server scripts | A plain JavaScript object of actions available for a record type.                                       |
| METHOD      | <b>action.execute(options)</b>         | Object                   | Client and server scripts | Executes the record action and returns action results in an object.                                     |
| METHOD      | <b>action.execute.promise(options)</b> | Promise                  | Client scripts            | Asynchronously executes the record action and returns the action results in an object.                  |
| METHOD      | <b>action.executeBulk(options)</b>     | string                   | Client and server scripts | Executes an asynchronous bulk record action and returns its task ID for later status inquiry.           |
| METHOD      | <b>action.find(options)</b>            | Object                   | Client and server scripts | Returns a plain JavaScript object of available record actions for the given record type.                |
| METHOD      | <b>action.find.promise(options)</b>    | Promise                  | Client scripts            | Asynchronously returns a plain JavaScript object of available record actions for the given record type. |
| METHOD      | <b>action.get(options)</b>             | action.Action            | Client and server scripts | Returns an executable record action for the given record type.                                          |
| METHOD      | <b>action.get.promise(options)</b>     | Promise                  | Client scripts            | Asynchronously returns an executable record action for the given record type.                           |

## Action Object Members

The following members are called on action.Action.

Return Type/Value Type

Supported Script Types

Client and server scripts

Executes the action and returns the action results in an object.

Action.promise(options)

| MEMBER TYPE | NAME                           | RETURN TYPE/VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                     |
|-------------|--------------------------------|------------------------|---------------------------|---------------------------------------------------------------------------------|
| METHOD      | <b>Action(options)</b>         | Object                 | Client and server scripts | Executes the action and returns the action results in an object.                |
| METHOD      | <b>Action.promise(options)</b> | Promise                | Client scripts            | Executes the action asynchronously and returns the action results in an object. |

| MEMBER TYPE | NAME                            | RETURN TYPE/VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                 |
|-------------|---------------------------------|------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------|
| METHOD      | Action.execute(options)         | Object                 | Client and server scripts | Executes the action and returns the action results in an object.                                            |
| METHOD      | Action.execute.promise(options) | Promise                | Client scripts            | Executes the action asynchronously and returns the action results in an object.                             |
| METHOD      | Action.executeBulk(options)     | string                 | Client and server scripts | Executes an asynchronous bulk record action and returns its task ID for later status inquiry.               |
| METHOD      | action.getBulkStatus(options)   | Object                 | Client and server scripts | Returns the current status of action.executeBulk(options) with the given task ID.                           |
| PROPERTY    | Action.description              | string                 | Client and server scripts | The action description.                                                                                     |
| PROPERTY    | Action.id                       | string                 | Client and server scripts | The ID of the action. For a list of action IDs, see Supported Record Actions.                               |
| PROPERTY    | Action.label                    | string                 | Client and server scripts | The action label.                                                                                           |
| PROPERTY    | Action.parameters               | Object                 | Client and server scripts | The action parameters.                                                                                      |
| PROPERTY    | Action.recordType               | string                 | Client and server scripts | The type of the record on which the action is to be performed. For a list of record types, see record.Type. |

### ► Official NetSuite Script Samples (3 Sample(s))

#### 📄 Sample 1: Locate and Execute an Action on a Timebill Record

```
/**
 * @NApiVersion 2.x
 */
require(['N/action', 'N/record'], function(action, record) {
    // create timebill record
    var rec = record.create({
        type: 'timebill',
        isDynamic: true
    });
    rec.setValue({
        fieldId: 'employee',
        value: 104
    });
    rec.setValue({
        fieldId: 'location',
        value: 312
    });
    rec.setValue({
        fieldId: 'hours',
        value: 5
    });
    var recordId = rec.save();

    var actions = action.find({
        recordType: 'timebill',
        recordId: recordId
    });

    log.debug("We've got the following actions: " + Object.keys(actions));
    if (actions.approve) {
        var result = actions.approve();
        log.debug("Timebill has been successfully approved");
    } else {
        log.debug("The timebill is already approved");
    }
});

// Outputs the following:
// We've got the following actions: approve, reject
// Timebill has been successfully approved
```

## Sample 2: Find Actions Available for the Timebill Record Asynchronously Using Promise Methods

```
/**
 * @NApiVersion 2.x
 * @NScriptType ClientScript
 */
require(['N/action', 'N/record'], function(action, record) {
    // create timebill record
    var rec = record.create({
        type: 'timebill',
        isDynamic: true
    });
    rec.setValue({
        fieldId: 'employee',
        value: 104
    });
    rec.setValue({
        fieldId: 'location',
        value: 312
    });
    rec.setValue({
        fieldId: 'hours',
        value: 5
    });
    var recordId = rec.save();

    // find all qualified actions and then execute approve if available
    action.find.promise({
        recordType: 'timebill',
        recordId: recordId
    }).then(function(actions) {
        console.log("We've got the following actions: " + Object.keys(actions));
        if (actions.approve) {
            actions.approve.promise().then(function(result) {
                console.log("Timebill has been successfully approved");
            });
        } else {
            console.log("The timebill is already approved");
        }
    });
});

// Outputs the following:
// We've got the following actions:
// The timebill has been successfully approved
```

### Sample 3: Execute a Bulk Action on a Timebill Record

```
/**
 * @NApiVersion 2.x
 */
require(['N/action', 'N/util'])function(action, util) {

    // 1a) Bulk execute the specified action on a provided list of record IDs.
    // The params property is an array of parameter objects where each object contains required recordId and arbitrary additional parameters.
    var handle = action.executeBulk({
        recordType: "timebill",
        id: "approve",
        params: [{
            recordId: 1,
            note: "this is a note for 1"
        },
        {
            recordId: 5,
            note: "this is a note for 5"
        },
        {
            recordId: 23,
            note: "this is a note for 23"
        }
    ]
    });

    // 1b) Bulk execute the specified action on a provided list of record IDs.
    // The parameters in the previous sample are similar and can be generated programmatically using the map function.
    var searchResults = /* result of a search, for example, [1, 5, 23] */ ;
    var handle = action.executeBulk({
        recordType: "timebill",
        id: "approve",
        params: searchResults.map(function(v) {
            return {
                recordId: v,
                note: "this is a note for " + v
            };
        })
    });

    // 2a) Bulk execute the specified action on a provided list of record IDs.
    // This time with homogenous parameters, that is, all parameter objects are equal except recordId.
    var handle = action.executeBulk({
        recordType: "timebill",
        id: "approve",
        params: searchResults.map(function(v) {
            return {
                recordId: v,
                foo: "bar",
                name: "John Doe"
            };
        })
    });

    // 2b) Bulk execute the specified action on a provided list of record IDs.
    // This time with homogenous parameters. Equivalent to the previous sample.
    var commonParams = {
        foo: "bar",
        name: "John Doe"
    };
    var handle = action.executeBulk({
        recordType: "timebill",
        id: "approve",
        params: searchResults.map(function(v) {
            return util.extend({
                recordId: v
            }, commonParams);
        })
    });

    // 3) Bulk execute the specified action on a provided list of record IDs.
    // This is the simplest usage with no extra parameters besides the record ID.
    var handle = action.executeBulk({
        recordType: "timebill",
        id: "approve",
        params: searchResults.map(function(v) {
            return {
                recordId: v
            }
        })
    });

    // 4) Bulk execute the specified action on all record instances that qualify.
    // Since we don't have a list of recordIds in hand, we only provide the callback
    // that will later be used to transform a recordId to the corresponding parameters object.
    var handle = action.executeBulk({
        recordType: "timebill",
        id: "approve",
        condition: action.ALL_QUALIFIED_INSTANCES,
        paramCallback: function(v) {
            return {

```

```

        recordId: v,
        note: "this is a note for " + v
    };
}
});

// 5) Get a particular action for a particular record type.
var approveTimebill = action.get({
    recordType: "timebill",
    id: "approve"
});

// 6) Bulk execute the previously obtained action on a provided list of record IDs.
// Params are generated the same way as above in action.executeBulk().
var handle = approveTimebill.executeBulk({
    params: searchResults.map(function(v) {
        return {
            recordId: v,
            note: "this is a note for " + v
        };
    })
});

// 7) Bulk execute the previously obtained action on all record instances that qualify.
var handle = approveTimebill.executeBulk({
    condition: action.ALL_QUALIFIED_INSTANCES,
    paramCallback: function(v) {
        return {
            recordId: v,
            note: "this is a note for " + v
        };
    }
});

// 8) Get status of a bulk action execution.
var res = action.getBulkStatus({
    taskId: handle
}); // returns a RecordActionTaskStatus object
log.debug(res.status);
});

```

## 2. N/auth Module

Load the N/auth module to change your NetSuite login credentials.

define(['N/auth'], (auth) => { ... })

✓ 1 Code Sample(s)

Scripts: Server scripts

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/auth'], (auth) => { // Access methods and properties on auth });
```

Use the N/auth module to change your NetSuite login credentials.  
N/auth Module Script Sample

### N/auth Module Members

Return Type / Value Type

Supported Script Types

auth.changeEmail(options)

Changes the current user's NetSuite email address (user name).

auth.changePassword(options)

Changes the current user's NetSuite password.

| MEMBER TYPE | NAME                         | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                    |
|-------------|------------------------------|--------------------------|------------------------|----------------------------------------------------------------|
| METHOD      | auth.changeEmail(options)    | void                     | Server scripts         | Changes the current user's NetSuite email address (user name). |
| METHOD      | auth.changePassword(options) | void                     | Server scripts         | Changes the current user's NetSuite password.                  |

### ► Official NetSuite Script Samples (1 Sample(s))

#### Sample 1: N/auth Module Script Sample

```
/**
 * @NApiVersion 2.x
 */

// This script changes the currently logged-in user's NetSuite email address and password.
require(['N/auth'],function(auth) {
  function changeEmailAndPassword() {
    var password = 'myCurrentPassword';
    auth.changeEmail({
      password: password,
      newEmail: 'auth_test@newemail.com'
    });
    auth.changePassword({
      currentPassword: password,
      newPassword: 'myNewPa55Word'
    });
  }
  changeEmailAndPassword();
});
```

### 3. N/cache Module

Load the N/cache module to enable the caching of needed data and improve performance.

```
define(['N/cache'], (cache) => { ... })
```

[✓ 3 Code Sample\(s\)](#)[Scripts: Server scripts](#)[Permissions: -](#)

```
/** * @NapiVersion 2.1 * @NScriptType UserEventScript */ define(['N/cache'], (cache) => { // Access methods and properties on cache });
```

Use the N/cache module to enable temporary, short-term storage of data. Using a cache improves performance by eliminating the need for scripts to repeatedly retrieve the same piece of data. You can use this module to build a cache to store and retrieve string values using a specific key.

You can create a cache that is available (1) to the current script only, (2) to all server scripts in the current bundle, or (3) to all server scripts in your NetSuite account. Data is stored in the cache according to its time to live (ttl) specified in the Cache.put(options) method.

#### N/cache Module Members

Return Type / Value Type

Supported Script Types

Encapsulates a cache which is a segment of memory that can be used to store data on a temporary, short-term basis.

cache.getCache(options)

Checks for a cache object with the specified name. If the cache object exists, this method returns it. If the cache object does not exist, the system creates and returns a new cache object.

Holds the string values that describe the availability of the cache. Use this enum to set the value of the Cache.scope property and the options.scope parameter of the cache.getCache(options) method.

| MEMBER TYPE | NAME                    | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                                            |
|-------------|-------------------------|--------------------------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT      | cache.Cache             | Object                   | Server scripts         | Encapsulates a cache which is a segment of memory that can be used to store data on a temporary, short-term basis.                                                                                     |
| METHOD      | cache.getCache(options) | cache.Cache              | Server scripts         | Checks for a cache object with the specified name. If the cache object exists, this method returns it. If the cache object does not exist, the system creates and returns a new cache object.          |
| ENUM        | cache.Scope             | enum                     | Server scripts         | Holds the string values that describe the availability of the cache. Use this enum to set the value of the Cache.scope property and the options.scope parameter of the cache.getCache(options) method. |

#### Cache Object Members

The following members are called on cache.Cache.

Return Type/Value Type

Supported Script Types

Cache.get(options)

Retrieves a value from the cache based on a key that you provide. If the requested value is not in the cache, the method calls the user-defined function identified by a method parameter.

Cache.put(options)

| MEMBER TYPE | NAME                  | RETURN TYPE/VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                                                           |
|-------------|-----------------------|------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | Cache.get(options)    | string                 | Server scripts         | Retrieves a value from the cache based on a key that you provide. If the requested value is not in the cache, the method calls the user-defined function identified by a method parameter.                            |
| METHOD      | Cache.put(options)    | string                 | Server scripts         | Puts a value into the cache.                                                                                                                                                                                          |
| METHOD      | Cache.remove(options) | string                 | Server scripts         | Removes a value from the cache.                                                                                                                                                                                       |
| PROPERTY    | Cache.name            | string                 | Server scripts         | The name of the cache.                                                                                                                                                                                                |
| PROPERTY    | Cache.scope           | string                 | Server scripts         | The availability of the cache. A cache can be made available to the current script only, to all scripts in the current bundle, or to all scripts in your NetSuite account. Set this value using the cache.Scope enum. |

 Sample 1: Look Up Folder IDs

```

const FOLDER_CACHE_NAME = 'folder_cache';

function folderCacheLoader(context) {
  const PARENT_FOLDER_ID = 0;
  const FOLDER_NAME = 1;
  const folderCacheKey = context.key.split('/');
  const parentFolderId = folderCacheKey[PARENT_FOLDER_ID];
  const folderName = folderCacheKey[FOLDER_NAME];

  var folderId = null;
  search.create ({
    type: search.Type.FOLDER,
    columns: ['internalId'],
    filters: [
      ['parent', search.Operator.ANYOF, parentFolderId],
      'AND',
      ['name', search.Operator.IS, folderName]
    ]
  }).run()
  .each(function(folder) {
    folderId = folder.id;
    return false;
  });

  if (!folderId) {
    var folder = record.create({
      type: record.Type.FOLDER
    });
    folder.setValue({
      fieldId: 'parent',
      value: parentFolderId
    });
    folder.setValue({
      fieldId: 'name',
      value: folderName
    });
    folderId = folder.save();
  }
  return folderId;
}

function getFolderCache() {
  return cache.getCache({
    name: FOLDER_CACHE_NAME
  });
}

function folderKey(folderName, parentFolderId) {
  return [parentFolderId, folderName].join('/');
}

function getFolder(folderName, parentFolderId) {
  return getFolderCache().get({
    key: folderKey(folderName, parentFolderId),
    loader: folderCacheLoader
  });
}

```

## Sample 2: Retrieve Name of a City Based on a ZIP Code Using Cache and a Custom Loader Function

```
/**
 * @NApiVersion 2.1
 * @NScriptType Suitelet
 */

// This script retrieves the name of a city based on a ZIP code from a cache.
define(['N/cache', '/SuiteScripts/zipToCityIndexCacheLoader'], function(cache, lib) {
    const ZIP_CODES_CACHE_NAME = 'ZIP_CODES_CACHE';
    const ZIP_TO_CITY_IDX_JSON = 'ZIP_TO_CITY_IDX_JSON';

    function getZipCodeToCityLookupObj() {
        const zipCache = cache.getCache({
            name: ZIP_CODES_CACHE_NAME
        });
        const zipCacheJson = zipCache.get({
            key: ZIP_TO_CITY_IDX_JSON,
            loader: lib.zipCodeDatabaseLoader
        });
        return JSON.parse(zipCacheJson);
    }

    function findCityByZipCode(options) {
        return getZipCodeToCityLookupObj()[String(options.zip)];
    }

    function onRequest(context) {
        const start = new Date();
        if (context.request.parameters.purgeZipCache === 'true') {
            const zipCache = cache.getCache({
                name: ZIP_CODES_CACHE_NAME
            });
            zipCache.remove({
                key: ZIP_TO_CITY_IDX_JSON
            });
        }
        const cityName = findCityByZipCode({
            zip: context.request.parameters.zipcode
        });

        context.response.writeLine(cityName || 'Unknown :(');

        if (context.request.parameters.auditPerf === 'true') {
            context.response.writeLine('Time Elapsed: ' + (new Date().getTime() - start.getTime()) + ' ms');
        }
    }
    return {
        onRequest: onRequest
    };
});
```

### Sample 3: Retrieve Name of a City Based on a ZIP Code Using Cache and a Custom Loader Function

```
/**
 * zipToCityIndexCacheLoader.js
 * @NApiVersion 2.1
 * @NModuleScope Public
 */

//This custom module is a loader function that uses a CSV file to retrieve a value that was missing from a cache.
define(['N/file', 'N/cache'], function(file, cache) {
  const ZIP_CODES_CSV_PATH = '/SuiteScripts/Resources/free-zipcode-CA-database-primary.csv';

  function trimOuterQuotes(str) {
    return (str || '').replace(/^"/, '').replace(/"$/, '');
  }

  function zipCodeDatabaseLoader(context) {
    log.audit({
      title: 'Loading Zip Codes',
      details: 'Loading Zip Codes for ZIP_CODES_CACHE'
    });
    const zipCodesCsvText = file.load({
      id: ZIP_CODES_CSV_PATH
    }).getContents();
    const zipToCityIndex = {};
    const csvLines = zipCodesCsvText.split('\n');
    util.each(csvLines.slice(1), function(el) {
      var cells = el.split(',');
      var key = trimOuterQuotes(cells[0]);
      var value = trimOuterQuotes(cells[2]);
      if (parseInt(key, 10))
        zipToCityIndex[String(key)] = value;
    });
    return zipToCityIndex;
  }

  return {
    zipCodeDatabaseLoader: zipCodeDatabaseLoader
  }
});
```

## 4. N/certificateControl Module

Load the N/certificateControl module to enable scripting access to the Digital Certificates list found at Setup > Company > Certificates. You can use this module to find the correct certificate for a subsidiary and check the file type. For more information, see Digital Signing and Uploading Digital Certificates.

```
define(['N/certificateControl'], (certificateControl) => { ... })
```

✓ 5 Code Sample(s)

Scripts: Server scripts

Permissions: Certificate Management

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/certificateControl'], (certificateControl) => { // Access methods and properties on certificateControl });
```

Use the N/certificateControl module to enable scripting access to the Digital Certificates list found in the UI at Setup > Company > Certificates. You can use this module to find, create, update, read and delete certificate records. For more information, see Digital Signing and Uploading Digital Certificates.

To access the N/certificateControl module, you must use the Execute As Role field on the script deployment record. Select either the Administrator role or a custom role with the Certificate Access permission. For more information, see Access to Digital Certificates.

The certificate record holds information for a digital certificate, but it is not a standard NetSuite record and cannot be accessed with the N/record.

### N/certificateControl Module Members

Return Type / Value Type

Supported Script Types

certificateControl.Certificate

Encapsulates a digital certificate record.

certificateControl.findCertificates(options)

Returns metadata about the certificate(s).

| MEMBER TYPE | NAME                                          | RETURN TYPE / VALUE TYPE       | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                              |
|-------------|-----------------------------------------------|--------------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT      | certificateControl.Certificate                | object                         | Server scripts         | Encapsulates a digital certificate record.                                                                                                                                               |
| METHOD      | certificateControl.findCertificates(options)  | object                         | Server scripts         | Returns metadata about the certificate(s).                                                                                                                                               |
| METHOD      | certificateControl.findUsages(options)        | object[]                       | Server scripts         | Returns an audit trail of how a certificate has been used. Includes operations performed with time stamps.                                                                               |
| METHOD      | certificateControl.createCertificate(options) | certificateControl.Certificate | Server scripts         | Creates a certificate record using a file from the File Cabinet. After saving with Certificate.save(), the certificate is accessible on the Certificates.                                |
| METHOD      | certificateControl.deleteCertificate(options) | string                         | Server scripts         | Deletes a certificate record that has been uploaded to the Certificates list in the UI or created using certificateControl.createCertificate(options) and saved with Certificate.save(). |
| METHOD      | certificateControl.loadCertificate(options)   | certificateControl.Certificate | Server scripts         | Loads a certificate record that has been uploaded to the Certificates list in the UI or created using certificateControl.createCertificate(options).                                     |
| METHOD      | certificateControl.lock(options)              | string                         | Server scripts         | Locks a certificate record so that it cannot be edited.                                                                                                                                  |
| METHOD      | certificateControl.unlock(options)            | string                         | Server scripts         | Unlocks a certificate record that has been locked with certificateControl.lock(options).                                                                                                 |
| ENUM        | certificateControl.Operation                  | enum                           | Server scripts         | Holds the values for the operation when searching for certificates with certificateControl.findUsages(options).                                                                          |
| ENUM        | certificateControl.Operator                   | enum                           | Server scripts         | Holds the values for search operators to use with the name and description parameters of the certificateControl.findCertificates(options) method.                                        |
| ENUM        | certificateControl.Type                       | enum                           | Server scripts         | Holds the values for the certificate file type to use with the type parameter of the certificateControl.findCertificates(options) method.                                                |

| MEMBER TYPE | NAME                    | RETURN TYPE / VALUE TYPE                                      | SUPPORTED SCRIPT TYPES | DESCRIPTION                       |
|-------------|-------------------------|---------------------------------------------------------------|------------------------|-----------------------------------|
| METHOD      | Certificate.save()      | object containing the script ID of the new certificate record | Server scripts         | Saves a certificate record.       |
| PROPERTY    | Certificate.description | string                                                        | Server scripts         | Describes the certificate record. |

| MEMBER TYPE | NAME                            | RETURN TYPE / VALUE TYPE   | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                                             |
|-------------|---------------------------------|----------------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PROPERTY    | Certificate.file                | File Object Members object | Server scripts         | Includes the properties of the file uploaded to create the certificate.                                                                                                                                 |
| PROPERTY    | Certificate.name                | string                     | Server scripts         | The name of the certificate record.                                                                                                                                                                     |
| PROPERTY    | Certificate.monthReminder       | boolean                    | Server scripts         | The setting of the Month box for Expiration Reminders on the certificate record.                                                                                                                        |
| PROPERTY    | Certificate.notifications       | number[]                   | Server scripts         | The internal IDs of the employees selected in the Copy Employees field on the certificate record.                                                                                                       |
| PROPERTY    | Certificate.password            | string (write-only)        | Server scripts         | The password for the digital certificate. You can create a GUID for the password using Form.addSecretKeyField(options) or you can create an API secret for the secret at Setup > Company > API Secrets. |
| PROPERTY    | Certificate.restrictions        | number[]                   | Server scripts         | The internal IDs of the employees selected in the Restrict to Employees field of the certificate record.                                                                                                |
| PROPERTY    | Certificate.scriptId            | string                     | Server scripts         | The ID of the certificate record.                                                                                                                                                                       |
| PROPERTY    | Certificate.subsidiaries        | number[]                   | Server scripts         | The internal IDs of the subsidiaries associated with the certificate record.                                                                                                                            |
| PROPERTY    | Certificate.threeMonthsReminder | boolean                    | Server scripts         | Indicates the setting of the 3 Months box for Expiration Reminders on the certificate record.                                                                                                           |
| PROPERTY    | Certificate.weekReminder        | boolean                    | Server scripts         | Indicates the setting of the Week box for Expiration Reminders on the certificate record.                                                                                                               |

### ► Official NetSuite Script Samples (5 Sample(s))

#### 📄 Sample 1: Filter the Digital Certificate List by Subsidiary and File Type

```
/**
 * @NApiVersion 2.1
 */
require(['N/certificateControl'], function(certificateControl){
    var all = certificateControl.findCertificates();
    var specificType = certificateControl.findCertificates({
        type: 'PFX'
    });
    var specificSub = certificateControl.findCertificates({
        subsidiary: 93
    });
    var specificTypeAndSub = certificateControl.findCertificates({
        type: 'PFX',
        subsidiary: 93
    });
});
```

#### 📄 Sample 2: Find the Audit Trail of POST Operations for a Certificate Record Based on ID

```
/**
 * @NApiVersion 2.1
 */
require(['N/certificateControl'], function(cc){
    var usages = cc.findUsages({
        id: 'custcertificate_china',
        operation: cc.Operation.POST
    });
});
```

### Sample 3: Create, Modify, and Save Certificate Record Based on a File in the File Cabinet

```
/**
 * @NApiVersion 2.1
 */

require(['N/certificateControl','N/file'],function(cc, file){
  var fileObj = file.load({
    id: 'SuiteScripts/dsa.p12'
  });
  var options = {
    file : fileObj,
    password : '022b490ad4334c7e86a8304f937ec68f',
    name : 'testCert',
    description : 'testDescription',
    scriptId : '_testid',
    subsidiaries : [1,3],
    weekReminder : false,
    monthReminder : true,
    threeMonthsReminder : false
  };
  var newCertificate = cc.createCertificate(options);
  newCertificate.save();

  var loadedCertificate = cc.loadCertificate({
    scriptId : 'custcertificate_testid'
  });
  fileObj = file.load({
    id: 'SuiteScripts/ecdsa.p12'
  });
  loadedCertificate.file = fileObj;
  loadedCertificate.password = '022b490ad4334c7e86a8304f937ec68f';
  loadedCertificate.save();
})
```

### Sample 4: Find and Use an Existing Certificate Record

```
/**
 * @NApiVersion 2.1
 */

require(['N/certificateControl','N/https/clientCertificate'],function(cc, cert){
  var yodlee = cc.findCertificates({
    name: 'Yodlee',
    description: 'Yodlee certificate'
  });
  cert.post({
    certId:yodlee[0].id,
    url:'url',
    body:'body',
    headers:'headers'
  });
})
```

## Sample 5: Establish an SFTP Connection Using an SSH Key

```
/**
 * @NApiVersion 2.1
 */

require(['N/file', 'N/sftp', 'N/certificateControl'], function(file, sftp, certificateControl) {
  var certPath = 'yyy/certificates'
  var certName = 'apiclient_cert.p12';

  // Establish SFTP connection
  var connection = sftp.createConnection({
    username: 'sftpuser',
    keyId: 'custkeysftp_nft_demo_key',
    url: 'my.sftp.example.com',
    port: 22,
    directory: 'inbound',
    hostKey: 'AAAAAB3NzaC1yc2EAAAADAQABAAQAC4gYD1K4LE9QnuYEgRRQChjrAM1+bTT95e71Xv8oQ60yVQIEiedhRqSMbP1CPPB4pjpBd0mPIQCCkug+3XwAQ6uNj3UM11zoGGmg86tyEJT6q
GB0SsrQJzHTb3EG38BSrB00WEz0WeJ8E8YODT3oAj1NrF8Ls3JbG0bRF+0uwJDILLrFkYS3kWCv27NhBnaytGe7iLBgrJdNVLitNqkxZfK0NsAYCaJWQjQLtz+GFFN5zTbKKNsDa6s/YW7oAMMOI3Q56QAqd
XtKY728WvxYTjr2FsYS/KM6nbq/cSTvZHWLE0z2TQtB2H0IIofvEP/QvXwmGEnCeVpCNgRwdHWQf'
  });
  // -----

  // Create new certificate
  var certScriptId = '_' + (Math.random().toString(36).substring(2, 10));
  var cert = certificateControl.createCertificate();
  cert.name = 'Test Certificate China API';
  cert.description = 'Test Certificate China created using API';
  // custcertificate prefix will be added automatically
  cert.scriptId = certScriptId;
  cert.file = connection.download({
    directory: certPath,
    filename: certName
  });
  //guid corresponding to the certificate's password';
  var pwd = '022b490ad4334c7e86a8304f937ec68f';
  cert.password = pwd;
  cert.save();
  /**/certScriptId = 'custcertificate' + certScriptId;
  log.debug('Certificate ' + cert.name + ' successfully created with id ' + certScriptId + '!');
  // -----
  // Rename certificate
  cert = certificateControl.loadCertificate({scriptId: certScriptId});
  cert.name = 'Test Certificate China API TEMP';
  cert.save();
  // Verify new certificate name
  /**/cert = certificateControl.loadCertificate({scriptId: certScriptId});
  log.debug('Certificate successfully renamed to ' + cert.name + '!');
  // -----
  // Delete certificate
  certificateControl.deleteCertificate(certScriptId);
  log.debug('Certificate deleted!');
  // -----
  // Load the deleted certificate
  // attempt to load the deleted certificate - this should error
  try {
    cert = certificateControl.loadCertificate({scriptId: certScriptId});
  }
  catch (e) {
    log.error(e.message);
  }
})
```

## 5. N/commerce Modules

Load modules in the N/commerce namespace to access different assets in the web store context, such as items and shopping cart. The modules within the N/commerce namespace are supported by the latest version of SuiteCommerce and by SuiteCommerce Advanced 2019.2 onwards.

define(['N/commerces'], (commerces) => { ... })

✓ 1 Code Sample(s)

Scripts: Client and server scripts

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/commerces'], (commerces) => { // Access methods and properties on commerces });
```

Modules in the N/commerce namespace have been designed for use with SSP applications written in SuiteScript 2.x. Their primary objective is to provide web developers a homogeneous development experience when working with Commerce and SuiteScript APIs. Note that N/commerce itself is not a module.

Developers can use modules in the N/commerce namespace to access different assets in the web store context, such as items and shopping cart. The modules within the N/commerce namespace are supported by the latest version of SuiteCommerce and by SuiteCommerce Advanced 2019.2 onwards.

Before you can load N/commerce modules, you must have an active shopping session.

The modules available within the N/commerce namespace are:

Module

Description

N/commerce/recordView Module

Use the N/commerce/recordView module to provide fast, cached, and public access to the item fields and website settings.

Commerce modules are not yet available for all web store interactions. To fully customize your web store on the SuiteCloud platform, you can use Commerce APIs with SSP applications written in SuiteScript 1.0. However, Commerce APIs are not available for use with SSP applications written in SuiteScript 2.x. For more information about the Commerce APIs, see Commerce API.

General Notices

### N/commerce/recordView Module Members

Provides fast, cached, and public access to item fields and website settings in commerce context.

| MEMBER TYPE | NAME                            | RETURN TYPE / VALUE TYPE                                                                                                           | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                  |
|-------------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------|---------------------------|------------------------------------------------------------------------------|
| METHOD      | recordView.viewItems(options)   | Object   array Returns one or more Items with requested items fields as an object with field:value pairs. See the Returns section. | Client and server scripts | Retrieves one or more Items with requested items fields from an Item Record. |
| METHOD      | recordView.viewWebsite(options) | Object Returns website and website fields as an object with field:value pairs. See the Returns section.                            | Client and server scripts | Retrieves the website details with requested website fields.                 |

### ► Official NetSuite Script Samples (1 Sample(s))

#### Sample 1: Retrieve Item and Website Data via N/commerce/recordView

```
/**
 * @NApiVersion 2.x
 * @NScriptType Suitelet
 */
define(['N/commerce/recordView'], function (recordView) {
    function service(context) {
        var result = {};
        try {
            // Retrieve item fields from web store cache
            var items = recordView.viewItems({
                ids: [101, 102],
                fields: ['itemid', 'displayname', 'salesdescription', 'baseprice']
            });

            // Retrieve website configuration and settings
            var website = recordView.viewWebsite({
                fields: ['displayname', 'entryform']
            });

            result.items = items;
            result.website = website;
            context.response.write(JSON.stringify(result));
        } catch (e) {
            log.error('Error viewing commerce records', e);
        }
    }
    return { onRequest: service };
});
```

## 6. N/compress Module

Load the N/compress module to compress, decompress, and archive files.

define(['N/compress'], (compress) => { ... })

✓ 2 Code Sample(s)

Scripts: Server scripts

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/compress'], (compress) => { // Access methods and properties on compress });
```

Use the N/compress module to compress and decompress files. You can also use this module to archive multiple files in a single file archive such as TAR or ZIP file.

You can compress and decompress individual files by using compress.gzip(options) and compress.gunzip(options).

You can create an archive by using compress.createArchiver() and add multiple files to the archive.

N/compress Module Script Sample

### N/compress Module Members

Return Type / Value Type

Supported Script Types

compress.Archiver

The functionality for creating archive files.

Use compress.createArchiver() to create this object.

compress.createArchiver()

| MEMBER TYPE | NAME                      | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                  |
|-------------|---------------------------|--------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT      | compress.Archiver         | Object                   | Server scripts         | The functionality for creating archive files. Use compress.createArchiver() to create this object.                                           |
| METHOD      | compress.createArchiver() | compress.Archiver        | Server scripts         | Creates a compress.Archiver object.                                                                                                          |
| METHOD      | compress.gunzip(options)  | file.File                | Server scripts         | Decompresses a file and returns it as a temporary file object.                                                                               |
| METHOD      | compress.gzip(options)    | file.File                | Server scripts         | Compresses a file and returns it as a temporary file object.                                                                                 |
| ENUM        | compress.Type             | enum                     | Server scripts         | Holds the string values for the archive types. Use this enum to set the value of the type parameter of the method Archiver.archive(options). |

### Archiver Object Members

The following members are called on the compress.Archiver object.

Return Type/Value Type

Supported Script Types

Archiver.add(options)

Adds a file to be archived.

Archiver.archive(options)

| MEMBER TYPE | NAME                      | RETURN TYPE/VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                        |
|-------------|---------------------------|------------------------|------------------------|------------------------------------------------------------------------------------|
| METHOD      | Archiver.add(options)     | void                   | Server scripts         | Adds a file to be archived.                                                        |
| METHOD      | Archiver.archive(options) | file.File              | Server scripts         | Creates an archive with the added files and returns it as a temporary file object. |

 **Sample 1: N/compress Module Script Sample**

```
require(['N/compress', 'N/file'], function(compress, file) {
    var unsavedTxtFile = file.create({
        fileType: 'PLAINTEXT',
        name: 'file.txt',
        contents: 'This is a sample content. This is a sample content. This is a sample content. This is only a sample.'
    });

    log.debug('Name, size, and contents of the original, uncompressed file are: ');
    log.debug('Name: ' + unsavedTxtFile.name);
    log.debug('Size: ' + unsavedTxtFile.size + 'b');
    log.debug('Contents: ' + unsavedTxtFile.getContents());

    // gzip the file with max compression (level = 9)
    var gzippedFile = compress.gzip({
        file: unsavedTxtFile,
        level: 9
    });

    log.debug('Name, size, and contents of the gzipped file are: ');
    log.debug('Name: ' + gzippedFile.name);
    log.debug('Size: ' + gzippedFile.size + 'b');
    log.debug('Contents: ' + gzippedFile.getContents().substring(0, 100));

    // gunzip the file
    var gunzippedFile = compress.gunzip({
        file: gzippedFile
    });

    log.debug('Name, size, and contents of the gunzipped file are: ');
    log.debug('Name: ' + gunzippedFile.name);
    log.debug('File size: ' + gunzippedFile.size + 'b');
    log.debug('Contents: ' + gunzippedFile.getContents());
});
```

 **Sample 2: N/compress Module Script Sample**

```
require(['N/compress', 'N/file'], function(compress, file) {
    // load/create files to be archived
    var binaryFile = file.load({
        id: 200
    });
    var textFile = file.create({
        name: 'file.txt',
        fileType: 'PLAINTEXT',
        contents: 'This is sample content.'
    });

    // create an archive as a temporary file object
    var archiver = compress.createArchiver();
    archiver.add({
        file: binaryFile
    });
    archiver.add({
        file: textFile,
        directory: 'txt/'
    });
    var zipFile = archiver.archive({
        name: 'myarchive.zip'
    });

    // save the archive to file cabinet
    zipFile.folder = 123;
    zipFile.save();
});
```

## 7. N/config Module

Load the N/config module to access NetSuite configuration settings. See config.Type for a list of supported configuration pages.

define(['N/config'], (config) => { ... })

✓ 1 Code Sample(s)

Scripts: Server scripts

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/config'], (config) => { // Access methods and properties on config });
```

Use the N/config module to access NetSuite configuration settings. The config.load(options) method returns a record.Record object. Use the record.Record object members to access configuration settings. You do not need to load the N/record module to do this.

See config.Type for a list of supported configuration objects.

N/config Module Script Sample

### N/config Module Members

Return Type / Value Type

Supported Script Types

config.load(options)

Loads a record.Record object that encapsulates the specified configuration page.

Holds the string values for supported configuration objects. Use this enum to set the value of the NetSuite configuration page you want to access in the config.load(options) method.

| MEMBER TYPE | NAME                 | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                           |
|-------------|----------------------|--------------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | config.load(options) | record.Record            | Server scripts         | Loads a record.Record object that encapsulates the specified configuration page.                                                                                                      |
| ENUM        | config.Type          | enum                     | Server scripts         | Holds the string values for supported configuration objects. Use this enum to set the value of the NetSuite configuration page you want to access in the config.load(options) method. |

### ► Official NetSuite Script Samples (1 Sample(s))

#### Sample 1: N/config Module Script Sample

```
/**
 * @NApiVersion 2.1
 */
require(['N/config'],
function(config) {
    function setTaxAndEmployerId() {
        var companyInfo = config.load({
            type: config.Type.COMPANY_INFORMATION
        });
        companyInfo.setValue({
            fieldId: 'taxid',
            value: '1122334455'
        });
        companyInfo.setValue({
            fieldId: 'employerid',
            value: '123456789'
        });
        companyInfo.save();
        companyInfo = config.load({
            type: config.Type.COMPANY_INFORMATION
        });
        var taxid = companyInfo.getValue({
            fieldId: 'taxid'
        });
    }
    setTaxAndEmployerId();
});
```

## 8. N/crypto Module

Load the N/crypto module to work with hashing, hash-based message authentication (hmac), and symmetrical encryption. You can access a set of wrappers for OpenSSL's hash, hmac, cipher, and decipher methods.

```
define(['N/crypto'], (crypto) => { ... })
```

✓ 2 Code Sample(s)

Scripts: Server scripts

Permissions: -

```
/** * @ApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/crypto'], (crypto) => { // Access methods and properties on crypto });
```

Use N/crypto module to perform hashing, hash-based message authentication (hmac), and symmetrical encryption functions.

When the N/crypto module is used, SuiteScript also loads N/encode Module.

### N/crypto Module Members

Return Type / Value Type

Supported Script Types

Encapsulates a cipher.

crypto.CipherPayload

Encapsulates a cipher payload.

Encapsulates a decipher.

| MEMBER TYPE | NAME                               | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                   |
|-------------|------------------------------------|--------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT      | crypto.Cipher                      | Object                   | Server scripts         | Encapsulates a cipher.                                                                                                                                                        |
| OBJECT      | crypto.CipherPayload               | Object                   | Server scripts         | Encapsulates a cipher payload.                                                                                                                                                |
| OBJECT      | crypto.Decipher                    | Object                   | Server scripts         | Encapsulates a decipher.                                                                                                                                                      |
| OBJECT      | crypto.Hash                        | Object                   | Server scripts         | Encapsulates a hash.                                                                                                                                                          |
| OBJECT      | crypto.Hmac                        | Object                   | Server scripts         | Encapsulates an hmac.                                                                                                                                                         |
| OBJECT      | crypto.SecretKey                   | Object                   | Server scripts         | Encapsulates a secret key handle.                                                                                                                                             |
| METHOD      | crypto.checkPasswordField(options) | boolean                  | Server scripts         | Checks whether a password in a record corresponds to the password entered by the user.                                                                                        |
| METHOD      | crypto.createCipher(options)       | Object                   | Server scripts         | Creates and returns a new crypto.Cipher Object.                                                                                                                               |
| METHOD      | crypto.createDecipher(options)     | Object                   | Server scripts         | Creates and returns a new crypto.Decipher object.                                                                                                                             |
| METHOD      | crypto.createHash(options)         | Object                   | Server scripts         | Creates and returns a new crypto.Hash Object.                                                                                                                                 |
| METHOD      | crypto.createHmac(options)         | Object                   | Server scripts         | Creates and returns a new crypto.Hmac Object.                                                                                                                                 |
| METHOD      | crypto.createSecretKey(options)    | Object                   | Server scripts         | Creates and returns a new crypto.SecretKey Object.                                                                                                                            |
| ENUM        | crypto.EncryptionAlg               | string (read-only)       | Server scripts         | Holds the string values for supported encryption algorithms. Use this enum to set the options.algorithm parameter for crypto.createCipher(options).                           |
| ENUM        | crypto.HashAlg                     | string (read-only)       | Server scripts         | Holds the string values for supported hashing algorithms. Use this enum to set the options.algorithm parameter for crypto.createHash(options) and crypto.createHmac(options). |
| ENUM        | crypto.Padding                     | string (read-only)       | Server scripts         | Holds the string values for supported cipher padding. Use this enum to set the options.padding parameter for crypto.createCipher(options) and crypto.createDecipher(options). |

### Cipher Object Members

The following members are called on crypto.Cipher.

Return Type / Value Type

Supported Script Types

Cipher.update(options)

Updates the clear data with the specified encoding.

Cipher.final(options)

| MEMBER TYPE | NAME                          | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                         |
|-------------|-------------------------------|--------------------------|------------------------|-----------------------------------------------------|
| METHOD      | <b>Cipher.update(options)</b> | void                     | Server scripts         | Updates the clear data with the specified encoding. |
| METHOD      | <b>Cipher.final(options)</b>  | Object                   | Server scripts         | Returns the cipher data.                            |

## CipherPayload Object Members

The following members are called on crypto.CipherPayload.

Return Type / Value Type

Supported Script Types

CipherPayload.ciphertext

The result of the ciphering process.

CipherPayload.iv

| MEMBER TYPE | NAME                            | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                          |
|-------------|---------------------------------|--------------------------|------------------------|--------------------------------------|
| PROPERTY    | <b>CipherPayload.ciphertext</b> | string                   | Server scripts         | The result of the ciphering process. |
| PROPERTY    | <b>CipherPayload.iv</b>         | number                   | Server scripts         | An initialization vector.            |

## Decipher Object Members

The following members are called on crypto.Decipher.

Return Type / Value Type

Supported Script Types

Decipher.final(options)

Returns the clear data.

Decipher.update(options)

| MEMBER TYPE | NAME                            | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                        |
|-------------|---------------------------------|--------------------------|------------------------|----------------------------------------------------|
| METHOD      | <b>Decipher.final(options)</b>  | string                   | Server scripts         | Returns the clear data.                            |
| METHOD      | <b>Decipher.update(options)</b> | void                     | Server scripts         | Updates decipher data with the specified encoding. |

## Hash Object Members

The following members are called on crypto.Hash.

Return Type / Value Type

Supported Script Types

Hash.digest(options)

Calculates the digest of the data to be hashed.

Hash.update(options)

| MEMBER TYPE | NAME                        | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                         |
|-------------|-----------------------------|--------------------------|------------------------|-----------------------------------------------------|
| METHOD      | <b>Hash.digest(options)</b> | string                   | Server scripts         | Calculates the digest of the data to be hashed.     |
| METHOD      | <b>Hash.update(options)</b> | void                     | Server scripts         | Updates the clear data with the encoding specified. |

## Hmac Object Members

The following members are called on crypto.Hmac.

Return Type / Value Type

Supported Script Types

Hmac.digest(options)

Gets the computed digest.

Hmac.update(options)

| MEMBER TYPE | NAME                        | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                         |
|-------------|-----------------------------|--------------------------|------------------------|-----------------------------------------------------|
| METHOD      | <b>Hmac.digest(options)</b> | string                   | Server scripts         | Gets the computed digest.                           |
| METHOD      | <b>Hmac.update(options)</b> | void                     | Server scripts         | Updates the clear data with the encoding specified. |

## SecretKey Object Members

The following members are called on `crypto.SecretKey`.

Return Type / Value Type

Supported Script Types

The GUID associated with the secret key.

`SecretKey.encoding`

The encoding used for the clear text value of the secret key.

| MEMBER TYPE | NAME                            | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                             |
|-------------|---------------------------------|--------------------------|------------------------|-------------------------------------------------------------------------|
| PROPERTY    | <code>SecretKey.guid</code>     | string                   | Server scripts         | The GUID associated with the secret key.                                |
| PROPERTY    | <code>SecretKey.encoding</code> | string                   | Server scripts         | The encoding used for the clear text value of the secret key.           |
| PROPERTY    | <code>SecretKey.secret</code>   | string                   | Server scripts         | The script ID of an API secret stored at Setup > Company > API Secrets. |

### ► Official NetSuite Script Samples (2 Sample(s))

#### Sample 1: Create a Secure Key Using SHA512

```
/**
 * @NApiVersion 2.1
 */
/**This sample demonstrates the APIs needed to generate a secure key using the SHA512 hashing algorithm.
 * The mySecret variable is a placeholder that must be replaced with a valid secret from your NetSuite account.
 * For information about secrets management page, see Account Administration > Authentication > Secrets Management
 * in the Help Center.
 */

require(['N/crypto', 'N/encode', 'N/runtime'], (crypto, encode, runtime) => {
  function createSecureKeyWithHash() {
    let mySecret = 'custsecret_my_secret';    //secret id take from secrets management page

    let sKey = crypto.createSecretKey({
      secret: mySecret,
      encoding: encode.Encoding.UTF_8
    });

    let hmacSHA512 = crypto.createHmac({
      algorithm: crypto.HashAlg.SHA512,
      key: sKey
    });

    hmacSHA512.update({
      input: inputString,
      inputEncoding: encode.Encoding.BASE_64
    });

    let digestSHA512 = hmacSHA512.digest({
      outputEncoding: encode.Encoding.HEX
    });
  }
  createSecureKeyWithHash();
});
```

## Sample 2: Create a Suitelet to Request User Credentials, Create a Secret Key, and Encode a Sample String

```
/**
 * @NApiVersion 2.1
 * @NScriptType Suitelet
 */
define(['N/ui/serverWidget', 'N/runtime', 'N/crypto', 'N/encode'], (serverWidget, runtime, crypto, encode) => {
  function onRequest(option) {
    if (option.request.method === 'GET') {
      let form = serverWidget.createForm({
        title: 'My Credential Form'
      });
      let skField = form.addSecretKeyField({
        id: 'mycredential',
        label: 'Credential',
        restrictToScriptIds: [runtime.getCurrentScript().id],
        restrictToCurrentUser: false
      });
      skField.maxLength = 200;

      form.addSubmitButton();

      option.response.writePage(form);
    } else {
      let form = serverWidget.createForm({
        title: 'My Credential Form'
      });

      const inputString = "YWJjZGVmZwo=";
      let myGuid = option.request.parameters.mycredential;

      // Create the key
      let sKey = crypto.createSecretKey({
        guid: myGuid,
        encoding: encode.Encoding.UTF_8
      });

      try {
        let hmacSha512 = crypto.createHmac({
          algorithm: 'SHA512',
          key: sKey
        });
        hmacSha512.update({
          input: inputString,
          inputEncoding: encode.Encoding.BASE_64
        });
        let digestSha512 = hmacSha512.digest({
          outputEncoding: encode.Encoding.HEX
        });
      } catch (e) {
        log.error({
          title: 'Failed to hash input',
          details: e
        });
      }

      form.addField({
        id: 'result',
        label: 'Your digested hash value',
        type: 'textarea'
      }).defaultValue = digestSha512;

      option.response.writePage(form);
    }
  }
  return {
    onRequest: onRequest
  };
});
```

## 9. N/crypto/certificate Module

Load the N/certificate module to sign XML documents or strings with digital certificates using asymmetric cryptography. In addition to signing XML documents, you can create signer and verifier objects and verify signed documents with this module.

```
define(['N/crypto/certificate'], (certificate) => { ... })
```

[✓ 2 Code Sample\(s\)](#)[Scripts: Server scripts](#)[Permissions: Certificate Access](#)

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/crypto/certificate'], (certificate) => { // Access methods and properties on certificate });
```

Use the N/crypto/certificate module to sign XML documents or strings with digital certificates using asymmetric cryptography. You can also use this module to create signer and verifier objects and verify signed documents.

### N/crypto/certificate Module Members

Return Type/Value Type

Supported Script Types

certificate.SignedXml

Encapsulates an XML string that has been digitally signed.

Use certificate.signXml(options) to create this object.

certificate.Signer

| MEMBER TYPE | NAME                                    | RETURN TYPE/VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                                                                       |
|-------------|-----------------------------------------|------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT      | certificate.SignedXml                   | Object                 | Server scripts         | Encapsulates an XML string that has been digitally signed. Use certificate.signXml(options) to create this object.                                                                                                                |
| OBJECT      | certificate.Signer                      | Object                 | Server scripts         | Encapsulates a created signature (signer) for plain strings. Use certificate.createSigner(options) to create this object.                                                                                                         |
| OBJECT      | certificate.Verifier                    | Object                 | Server scripts         | Encapsulates a created verifier for verifying plain string signatures. Use certificate.createVerifier(options) to create this object.                                                                                             |
| METHOD      | certificate.createSigner(options)       | certificate.Signer     | Server scripts         | Creates a certificate.Signer object for signing plain strings.                                                                                                                                                                    |
| METHOD      | certificate.createVerifier(options)     | certificate.Verifier   | Server scripts         | Creates a certificate.Verifier object for verifying signatures of plain strings.                                                                                                                                                  |
| METHOD      | certificate.signXml(options)            | certificate.SignedXml  | Server scripts         | Signs the input XML string using the Certificate ID. Returns the certificate.SignedXml as a string. Note: Formatting, such as line breaks, is disabled in signatures.                                                             |
| METHOD      | certificate.verifyXmlSignature(options) | void                   | Server scripts         | Verifies the signature in the SignedXml.asFile() file.                                                                                                                                                                            |
| ENUM        | certificate.HashAlg                     | enum                   | Server scripts         | Holds the string values for hash algorithm types. Use this enum to set the option.algorithm property values for the certificate.createSigner(options), certificate.createVerifier(options), certificate.signXml(options) methods. |

### SignedXml Object Members

The following members are called on the certificate.SignedXml object.

Return Type / Value Type

Supported Script Types

SignedXml.asFile()

Returns the signed XML as a file object.

SignedXml.asString()

| MEMBER TYPE | NAME                 | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                              |
|-------------|----------------------|--------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | SignedXml.asFile()   | file.File                | Server scripts         | Returns the signed XML as a file object.                                                                                                 |
| METHOD      | SignedXml.asString() | string                   | Server scripts         | Returns the signed XML as a string.                                                                                                      |
| METHOD      | SignedXml.asXml()    | xml.Document             | Server scripts         | Returns the signed XML as an XML document. You can use the N/xml Module with this document to access elements and attributes in the XML. |

### Signer Object Members

The following members are called on the certificate.Signer object.

Return Type / Value Type

Supported Script Types

Signer.sign(options)

Signs the string and returns the signature.

Signer.update(options)

| MEMBER TYPE | NAME                   | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                       |
|-------------|------------------------|--------------------------|------------------------|-------------------------------------------------------------------|
| METHOD      | Signer.sign(options)   | void                     | Server scripts         | Signs the string and returns the signature.                       |
| METHOD      | Signer.update(options) | void                     | Server scripts         | Updates the input string to be signed. The string can be encoded. |

Verifier Object Members

The following members are called on the certificate.Verifier object.

Return Type / Value Type

Supported Script Types

Verifier.update(options)

Updates the string to be verified against specified certificate.

Verifier.verify(options)

| MEMBER TYPE | NAME                     | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                   |
|-------------|--------------------------|--------------------------|------------------------|-------------------------------------------------------------------------------|
| METHOD      | Verifier.update(options) | void                     | Server scripts         | Updates the string to be verified against specified certificate.              |
| METHOD      | Verifier.verify(options) | void                     | Server scripts         | Verifies the string against a provided signature using specified certificate. |

► Official NetSuite Script Samples (2 Sample(s))

📄 Sample 1: Load an XML File from the File Cabinet and Sign It Using a Digital Certificate

```
/**
 * @NApiVersion 2.1
 */

require(['N/crypto/certificate','N/file'],(certificate, file) => {
  // Load the file from the File Cabinet.
  // Note that the id value is hard-coded in this sample, and you should use
  // a valid file id from your account.
  let infNFe = file.load({
    id: 922
  });
  let signedXml = certificate.signXml({
    algorithm: certificate.HashAlg.SHA256,
    certId: 'custcertificate1',
    rootTag: 'infNFe',
    xmlString: infNFe.getContents()
  });
  certificate.verifyXMLSignature({
    signedXml:signedXml,
    rootTag: 'infNFe'
  });
});
```

## Sample 2: Create Signer and Verifier Objects

```
/**
 * @NApiVersion 2.1
 */

require(['N/crypto/certificate'], (certificate) => {
  let signer = certificate.createSigner({
    certId: 'custcertificate1',
    algorithm: certificate.HashAlg.SHA256
  });
  signer.update('test');

  let result = signer.sign();
  let verifier = certificate.createVerifier({
    certId: 'custcertificate1',
    algorithm: certificate.HashAlg.SHA256
  });
  verifier.update('test');
  verifier.verify(result);
})
```

# 10. N/crypto/random Module

Load the N/crypto/random module to work with cryptographically-secure pseudo-random generator methods.

define(['N/crypto/random'], (random) => { ... })

✓ 1 Code Sample(s)

Scripts: Client and server scripts; server scripts support SuiteScript 2.1 only

Permissions: -

```
/** * @ApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/crypto/random'], (random) => { // Access methods and properties on random });
```

The N/crypto/random module is available for both client and server scripts, but server scripts need to use SuiteScript 2.1.

If you cannot update your server script code to SuiteScript 2.1, consider implementing a small RESTlet in SuiteScript 2.1 that uses N/crypto/random module and consuming it from your script with https.requestRestlet(options)

N/crypto/random Module Samples

## N/crypto/random Module Members

Return Type / Value Type

Supported Script Types

random.generateBytes(options)

Client and server scripts

Generates cryptographically strong pseudorandom set of bytes.

random.generateInt(options)

| MEMBER TYPE | NAME                          | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                          |
|-------------|-------------------------------|--------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------|
| METHOD      | random.generateBytes(options) | Uint8Array               | Client and server scripts | Generates cryptographically strong pseudorandom set of bytes.                                                        |
| METHOD      | random.generateInt(options)   | number                   | Client and server scripts | Method used to generate cryptographically strong pseudorandom number.                                                |
| METHOD      | random.generateUUID()         | string                   | Client and server scripts | Method used to generate a v4 Universally Unique Identifier using a cryptographically secure random number generator. |

## ► Official NetSuite Script Samples (1 Sample(s))

### Sample 1: N/crypto/random Module Samples

```
/**
 * @ApiVersion 2.1
 * @NScriptType restlet
 */
define(['N/crypto/random'], function(random) {
  return {
    get : function() {
      return JSON.stringify({
        number: random.generateInt({min: 1, max:7})
      })
    }
  };
});
```

# 11. N/currency Module

Load the N/currency module to work with exchange rates within your NetSuite account. You can use this module to find the exchange rate between two currencies based on a certain date.

define(['N/currency'], (currency) => { ... })

✓ 1 Code Sample(s)

Scripts: Client and server scripts

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/currency'], (currency) => { // Access methods and properties on currency });
```

Use the N/currency module to work with exchange rates within your NetSuite account. You can use this module to find the exchange rate between two currencies based on a certain date.

To use multiple currencies, the Multiple Currencies feature must be enabled. For information about enabling this feature, see Enabling the Multiple Currencies Feature.

N/currency Module Script Sample

Currency formatting is handled by the N/format Module.

## N/currency Module Member

Return Type / Value Type

Supported Script Types

currency.exchangeRate(options)

Client and server scripts

Returns an exchange rate between two currencies.

| MEMBER TYPE | NAME                           | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                      |
|-------------|--------------------------------|--------------------------|---------------------------|--------------------------------------------------|
| METHOD      | currency.exchangeRate(options) | number                   | Client and server scripts | Returns an exchange rate between two currencies. |

### ► Official NetSuite Script Samples (1 Sample(s))

#### Sample 1: N/currency Module Script Sample

```
/**
 * @NApiVersion 2.1
 */

require(['N/currency'], function(currency) {
  function getUSDFromCAD() {
    var canadianAmount = 100;
    var rate = currency.exchangeRate({
      source: 'CAD',
      target: 'USD',
      date: new Date('7/28/2015')
    });

    var usdAmount = canadianAmount * rate;
  }

  getUSDFromCAD();
});
```

## 12. N/currentRecord Module

Load the N/currentRecord module to access the record instance that you are currently working on. You can then use the record instance in a client context.

```
define(['N/currentRecord'], (currentRecord) => { ... })
```

✓ 2 Code Sample(s)

Scripts: Client scripts

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/currentRecord'], (currentRecord) => { // Access methods and properties on currentRecord });
```

Use the N/currentRecord module to access the record that is active in the current client context. This module is always a dynamic object and mode of work is always dynamic, not deferred dynamic/standard. For more information, see SuiteScript 2.x Standard and Dynamic Modes. Be aware that when the current record is in view mode it cannot be edited; it is a read-only record when in view mode. As such, any set APIs do not work on the current record in view mode.

You can use the currentRecord module in the following types of scripts:

**Entry point client scripts** - These scripts use the @NScriptType ClientScript annotation. (For details, see SuiteScript 2.1 JSDoc Validation.) The system automatically provides this type of script with a currentRecord.CurrentRecord object that represents the current record. For this reason, an entry point client script does not have to explicitly load the N/currentRecord module. To access the currentRecord object, create a variable and initialize it to the value of the scriptContext.currentRecord property, which is available in each of the SuiteScript 2.1 Client Script Entry Points and API. For an example, see SuiteScript Client Script Sample.

**Client custom modules** - These scripts do not use an @NScriptType annotation (see SuiteScript 2.1 Custom Modules). For these scripts, you must manually load the N/currentRecord module by naming it in the script's define statement. Additionally, you must actively retrieve a currentRecord.CurrentRecord object by using the currentRecord.get() or currentRecord.get.promise() method. For an example, see N/currentRecord Module Script Samples.

Like the N/record Module, the currentRecord module provides access to body and sublist fields. However, you should use the record module for server scripts and for cases where a client script needs to interact with a record other than the currently active record. You should use the currentRecord module for client scripts that need to interact with the currently active record.

Additionally, the functionality of the two modules varies slightly. For example, the currentRecord module does not permit the editing of subrecords, although subrecords can be retrieved in view mode. For additional details, see the following topics:

### N/currentRecord Module Members

Return Type / Value Type

Supported Script Types

currentRecord.Column

Encapsulates a column of a sublist on the current record.

currentRecord.CurrentRecord

Represents the record active on the current page.

| MEMBER TYPE | NAME                        | RETURN TYPE / VALUE TYPE    | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                           |
|-------------|-----------------------------|-----------------------------|------------------------|-----------------------------------------------------------------------|
| OBJECT      | currentRecord.Column        | Object                      | Client scripts         | Encapsulates a column of a sublist on the current record.             |
| OBJECT      | currentRecord.CurrentRecord | Object                      | Client scripts         | Represents the record active on the current page.                     |
| OBJECT      | currentRecord.Field         | Object                      | Client scripts         | Represents a body or sublist field.                                   |
| OBJECT      | currentRecord.Sublist       | Object                      | Client scripts         | Represents a sublist.                                                 |
| METHOD      | currentRecord.get()         | currentRecord.CurrentRecord | Client scripts         | Retrieves a record object that represents the current record.         |
| METHOD      | currentRecord.get.promise() | Promise                     | Client scripts         | Retrieves a promise for an object that represents the current record. |

### Column Object Members

The following members are called on the currentRecord.Column object.

Return Type / Value Type

Supported Script Types

string (read-only)

Returns the internal ID of the column.

Column.isDisabled

| MEMBER TYPE | NAME               | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                               |
|-------------|--------------------|--------------------------|------------------------|-------------------------------------------|
| PROPERTY    | Column.id          | string (read-only)       | Client scripts         | Returns the internal ID of the column.    |
| PROPERTY    | Column.isDisabled  | boolean                  | Client scripts         | Indicates whether the column is disabled. |
| PROPERTY    | Column.isMandatory | boolean                  | Client scripts         | Indicates whether the column is required. |

| MEMBER TYPE | NAME             | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                         |
|-------------|------------------|--------------------------|------------------------|-------------------------------------------------------------------------------------|
| PROPERTY    | Column.label     | string (read-only)       | Client scripts         | Returns the UI label for the column.                                                |
| PROPERTY    | Column.sublistId | string (read-only)       | Client scripts         | Returns the internal ID of the standard or custom sublist that contains the column. |
| PROPERTY    | Column.type      | string (read-only)       | Client scripts         | Returns the column type.                                                            |

## CurrentRecord Object Members

The following members are called on the `currentRecord.CurrentRecord` object.

Return Type / Value Type

Supported Script Types

`CurrentRecord.cancelLine(options)`

`currentRecord.CurrentRecord`

Cancels the changes made to the currently selected line.

| MEMBER TYPE | NAME                                                               | RETURN TYPE / VALUE TYPE                 | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                         |
|-------------|--------------------------------------------------------------------|------------------------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | <code>CurrentRecord.cancelLine(options)</code>                     | <code>currentRecord.CurrentRecord</code> | Client scripts         | Cancels the changes made to the currently selected line.                                                                            |
| METHOD      | <code>CurrentRecord.commitLine(options)</code>                     | <code>currentRecord.CurrentRecord</code> | Client scripts         | Commits the currently selected line.                                                                                                |
| METHOD      | <code>CurrentRecord.findMatrixSublistLineWithValue(options)</code> | number                                   | Client scripts         | Returns the line number of the first line that contains the specified value in the matrix column.                                   |
| METHOD      | <code>CurrentRecord.findSublistLineWithValue(options)</code>       | number                                   | Client scripts         | Gets the line number for the first occurrence of a field value in a sublist.                                                        |
| METHOD      | <code>CurrentRecord.getCurrentMatrixSublistValue(options)</code>   | number   Date   string   array   boolean | Client scripts         | Gets the value for the currently selected line in the matrix.                                                                       |
| METHOD      | <code>CurrentRecord.getCurrentSublistIndex(options)</code>         | number                                   | Client scripts         | Gets the line number of the currently selected line.                                                                                |
| METHOD      | <code>CurrentRecord.getCurrentSublistSubrecord(options)</code>     | <code>currentRecord.CurrentRecord</code> | Client scripts         | Gets the subrecord for the associated sublist field on the current line. The subrecord object is retrieved in view mode.            |
| METHOD      | <code>CurrentRecord.getCurrentSublistText(options)</code>          | number   Date   string   array   boolean | Client scripts         | Gets the value of the field in the currently selected line by text representation.                                                  |
| METHOD      | <code>CurrentRecord.getCurrentSublistValue(options)</code>         | number   Date   string   array   boolean | Client scripts         | Gets the value of the field in the currently selected line.                                                                         |
| METHOD      | <code>CurrentRecord.getField(options)</code>                       | <code>currentRecord.Field</code>         | Client scripts         | Gets a field object from the record.                                                                                                |
| METHOD      | <code>CurrentRecord.getLineCount(options)</code>                   | number                                   | Client scripts         | Returns the number of lines in the sublist.                                                                                         |
| METHOD      | <code>CurrentRecord.getMatrixHeaderCount(options)</code>           | number                                   | Client scripts         | Returns the number of columns for the specified matrix.                                                                             |
| METHOD      | <code>CurrentRecord.getMatrixHeaderField(options)</code>           | <code>currentRecord.Field</code>         | Client scripts         | Gets the field for the specified header in the matrix.                                                                              |
| METHOD      | <code>CurrentRecord.getMatrixHeaderValue(options)</code>           | number   Date   string   array   boolean | Client scripts         | Gets the value for the associated header in the matrix.                                                                             |
| METHOD      | <code>CurrentRecord.getMatrixSublistField(options)</code>          | <code>currentRecord.Field</code>         | Client scripts         | Gets the field for the specified sublist in the matrix.                                                                             |
| METHOD      | <code>CurrentRecord.getMatrixSublistValue(options)</code>          | number   Date   string   array   boolean | Client scripts         | Gets the value for the associated field in the matrix.                                                                              |
| METHOD      | <code>CurrentRecord.getSublist(options)</code>                     | <code>currentRecord.Sublist</code>       | Client scripts         | Gets the specified sublist object.                                                                                                  |
| METHOD      | <code>CurrentRecord.getSublistField(options)</code>                | <code>currentRecord.Field</code>         | Client scripts         | Gets the specified field object from the sublist.                                                                                   |
| METHOD      | <code>CurrentRecord.getSublistText(options)</code>                 | string                                   | Client scripts         | Gets the value of the field in a sublist by a string representation.                                                                |
| METHOD      | <code>CurrentRecord.getSublistValue(options)</code>                | number   Date   string   array   boolean | Client scripts         | Gets the value of the field in a sublist.                                                                                           |
| METHOD      | <code>CurrentRecord.getSubrecord(options)</code>                   | <code>currentRecord.CurrentRecord</code> | Client scripts         | Gets the subrecord associated with the field. The subrecord object is retrieved in view mode.                                       |
| METHOD      | <code>CurrentRecord.getText(options)</code>                        | string                                   | Client scripts         | Gets the value of the field by a string representation.                                                                             |
| METHOD      | <code>CurrentRecord.getValue(options)</code>                       | number   Date   string   array   boolean | Client scripts         | Gets the value of the field. Not to be used on custom password fields. Use <code>crypto.checkPasswordField(options)</code> instead. |

| MEMBER TYPE | NAME                                                 | RETURN TYPE / VALUE TYPE    | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                          |
|-------------|------------------------------------------------------|-----------------------------|------------------------|------------------------------------------------------------------------------------------------------|
| METHOD      | CurrentRecord.hasCurrentSublistSubrecord(options)    | boolean                     | Client scripts         | Returns a value indicating whether the associated sublist field has a subrecord on the current line. |
| METHOD      | CurrentRecord.hasSublistSubrecord(options)           | boolean                     | Client scripts         | Returns a value indicating whether the associated sublist field contains a subrecord.                |
| METHOD      | CurrentRecord.hasSubrecord(options)                  | boolean                     | Client scripts         | Indicates whether the field has a subrecord.                                                         |
| METHOD      | CurrentRecord.insertLine(options)                    | currentRecord.CurrentRecord | Client scripts         | Inserts a new line in a sublist.                                                                     |
| METHOD      |                                                      |                             |                        |                                                                                                      |
| METHOD      | CurrentRecord.removeCurrentSublistSubrecord(options) | currentRecord.CurrentRecord | Client scripts         | Removes the subrecord for the associated sublist field on the current line.                          |
| METHOD      | CurrentRecord.removeLine(options)                    | currentRecord.CurrentRecord | Client scripts         | Removes a line from a sublist.                                                                       |
| METHOD      | CurrentRecord.removeSubrecord(options)               | currentRecord.CurrentRecord | Client scripts         | Removes the subrecord associated with the field.                                                     |
| METHOD      | CurrentRecord.selectLine(options)                    | void                        | Client scripts         | Selects a line item in a sublist.                                                                    |
| METHOD      | CurrentRecord.selectNewLine(options)                 | currentRecord.CurrentRecord | Client scripts         | Selects a new line at the end of the sublist.                                                        |
| METHOD      | CurrentRecord.setCurrentMatrixSublistValue(options)  | currentRecord.CurrentRecord | Client scripts         | Sets the value for the currently selected line in the matrix.                                        |
| METHOD      | CurrentRecord.setCurrentSublistText(options)         | currentRecord.CurrentRecord | Client scripts         | Sets the value of the field in the currently selected line using a string representation.            |
| METHOD      | CurrentRecord.setCurrentSublistValue(options)        | currentRecord.CurrentRecord | Client scripts         | Sets the value of the field in the currently selected line.                                          |
| METHOD      | CurrentRecord.setMatrixHeaderValue(options)          | currentRecord.CurrentRecord | Client scripts         | Sets the value for the associated header in the matrix.                                              |
| METHOD      | CurrentRecord.setMatrixSublistValue(options)         | currentRecord.CurrentRecord | Client scripts         | Sets the value for the associated field in the matrix.                                               |
| METHOD      | CurrentRecord.setText(options)                       | currentRecord.CurrentRecord | Client scripts         | Sets the value of the field using a string representation.                                           |
| METHOD      | CurrentRecord.setValue(options)                      | currentRecord.CurrentRecord | Client scripts         | Sets the value of the field.                                                                         |
| PROPERTY    | CurrentRecord.id                                     | number (read-only)          | Client scripts         | Returns the internal record ID.                                                                      |
| PROPERTY    | CurrentRecord.isDynamic                              | boolean (read-only)         | Client scripts         | Indicates whether the record is dynamic.                                                             |
| PROPERTY    | CurrentRecord.type                                   | string (read-only)          | Client scripts         | Returns the record type.                                                                             |

## Field Object Members

The following members are called on the currentRecord.Field object.

Return Type / Value Type

Supported Script Types

Field.getSelectOptions(options)

Returns an array of available options on a standard or custom select, multiselect, or radio field as key-value pairs. Only the first 1,000 available options are returned.

Field.insertSelectOption(options)

| MEMBER TYPE | NAME                              | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                                                                  |
|-------------|-----------------------------------|--------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | Field.getSelectOptions(options)   | array                    | Client scripts         | Returns an array of available options on a standard or custom select, multiselect, or radio field as key-value pairs. Only the first 1,000 available options are returned.                                                   |
| METHOD      | Field.insertSelectOption(options) | void                     | Client scripts         | Inserts an option into certain types of select and multiselect fields. This method is usable only in fields that were added by a front-end Suitelet or beforeLoad user event script.                                         |
| METHOD      | Field.removeSelectOption(options) | void                     | Client scripts         | Removes an option from certain types of select and multiselect fields. This method is usable only in fields that were added by a front-end Suitelet or beforeLoad user event script. It is supported only in client scripts. |
| OBJECT      | Field.id                          | string (read-only)       | Client scripts         | Returns the internal ID of a standard or custom body or sublist field.                                                                                                                                                       |
| OBJECT      | Field.isDisabled                  | boolean                  | Client scripts         | Returns true if the standard or custom field is disabled on the record form, or false otherwise.                                                                                                                             |
| OBJECT      | Field.isDisplay                   | boolean                  | Client scripts         | Returns true if the field is set to display on the record form, or false otherwise. This property is read-only for sublist fields.                                                                                           |
| OBJECT      | Field.isMandatory                 | boolean                  | Client scripts         | Returns true if the standard or custom field is mandatory on the record form, or false otherwise.                                                                                                                            |

| MEMBER TYPE | NAME             | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                                        |
|-------------|------------------|--------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT      | Field.isPopup    | boolean (read-only)      | Client scripts         | Returns true if the field is a popup list field, or false otherwise.                                                                                                                               |
| OBJECT      | Field.isReadOnly | boolean                  | Client scripts         | Returns true if the field on the record form cannot be edited, or false otherwise. For textarea fields, this property can be read or written to. For all other fields, this property is read-only. |
| OBJECT      | Field.isVisible  | boolean (read-only)      | Client scripts         | Returns true if the field is visible on the record form, or false otherwise.                                                                                                                       |
| OBJECT      | Field.label      | string (read-only)       | Client scripts         | Returns the UI label for a standard or custom field body or sublist field.                                                                                                                         |
| OBJECT      | Field.sublistId  | string (read-only)       | Client scripts         | Returns the ID of the sublist associated with the specified sublist field.                                                                                                                         |
| OBJECT      | Field.type       | string (read-only)       | Client scripts         | Returns the type of a body or sublist field.                                                                                                                                                       |

### Sublist Object Members

The following members are called on the `currentRecord.Sublist` object.

Return Type / Value Type

Supported Script Types

Sublist.getColumn(options)

currentRecord.Column

Returns a column in the sublist.

| MEMBER TYPE | NAME                       | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                            |
|-------------|----------------------------|--------------------------|------------------------|------------------------------------------------------------------------|
| METHOD      | Sublist.getColumn(options) | currentRecord.Column     | Client scripts         | Returns a column in the sublist.                                       |
| PROPERTY    | Sublist.id                 | string (read-only)       | Client scripts         | Returns the internal ID of the sublist.                                |
| PROPERTY    | Sublist.isChanged          | boolean (read-only)      | Client scripts         | Indicates whether the sublist has changed on the current record form.  |
| PROPERTY    | Sublist.isDisplay          | boolean (read-only)      | Client scripts         | Indicates whether the sublist is displayed on the current record form. |
| PROPERTY    | Sublist.type               | string (read-only)       | Client scripts         | Returns the sublist type.                                              |

### ► Official NetSuite Script Samples (2 Sample(s))

#### Sample 1: Perform Field Sourcing Synchronously

```
/**
 * @NApiVersion 2.1
 * @NScriptType ClientScript
 */

define(['N/currentRecord'], (currentRecord) => {
  const pageInit = () => {
    const rec = currentRecord.get();

    rec.selectNewLine({
      sublistId: 'item'
    });

    rec.setCurrentSublistValue({
      sublistId: 'item',
      fieldId: 'item',
      value: 39,
      forceSyncSourcing: true
    });

    rec.setCurrentSublistValue({
      sublistId: 'item',
      fieldId: 'quantity',
      value: 1,
      forceSyncSourcing: true
    });

    rec.commitLine({
      sublistId: 'item'
    });
  };

  return {
    pageInit
  };
});
```

## Sample 2: SuiteScript Client Script Sample

```
/**
 * @ApiVersion 2.x
 * @NScriptType ClientScript
 */
define(['N/error'], function(error) {
    function pageInit(context) {
        if (context.mode !== 'create')
            return;
        var currentRecord = context.currentRecord;
        currentRecord.setValue({
            fieldId: 'entity',
            value: 107
        });
    }
    function saveRecord(context) {
        var currentRecord = context.currentRecord;
        if (!currentRecord.getValue({fieldId: 'entity'}) || currentRecord.getLineCount({sublistId: 'item'}) < 1)
            throw error.create({
                name: 'MISSING_REQ_ARG',
                message: 'Please enter all the necessary fields on the salesorder before saving'
            });
        return true;
    }
    function validateField(context) {
        var currentRecord = context.currentRecord;
        var sublistName = context.sublistId;
        var sublistFieldName = context.fieldId;
        var line = context.line;
        if (sublistName === 'item') {
            if (sublistFieldName === 'quantity') {
                if (currentRecord.getCurrentSublistValue({
                    sublistId: sublistName,
                    fieldId: sublistFieldName
                }) < 3)
                    currentRecord.setValue({
                        fieldId: 'otherrefnum',
                        value: 'Quantity is less than 3'
                    });
            } else
                currentRecord.setValue({
                    fieldId: 'otherrefnum',
                    value: 'Quantity accepted'
                });
        }
    }
    return true;
}
function fieldChanged(context) {
    var currentRecord = context.currentRecord;
    var sublistName = context.sublistId;
    var sublistFieldName = context.fieldId;
    var line = context.line;
    if (sublistName === 'item' && sublistFieldName === 'item')
        currentRecord.setValue({
            fieldId: 'memo',
            value: 'Item: ' + currentRecord.getCurrentSublistValue({
                sublistId: 'item',
                fieldId: 'item'
            }) + ' is selected'
        });
}
function postSourcing(context) {
    var currentRecord = context.currentRecord;
    var sublistName = context.sublistId;
    var sublistFieldName = context.fieldId;
    var line = context.line;
    if (sublistName === 'item' && sublistFieldName === 'item')
        if (currentRecord.getCurrentSublistValue({
            sublistId: sublistName,
            fieldId: sublistFieldName
        }) === '39')
            if (currentRecord.getCurrentSublistValue({
                sublistId: sublistName,
                fieldId: 'pricelevels'
            }) === '1-1')
                currentRecord.setCurrentSublistValue({
                    sublistId: sublistName,
                    fieldId: 'pricelevels',
                    value: '1-1'
                });
}
function lineInit(context) {
    var currentRecord = context.currentRecord;
    var sublistName = context.sublistId;
    if (sublistName === 'partners')
        currentRecord.setCurrentSublistValue({
            sublistId: sublistName,
            fieldId: 'partner',
            value: '55'
        });
}
```

```

function validateDelete(context) {
  var currentRecord = context.currentRecord;
  var sublistName = context.sublistId;
  if (sublistName === 'partners')
    if (currentRecord.getCurrentSublistValue({
      sublistId: sublistName,
      fieldId: 'partner'
    }) === '55')
      currentRecord.setValue({
        fieldId: 'memo',
        value: 'Removing partner sublist'
      });
  return true;
}

function validateInsert(context) {
  var currentRecord = context.currentRecord;
  var sublistName = context.sublistId;
  if (sublistName === 'partners')
    if (currentRecord.getCurrentSublistValue({
      sublistId: sublistName,
      fieldId: 'contribution'
    }) === '100.0%')
      currentRecord.setCurrentSublistValue({
        sublistId: sublistName,
        fieldId: 'contribution',
        value: '100.0%'
      });
  return true;
}

function validateLine(context) {
  var currentRecord = context.currentRecord;
  var sublistName = context.sublistId;
  if (sublistName === 'partners')
    if (currentRecord.getCurrentSublistValue({
      sublistId: sublistName,
      fieldId: 'contribution'
    }) === '100.0%')
      currentRecord.setCurrentSublistValue({
        sublistId: sublistName,
        fieldId: 'contribution',
        value: '100.0%'
      });
  return true;
}

function sublistChanged(context) {
  var currentRecord = context.currentRecord;
  var sublistName = context.sublistId;
  var op = context.operation;
  if (sublistName === 'item')
    currentRecord.setValue({
      fieldId: 'memo',
      value: 'Total has changed to ' + currentRecord.getValue({fieldId: 'total'})
    });
}

return {
  pageInit: pageInit,
  fieldChanged: fieldChanged,
  postSourcing: postSourcing,
  sublistChanged: sublistChanged,
  lineInit: lineInit,
  validateField: validateField,
  validateLine: validateLine,
  validateInsert: validateInsert,
  validateDelete: validateDelete,
  saveRecord: saveRecord
};
};

```

### 13. N/dataset Module

Load the N/dataset module to create and manage datasets in SuiteAnalytics Workbook. Use this module with the N/workbook module to manage all aspects of your datasets and workbooks in SuiteAnalytics Workbook.

define(['N/dataset'], (dataset) => { ... })

✓ 2 Code Sample(s)

Scripts: Server scripts

Permissions: SuiteAnalytics Workbook

```
/** * @ApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/dataset'], (dataset) => { // Access methods and properties on dataset });
```

Use the N/dataset module to create, load, list, or save datasets. You can only use this module in server scripts.

With this module, you can do things like:

- Create columns, joins, and conditions within a dataset.
- Delete a dataset using the N/query Module.
- Execute a dataset and obtain results-similar to running a query definition with the N/query Module.
- Use string aliasing to identify columns, which is helpful when linking dataset columns to a workbook.
- Apply column labels, which are the string descriptions shown in the UI.
- Save a dataset.

Datasets are the foundation for workbooks and their components. In a dataset, you combine record type fields and filters to make a query. You can use the results as source data for any workbook, and you can use one dataset in several workbooks.

To learn more about datasets in SuiteAnalytics, see Defining a Dataset. For details on workbooks, see N/workbook Module.

The N/dataset module doesn't work in unauthenticated client-side contexts.

For details, see the SuiteAnswers Outbound HTTPs in an unauthenticated client-side context.

#### N/dataset Module Members

- Return Type / Value Type
- Supported Script Types
- A column in the dataset, which usually represents a record field.
- dataset.Condition
- A condition or set of conditions to apply to a column.
- A representation of the entire dataset, including columns, conditions, and joins.

| MEMBER TYPE | NAME                               | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                    |
|-------------|------------------------------------|--------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT      | dataset.Column                     | Object                   | Server scripts         | A column in the dataset, which usually represents a record field.                                                                                              |
| OBJECT      | dataset.Condition                  | Object                   | Server scripts         | A condition or set of conditions to apply to a column.                                                                                                         |
| OBJECT      | dataset.Dataset                    | Object                   | Server scripts         | A representation of the entire dataset, including columns, conditions, and joins.                                                                              |
| OBJECT      | dataset.Join                       | Object                   | Server scripts         | A joined record used in the dataset.                                                                                                                           |
| METHOD      | dataset.create(options)            | dataset.Dataset          | Server scripts         | Creates a dataset.                                                                                                                                             |
| METHOD      | dataset.createColumn(options)      | dataset.Column           | Server scripts         | Creates a dataset column.                                                                                                                                      |
| METHOD      | dataset.createCondition(options)   | dataset.Condition        | Server scripts         | Creates a dataset condition (criteria). A condition is applied to a dataset column and includes an operator.                                                   |
| METHOD      | dataset.createJoin(options)        | dataset.Join             | Server scripts         | Creates a dataset join.                                                                                                                                        |
| METHOD      | dataset.createTranslation(options) | workbook.Expression      | Server scripts         | Creates a translation expression based on a Translation Collection.                                                                                            |
| METHOD      | dataset.describe(options)          | Object[]                 | Server scripts         | Retrieves descriptive information about a dataset, including name, description, and a list of columns or formulas with their labels and types.                 |
| METHOD      | dataset.describe.promise(options)  | Promise Object           | Server scripts         | Asynchronously retrieves descriptive information about a dataset, including name, description, and a list of columns or formulas with their labels and types.. |
| METHOD      | dataset.list()                     | Object[]                 | Server scripts         | Lists all existing datasets.                                                                                                                                   |
| METHOD      | dataset.listPaged(options)         | PagedInfoData            | Server scripts         | Returns metadata about datasets as a set of paged results.                                                                                                     |
| METHOD      | dataset.load(options)              | dataset.Dataset          | Server scripts         | Loads an existing dataset.                                                                                                                                     |

| MEMBER TYPE   | NAME                                 | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                               |
|---------------|--------------------------------------|--------------------------|------------------------|-------------------------------------------|
| <b>METHOD</b> | <b>dataset.load.promise(options)</b> | Promise Object           | Server scripts         | Asynchronously loads an existing dataset. |

## Column Object Members

This object encapsulates the record fields in the dataset. Columns are equivalent to the fields you use when you build a dataset in SuiteAnalytics. For more information about datasets in SuiteAnalytics, see Custom Workbooks and Datasets.

The following members are available for a dataset.Column object.

Return Type / Value Type

Supported Script Types

string (read-only)

The alias of the column.

| MEMBER TYPE     | NAME                  | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                 |
|-----------------|-----------------------|--------------------------|------------------------|-----------------------------------------------------------------------------|
| <b>PROPERTY</b> | <b>Column.alias</b>   | string (read-only)       | Server scripts         | The alias of the column.                                                    |
| <b>PROPERTY</b> | <b>Column.fieldId</b> | string (read-only)       | Server scripts         | The ID of the record field associated with the column.                      |
| <b>PROPERTY</b> | <b>Column.formula</b> | string (read-only)       | Server scripts         | The formula of the column.                                                  |
| <b>PROPERTY</b> | <b>Column.id</b>      | number (read-only)       | Server Scripts         | The ID of the column.                                                       |
| <b>PROPERTY</b> | <b>Column.join</b>    | dataset.Join (read-only) | Server scripts         | The join for the column. Used only when the column is from a joined record. |
| <b>PROPERTY</b> | <b>Column.label</b>   | string (read-only)       | Server scripts         | The label of the column.                                                    |
| <b>PROPERTY</b> | <b>Column.type</b>    | string (read-only)       | Server scripts         | The return type of the formula.                                             |

## Condition Object Members

This object encapsulates the criteria in the dataset. Conditions are equivalent to the criteria you use when you build a dataset in SuiteAnalytics. For more information about criteria used in datasets in SuiteAnalytics, see Dataset Criteria Filters.

The following members are available for a dataset.Condition object.

Return Type / Value Type

Supported Script Types

Condition.caseSensitive

Indicates whether the condition in a sort is case sensitive.

| MEMBER TYPE     | NAME                           | RETURN TYPE / VALUE TYPE                                        | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                               |
|-----------------|--------------------------------|-----------------------------------------------------------------|------------------------|---------------------------------------------------------------------------|
| <b>PROPERTY</b> | <b>Condition.caseSensitive</b> | boolean                                                         | Server scripts         | Indicates whether the condition in a sort is case sensitive.              |
| <b>PROPERTY</b> | <b>Condition.children</b>      | dataset.Condition[] (read-only)                                 | Server scripts         | The children of the condition (for example, subconditions AND'd or OR'd). |
| <b>PROPERTY</b> | <b>Condition.column</b>        | dataset.Column (read-only)                                      | Server scripts         | The column on which the condition is placed.                              |
| <b>PROPERTY</b> | <b>Condition.operator</b>      | string (read-only)                                              | Server scripts         | The operator of the condition.                                            |
| <b>PROPERTY</b> | <b>Condition.values</b>        | string[]   number[]   boolean[]   Date[]   Object[] (read-only) | Server scripts         | The values for the condition.                                             |

## Dataset Object Members

This object encapsulates the entire dataset. For more information about datasets in SuiteAnalytics, see Custom Workbooks and Datasets.

The following members are available for a dataset.Dataset object.

Return Type / Value Type

Supported Script Types

Dataset.getExpressionFromColumn(options)

workbook.Expression

| MEMBER TYPE   | NAME                                            | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                      |
|---------------|-------------------------------------------------|--------------------------|------------------------|----------------------------------------------------------------------------------|
| <b>METHOD</b> | <b>Dataset.getExpressionFromColumn(options)</b> | workbook.Expression      | Server scripts         | Returns an expression which can be used in a workbook.                           |
| <b>METHOD</b> | <b>Dataset.run()</b>                            | query.ResultSet          | Server scripts         | Executes the dataset and returns the result set (the same as in N/query Module). |

| MEMBER TYPE | NAME                                   | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                             |
|-------------|----------------------------------------|--------------------------|------------------------|---------------------------------------------------------------------------------------------------------|
| METHOD      | <code>Dataset.run.promise()</code>     | Promise Object           | Server scripts         | Asynchronously executes the dataset and returns the result set (the same as in N/query Module).         |
| METHOD      | <code>Dataset.runPaged(options)</code> | query.PagedData          | Server scripts         | Executes the dataset and returns the result set in paginated data form (the same as in N/query Module). |
| METHOD      | <code>Dataset.save(options)</code>     | Object                   | Server scripts         | Saves a dataset.                                                                                        |
| PROPERTY    | <code>Dataset.columns</code>           | dataset.Column[]         | Server scripts         | The columns in the dataset.                                                                             |
| PROPERTY    | <code>Dataset.condition</code>         | dataset.Condition        | Server scripts         | The condition (criteria) for the entire dataset.                                                        |
| PROPERTY    | <code>Dataset.description</code>       | string                   | Server scripts         | The description of the dataset.                                                                         |
| PROPERTY    | <code>Dataset.id</code>                | string                   | Server scripts         | The ID of the dataset.                                                                                  |
| PROPERTY    | <code>Dataset.name</code>              | string                   | Server scripts         | The name of the dataset.                                                                                |
| PROPERTY    | <code>Dataset.type</code>              | string                   | Server scripts         | The internal ID for the base record type for the dataset.                                               |

## Join Object Members

Encapsulates a joined record used in the dataset. For more information about using joins in a dataset, see [Joining Record Types in a Dataset](#).

The following members are available for a `dataset.Join` object.

Return Type / Value Type

Supported Script Types

string (read-only)

The ID of the field on which the join is performed.

| MEMBER TYPE | NAME                      | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                             |
|-------------|---------------------------|--------------------------|------------------------|---------------------------------------------------------|
| PROPERTY    | <code>Join.fieldId</code> | string (read-only)       | Server scripts         | The ID of the field on which the join is performed.     |
| PROPERTY    | <code>Join.join</code>    | dataset.Join (read-only) | Server scripts         | The child join, if the join is a multilevel join.       |
| PROPERTY    | <code>Join.source</code>  | string (read-only)       | Server scripts         | The internal ID for the source record type of the join. |
| PROPERTY    | <code>Join.target</code>  | string (read-only)       | Server scripts         | The polymorphic target of the join.                     |

 **Sample 1: Create a Dataset, Run the Dataset, and List All Existing Datasets**

```

/**
 * @NApiVersion 2.x
 */

// The following script creates a dataset with three columns, one of which is based on a join, and a condition. Then, all datasets are listed and the newly c
reated dataset is loaded and reviewed.

require(['N/dataset', 'N/query'], function(dataset, query){
  var myTransactionDateColumn = dataset.createColumn({
    fieldId: 'trandate',
    alias: 'date'
  });
  var myJoin = dataset.createJoin({
    fieldId: 'createdby',
    target: 'entity'
  });
  var myNameColumn = dataset.createColumn({
    fieldId: 'lastname',
    alias: 'name',
    join: myJoin
  });
  var myTransactionIdColumn = dataset.createColumn({
    fieldId: 'tranid',
    alias: 'id'
  });
  var myTotalColumn = dataset.createColumn({
    fieldId: 'foreigntotal',
    alias: 'total'
  });
  var myColumns = [myTransactionIdColumn, myNameColumn, myTransactionDateColumn, myTotalColumn];
  var myCondition = dataset.createCondition({
    column: myNameColumn,
    operator: query.Operator.EMPTY
  });
  var myDataset = dataset.create({
    type: 'transaction',
    columns: myColumns,
    condition: myCondition
  });

  // List all created datasets
  var allDatasets = dataset.list();
  log.audit({
    title: 'All datasets:',
    details: allDatasets
  });

  // Review the newly created dataset components (in the log)
  log.audit({
    title: 'My Dataset = ',
    details: myDataset
  });

  // Now run the dataset
  var runResult = myDataset.run();
  var runPagedResult = myDataset.runPaged({
    pageSize: 2
  });
  log.audit({
    title: 'myDataset runResult: ',
    details: runResult
  });
  log.audit({
    title: 'myDataset runPagedResult: ',
    details: runPagedResult
  });
});

```

## Sample 2: List All Datasets and Load the First Dataset

```
/**
 * @NApiVersion 2.x
 */

// The following script lists all existing datasets and then loads the first dataset.

require(['N/dataset'], function(dataset){
  // List all created datasets
  var allDatasets = dataset.list();
  log.audit({
    title: 'All datasets:',
    details: allDatasets
  });

  // Load the first dataset
  var myFirstDataset = dataset.load({
    id: allDatasets[0].id
  });
  log.audit('myFirstDataset:', myFirstDataset);
});
```

# 14. N/documentCapture Module

Load the N/documentCapture module to extract text content from supported documents. This module lets you programmatically extract structured content and key information from a variety of document types (such as invoices, receipts, contracts, and so on).

define(['N/documentCapture'], (documentCapture) => { ... })

✓ 4 Code Sample(s)

Scripts: Server scripts SuiteScript 2.1 only

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/documentCapture'], (documentCapture) => { // Access methods and properties on documentCapture });
```

The content in this help topic pertains to SuiteScript 2.1.

Load the N/documentCapture module to extract text content from supported documents.

The N/documentCapture module lets you programmatically extract structured content and key information from a variety of document types (such as invoices, receipts, contracts, and so on) directly within NetSuite. This module uses the AI-driven capabilities of the Oracle Cloud Infrastructure (OCI) Document Understanding service and can automate document processing, reduce manual data entry, and enhance business workflows. For more information about the OCI Document Understanding service, refer to Document Understanding in the Oracle Cloud Infrastructure Documentation.

This module provides the following features and benefits:

- Automated content extraction - Extracts text, tables, and key-value pairs from scanned documents, PDFs, and images.
- AI-powered data recognition - Uses advanced machine learning models from OCI to accurately identify and extract relevant information.
- Support for multiple document types - Works with invoices, receipts, tax forms, and other business documents in PDF, PNG, JPG, and TIFF formats.
- Synchronous and asynchronous requests - Supports synchronous requests (for documents up to five pages in length) and asynchronous requests (for documents longer than five pages).
- Document classification - Automatically classifies documents by type, enabling use cases such as intelligent routing and processing.
- Support for multiple languages and layouts - Supports documents in multiple languages and using various layouts, increasing flexibility.
- Error handling and confidence scores - Provides confidence scores for extracted data and error handling for improved reliability.

This module is available in NetSuite by default when the Server SuiteScript feature is enabled. For more information, see Enabling Features.

To learn how to get started with the N/documentCapture module, see Getting Started with the N/documentCapture Module.

## N/documentCapture Module Members

Return Type / Value Type

Supported Script Types

documentCapture.Cell

An extracted table cell from a document.

documentCapture.Document

The extracted data from a document.

| MEMBER TYPE | NAME                                                 | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                      |
|-------------|------------------------------------------------------|--------------------------|------------------------|--------------------------------------------------|
| OBJECT      | documentCapture.Cell                                 | Object                   | Server scripts         | An extracted table cell from a document.         |
| OBJECT      | documentCapture.Document                             | Object                   | Server scripts         | The extracted data from a document.              |
| OBJECT      | documentCapture.Field                                | Object                   | Server scripts         | An extracted field from a document.              |
| OBJECT      | documentCapture.FieldLabel                           | Object                   | Server scripts         | An extracted field label from a document.        |
| OBJECT      | documentCapture.FieldValue                           | Object                   | Server scripts         | An extracted field value from a document.        |
| OBJECT      | documentCapture.Line                                 | Object                   | Server scripts         | An extracted line of text from a document.       |
| OBJECT      | documentCapture.Page                                 | Object                   | Server scripts         | An extracted page from a document.               |
| OBJECT      | documentCapture.Table                                | Object                   | Server scripts         | An extracted table from a document.              |
| OBJECT      | documentCapture.TableRow                             | Object                   | Server scripts         | An extracted table row from a document.          |
| OBJECT      | documentCapture.Word                                 | Object                   | Server scripts         | An extracted word from a document.               |
| METHOD      | documentCapture.documentToStructure(options)         | documentCapture.Document | Server scripts         | Extracts content from a document.                |
| METHOD      | documentCapture.documentToStructure.promise(options) | Promise                  | Server scripts         | Asynchronously extracts content from a document. |
| METHOD      | documentCapture.documentToText(options)              | string                   | Server scripts         | Extracts text content from a PDF file.           |

| MEMBER TYPE | NAME                                              | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                   |
|-------------|---------------------------------------------------|--------------------------|------------------------|-------------------------------------------------------------------------------|
| METHOD      | documentCapture.documentToText.promise(options)   | Promise                  | Server scripts         | Asynchronously extracts text content from a PDF file.                         |
| METHOD      | documentCapture.getRemainingConcurrency()         | number                   | Server scripts         | Returns the number of available concurrent requests remaining.                |
| METHOD      | documentCapture.getRemainingConcurrency.promise() | Promise                  | Server scripts         | Asynchronously returns the number of available concurrent requests remaining. |
| METHOD      | documentCapture.getRemainingFreeUsage()           | number                   | Server scripts         | Deprecated. Calls llm.getRemainingUsage().                                    |
| METHOD      | documentCapture.getRemainingFreeUsage.promise()   | Promise                  | Server scripts         | Deprecated. Calls llm.getRemainingUsage.promise().                            |
| METHOD      | documentCapture.parseResult(options)              | documentCapture.Document | Server scripts         | Converts a JSON file into a documentCapture.Document object.                  |
| ENUM        | documentCapture.DocumentType                      | enum                     | Server scripts         | Holds values for the document type.                                           |
| ENUM        | documentCapture.Feature                           | enum                     | Server scripts         | Holds values for the feature to extract.                                      |
| ENUM        | documentCapture.FieldType                         | enum                     | Server scripts         | Holds values for the type of a field.                                         |
| ENUM        | documentCapture.Language                          | enum                     | Server scripts         | Holds values for the language of a document.                                  |

Cell Object Members

Return Type / Value Type

Supported Script Types

The confidence level for the cell.

The extracted text of the cell.

| MEMBER TYPE | NAME            | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                        |
|-------------|-----------------|--------------------------|------------------------|------------------------------------|
| PROPERTY    | Cell.confidence | number                   | Server scripts         | The confidence level for the cell. |
| PROPERTY    | Cell.text       | string                   | Server scripts         | The extracted text of the cell.    |

Document Object Members

Return Type / Value Type

Supported Script Types

Document.mimeType

The MIME type of the document.

documentCapture.Page[]

The pages of the document.

| MEMBER TYPE | NAME               | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                              |
|-------------|--------------------|--------------------------|------------------------|------------------------------------------|
| PROPERTY    | Document.mimeType  | string                   | Server scripts         | The MIME type of the document.           |
| PROPERTY    | Document.pages     | documentCapture.Page[]   | Server scripts         | The pages of the document.               |
| METHOD      | Document.getText() | string                   | Server scripts         | Returns the entire text of the document. |

Field Object Members

Return Type / Value Type

Supported Script Types

documentCapture.FieldLabel

The label (name) of the field.

The type of the field.

documentCapture.FieldValue

| MEMBER TYPE | NAME        | RETURN TYPE / VALUE TYPE   | SUPPORTED SCRIPT TYPES | DESCRIPTION                    |
|-------------|-------------|----------------------------|------------------------|--------------------------------|
| PROPERTY    | Field.label | documentCapture.FieldLabel | Server scripts         | The label (name) of the field. |
| PROPERTY    | Field.type  | string                     | Server scripts         | The type of the field.         |
| PROPERTY    | Field.value | documentCapture.FieldValue | Server scripts         | The value of the field.        |

FieldLabel Object Members

Return Type / Value Type

Supported Script Types

FieldLabel.confidence

The confidence level for the field label.

The name of the field label.

| MEMBER TYPE | NAME                  | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                               |
|-------------|-----------------------|--------------------------|------------------------|-------------------------------------------|
| PROPERTY    | FieldLabel.confidence | number                   | Server scripts         | The confidence level for the field label. |
| PROPERTY    | FieldLabel.name       | string                   | Server scripts         | The name of the field label.              |

FieldValue Object Members

Return Type / Value Type

Supported Script Types

FieldValue.confidence

The confidence level for the field value.

The text of the field value.

| MEMBER TYPE | NAME                  | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                               |
|-------------|-----------------------|--------------------------|------------------------|-------------------------------------------|
| PROPERTY    | FieldValue.confidence | number                   | Server scripts         | The confidence level for the field value. |
| PROPERTY    | FieldValue.text       | string                   | Server scripts         | The text of the field value.              |

Line Object Members

Return Type / Value Type

Supported Script Types

The confidence level for the line.

The text of the line.

| MEMBER TYPE | NAME            | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                        |
|-------------|-----------------|--------------------------|------------------------|------------------------------------|
| PROPERTY    | Line.confidence | number                   | Server scripts         | The confidence level for the line. |
| PROPERTY    | Line.text       | string                   | Server scripts         | The text of the line.              |

Page Object Members

Return Type / Value Type

Supported Script Types

Page.detectedDocumentTypes

A set of confidence levels indicating whether the page represents a particular type of document.

documentCapture.Field[]

The extracted fields from the page of a document.

| MEMBER TYPE | NAME                       | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                      |
|-------------|----------------------------|--------------------------|------------------------|--------------------------------------------------------------------------------------------------|
| PROPERTY    | Page.detectedDocumentTypes | Object[]                 | Server scripts         | A set of confidence levels indicating whether the page represents a particular type of document. |
| PROPERTY    | Page.fields                | documentCapture.Field[]  | Server scripts         | The extracted fields from the page of a document.                                                |
| PROPERTY    | Page.lines                 | documentCapture.Line[]   | Server scripts         | The extracted lines from the page of a document.                                                 |
| PROPERTY    | Page.tables                | documentCapture.Table[]  | Server scripts         | The extracted tables from the page of a document.                                                |
| PROPERTY    | Page.words                 | documentCapture.Word[]   | Server scripts         | The extracted words from the page of a document.                                                 |
| METHOD      | Page.getText()             | string                   | Server scripts         | Returns the entire text of the page.                                                             |

Table Object Members

Return Type / Value Type

Supported Script Types

documentCapture.TableRow[]

The extracted body rows from the table in a document.

Table.columnCount

The number of extracted columns from the table in a document.

| MEMBER TYPE | NAME              | RETURN TYPE / VALUE TYPE   | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                   |
|-------------|-------------------|----------------------------|------------------------|---------------------------------------------------------------|
| PROPERTY    | Table.bodyRows    | documentCapture.TableRow[] | Server scripts         | The extracted body rows from the table in a document.         |
| PROPERTY    | Table.columnCount | number                     | Server scripts         | The number of extracted columns from the table in a document. |
| PROPERTY    | Table.confidence  | number                     | Server scripts         | The confidence level for the table.                           |
| PROPERTY    | Table.footerRows  | documentCapture.TableRow[] | Server scripts         | The extracted footer rows from the table in a document.       |
| PROPERTY    | Table.headerRows  | documentCapture.TableRow[] | Server scripts         | The extracted header rows from the table in a document.       |
| PROPERTY    | Table.rowCount    | number                     | Server scripts         | The number of extracted rows from the table in a document.    |

TableRow Object Members

Return Type / Value Type

Supported Script Types

documentCapture.Cell[]

The extracted cells in the table row.

| MEMBER TYPE | NAME           | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                           |
|-------------|----------------|--------------------------|------------------------|---------------------------------------|
| PROPERTY    | TableRow.cells | documentCapture.Cell[]   | Server scripts         | The extracted cells in the table row. |

Word Object Members

Return Type / Value Type

Supported Script Types

The confidence level for the word.

The extracted text of the word.

| MEMBER TYPE | NAME            | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                        |
|-------------|-----------------|--------------------------|------------------------|------------------------------------|
| PROPERTY    | Word.confidence | number                   | Server scripts         | The confidence level for the word. |
| PROPERTY    | Word.text       | string                   | Server scripts         | The extracted text of the word.    |

► Official NetSuite Script Samples (4 Sample(s))

Sample 1: Extract Text from a PDF File

```
require(['N/file', 'N/documentCapture', 'N/llm'],
function(file, documentCapture, llm) {
  // "14" is the unique ID of a PDF stored in the NetSuite File Cabinet
  const fileObj = file.load({
    id: "14"
  });
  const extractedData = documentCapture.documentToText({
    file: fileObj
  });

  const response = llm.generateText({
    prompt: "What is this invoice for?",
    documents: [{
      id: '14',
      data: extractedData
    }]
  });

  log.debug("Answer: ", response.text);
  log.debug("Citations: ", response.citations);
}
);
```

### Sample 2: Extract Feature Content from a Document Synchronously

```
require(['N/file', 'N/documentCapture', 'N/llm'],
function(file, documentCapture, llm) {
    // "14" is the internal ID of a file stored in the NetSuite File Cabinet
    const extractedData = documentCapture.documentToStructure({
        file: file.load(
            id: "14"
        ),
        features: [
            documentCapture.Feature.TEXT_EXTRACTION,
            documentCapture.Feature.TABLE_EXTRACTION,
            documentCapture.Feature.FIELD_EXTRACTION
        ],
        documentType: documentCapture.DocumentType.INVOICE
    });

    const documentText = extractedData.getText();

    const response = llm.chat({
        preamble: "Your task is to parse the provided document and answer questions about that document.",
        prompt: "What is this invoice for?",
        documents: [{
            id: "1",
            data: documentText
        }]
    });

    log.debug("Answer: ", response.text);
    log.debug("Citations: ", response.citations);
}
});
```

### Sample 3: Extract Content from a Document Asynchronously

```
require(['N/task', 'N/file', 'N/documentCapture'],
function(task, file, documentCapture) {
    // Create the document capture task
    var docTask = task.create(task.TaskType.DOCUMENT_CAPTURE);

    // Specify task parameters
    docTask.inputFile = file.load("443");
    docTask.outputFilePath="SuiteScripts/result.json";
    docTask.documentType = documentCapture.DocumentType.INVOICE;

    // Submit the task
    const submissionId = docTask.submit();
    log.debug("Submission ID: ", submissionId);
});
```

### Sample 4: Extract Content from a Document Asynchronously

```
/**
 * @NApiVersion 2.1
 * @NScriptType ScheduledScript
 * @NModuleScope SameAccount
 */
define(['N/documentCapture', 'N/file', 'N/llm'],
function(documentCapture, file, llm) {
    function execute(scriptContext) {
        const resultFile = file.load("SuiteScripts/result.json");
        const resultDocument = documentCapture.parseResult(resultFile);
        const resultText = resultDocument.getText();

        const response = llm.chat({
            preamble: "Your task is to parse the provided document and answer questions about that document.",
            prompt: "What is this invoice for?",
            documents: [{
                id: "1",
                data: resultText
            }]
        });
        log.debug("Response: ", response.text);
    }
    return {
        execute: execute
    };
}
);
```

# 15. N/email Module

Load the N/email module to send email messages from within NetSuite. You can use the email module to send regular, bulk, and campaign email.

define(['N/email'], (email) => { ... })

✓ 1 Code Sample(s)

Scripts: Client and server scripts

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/email'], (email) => { // Access methods and properties on email });
```

Use the N/email module to send email messages from within NetSuite. You can use the N/email module to send regular, bulk, and campaign email.

N/email Module Script Sample

## N/email Module Members

Return Type / Value Type

Supported Script Types

email.send(options)

Client and server scripts

Sends transactional email to an individual or group of recipients and receives bounceback notifications.

To send email on another user's behalf, the user triggering the email send must have a role with the Vicarious Email (ADML\_VICARIOUS\_EMAIL) permission. If a user without this permission executes SuiteScript that sends email, a Permission Violation error is returned.

| MEMBER TYPE | NAME                                     | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                               |
|-------------|------------------------------------------|--------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | email.send(options)                      | void                     | Client and server scripts | Sends transactional email to an individual or group of recipients and receives bounceback notifications. Note: To send email on another user's behalf, the user triggering the email send must have a role with the Vicarious Email (ADML_VICARIOUS_EMAIL) permission. If a user without this permission executes SuiteScript that sends email, a Permission Violation error is returned. |
| METHOD      | email.send.promise(options)              | void                     | Client scripts            | Sends transactional email asynchronously to an individual or group of recipients and receives bounceback notifications.                                                                                                                                                                                                                                                                   |
| METHOD      | email.sendBulk(options)                  | void                     | Client and server scripts | Sends bulk email (for use when a bounceback notification is not required).                                                                                                                                                                                                                                                                                                                |
| METHOD      | email.sendBulk.promise(options)          | void                     | Client scripts            | Sends bulk email asynchronously (for use when a bounceback notification is not required).                                                                                                                                                                                                                                                                                                 |
| METHOD      | email.sendCampaignEvent(options)         | number                   | Client and server scripts | Sends a single 'on-demand' campaign email to a specified recipient and return a campaign response ID.                                                                                                                                                                                                                                                                                     |
| METHOD      | email.sendCampaignEvent.promise(options) | number                   | Client and server scripts | Sends a single 'on-demand' campaign email asynchronously to a specified recipient and return a campaign response ID.                                                                                                                                                                                                                                                                      |

 Sample 1: N/email Module Script Sample

```
/**
 * @NApiVersion 2.1
 */

// This script sends an email with an attachment.

require(['N/email', 'N/record', 'N/file'], (email, record, file) => {
  const senderId = -515;
  const recipientEmail = 'notify@myCompany.com';
  let timeStamp = new Date().getUTCMilliseconds();

  let recipient = record.create({
    type: record.Type.CUSTOMER,
    isDynamic: true
  });
  recipient.setValue({
    fieldId: 'subsidiary',
    value: '1'
  });
  recipient.setValue({
    fieldId: 'companyname',
    value: 'Test Company' + timeStamp
  });
  recipient.setValue({
    fieldId: 'email',
    value: recipientEmail
  });

  let recipientId = recipient.save();

  let fileObj = file.load({
    id: 88
  });

  email.send({
    author: senderId,
    recipients: recipientId,
    subject: 'Test Sample Email Module',
    body: 'email body',
    attachments: [fileObj],
    relatedRecords: {
      entityId: recipientId,
      customRecord: {
        id: recordId,
        recordType: recordTypeId
      }
    }
  });
});
```

# 16. N/encode Module

Load the N/encode module to convert a string to another type of encoding. See encode.Encoding for a list of supported character set encoding.

define(['N/encode'], (encode) => { ... })

✓ 1 Code Sample(s)

Scripts: Server scripts

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/encode'], (encode) => { // Access methods and properties on encode });
```

Use the N/encode module to convert a string to another type of encoding. This module exposes string encoding and decoding functionality.

## N/encode Module Members

Return Type / Value Type

Supported Script Types

encode.convert(options)

Converts a string to another type of encoding and returns the re-encoded string.

Holds the string values for the supported character set encoding. Use this enum to set the inputEncoding and outputEncoding parameter values in N/crypto Module or N/encode Module.

| MEMBER TYPE | NAME                    | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                         |
|-------------|-------------------------|--------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | encode.convert(options) | string                   | Server scripts         | Converts a string to another type of encoding and returns the re-encoded string.                                                                                                    |
| ENUM        | encode.Encoding         | enum                     | Server scripts         | Holds the string values for the supported character set encoding. Use this enum to set the inputEncoding and outputEncoding parameter values in N/crypto Module or N/encode Module. |

### ► Official NetSuite Script Samples (1 Sample(s))

#### Sample 1: N/encode Module Script Samples

```
/**
 * @NApiVersion 2.x
 */

require(['N/encode'], function(encode) {
  function convertStringToDifferentEncoding() {
    var stringInput = "TÃfÃ@st StriÃfÃ±g Input";
    var base64EncodedString = encode.convert({
      string: stringInput,
      inputEncoding: encode.Encoding.UTF_8,
      outputEncoding: encode.Encoding.BASE_64
    });
    var hexEncodedString = encode.convert({
      string: stringInput,
      inputEncoding: encode.Encoding.UTF_8,
      outputEncoding: encode.Encoding.HEX
    });
  }

  convertStringToDifferentEncoding();
});
```

## 17. N/error Module

Load the N/error module to create your own custom SuiteScript errors. Use these custom errors in try-catch statements to abort script execution.

```
define(['N/error'], (error) => { ... })
```

[✓ 2 Code Sample\(s\)](#)[Scripts: Server scripts](#)[Permissions: -](#)

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/error'], (error) => { // Access methods and properties on error });
```

Use the N/error module to create custom SuiteScript errors that you can use in try-catch statements to abort script execution. Note that this module doesn't provide functionality to throw custom errors. However, you can include logic such as try-catch statements in your script to throw custom SuiteScript errors.

### N/error Module Members

Return Type / Value Type

Supported Script Types

error.SuiteScriptError

Encapsulates a custom SuiteScript error for any server script type.

error.create(options)

error.SuiteScriptError

| MEMBER TYPE | NAME                          | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                                                     |
|-------------|-------------------------------|--------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT      | <b>error.SuiteScriptError</b> | Object                   | Server scripts         | Encapsulates a custom SuiteScript error for any server script type.                                                                                                                                             |
| METHOD      | <b>error.create(options)</b>  | error.SuiteScriptError   | Server scripts         | Creates a new error.SuiteScriptError object.                                                                                                                                                                    |
| ENUM        | <b>error.Type</b>             | enum                     | Server scripts         | Holds the string values for error types. Use this enum to set the value for the SuiteScriptError.name parameter of the error.create(options) method. This sets the value of the SuiteScriptError.type property. |

### SuiteScriptError Object Members

Return Type / Value Type

Supported Script Types

SuiteScriptError.cause

string (read-only)

Cause of the error.

SuiteScriptError.id

| MEMBER TYPE | NAME                              | RETURN TYPE / VALUE TYPE     | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                             |
|-------------|-----------------------------------|------------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PROPERTY    | <b>SuiteScriptError.cause</b>     | string (read-only)           | Server scripts         | Cause of the error.                                                                                                                                                     |
| PROPERTY    | <b>SuiteScriptError.id</b>        | string (read-only)           | Server scripts         | Error ID that is automatically generated when a new error is created.                                                                                                   |
| PROPERTY    | <b>SuiteScriptError.message</b>   | string (read-only)           | Server scripts         | Error message text displayed in the Details column of the Execution Log. Set from the options.message parameter when you create a new error using error.create(options) |
| PROPERTY    | <b>SuiteScriptError.name</b>      | string (read-only)           | Server scripts         | Error name or error code. Set from the options.name parameter when you create a new error using error.create(options).                                                  |
| PROPERTY    | <b>SuiteScriptError.notifyOff</b> | boolean (read-only)          | Server scripts         | Suppresses email notification when set to true. Set from the options.notifyOff parameter when you create a new error using error.create(options)                        |
| PROPERTY    | <b>SuiteScriptError.stack</b>     | Array of strings (read-only) | Server scripts         | List of method calls that the script is executing when the error is thrown.                                                                                             |
| PROPERTY    | <b>SuiteScriptError.type</b>      | error.Type (read-only)       | Server scripts         | Error type (error.SuiteScriptError).                                                                                                                                    |

 **Sample 1: Create a Custom Error**

```
/**
 * @NApiVersion 2.x
 */

// This script creates a custom error.
require(['N/error'], function(error) {
    function createError() {
        var myCustomError = error.create({
            name: 'MY_ERROR_CODE',
            message: 'My custom error details',
            notifyOff: true
        });
    }
    createError();
});
```

 **Sample 2: Create an Error Based on a Condition**

```
/**
 * @NApiVersion 2.x
 */

// This script conditionally creates and throws an error.
require(['N/error'], function(error) {
    function showError() {
        var someVariable = false;

        if (!someVariable) {
            var myCustomError = error.create({
                name: 'WRONG_PARAMETER_TYPE',
                message: 'Wrong parameter type selected.',
                notifyOff: false
            });

            // This will write 'Error: WRONG_PARAMETER_TYPE Wrong parameter type selected' to the log
            log.error('Error: ' + myCustomError.name + myCustomError.message);
            throw myCustomError;
        }
    }
    showError();
});
```

# 18. N/file Module

Load the N/file module to work with files in NetSuite.

define(['N/file'], (file) => { ... })

✓ 6 Code Sample(s)

Scripts: Server scripts

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/file'], (file) => { // Access methods and properties on file });
```

Use the N/file module to work with files within NetSuite. You can use this module to upload files to the NetSuite File Cabinet, as well as send files as attachments without uploading them to the File Cabinet. You can also use this module to copy files in the File Cabinet, and you can use conflict resolution options to handle conflicts (such as when a file with the same name already exists in the same folder in the File Cabinet).

When dealing with file naming conventions, especially when uploading files to the File Cabinet using SuiteScript, it is important to follow best practices to prevent file duplication. If your system works with multiple files simultaneously, consider including timestamps with millisecond precision in the file names. Additionally, generating a random number as part of the file name can further reduce the chances of duplicates.

A file.Reader object, which is returned by File.getReader(), can be used for special read operations. Use File.getSegments(options) to retrieve an iterator of custom segments of a file.

Methods that load content in memory, such as File.getContents(), have a 10 MB size limit. This limit does not apply when content is streamed, such as when File.save() is called.

The File.appendLine(options) method inserts a line to the end of a CSV or text file, which increases the file size accordingly. All files are screened for malicious content when they are uploaded or updated, and increased file size can slow down the process. Consider splitting large files into several smaller ones if you experience performance issues.

## N/file Module Members

- Return Type / Value Type
- Supported Script Types
- Encapsulates a file within NetSuite.
- Encapsulates a reader that you can use to perform special read operations.
- file.copy(options)  
Copies an existing file in the File Cabinet.

| MEMBER TYPE | NAME                        | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                                                           |
|-------------|-----------------------------|--------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT      | file.File                   | Object                   | Server scripts         | Encapsulates a file within NetSuite.                                                                                                                                                                                  |
| OBJECT      | file.Reader                 | Object                   | Server scripts         | Encapsulates a reader that you can use to perform special read operations.                                                                                                                                            |
| METHOD      | file.copy(options)          | file.File                | Server scripts         | Copies an existing file in the File Cabinet.                                                                                                                                                                          |
| METHOD      | file.create(options)        | file.File                | Server scripts         | Creates a new file.File.                                                                                                                                                                                              |
| METHOD      | file.delete(options)        | void                     | Server scripts         | Deletes an existing file.File from the NetSuite File Cabinet.                                                                                                                                                         |
| METHOD      | file.load(options)          | file.File                | Server scripts         | Loads an existing file.File from the NetSuite File Cabinet.                                                                                                                                                           |
| ENUM        | file.Encoding               | enum                     | Server scripts         | Holds character encoding values for file contents. Use this enum to set the value of the File.encoding property.                                                                                                      |
| ENUM        | file.NameConflictResolution | enum                     | Server scripts         | Holds conflict resolution values that apply when copying a file. Use this enum to specify how to resolve conflicts when copying files and to set the value of the conflict resolution parameter in file.copy(options) |
| ENUM        | file.Type                   | enum                     | Server scripts         | Holds file type values. Use this enum to set the value of the File.fileType property.                                                                                                                                 |

## File Object Members

- The following members are available for a file.File object.
- Return Type / Value Type
  - Supported Script Types
  - File.appendLine(options)  
Inserts a line to the end of a CSV or text file.
  - File.getContents()  
Returns the content of a file in string format.

| MEMBER TYPE | NAME                     | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                      |
|-------------|--------------------------|--------------------------|------------------------|--------------------------------------------------|
| METHOD      | File.appendLine(options) | file.File                | Server scripts         | Inserts a line to the end of a CSV or text file. |
| METHOD      | File.getContents()       | string                   | Server scripts         | Returns the content of a file in string format.  |

| MEMBER TYPE | NAME                                   | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                          |
|-------------|----------------------------------------|--------------------------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | <code>File.lines.iterator()</code>     | boolean                  | Server scripts         | Calls a developer-defined function for each line. Returns false when line processing stops.                                          |
| METHOD      | <code>File.resetStream()</code>        | void                     | Server scripts         | Resets the file stream to its previous state.                                                                                        |
| METHOD      | <code>File.save()</code>               | number                   | Server scripts         | Saves a new or updated file to the File Cabinet.                                                                                     |
| METHOD      | <code>File.getReader()</code>          | Object                   | Server scripts         | Returns reader object for read operations.                                                                                           |
| METHOD      | <code>File.getSegments(options)</code> | Object                   | Server scripts         | Returns an iterator of segments that are delimited by the specified separator.                                                       |
| PROPERTY    | <code>File.description</code>          | string                   | Server scripts         | Description of a file.                                                                                                               |
| PROPERTY    | <code>File.encoding</code>             | string                   | Server scripts         | Character encoding on a file.                                                                                                        |
| PROPERTY    | <code>File.fileType</code>             | enum                     | Server scripts         | File type of a file.                                                                                                                 |
| PROPERTY    | <code>File.folder</code>               | number                   | Server scripts         | Internal ID of the folder that houses a file within the NetSuite File Cabinet.                                                       |
| PROPERTY    | <code>File.id</code>                   | number (read-only)       | Server scripts         | Internal ID of a file in the NetSuite File Cabinet.                                                                                  |
| PROPERTY    | <code>File.isInactive</code>           | boolean                  | Server scripts         | Inactive status of a file. If set to true, the file is inactive.                                                                     |
| PROPERTY    | <code>File.isOnline</code>             | boolean                  | Server scripts         | 'Available without Login' status of a file. If set to true, users can download the file outside of a current NetSuite login session. |
| PROPERTY    | <code>File.isText</code>               | boolean (read-only)      | Server scripts         | Indicates whether a file type is text-based.                                                                                         |
| PROPERTY    | <code>File.name</code>                 | string                   | Server scripts         | Name of a file.                                                                                                                      |
| PROPERTY    | <code>File.path</code>                 | string (read-only)       | Server scripts         | Relative path to a file in the NetSuite File Cabinet.                                                                                |
| PROPERTY    | <code>File.size</code>                 | number (read-only)       | Server scripts         | Size of a file in bytes.                                                                                                             |
| PROPERTY    | <code>File.url</code>                  | string (read-only)       | Server scripts         | URL of a file.                                                                                                                       |

## Reader Object Members

The following members are available for a `file.Reader` object.

Return Type / Value Type

Supported Script Types

`Reader.readChars(options)`

Returns the next `options.number` characters from the current position.

`Reader.readUntil(options)`

| MEMBER TYPE | NAME                                   | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                               |
|-------------|----------------------------------------|--------------------------|------------------------|-------------------------------------------------------------------------------------------|
| METHOD      | <code>Reader.readChars(options)</code> | string                   | Server scripts         | Returns the next <code>options.number</code> characters from the current position.        |
| METHOD      | <code>Reader.readUntil(options)</code> | string                   | Server scripts         | Returns string from current position to the next occurrence of <code>options.tag</code> . |

 **Sample 1: Create and Save a File to the File Cabinet**

```
/**
 * @NApiVersion 2.1
 */
require(['N/file'], file => {
  // Create a file containing text
  let fileObj = file.create({
    name: 'testHelloWorld.txt',
    fileType: file.Type.PLAINTEXT,
    contents: 'Hello World\nHello World'
  });

  // Set the folder for the file
  // Note that this value is hard-coded in this sample, and you should use
  // a valid folder ID from your account
  fileObj.folder = -15;

  // Save the file
  let id = fileObj.save();

  // Load the same file to ensure it was saved correctly
  fileObj = file.load({
    id: id
  });
});
```

 **Sample 2: Create a File, Set Property Values, and Save It to the File Cabinet**

```
/**
 * @NApiVersion 2.1
 */
require(['N/file'], file => {
  // Create a file containing text
  // Note that the folder value is hard-coded in this sample, and you should
  // use a valid folder ID from your account
  let fileObj = file.create({
    name: 'testHelloWorld3.txt',
    fileType: file.Type.PLAINTEXT,
    contents: 'Hello World\nHello World',
    folder: -15,
    isOnline: true
  });

  // Save the file
  let id = fileObj.save();

  // Load the same file to ensure it was saved correctly
  fileObj = file.load({
    id: id
  });
});
```

### Sample 3: Create and Save a CSV File then Reload the File and Parse Its Contents

```
/**
 * @NApiVersion 2.x
 */
require(['N/file', 'N/error', 'N/log'], function (file, error, log) {
  // This sample calculates the total for the
  // second column value in a CSV file.
  //
  // Each line in the CSV file has the following format:
  // date,amount
  //
  // Here is the data that the script adds to the file:
  // 10/21/14,200.0
  // 10/21/15,210.2
  // 10/21/16,250.3

  // Create the CSV file
  var csvFile = file.create({
    name: 'data.csv',
    contents: 'date,amount\n',
    folder: 39,
    fileType: 'CSV'
  });

  // Add the data
  csvFile.appendLine({
    value: '10/21/14,200.0'
  });
  csvFile.appendLine({
    value: '10/21/15,210.2'
  });
  csvFile.appendLine({
    value: '10/21/16,250.3'
  });

  // Save the file
  var csvFileId = csvFile.save();

  // Create a variable to store the calculated total
  var total = 0.0;

  // Load the file
  var invoiceFile = file.load({
    id: csvFileId
  });

  // Obtain an iterator to process each line in the file
  var iterator = invoiceFile.lines.iterator();

  // Skip the first line, which is the CSV header line
  iterator.each(function () {return false;});

  // Process each line in the file
  iterator.each(function (line) {
    // Update the total based on the line value
    var lineValues = line.value.split(',');
    var lineAmount = parseFloat(lineValues[1]);
    if (!lineAmount) {
      throw error.create({
        name: 'INVALID_INVOICE_FILE',
        message: 'Invoice file contained non-numeric value for total: ' + lineValues[1]
      });
    }
    total += lineAmount;
    return true;
  });

  // At the completion of the iteration.each function, the total is 660.5
});
```

#### Sample 4: Read and Log File Contents Using Commas and New Lines as Separators

```
/**
 * @NApiVersion 2.x
 * @NScriptType bankStatementParserPlugin
 */

define(['N/file', 'N/log'], function(file, log) {
  return {
    parseBankStatement: function(context) {
      var reader = context.input.file.getReader();

      var textUntilFirstComma = reader.readUntil(',');
      var next10Characters = reader.readChars(10);
      var textUntilNextNewLine = reader.readUntil('\n');
      var next100Characters = reader.readChars(100);

      log.debug({
        title: 'STATEMENT TEXT',
        details: textUntilFirstComma
      });

      log.debug({
        title: 'STATEMENT TEXT',
        details: next10Characters
      });

      log.debug({
        title: 'STATEMENT TEXT',
        details: textUntilNextNewLine
      });

      log.debug({
        title: 'STATEMENT TEXT',
        details: next100Characters
      })
    }
  }
});
```

#### Sample 5: Read and Log Segments of a File Using a Set of Characters as Separators

```
/**
 * @NApiVersion 2.x
 * @NScriptType bankStatementParserPlugin
 */

define(['N/file', 'N/log'], function(file, log) {
  return {
    parseBankStatement: function(context) {
      var statementFile = context.input.file;

      var statementSegmentIterator = statementFile.getSegments({separator: '\\|_|/'}).iterator();
      statementSegmentIterator.each(function (segment) {
        log.debug({
          title: 'STATEMENT TEXT',
          details: segment.value
        });
        return true;
      });
    }
  }
});
```

#### Sample 6: Copy a File Using Conflict Resolution

```
/**
 * @NApiVersion 2.x
 */
require(['N/file'], function(file) {
  var fileObj = file.create({
    name: 'test.txt',
    fileType: file.Type.PLAINTEXT,
    contents: 'Hello World\nHello World'
  });
  fileObj.folder = 2667;
  var id = fileObj.save();

  fileObj = file.copy({
    id: id,
    folder: 2670,
    conflictResolution: file.NameConflictResolution.RENAME_TO_UNIQUE
  });
});
```

# 19. N/format Module

Load the N/format module to convert strings into a specified format and to parse formatted data into strings.

`define(['N/format'], (format) => { ... })`

✓ 4 Code Sample(s)

Scripts: Client and server scripts

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/format'], (format) => { // Access methods and properties on format });
```

Use the format module to parse formatted data into strings and to convert strings into a specified format. The format module formats data according to personal preferences set on the Set Preferences page, accessible from Home &gt; Set Preferences. See Setting Personal Preferences.

## N/format Module Members

Return Type / Value Type

Supported Script Types

format.format(options)

Client and server scripts

Takes a raw value and returns a formatted value.

This method is overloaded when you format a datetime or datetimetz value.

| MEMBER TYPE | NAME                   | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                                                                       |
|-------------|------------------------|--------------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | format.format(options) | string   Date            | Client and server scripts | Takes a raw value and returns a formatted value. Note: This method is overloaded when you format a datetime or datetimetz value.                                                  |
| METHOD      | format.parse(options)  | Date   string   number   | Client and server scripts | Takes a formatted value and returns a raw value. Note: This method is overloaded when you format a datetime or datetimetz value.                                                  |
| ENUM        | format.Timezone        | enum                     | Client and server scripts | Holds the string values for supported time zone formats. Use this enum to set the value of the options.timezone parameter.                                                        |
| ENUM        | format.Type            | enum                     | Client and server scripts | Holds the string values for the supported field types. Use this enum to set the value of the options.type parameter when calling format.format(options) or format.parse(options). |

### Sample 1: Parse a String to a Date Object

```
/**
 * @NApiVersion 2.1
 * @NScriptType Suitelet
 */
define(['N/ui/serverWidget', 'N/format'], function(serverWidget, format) {
    function parseAndFormatDateString() {
        // Assuming Date format is MM/DD/YYYY
        var initialFormattedDateString = "07/28/2015";
        var parsedDateStringAsRawDateObject = format.parse({
            value: initialFormattedDateString,
            type: format.Type.DATE
        });
        var formattedDateString = format.format({
            value: parsedDateStringAsRawDateObject,
            type: format.Type.DATE
        });
        return [parsedDateStringAsRawDateObject, formattedDateString];
    }
    function onRequest(context) {
        var data = parseAndFormatDateString();

        var form = serverWidget.createForm({
            title: "Date"
        });

        var fldDate = form.addField({
            type: serverWidget.FieldType.DATE,
            id: "date",
            label: "Date"
        });
        fldDate.defaultValue = data[0];

        var fldString = form.addField({
            type: serverWidget.FieldType.TEXT,
            id: "dateastext",
            label: "Date as text"
        });
        fldString.defaultValue = data[1];

        context.response.writePage(form);
    }
    return {
        onRequest: onRequest
    };
});
```

### Sample 2: Parse a String to a Number

```
/**
 * @NApiVersion 2.x
 */
require(['N/format'],
    function(format){
        function parseToValue() {
            // Assume number format is 1.000.000,00 and negative format is -100
            var formattedNum = "-20.000,25"
            return format.parse({value:formattedNum, type: format.Type.FLOAT})
        }
        var rawNum = parseToValue(); // -20000.25 -- a number
    });
```

### Sample 3: Format a Number as a String

```
/**
 * @NApiVersion 2.x
 */
require(['N/format'],
    function(format){
        function formatToString() {
            // Assume number format is 1.000.000,00 and negative format is (100)
            var rawNum2 = -44444.44
            return format.format({value:rawNum2, type: format.Type.FLOAT})
        }
        var formattedNum2 = formatToString(); // "44.444,44" -- a string
    });
```

#### Sample 4: Format Time of Day as a String

```
/**
 * @NApiVersion 2.1
 */
require(['N/format'],
function(format){
  function formatTimeOfDay() {
    // Instantiate a new Date object assuming it's now 7:01PM.
    const now = new Date();
    // Select a format from the format.Type enum. In this case TIMEOFDAY
    return format.format({value: now, type: format.Type.TIMEOFDAY})
  }
  const formattedTime = formatTimeOfDay(); // "7:01 pm" -- a string
}); const formattedTime = formatTimeOfDay(); // "7:01 pm" -- a string
});
```

## 20. N/format/i18n Module

Load the N/format/i18n module to format currency.

```
define(['N/format/i18n'], (i18n) => { ... })
```

✓ 8 Code Sample(s)

Scripts: Client and server scripts

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/format/i18n'], (i18n) => { // Access methods and properties on i18n });
```

Use the N/format/i18n module to format strings in international context and format numbers to currency or number strings. You can also use this module to format phone number to strings and parse strings to phone number.

### N/format/i18n Module Members

Return Type / Value Type

Supported Script Types

format.CurrencyFormatter

Client and server scripts

Represents the object that formats the number to currency string.

format.NumberFormatter

| MEMBER TYPE | NAME                                    | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                                                        |
|-------------|-----------------------------------------|--------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT      | format.CurrencyFormatter                | Object                   | Client and server scripts | Represents the object that formats the number to currency string.                                                                                                  |
| OBJECT      | format.NumberFormatter                  | Object                   | Client and server scripts | Represents the object that formats the number to string.                                                                                                           |
| OBJECT      | format.PhoneNumberFormatter             | Object                   | Client and server scripts | Represents the object that formats the phone number to string.                                                                                                     |
| OBJECT      | format.PhoneNumberParser                | Object                   | Client and server scripts | Represents the object that parses the string to phone number.                                                                                                      |
| METHOD      | format.spellOut(options)                | string                   | Client and server scripts | Creates a string containing the spelled-out version of the specified number in a specified locale.                                                                 |
| METHOD      | format.getCurrencyFormatter(options)    | object                   | Client and server scripts | Creates a format.CurrencyFormatter object to format numbers to currency strings.                                                                                   |
| METHOD      | format.getNumberFormatter(options)      | object                   | Client and server scripts | Creates a format.NumberFormatter object to format numbers to strings.                                                                                              |
| METHOD      | format.getPhoneNumberFormatter(options) | object                   | Client and server scripts | Creates a format.PhoneNumberFormatter object to format phone numbers to strings.                                                                                   |
| METHOD      | format.getPhoneNumberParser(options)    | object                   | Client and server scripts | Parses a phone number from a string. Returns a format.PhoneNumberParser object.                                                                                    |
| ENUM        | format.NegativeNumberFormat             | enum                     | Client and server scripts | Holds the values for the negative number format. Used to set the value of the options.negativeNumberFormatter parameter of the format.getNumberFormatter(options). |
| ENUM        | format.Currency                         | enum                     | Client and server scripts | Holds the values for the currency code. Used to set the value of the options.currency parameter of the format.getCurrencyFormatter(options) method.                |

### Currency Formatter Object Members

The following members are called on the format.CurrencyFormatter object.

Return Type / Value Type

Supported Script Types

CurrencyFormatter.currency

Client and server scripts

Indicates the currency code.

| MEMBER TYPE | NAME                       | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                           |
|-------------|----------------------------|--------------------------|---------------------------|---------------------------------------|
| PROPERTY    | CurrencyFormatter.currency | string                   | Client and server scripts | Indicates the currency code.          |
| PROPERTY    | CurrencyFormatter.locale   | string                   | Client and server scripts | The locale of the currency formatter. |

| MEMBER TYPE | NAME                              | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                                          |
|-------------|-----------------------------------|--------------------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| PROPERTY    | CurrencyFormatter.symbol          | string                   | Client and server scripts | Indicates the currency symbol.                                                                                                                       |
| PROPERTY    | CurrencyFormatter.numberFormatter | object                   | Client and server scripts | Contains the format.NumberFormatter object derived from format.CurrencyFormatter with the same number formatting parameters without currency symbol. |
| PROPERTY    | CurrencyFormatter.format(options) | string                   | Client and server scripts | Formats the number to the currency string.                                                                                                           |

## Number Formatter Object Members

The following members are called on the format.NumberFormatter object.

Return Type / Value Type

Supported Script Types

NumberFormatter.groupSeparator

Client and server scripts

Indicates the group separator.

| MEMBER TYPE | NAME                             | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                           |
|-------------|----------------------------------|--------------------------|---------------------------|---------------------------------------|
| PROPERTY    | NumberFormatter.groupSeparator   | string                   | Client and server scripts | Indicates the group separator.        |
| PROPERTY    | NumberFormatter.decimalSeparator | string                   | Client and server scripts | Indicates the decimal separator.      |
| PROPERTY    | NumberFormatter.locale           | string                   | Client and server scripts | The locale of the number formatter.   |
| PROPERTY    | NumberFormatter.precision        | number                   | Client and server scripts | Indicates the precision.              |
| PROPERTY    | format.NegativeNumberFormat      | enum                     | Client and server scripts | Indicates the negative number format. |
| PROPERTY    | NumberFormatter.format(options)  | string                   | Client and server scripts | Formats the number to string.         |

## Phone Number Formatter Object Members

The following members are called on the format.PhoneNumberFormatter object.

Return Type / Value Type

Supported Script Types

format.PhoneNumberFormatter

Client and server scripts

The object that formats the phone number to string.

| MEMBER TYPE | NAME                                 | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                                                         |
|-------------|--------------------------------------|--------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT      | format.PhoneNumberFormatter          | Object                   | Client and server scripts | The object that formats the phone number to string.                                                                                                                 |
| METHOD      | PhoneNumberFormatter.format(options) | string                   | Client and server scripts | Formats phone number object to string.                                                                                                                              |
| ENUM        | format.PhoneNumberFormatType         | enum                     | Client and server scripts | Holds the values for the phone number format type. Used to set the value of the options.formatType parameter of the format.getPhoneNumberFormatter(options) method. |
| ENUM        | format.Country                       | enum                     | Client and server scripts | Hold the values for the countries. Used to set the value of the options.defaultCountry parameter in the format.getPhoneNumberParser(options) method.                |

| MEMBER TYPE | NAME                             | RETURN TYPE / VALUE TYPE   | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                        |
|-------------|----------------------------------|----------------------------|---------------------------|----------------------------------------------------|
| OBJECT      | format.PhoneNumberParser         | Object                     | Client and server scripts | The object that parses the string to phone number. |
| METHOD      | PhoneNumberParser.parse(options) | Object of type PhoneNumber | Client and server scripts | Parses string to phone number.                     |

 **Sample 1: Format 12345 as a German String**

```
/**
 * @NApiVersion 2.x
 */
require(['N/format/i18n'], function(format) {
    var spellOut = format.spellOut({
        number: 12345,
        locale: "DE"
    });
    // spellOut is 'zwölftausenddreihundertfünfundvierzig'
});
```

 **Sample 2: Format a Number as a String Using N/format/i18n**

```
/**
 * @NApiVersion 2.x
 */
require(['N/format/i18n'], function(format) {
    var numberFormatter = format.getNumberFormatter();

    var gs = numberFormatter.groupSeparator;    // gs is ','
    var ds = numberFormatter.decimalSeparator;  // ds is '.'
    var precision = numberFormatter.precision;  // precision is '2'
    var nnf = numberFormatter.negativeNumberFormat; // nnf is 'BRACKETS'

    var formatNum1 = numberFormatter.format({
        number: 12.53
    }); // formatNum1 is '12.53'

    var formatNum2 = numberFormatter.format({
        number: 12845.22
    }); // formatNum2 is '12,845.22'

    var formatNum3 = numberFormatter.format({
        number: -5421
    }); // formatNum3 is '(5,421.00)'

    var formatNum4 = numberFormatter.format({
        number: 0.00
    }); // formatNum4 is '0.00'

    var formatNum5 = numberFormatter.format({
        number: 0.3456789
    }); // formatNum5 is '0.35'
});
```

### Sample 3: Format Numbers as Currency Strings

```
/**
 * @NApiVersion 2.1
 */
require(['N/Format/i18n'], function(format) {
  var curFormatter = format.getCurrencyFormatter({
    currency: "EUR"
  });

  var curCur = curFormatter.currency;           // curCur is 'EUR'
  var numberFormat = curFormatter.numberFormatter;

  var curSymbol = curFormatter.symbol;          // curSymbol is '€'
  var curSeparator = numberFormat.groupSeparator; // curSeparator is ' '
  var curDecimalSep = numberFormat.decimalSeparator; // curDecimalSep is ','
  var curPrecision = numberFormat.precision;      // curPrecision is 2
  var curNNF = numberFormat.negativeNumberFormat; // curNNF is MINUS

  curFormatNum1 = curFormatter.format({
    number: 12.53
  }); // curFormat1 is '€12,53'

  curFormatNum2 = curFormatter.format({
    number: -5421
  }); // curFormat2 is '€-5 421,00'

  curFormatNum3 = curFormatter.format({
    number: 0.00
  }); // curFormat3 is '€0,00'

  curFormatNum4 = curFormatter.format({
    number: 0.3456789
  }); // curFormat4 is '€0,35'
});
```

### Sample 4: Parse a Czech Phone Number

```
/**
 * @NApiVersion 2.x
 */
require(['N/Format/i18n'], function(format) {
  var origNumberStr = "602547854ext.154";
  var pnParser = format.getPhoneNumberParser({
    defaultCountry: format.Country.CZECH_REPUBLIC
  });
  var phoneNumber = pnParser.parse({
    number: origNumberStr
  });

  /* after parsing:
   phoneNumber.countryCode is '420'
   phoneNumber.extension is '154'
   phoneNumber.nationalNumber is '602547854'
   phoneNumber.numberOfLeadingZeros is 1
   phoneNumber.carrierCode is ' '
   phoneNumber.rawInput is '602547854ext.154'
  */

  var pnFormatter = format.getPhoneNumberFormatter({});
  var strNumber = pnFormatter.format({
    number: phoneNumber
  }); // strNumber is '+420 602 547 854 ext. 154'
});
```

### Sample 5: Parse a U.S. Phone Number

```
/**
 * @NApiVersion 2.x
 */
require(['N/format/i18n'], function(format) {
  var origNumberStr = "7524105210ext.154";
  var pnParser = format.getPhoneNumberParser({
    defaultCountry: format.Country.UNITED_STATES
  });
  var phoneNumber = pnParser.parse({
    number: origNumberStr
  });

  /* after parsing:
    phoneNumber.countryCode is '1'
    phoneNumber.extension is '154'
    phoneNumber.nationalNumber is '7524105210'
    phoneNumber.numberofLeadingZeros is 1
    phoneNumber.carrierCode is ' '
    phoneNumber.rawInput is '7524105210ext.154'
  */

  var pnFormatter = format.getPhoneNumberFormatter({
    formatType: format.PhoneNumberFormatType.NATIONAL
  });
  var strNumber = pnFormatter.format({
    number: phoneNumber
  }); // strNumber is '(752) 410-5210 ext. 154'
});
```

### Sample 6: Format Numbers Based on the Locale Parameter

```
/**
 * @NApiVersion 2.x
 */
require(['N/format/i18n'], function(format) {
  var numberFormatter = format.getNumberFormatter({
    locale: "fr_FR"
  });

  var gs = numberFormatter.groupSeparator; // gs is ' '
  var ds = numberFormatter.decimalSeparator; // ds is ','
  var prec = numberFormatter.precision; // prec is 2
  var nnf = numberFormatter.negativeNumberFormat; // nnf is MINUS

  var number1 = numberFormatter.format({
    number: 123456.55
  }); // number1 is '123 456,55'

  var number2 = numberFormatter.format({
    number: -123456.55
  }); // number2 is '-123 456,55'
});
```

### Sample 7: Format Currency Based on the Locale Parameter

```
/**
 * @NApiVersion 2.1
 */
require(['N/format/i18n'], function(format) {
  var curFormatter = format.getCurrencyFormatter({
    locale: "IT_IT"
  });

  var curCur = curFormatter.currency; // curCur is 'EUR'
  var sym = curFormatter.symbol; // sym is '€'

  var numberFormat = curFormatter.numberFormatter;

  var gs = numberFormat.groupSeparator; // gs is ','
  var ds = numberFormat.decimalSeparator; // ds is '.'
  var prec = numberFormat.precision; // prec is 2
  var nnf = numberFormat.negativeNumberFormat; // nnf is MINUS

  var currency1 = curFormatter.format({
    number: 123456.55
  }); // currency1 is '€123.456,55'

  var currency2 = curFormatter.format({
    number: -123456.55
  }); // currency2 is '€-123.456,55'
});
```

## Sample 8: Format Numbers and Currencies Based on the English-India Locale Parameter

```
/**
 * @NApiVersion 2.1
 */
require(['N/format/i18n'], function(format) {
  var curFormatter = format.getCurrencyFormatter({
    locale: "en_IN"
  });

  var curCur = curFormatter.currency; // curCur is 'INR'

  var numberFormat = curFormatter.numberFormatter;
  var sym = curFormatter.symbol;           // sym is '₹'
  var gs = numberFormat.groupSeparator;    // gs is ','
  var ds = numberFormat.decimalSeparator;  // ds is '.'
  var prec = numberFormat.precision;       // prec is 2
  var nnf = numberFormat.negativeNumberFormat; // nnf is MINUS

  var currency1 = curFormatter.format({
    number: 12345678.55
  }); // currency1 is '₹1,23,45,678.55'

  var currency2 = curFormatter.format({
    number: -345678.55
  }); // currency2 is '₹-3,45,678.55'
});
```

## 21. N/http Module

Load the N/http module to make http calls.

define(['N/http'], (http) => { ... })

✓ 2 Code Sample(s)

Scripts: Client and server scripts

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/http'], (http) => { // Access methods and properties on http });
```

Use the N/http module to make HTTP calls from server or client scripts. For client scripts, this module also provides the ability to make cross-domain HTTP requests using NetSuite servers as proxies.

All HTTP content types are supported.

The N/http module does not accept the HTTPS protocol. Use the N/https Module for that purpose.

### HTTP Header Information

HTTP headers can be used to pass additional information with an HTTP request or response. Each HTTP header consists of its case-insensitive name followed by a colon (:), then by its value (without line breaks). If you use custom headers, make sure the names of these headers do not contain underscores. For a general list of all HTTP headers, visit <http://developer.mozilla.org/en-US/docs/Web/HTTP/Headers>.

Some headers are not supported in NetSuite and are blocked. These are listed below as either general HTTP headers or Suitelet response headers.

Be aware that certain headers cannot be set manually when using the N/http module methods. If a script attempts to set values for any of the following headers, the values are discarded. These headers are listed in the following table.

Transfer-Encoding

In addition to the headers described in General Blocked HTTP Headers, certain headers cannot be set manually when interacting with the http.ServerResponse Objects sent by Suitelets. If a script attempts to set values for any of these headers, the system throws an SSS\_INVALID\_HEADER error. These headers are listed in the following table.

Content-Location

|                                                       |                                                                                     |                                                                                     |  |
|-------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--|
| CONNECTION CONTENT-LENGTH HOST JSESSIONID             |                                                                                     | TRAILER TRANSFER-ENCODING UPGRADE VIA                                               |  |
| ALLOW CONTENT-LOCATION CONTENT-MD5 CONTENT-RANGE DATE | LOCATION PROXY-AUTHENTICATE PUBLIC-KEY-PINS PUBLIC-KEY-PINS-REPORT-ONLY RETRY-AFTER | SERVER STRICT-TRANSPORT-SECURITY UPGRADE-INSECURE-REQUESTS WARNING WWW-AUTHENTICATE |  |

### N/http Module Members

Return Type / Value Type

Supported Script Types

http.ClientResponse

Object (read-only)

The response from the server to an HTTP client request (for example, http.get(options)).

http.ServerRequest

| MEMBER TYPE | NAME                         | RETURN TYPE / VALUE TYPE                   | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                        |
|-------------|------------------------------|--------------------------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------|
| OBJECT      | http.ClientResponse          | Object (read-only)                         | Server scripts            | The response from the server to an HTTP client request (for example, http.get(options)).                           |
| OBJECT      | http.ServerRequest           | Object (read-only)                         | Server scripts            | HTTP request information sent to an HTTP server. For example, a request sent to/received by a Suitelet or RESTlet. |
| OBJECT      | http.ServerResponse          | Object                                     | Server scripts            | The response from an HTTP server to an HTTP request. For example, a response from a Suitelet or RESTlet.           |
| METHOD      | http.delete(options)         | http.ClientResponse or http.ServerResponse | Client and server scripts | Sends an HTTP DELETE request and returns the response.                                                             |
| METHOD      | http.delete.promise(options) | Promise Object                             | Client and server scripts | Sends an HTTP DELETE request asynchronously and returns the response.                                              |
| METHOD      | http.get(options)            | http.ClientResponse or http.ServerResponse | Client and server scripts | Sends an HTTP GET request and returns the response.                                                                |
| METHOD      | http.get.promise(options)    | Promise Object                             | Client and server scripts | Sends an HTTP GET request asynchronously and returns the response.                                                 |
| METHOD      | http.post(options)           | http.ClientResponse or http.ServerResponse | Client and server scripts | Sends an HTTP POST request and returns the response.                                                               |
| METHOD      | http.post.promise(options)   | Promise Object                             | Client and server scripts | Sends an HTTP POST request asynchronously and returns the response.                                                |

| MEMBER TYPE | NAME                                       | RETURN TYPE / VALUE TYPE                                             | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                                                                                     |
|-------------|--------------------------------------------|----------------------------------------------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | <code>http.put(options)</code>             | <code>http.ClientResponse</code> or <code>http.ServerResponse</code> | Client and server scripts | Sends an HTTP PUT request and returns the response.                                                                                                                                             |
| METHOD      | <code>http.put.promise(options)</code>     | Promise Object                                                       | Client and server scripts | Sends an HTTP PUT request asynchronously and returns the response.                                                                                                                              |
| METHOD      | <code>http.request(options)</code>         | <code>http.ClientResponse</code> or <code>http.ServerResponse</code> | Client and server scripts | Sends an HTTP request and returns the response.                                                                                                                                                 |
| METHOD      | <code>http.request.promise(options)</code> | Promise Object                                                       | Client and server scripts | Sends an HTTP request asynchronously and returns the response.                                                                                                                                  |
| ENUM        | <code>http.CacheDuration</code>            | enum                                                                 | Server scripts            | Holds the string values for supported cache durations. Use this enum to set the value of the type parameter in <code>ServerResponse.setCdnCacheable(options)</code> .                           |
| ENUM        | <code>http.Method</code>                   | enum                                                                 | Server scripts            | Holds the string values for supported HTTP requests. Use this enum to set the value of method parameter in <code>http.request(options)</code> .                                                 |
| ENUM        | <code>http.RedirectType</code>             | enum                                                                 | Server scripts            | Holds the string values for supported NetSuite resources that you can redirect to. Use this enum to set the value of the type parameter for <code>ServerResponse.sendRedirect(options)</code> . |

## ClientResponse Object Members

The following members are called on the `http.ClientResponse` Object.

Return Type / Value Type

Supported Script Types

`ClientResponse.body`

string (read-only)

The client response body.

| MEMBER TYPE | NAME                                | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                  |
|-------------|-------------------------------------|--------------------------|------------------------|------------------------------|
| PROPERTY    | <code>ClientResponse.body</code>    | string (read-only)       | Server scripts         | The client response body.    |
| PROPERTY    | <code>ClientResponse.code</code>    | number (read-only)       | Server scripts         | The client response code.    |
| PROPERTY    | <code>ClientResponse.headers</code> | Object (read-only)       | Server scripts         | The client response headers. |

## ServerRequest Object Members

The following members are called on the `http.ServerRequest` Object.

Return Type / Value Type

Supported Script Types

`ServerRequest.getLineCount(options)`

Returns the number of lines in a sublist.

`ServerRequest.getSublistValue(options)`

| MEMBER TYPE | NAME                                                | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                               |
|-------------|-----------------------------------------------------|--------------------------|------------------------|-------------------------------------------|
| METHOD      | <code>ServerRequest.getLineCount(options)</code>    | number                   | Server scripts         | Returns the number of lines in a sublist. |
| METHOD      | <code>ServerRequest.getSublistValue(options)</code> | string                   | Server scripts         | Returns the value of a sublist line item. |
| PROPERTY    | <code>ServerRequest.body</code>                     | string (read-only)       | Server scripts         | The server request body.                  |
| PROPERTY    | <code>ServerRequest.files</code>                    | Object (read-only)       | Server scripts         | The server request files.                 |
| PROPERTY    | <code>ServerRequest.headers</code>                  | Object (read-only)       | Server scripts         | The server request headers.               |
| PROPERTY    | <code>ServerRequest.clientIpAddress</code>          | String (read-only)       | Server scripts         | The remote client IP address.             |
| PROPERTY    | <code>ServerRequest.method</code>                   | <code>http.Method</code> | Server scripts         | The server request HTTP method.           |
| PROPERTY    | <code>ServerRequest.parameters</code>               | Object (read-only)       | Server scripts         | The server request parameters.            |
| PROPERTY    | <code>ServerRequest.url</code>                      | string (read-only)       | Server scripts         | The server request URL.                   |

## ServerResponse Object Members

The following members are called on the `http.ServerResponse` Object.

Return Type / Value Type

Supported Script Types

`ServerResponse.addHeader(options)`

Adds a header to the response.

ServerResponse.getHeader(options)

| MEMBER TYPE | NAME                                    | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                |
|-------------|-----------------------------------------|--------------------------|------------------------|------------------------------------------------------------|
| METHOD      | ServerResponse.addHeader(options)       | void                     | Server scripts         | Adds a header to the response.                             |
| METHOD      | ServerResponse.getHeader(options)       | string   string[]        | Server scripts         | Returns the value of a response header.                    |
| METHOD      | ServerResponse.renderPdf(options)       | void                     | Server scripts         | Generates and renders a PDF directly to the response.      |
| METHOD      | ServerResponse.sendRedirect(options)    | void                     | Server scripts         | Sets the redirect URL by resolving to a NetSuite resource. |
| METHOD      | ServerResponse.setCdnCacheable(options) | void                     | Server scripts         | Sets CDN caching for a period of time.                     |
| METHOD      | ServerResponse.setHeader(options)       | void                     | Server scripts         | Sets the value of a response header.                       |
| METHOD      | ServerResponse.write(options)           | void                     | Server scripts         | Writes information (text, xml, html) to the response.      |
| METHOD      | ServerResponse.writeFile(options)       | void                     | Server scripts         | Writes a file to the response.                             |
| METHOD      | ServerResponse.writeLine(options)       | void                     | Server scripts         | Writes line information (text, xml, html) to the response. |
| METHOD      | ServerResponse.writePage(options)       | void                     | Server scripts         | Generates a page.                                          |
| PROPERTY    | ServerResponse.headers                  | Object (read-only)       | Server scripts         | The server response headers.                               |

## ► Official NetSuite Script Samples (2 Sample(s))

### Sample 1: Request a URL

```
/**
 * @NApiVersion 2.1
 */

// This script uses an HTTP GET request for a URL.
require(['N/http'], (http)=> {
  function sendGetRequest() {
    let response = http.get({
      url: 'http://www.google.com'
    });
  }
  sendGetRequest();
});
```

### Sample 2: Redirect a record

```
/**
 * @NApiVersion 2.1
 * @NScriptType Suitelet
 */

// This script redirects a new sales order record and sets the entity.
define(['N/record', 'N/http'], (record, http)=> {
  function onRequest(context) {
    context.response.sendRedirect({
      type: http.RedirectType.RECORD,
      identifier: record.Type.SALES_ORDER,
      parameters: {
        entity: 6
      }
    });
  }
  return {
    onRequest: onRequest
  };
});
```

## 22. N/https Module

Load the N/https module to make https calls. You can also use this module to encode binary content or securely access a handle to the value in a NetSuite credential field.

define(['N/https'], (https) => { ... })

✓ 10 Code Sample(s)

Scripts: Client and server scripts

Permissions: -

```
/** * @ApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/https'], (https) => { // Access methods and properties on https });
```

Use the N/https module to manage content sent to a third party using HTTPS calls. This module encapsulates all the functionality of the N/http Module, but does not allow the HTTP protocol. You can make HTTPS calls from client and server scripts.

You can use the N/https module to encode binary content or access a handle to the value in a NetSuite credential field.

You can use the N/https module to communicate between SuiteScript scripts, RESTlets, and SuiteTalk REST APIs without having to reauthenticate, using the https.requestRestlet(options) and https.requestSuiteTalkRest(options) methods.

When the N/https module is used, SuiteScript also loads the N/crypto Module and N/encode Module.

Use TLS 1.2 for HTTPS requests. SuiteScript 2.0 requests such https.delete(options), https.get(options), https.post(options), https.put(options), and https.request(options) usually go to third-party servers. Management of these servers is not within the control of your company. These HTTPS requests now fail the handshake when they attempt to connect to servers that do not support TLS 1.2. You should communicate with those who manage any third-party servers to which you connect, and ensure their servers support the TLS 1.2 protocol.

NetSuite supports the same list of trusted third-party certificate authorities (CAs) as the Mozilla Included CA Certificate List.

The target endpoint, domain, or server must use one of these trusted third-party CAs, or the connection cannot be established. Oracle NetSuite requires that the endpoints that you are connecting to from NetSuite provide a full certification chain, including intermediate certificates.

For a list of certificate authorities, see [https://wiki.mozilla.org/CA/Included\\_Certificates](https://wiki.mozilla.org/CA/Included_Certificates).

Using plain text or other unencrypted user credentials is unsafe and can pose a security threat. Whenever possible, use Token-based Authentication (TBA) or OAuth 2.0 to specify user credentials.

### HTTPS Header Information

HTTP headers add extra information to a request or response. A header has a case-insensitive name, a colon (:), and a value on the same line. When you define custom headers, don't include underscores in the name. For a list of standard HTTP and HTTPS headers, see <https://developer.mozilla.org/docs/Web/HTTP/Headers>.

If you call https.post(options) without a Content-Type header, NetSuite sets a default value:

POST Request Body Type

Default Content-Type Header

application/octet-stream

application/x-www-form-urlencoded; charset=UTF-8

| POST REQUEST BODY TYPE | DEFAULT CONTENT-TYPE HEADER                      |
|------------------------|--------------------------------------------------|
| OBJECT                 | application/octet-stream                         |
| UINT8ARRAY             | application/x-www-form-urlencoded; charset=UTF-8 |
| OTHER TYPE             | text/xml; charset=UTF-8                          |

| CONNECTION CONTENT-LENGTH HOST JSESSIONID             |                                                                                     | TRAILER TRANSFER-ENCODING UPGRADE VIA                                               |
|-------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| ALLOW CONTENT-LOCATION CONTENT-MD5 CONTENT-RANGE DATE | LOCATION PROXY-AUTHENTICATE PUBLIC-KEY-PINS PUBLIC-KEY-PINS-REPORT-ONLY RETRY-AFTER | SERVER STRICT-TRANSPORT-SECURITY UPGRADE-INSECURE-REQUESTS WARNING WWW-AUTHENTICATE |

### N/https Module Members

Return Type / Value Type

Supported Script Types

https.SecureString

Encapsulates data that may be sent to a third-party using an HTTPS call.

https.ClientResponse

Object (read-only)

| MEMBER TYPE | NAME                                                | RETURN TYPE / VALUE TYPE                                               | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                                                                                                                                                   |
|-------------|-----------------------------------------------------|------------------------------------------------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT      | <code>https.SecureString</code>                     | Object                                                                 | Server scripts            | Encapsulates data that may be sent to a third-party using an HTTPS call.                                                                                                                                                                                      |
| OBJECT      | <code>https.ClientResponse</code>                   | Object (read-only)                                                     | Server scripts            | Encapsulates the response to an HTTPS client request.                                                                                                                                                                                                         |
| OBJECT      | <code>https.ServerRequest</code>                    | Object (read-only)                                                     | Server scripts            | Encapsulates the HTTPS request information sent to an HTTPS server. For example, a request received by a Suitelet or RESTlet.                                                                                                                                 |
| OBJECT      | <code>https.ServerResponse</code>                   | Object                                                                 | Server scripts            | Encapsulates the response from an HTTPS server to an HTTPS request. For example, a response from a Suitelet or RESTlet.                                                                                                                                       |
| METHOD      | <code>https.createSecretKey(options)</code>         | Object                                                                 | Server scripts            | Creates a key for the contents of a credential field.                                                                                                                                                                                                         |
| METHOD      | <code>https.createSecureString(options)</code>      | Object                                                                 | Server scripts            | Creates an <code>https.SecureString</code> Object.                                                                                                                                                                                                            |
| METHOD      | <code>https.delete(options)</code>                  | <code>https.ClientResponse</code> or <code>https.ServerResponse</code> | Client and server scripts | Sends an HTTPS DELETE request and returns the response.                                                                                                                                                                                                       |
| METHOD      | <code>https.delete.promise(options)</code>          | Promise Object                                                         | Client and server scripts | Sends an HTTPS DELETE request asynchronously and returns the response.                                                                                                                                                                                        |
| METHOD      | <code>https.get(options)</code>                     | <code>https.ClientResponse</code> or <code>https.ServerResponse</code> | Client and server scripts | Sends an HTTPS GET request and returns the response.                                                                                                                                                                                                          |
| METHOD      | <code>https.get.promise(options)</code>             | Promise Object                                                         | Client and server scripts | Sends an HTTPS GET request asynchronously and returns the response.                                                                                                                                                                                           |
| METHOD      | <code>https.post(options)</code>                    | <code>https.ClientResponse</code> or <code>https.ServerResponse</code> | Client and server scripts | Sends an HTTPS POST request and returns the response.                                                                                                                                                                                                         |
| METHOD      | <code>https.post.promise(options)</code>            | Promise Object                                                         | Client and server scripts | Sends an HTTPS POST request asynchronously and returns the response.                                                                                                                                                                                          |
| METHOD      | <code>https.put(options)</code>                     | <code>https.ClientResponse</code> or <code>https.ServerResponse</code> | Client and server scripts | Sends an HTTPS PUT request and returns the response.                                                                                                                                                                                                          |
| METHOD      | <code>https.put.promise(options)</code>             | Promise Object                                                         | Client and server scripts | Sends an HTTPS PUT asynchronously request and returns the response.                                                                                                                                                                                           |
| METHOD      | <code>https.request(options)</code>                 | <code>https.ClientResponse</code> or <code>https.ServerResponse</code> | Client and server scripts | Sends an HTTPS request and returns the response. If a request fails, an <code>error.SuiteScriptError</code> is thrown.                                                                                                                                        |
| METHOD      | <code>https.request.promise(options)</code>         | Promise Object                                                         | Client and server scripts | Sends an HTTPS request asynchronously and returns the response. If a request fails, a <code>Promise.reject</code> is thrown with a parameter <code>Error</code> .                                                                                             |
| METHOD      | <code>https.requestRestlet(options)</code>          | <code>https.ClientResponse</code>                                      | Server scripts            | Sends an HTTPS request to a RESTlet and returns the response. Authentication headers are automatically added. The RESTlet will run with the same privileges as the calling script.                                                                            |
| METHOD      | <code>https.requestRestlet.promise(options)</code>  | Promise Object                                                         | Client and server scripts | Sends an HTTPS request to a Restlet and returns the response.                                                                                                                                                                                                 |
| METHOD      | <code>https.requestSuitelet(options)</code>         | <code>https.ClientResponse</code>                                      | Client and server scripts | Sends an HTTPS request to a Suitelet and returns the response.                                                                                                                                                                                                |
| METHOD      | <code>https.requestSuitelet.promise(options)</code> | Promise Object                                                         | Client and server scripts | Sends an HTTPS request asynchronously to a Suitelet and returns the response.                                                                                                                                                                                 |
| METHOD      | <code>https.requestSuiteTalkRest(options)</code>    | <code>https.ClientResponse</code>                                      | Server scripts            | Sends an HTTPS request to a SuiteTalk REST endpoint and returns the response. Authentication headers are automatically added.                                                                                                                                 |
| ENUM        | <code>https.CacheDuration</code>                    | enum                                                                   | Server scripts            | Holds the string values for supported cache durations. Use this enum to set the value of the type parameter in <code>ServerResponse.setCdnCacheable(options)</code> .                                                                                         |
| ENUM        | <code>https.Encoding</code>                         | enum                                                                   | Server scripts            | Holds the string values for supported encoding types. Use this enum to set the value of parameters in <code>SecureString.appendString(options)</code> , <code>SecureString.convertEncoding(options)</code> , <code>https.createSecureString(options)</code> . |
| ENUM        | <code>https.HashAlg</code>                          | enum                                                                   | Server scripts            | Holds the string values for supported hashing algorithms. Use this enum to set the value of parameters in <code>SecureString.hash(options)</code> and <code>SecureString.hmac(options)</code> .                                                               |
| ENUM        | <code>https.Method</code>                           | enum                                                                   | Server scripts            | Holds the string values for supported HTTPS requests. Use this enum to set the value of method parameter in <code>https.request(options)</code> .                                                                                                             |
| ENUM        | <code>https.RedirectType</code>                     | enum                                                                   | Server scripts            | Holds the string values for supported NetSuite resources that you can redirect to. Use this enum to set the value of the type parameter for <code>ServerResponse.sendRedirect(options)</code> .                                                               |

## SecureString Object Members

SecureString functionality is supported only in server scripts.

The following members are called on the `https.SecureString` Object.

Return Type / Value Type

Supported Script Types

`SecureString.appendSecureString(options)`

`https.SecureString`

| MEMBER TYPE | NAME                                                  | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                         |
|-------------|-------------------------------------------------------|--------------------------|------------------------|-----------------------------------------------------------------------------------------------------|
| METHOD      | <code>SecureString.appendSecureString(options)</code> | https.SecureString       | Server scripts         | Appends one https.SecureString to another https.SecureString.                                       |
| METHOD      | <code>SecureString.appendString(options)</code>       | https.SecureString       | Server scripts         | Appends a string to a https.SecureString.                                                           |
| METHOD      | <code>SecureString.convertEncoding(options)</code>    | https.SecureString       | Server scripts         | Converts the content of a https.SecureString between two encodings.                                 |
| METHOD      | <code>SecureString.hash(options)</code>               | https.SecureString       | Server scripts         | Creates a hash for a https.SecureString.                                                            |
| METHOD      | <code>SecureString.hmac(options)</code>               | https.SecureString       | Server scripts         | Creates an hmac for a https.SecureString.                                                           |
| METHOD      | <code>SecureString.replaceString(options)</code>      | https.SecureString       | Server Scripts         | Replaces all occurrences of a pattern string inside a https.SecureString with a replacement string. |

## ClientResponse Object Members

The following members are called on the https.ClientResponse Object.

Return Type / Value Type

Supported Script Types

ClientResponse.body

string (read-only)

The response body.

| MEMBER TYPE | NAME                                | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION        |
|-------------|-------------------------------------|--------------------------|------------------------|--------------------|
| PROPERTY    | <code>ClientResponse.body</code>    | string (read-only)       | Server scripts         | The response body. |
| PROPERTY    | <code>ClientResponse.code</code>    | number (read-only)       | Server scripts         | The response code. |
| PROPERTY    | <code>ClientResponse.headers</code> | Object (read-only)       | Server scripts         | The response body. |

## ServerRequest Object Members

The following members are called on the https.ServerRequest Object.

Return Type / Value Type

Supported Script Types

ServerRequest.getLineCount(options)

Returns the number of lines in a sublist.

ServerRequest.getSublistValue(options)

| MEMBER TYPE | NAME                                                | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                          |
|-------------|-----------------------------------------------------|--------------------------|------------------------|----------------------------------------------------------------------|
| METHOD      | <code>ServerRequest.getLineCount(options)</code>    | number                   | Server scripts         | Returns the number of lines in a sublist.                            |
| METHOD      | <code>ServerRequest.getSublistValue(options)</code> | string                   | Server scripts         | Returns the value of a sublist line item.                            |
| PROPERTY    | <code>ServerRequest.body</code>                     | string (read-only)       | Server scripts         | The server request body                                              |
| PROPERTY    | <code>ServerRequest.files</code>                    | Object (read-only)       | Server scripts         | The server request files represented as object in ID-file.File pair. |
| PROPERTY    | <code>ServerRequest.headers</code>                  | Object (read-only)       | Server scripts         | The server request headers.                                          |
| PROPERTY    | <code>ServerRequest.method</code>                   | https.Method             | Server scripts         | The HTTPS method for the server request.                             |
| PROPERTY    | <code>ServerRequest.parameters</code>               | Object (read-only)       | Server scripts         | The server request parameters.                                       |
| PROPERTY    | <code>ServerRequest.url</code>                      | string (read-only)       | Server scripts         | The server request URL.                                              |

## ServerResponse Object Members

The following members are called on the https.ServerResponse Object.

Return Type / Value Type

Supported Script Types

ServerResponse.addHeader(options)

Adds a header to the response.

ServerResponse.getHeader(options)

| MEMBER TYPE | NAME                                    | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                |
|-------------|-----------------------------------------|--------------------------|------------------------|------------------------------------------------------------|
| METHOD      | ServerResponse.addHeader(options)       | void                     | Server scripts         | Adds a header to the response.                             |
| METHOD      | ServerResponse.getHeader(options)       | string   string[]        | Server scripts         | Returns the value of a response header.                    |
| METHOD      | ServerResponse.renderPdf(options)       | void                     | Server scripts         | Generates and renders a PDF directly to the response.      |
| METHOD      | ServerResponse.sendRedirect(options)    | void                     | Server scripts         | Sets the redirect URL by resolving to a NetSuite resource. |
| METHOD      | ServerResponse.setCdnCacheable(options) | void                     | Server scripts         | Sets CDN caching for a period of time.                     |
| METHOD      | ServerResponse.setHeader(options)       | void                     | Server scripts         | Sets the value of a response header.                       |
| METHOD      | ServerResponse.write(options)           | void                     | Server scripts         | Writes information (text/xml/html) to the response.        |
| METHOD      | ServerResponse.writeFile(options)       | void                     | Server scripts         | Writes a file to the response.                             |
| METHOD      | ServerResponse.writeLine(options)       | void                     | Server scripts         | Writes line information (text/xml/html) to the response.   |
| METHOD      | ServerResponse.writePage(options)       | void                     | Server scripts         | Generates a page.                                          |
| PROPERTY    | ServerResponse.headers                  | Object (read-only)       | Server scripts         | The server response headers.                               |

### ► Official NetSuite Script Samples (10 Sample(s))

#### Sample 1: Generate a Secure Token and a Secret Key

```
/**
 * @NApiVersion 2.1
 */

// This script uses a GUID to generate a secure token and a secret key.
require(['N/https', 'N/runtime'], (https, runtime) => {
  function createSecureString() {
    const passwordGuid = '{284CFB2D225B1076FB94D150207E49DF}';
    let secureToken = https.createSecureString({
      input: passwordGuid
    });
    let secretKey = https.createSecretKey({
      input: passwordGuid
    });
    secureToken = secureToken.hmac({
      algorithm: https.HashAlg.SHA256,
      key: secretKey
    });
  }
  createSecureString();
});
```

## Sample 2: Create a Form with a Credential Field

```
/**
 * @NApiVersion 2.1
 * @NScriptType Suitelet
 */

// This script creates a form with a credential field.
define(['N/ui/serverWidget', 'N/https', 'N/url'], (serverWidget, https, url) => {
  function onRequest(context) {
    if (context.request.method === 'GET') {
      const form = serverWidget.createForm({
        title: 'Password Form'
      });

      const credField = form.addCredentialField({
        id: 'password',
        label: 'Password',
        restrictToDomains: ['&lt;accountID&gt;.app.netsuite.com'],
        restrictToCurrentUser: false,
        restrictToScriptIds: 'customscript_my_script'
      });

      credField.maxLength = 32;

      form.addSubmitButton();

      context.response.writePage({
        pageObject: form
      });
    }
    else {
      // Request to an existing Suitelet with credentials
      let passwordGuid = context.request.parameters.password;

      // Replace SCRIPTID and DEPLOYMENTID with the internal ID of the suitelet script and deployment in your account
      let baseUrl = url.resolveScript({
        scriptId: SCRIPTID,
        deploymentId: DEPLOYMENTID,
        returnExternalURL: true
      });

      let authUrl = baseUrl + '&pwd={ ' + passwordGuid + ' }';

      let secureStringUrl = https.createSecureString({
        input: authUrl
      });

      let headers = ({
        'pwd': passwordGuid
      });

      let response = https.post({
        credentials: [passwordGuid],
        url: secureStringUrl,
        body: {authorization: ' ' + passwordGuid + ', data: anything can be here'},
        headers: headers
      });
    }
  }
  return {
    onRequest: onRequest
  };
});
```

### Sample 3: Create an Authentication Header Using a Secure String

```
/**
 * @NApiVersion 2.1
 * @NScriptType Suitelet
 */

// This script creates an authentication header using an https.SecureString.
define(['N/https', 'N/encode'], (https, encode) => {
    function onRequest(context) {

        // Secrets with these two Script IDs must be existing and allowed for this script
        const nameToken = "custsecret_myName";
        const passwordToken = "custsecret_mypPassword";

        // Create BASE-64 encoded name:password pair
        const secStringKeyInBase64 = https.createSecureString({
            input: "{" + nameToken + "}:{ " + passwordToken + "}"
        });

        secStringKeyInBase64.convertEncoding({
            toEncoding: encode.Encoding.BASE_64,
            fromEncoding: encode.Encoding.UTF_8
        });

        // Construct the Authorization header
        const secStringBasicAuthHeader = https.createSecureString({
            input: "Basic "
        });

        secStringBasicAuthHeader.appendSecureString({
            secureString: secStringKeyInBase64,
            keepEncoding: true
        });

        // Send the request to third party with the Authorization header
        const resp = https.get({
            url: "myUrl",
            headers: {
                "Authorization": secStringBasicAuthHeader
            }
        });
    };
    return {
        onRequest: onRequest
    };
});
```

### Sample 4: Concatenate API Secrets with Strings

```
/**
 * @NApiVersion 2.1
 */

// This script uses appendSecureString to concatenate strings to use as an API secret.
require(['N/https', 'N/runtime'], (https, runtime) => {
    function concatToCreateSecureString() {
        let baseUrl = https.createSecureString({
            input: 'www.someurl.com/add?apikey='
        });
        let apiKey = https.createSecureString({
            input: '{CUSTSECRET_SOME_INTEGRATION}'
        });
        let url = baseUrl.appendSecureString({
            secureString: apiKey
        });
    }
    concatToCreateSecureString();
});
```

## Sample 5: Create a JWT Token Using a SecureString

```
/**
 * @NApiVersion 2.1
 * @NScriptType Suitelet
 */

// This script creates a JWT token using https.SecureString.
define(['N/https', 'N/encode'], (https, encode) => {
  function onRequest(context) {
    let nameToken = "custsecret_myName";
    let passwordToken = "custsecret_myPassword";
    let headerObj = {
      "alg": "HS256",
      "typ": "JWT"
    }
    let payloadObj = {
      "sub": "1234567890",
      "name": "John Doe",
      "iat": 1516239002
    }

    let headerJSON = JSON.stringify(headerObj);
    let payloadJSON = JSON.stringify(payloadObj);
    let headerBASE64 = encode.convert({
      string: headerJSON,
      inputEncoding: encode.Encoding.UTF_8,
      outputEncoding: encode.Encoding.BASE_64_URL_SAFE
    });

    let payloadBASE64 = encode.convert({
      string: payloadJSON,
      inputEncoding: encode.Encoding.UTF_8,
      outputEncoding: encode.Encoding.BASE_64_URL_SAFE
    });

    headerBASE64 = headerBASE64.replace(/=/g, ""); // remove = padding as per JWT spec 'base64UrlEncode' - URL-safe BASE-64 without padding
    payloadBASE64 = payloadBASE64.replace(/=/g, ""); // remove = padding as per JWT spec 'base64UrlEncode' - URL-safe BASE-64 without padding

    let secStringJwtSignature = https.createSecureString({
      input: headerBASE64 + "." + payloadBASE64
    })
    .hmac({
      algorithm: https.HashAlg.SHA256,
      key: https.createSecretKey({
        secret: passwordToken,
        encoding: encode.Encoding.UTF_8
      }),
      resultEncoding: encode.Encoding.BASE_64_URL_SAFE
    })
    .replaceString({ // remove = padding as per JWT spec 'base64UrlEncode' - URL-safe BASE-64 without padding
      pattern: "=",
      replacement: ""
    })

    let secStringJwtAuthHeader = https.createSecureString({
      input: "Bearer " + headerBASE64 + "." + payloadBASE64 + "."
    })
    .appendSecureString({
      secureString: secStringJwtSignature,
      keepEncoding: true
    })

    // Reflect the response using a echo-request suitelet
    let resp = https.get({
      url: "myURL",
      headers: {
        "Authorization": secStringJwtAuthHeader
      }
    });

    {
      log.debug("resp-code", resp.code);
      log.debug("resp-body", resp.body);

      let respAuth = JSON.parse(resp.body)["headers"]["Authorization"];

      log.debug("reps-head-auth", respAuth);
      log.debug("reps-head-auth-expected",
        "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiIxMjMONTY3ODkwIiwibmFtZSI6IkpvaG4gR69LIiwiaWF0IjoxNTE2MjM5MDIyfQ.ue13RLILSJ9Q9W2Gomh8vAJQAgdbnd6TS4b7plyF0tA" ); // see https://jwt.io/#debugger-io
    }
  }
  return {
    onRequest: onRequest
  };
});
```

## Sample 6: Retrieve Employee Information Using a Suitelet and a RESTlet Script with a Defined Content-Type Header

```
/**
 * @NApiVersion 2.1
 * @NScriptType Suitelet
 */
define(['N/https'],

    (https) => {
        /**
         * Defines the Suitelet script trigger point.
         * @param {Object} scriptContext
         * @param {ServerRequest} scriptContext.request - Incoming request
         * @param {ServerResponse} scriptContext.response - Suitelet response
         * @since 2015.2
         */
        const onRequest = (scriptContext) => {
            // Get the employee ID from the parameters
            const employeeId = scriptContext.request.parameters.employeeId || '0';

            // Define output variable
            let output = '';

            // Retrieve the employee name
            const restletResponse = https.requestRestlet({
                body: JSON.stringify({ employeeId }),
                deploymentId: 'customdeploy_samples_rs_requestrestlet',
                scriptId: 'customscript_samples_rs_requestrestlet',
                headers: { 'Content-Type': 'application/json' },
                method: 'POST'
            });

            // Check if the RESTlet call is successful
            if (restletResponse.code === 200) {
                if (restletResponse.body === '') {
                    // Parse the RESTlet response
                    const restletBody = JSON.parse(restletResponse.body);

                    // Write the suitelet output
                    output = 'Employee Name is: ${restletBody.name}';
                } else {
                    // Write error message.
                    output = 'No employee found.';
                }
            } else {
                // Write error message.
                output = 'Unexpected error. Please check script logs.';
            }

            // Write output to response object.
            scriptContext.response.write(output);
        }

        return {onRequest}
    });
```

### Sample 7: Retrieve Employee Information Using a Suitelet and a RESTlet Script with a Defined Content-Type Header

```
/*
 * Copyright (c) 2022, Oracle and/or its affiliates.
 */

/**
 * @NApiVersion 2.1
 * @NScriptType Restlet
 */
define(['/N/scriptTypes/restlet', 'N/query'], function (restlet, query) {
    // Restlet entry point
    const post = function (requestBody) {
        // Define the output variable
        let returnValue = '';

        // Create the Employee Query
        const qObj = query.create({
            type: query.Type.EMPLOYEE
        });

        // Check if the requestBody and the employeeId field is defined
        if (!!requestBody && !!requestBody.employeeId) {
            // Define the query condition for employee ID field in Employee record
            qObj.condition = qObj.createCondition({
                fieldId: 'id',
                operator: query.Operator.EQUAL,
                values: requestBody.employeeId
            });

            // Add the desired columns in the Query
            qObj.columns = [
                qObj.createColumn({
                    fieldId: 'firstname'
                }),
                qObj.createColumn({
                    fieldId: 'lastname'
                })
            ];

            // Retrieve the results
            const results = qObj.run().asMappedResults();
            for (let i = 0; i < results.length; i++) {
                const result = results[i];
                returnValue = {
                    name: result.firstname + ' ' + (result.lastname || '')
                };
            }
        }

        return returnValue;
    }

    return { post };
});
```

### Sample 8: Call a Suitelet from a Client Script

```
/**
 * @NApiVersion 2.1
 * @NScriptType UserEventScript
 */

// This script adds a button to a form and calls a custom function defined in a client script.
define(['/N/ui/serverWidget'],

function (serverWidget) {
    function beforeLoad(scriptContext) {

        // Add a button that calls the callSuitelet function
        scriptContext.form.addButton({
            id: 'custpage_call_suitelet_button',
            label: 'Click to Open Suitelet',
            functionName: 'callSuitelet'
        });

        // Specify the path to the client script where the callSuitelet function is defined
        scriptContext.form.clientScriptModulePath = './cs_request_suitelet.js' // Replace with the path to your client script
    }

    return {beforeLoad}
});
```

#### Sample 9: Call a Suitelet from a Client Script

```
/**
 * @NApiVersion 2.x
 * @NScriptType ClientScript
 * @NModuleScope SameAccount
 */

// This script defines a custom function to call a Suitelet.
define(['N/https', 'N/ui/dialog'],

function(https, dialog) {
    function pageInit(scriptContext) {}

    // Create a custom function that calls a Suitelet and displays the response in a dialog
    function callSuitelet() {
        const response = https.requestSuitelet({
            scriptId: 'customscript_sl_plf_requestsuitelet', // Replace with your Suitelet script ID
            deploymentId: 'customdeploy_sl_plf_requestsuitelet', // Replace with your Suitelet deployment ID
        });
        dialog.alert({
            title: 'Suitelet Response',
            message: response.body
        });
    }

    return {
        pageInit: pageInit,
        callSuitelet: callSuitelet
    };
});
```

#### Sample 10: Call a Suitelet from a Client Script

```
/**
 * @NApiVersion 2.1
 * @NScriptType Suitelet
 */

// This Suitelet shows the text 'Hello World!' in plain text.
define([],

function() {
    function onRequest(context) {
        context.response.setHeader({name: 'Content-Type', value: 'text/plain'});
        context.response.write('Hello World!');
    }

    return {onRequest};
});
```

## 23. N/https/clientCertificate Module

Load the N/https/clientCertificate module to send SSL requests with a digital certificate.

define(['N/https/clientCertificate'], (clientCertificate) => { ... })

✓ 1 Code Sample(s)

Scripts: Server scripts

Permissions: Certificate Access

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/https/clientCertificate'], (clientCertificate) => { // Access methods and properties on clientCertificate });
```

Use the clientCertificate module to send SSL requests with a digital certificate.

N/https/clientCertificate Module Script Sample

### N/https/clientCertificate Module Members

Return Type/Value Type

Supported Script Types

clientCertificate.post(options)

https.ClientResponse

Sends a SSL secured POST request to a remote server.

clientCertificate.get(options)

| MEMBER TYPE | NAME                               | RETURN TYPE/VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                             |
|-------------|------------------------------------|------------------------|------------------------|---------------------------------------------------------|
| METHOD      | clientCertificate.post(options)    | https.ClientResponse   | Server scripts         | Sends a SSL secured POST request to a remote server.    |
| METHOD      | clientCertificate.get(options)     | https.ClientResponse   | Server scripts         | Sends a SSL secured GET request to a remote server.     |
| METHOD      | clientCertificate.put(options)     | https.ClientResponse   | Server scripts         | Sends a SSL secured PUT request to a remote server.     |
| METHOD      | clientCertificate.delete(options)  | https.ClientResponse   | Server scripts         | Sends a SSL secured DELETE request to a remote server.  |
| METHOD      | clientCertificate.request(options) | https.ClientResponse   | Server scripts         | Sends a SSL secured REQUEST request to a remote server. |

### ► Official NetSuite Script Samples (1 Sample(s))

#### Sample 1: N/https/clientCertificate Module Script Sample

```
/**
 * @NApiVersion 2.1
 */

require(['N/https/clientCertificate'], (cert) => {
  // Set the URL
  const url = "https://nfe.fazenda.sp.gov.br/ws/cadconsultacadaastro4.asmx";

  let data = "<?xml version='1.0' encoding='utf-8'><soapenv:Envelope xmlns:soapenv='http://www.w3.org/2003/05/soap-envelope'><soapen
v:Body><ns1:nfeDadosMsg xmlns:ns1='http://www.portalfiscal.inf.br/nfe/wsd/CadConsultaCadaastro4'><ConsCad xmlns='http://www.portalfiscal.inf.br/nfe' versao='2.00'><infCons><xServ>CONS-CAD</xServ><UF>SP</UF><CNPJ>47508411000156</CNPJ></infCons> &
lt;/ConsCad></ns1:nfeDadosMsg></soapenv:Body></soapenv:Envelope>";

  const key = "custcertificate1";

  let headers = {
    "Content-Type": "application/soap+xml"
  };

  let response = cert.post({
    url: url,
    certId: key,
    body: data,
    headers: headers
  });
});
```

## 24. N/keyControl Module

Load the N/keyControl module to access key storage.

```
define(['N/keyControl'], (keyControl) => { ... })
```

✓ 2 Code Sample(s)

Scripts: Server scripts

Permissions: Key Management

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/keyControl'], (keyControl) => { // Access methods and properties on keyControl });
```

Use the N/keyControl module to use SSH keys and access key storage. You can also access keys in the UI at Setup > Company > Preferences > Keys.

By using the SSH keys, you can manage files and directories by using the SSH file transfer (SFTP) protocol. For more information, see SSH Keys for SFTP. For more information about SFTP, see N/sftp Module.

### N/keyControl Module Members

Return Type / Value Type

Supported Script Types

Represents the key object.

keyControl.findKeys(options)

Searches and returns a list of keys based on criteria set. If no options are set for criteria, the full list of keys stored in NetSuite is returned.

keyControl.createKey(options)

| MEMBER TYPE | NAME                          | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                          |
|-------------|-------------------------------|--------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT      | keyControl.Key                | Object                   | Server scripts         | Represents the key object.                                                                                                                           |
| METHOD      | keyControl.findKeys(options)  | Object                   | Server scripts         | Searches and returns a list of keys based on criteria set. If no options are set for criteria, the full list of keys stored in NetSuite is returned. |
| METHOD      | keyControl.createKey(options) | keyControl.Key           | Server scripts         | Creates a key.                                                                                                                                       |
| METHOD      | keyControl.deleteKey(options) | Object                   | Server scripts         | Marks the key as deleted in database. The history is retained.                                                                                       |
| METHOD      | keyControl.loadKey(options)   | keyControl.Key           | Server scripts         | Loads a key.                                                                                                                                         |
| METHOD      | keyControl.lock(options)      | string                   | Server scripts         | Locks a key so that it cannot be edited in the UI.                                                                                                   |
| METHOD      | keyControl.unlock(options)    | string                   | Server scripts         | Unlocks a key that has been locked by keyControl.lock(options).                                                                                      |
| ENUM        | keyControl.Operator           | enum                     | Server scripts         | Holds the values for the key operators of keyControl.findKeys(options).                                                                              |

### Key Object Members

The following members are called on the keyControl.Key object.

Return Type / Value Type

Supported Script Types

File object of the key.

Password of the key (write-only).

You can create a GUID using Form.addSecretKeyField(options). This property also accepts the script ID of an API secret stored at Setup > Company > API Secrets.

| MEMBER TYPE | NAME             | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                                       |
|-------------|------------------|--------------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PROPERTY    | Key.file         | file.File                | Server scripts         | File object of the key.                                                                                                                                                                           |
| PROPERTY    | Key.password     | string                   | Server scripts         | Password of the key (write-only). You can create a GUID using Form.addSecretKeyField(options). This property also accepts the script ID of an API secret stored at Setup > Company > API Secrets. |
| PROPERTY    | Key.scriptId     | string                   | Server scripts         | Script ID of the key. NetSuite prepends this ID with custkey.                                                                                                                                     |
| PROPERTY    | Key.name         | string                   | Server scripts         | Name of the key.                                                                                                                                                                                  |
| PROPERTY    | Key.description  | string                   | Server scripts         | Description of the key.                                                                                                                                                                           |
| PROPERTY    | Key.restrictions | string                   | Server scripts         | The internal IDs of the employees selected in the Restrict to Employees field of the key record.                                                                                                  |
| METHOD      | Key.save()       | Object                   | Server scripts         | Saves the key.                                                                                                                                                                                    |

 **Sample 1: Create a Secret Key**

```
/**
 * @NApiVersion 2.x
 */
require(['N/keyControl','N/file'],function(keyControl,file){
    var key = keyControl.createKey();
    key.file = file.load(422);
    //id of file containing private key (id_ecdsa or id_rsa)
    key.name = "SFTP key";
    key.save();
})
```

 **Sample 2: Add a Secret Key Field to a Form**

```
/**
 * @NApiVersion 2.1
 * @NScriptType Suitelet
 */
define(['N/ui/serverWidget', 'N/file', 'N/keyControl', 'N/runtime'], function(serverWidget, file, keyControl, runtime) {
    function onRequest(context) {
        var request = context.request;
        var response = context.response;

        if (request.method === 'GET') {
            var form = serverWidget.createForm({
                title: 'Enter Password'
            });

            var credField = form.addSecretKeyField({
                id: 'custfield_password',
                label: 'Password',
                restrictToScriptIds: [runtime.getCurrentScript().id],
                restrictToCurrentUser: true //Depends on use case
            });
            credField.maxLength = 64;

            form.addSubmitButton();
            response.writePage(form);
        } else {
            // Read the request parameter matching the field ID we specified in the form
            var passwordToken = request.parameters.custfield_password;

            var pem = file.load({
                id: 422
            });

            var key = keyControl.createKey();
            key.file = pem;
            key.name = 'Test';
            key.password = passwordToken;
            key.save();
        }
    }
    return {
        onRequest: onRequest
    };
});
```

## 25. N/llm Module

Load the N/llm module to use generative artificial intelligence (AI) capabilities. You can use this module to send requests to the large language models (LLMs) supported by NetSuite and to receive LLM responses to use in your scripts.

```
define(['N/llm'], (llm) => { ... })
```

✓ 9 Code Sample(s)

Scripts: Server scripts SuiteScript 2.1 only

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/llm'], (llm) => { // Access methods and properties on llm });
```

The content in this help topic pertains to SuiteScript 2.1.

The N/llm module supports generative artificial intelligence (AI) capabilities in SuiteScript. You can use this module to send requests to the large language models (LLMs) supported by NetSuite and receive responses to use in your scripts.

Methods in this module that send requests to an LLM consume AI Units. For more information, see [NetSuite AI Units](#) and [NetSuite Features and AI Units FAQ](#).

If you're new to using generative AI in SuiteScript, see [SuiteScript 2.1 Generative AI APIs](#). That topic contains essential information about this feature.

The following list summarizes the main features that are available using the N/llm module:

**Content generation** - You can request generative AI content from a supported LLM using `llm.generateText(options)`. You can provide a prompt that describes the content you want to generate, and the module sends the request to the Oracle Cloud Infrastructure (OCI) Generative AI service to generate a response.

**Prompt evaluation** - If you use Prompt Studio to manage existing prompts in your NetSuite account, you can use `llm.evaluatePrompt(options)` to send a prompt from Prompt Studio to the LLM for evaluation. This method uses the information from the prompt definition in Prompt Studio (such as the model and model parameters), and it lets you provide values for any variables the prompt uses before sending it for evaluation. For more information about Prompt Studio, see [Prompt Studio](#).

**Prompt and Text Enhance action management** - By using the N/record module, you can create, update, and delete prompts and Text Enhance actions in your scripts. For more information, see [Managing Prompts and Text Enhance Actions Using the N/llm Module](#).

**Retrieval-augmented generation (RAG) support** - You can give source documents to the LLM when calling `llm.generateText(options)`. The LLM uses information from the source documents to augment its response. The LLM also returns citations that identify which source documents it used. For an example of how to implement a RAG use case using the N/llm module, see [Provide Source Documents When Calling the LLM](#).

**Image support** - When using the Cohere Command A Vision model, you can send an image to the LLM when calling `llm.generateText(options)` or `llm.generateTextStreamed(options)`. You can ask questions about the image and receive information such as advanced image captions, a detailed description of the image, or information about charts and graphs in the image.

**Embedding support** - The `llm.embed(options)` method converts text to vector embeddings. Your SuiteScript applications can use vector embeddings for use cases such as semantic searches, recommender systems, text classification, or text clustering. For an example of how to generate and use embeddings, see [Find Similar Items Using Embeddings](#). For more information about embedding models, refer to [Offered Pretrained Foundational Models in Generative AI in the Oracle Cloud Infrastructure Documentation](#).

Embeddings include the semantic information of the original data and should be treated with the same level of sensitivity. Embeddings created from sensitive data must be protected, stored, logged, shared, retained, and deleted according to the same rules as the original data.

**Streaming support** - Using the `llm.generateTextStreamed(options)` and `llm.evaluatePromptStreamed(options)` methods, your code gets content as the LLM generates it, instead of waiting to receive it all at the same time. For an example of how to work with streamed content, see [Receive a Partial Response from the LLM](#).

**Tooling support** - You can define custom tools using `llm.createTool(options)` that let you extend and enhance LLM interactions. Tools can run business-specific logic in your scripts (such as SuiteQL queries), receive inputs from the LLM, and return structured outputs. The LLM can request tool calls as needed and integrate tool results into its generated responses, which can provide richer and more dynamic responses.

For more information, see [Tooling in the N/llm Module](#). For an example of how to create tools and provide them to the LLM, see [Use Tools to Enhance the Response from the LLM](#).

**Method aliases** - You can use the following aliases in your code instead of the method names:

`llm.chat(options)` is an alias for `llm.generateText(options)`.

`llm.executePrompt(options)` is an alias for `llm.evaluatePrompt(options)`.

`llm.chatStreamed(options)` is an alias for `llm.generateTextStreamed(options)`.

`llm.executePromptStreamed(options)` is an alias for `llm.evaluatePromptStreamed(options)`.

Promise versions are also available for these methods.

When aliases are available for methods, you'll see them listed in the main table of the method's help topic. For an example, see `llm.generateTextStreamed(options)`.

This module is available in NetSuite by default when the Server SuiteScript feature is enabled. For more information, see [Enabling Features](#).

As you work with this module, keep the following considerations in mind:

Generative AI features, such as the N/llm module, use creativity in their responses. Make sure you validate the AI-generated responses for accuracy and quality. Oracle NetSuite isn't responsible or liable for the use or interpretation of AI-generated content.

### N/llm Module Members

Return Type / Value Type

Supported Script Types

The chat message.

A citation returned from the LLM.

A document to be used as source content when calling the LLM.

| MEMBER TYPE | NAME                                        | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                             |
|-------------|---------------------------------------------|--------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT      | llm.ChatMessage                             | Object                   | Server scripts         | The chat message.                                                                                                                                                                       |
| OBJECT      | llm.Citation                                | Object                   | Server scripts         | A citation returned from the LLM.                                                                                                                                                       |
| OBJECT      | llm.Document                                | Object                   | Server scripts         | A document to be used as source content when calling the LLM.                                                                                                                           |
| OBJECT      | llm.EmbedResponse                           | Object                   | Server scripts         | The embeddings response returned from the LLM.                                                                                                                                          |
| OBJECT      | llm.Response                                | Object                   | Server scripts         | The response returned from the LLM.                                                                                                                                                     |
| OBJECT      | llm.StreamedResponse                        | Object                   | Server scripts         | The streamed response returned from the LLM.                                                                                                                                            |
| OBJECT      | llm.Tool                                    | Object                   | Server scripts         | A tool the LLM can request.                                                                                                                                                             |
| OBJECT      | llm.ToolCall                                | Object                   | Server scripts         | A tool call request from the LLM.                                                                                                                                                       |
| OBJECT      | llm.ToolParameter                           | Object                   | Server scripts         | A parameter for a tool.                                                                                                                                                                 |
| OBJECT      | llm.ToolResult                              | Object                   | Server scripts         | A tool result to send back to the LLM.                                                                                                                                                  |
| OBJECT      | llm.Usage                                   | Object                   | Server scripts         | The token usage for a request to the LLM.                                                                                                                                               |
| METHOD      | llm.createChatMessage(options)              | llm.ChatMessage          | Server scripts         | Creates a chat message based on a specified role and text.                                                                                                                              |
| METHOD      | llm.createDocument(options)                 | llm.Document             | Server scripts         | Creates a document to be used as source content when calling the LLM.                                                                                                                   |
| METHOD      | llm.createTool(options)                     | llm.Tool                 | Server scripts         | Creates a tool the LLM can request.                                                                                                                                                     |
| METHOD      | llm.createToolParameter(options)            | llm.ToolParameter        | Server scripts         | Creates a tool parameter for a tool.                                                                                                                                                    |
| METHOD      | llm.createToolResult(options)               | llm.ToolResult           | Server scripts         | Creates a tool result to send back to the LLM.                                                                                                                                          |
| METHOD      | llm.embed(options)                          | llm.EmbedResponse        | Server scripts         | Returns the embeddings from the LLM for a given input.                                                                                                                                  |
| METHOD      | llm.embed.promise(options)                  | Promise                  | Server scripts         | Asynchronously returns the embeddings from the LLM for a given input.                                                                                                                   |
| METHOD      | llm.evaluatePrompt(options)                 | llm.Response             | Server scripts         | Takes the ID of an existing prompt and values for variables used in the prompt and returns the response from the LLM.                                                                   |
| METHOD      | llm.evaluatePrompt.promise(options)         | Promise                  | Server scripts         | Takes the ID of an existing prompt and values for variables used in the prompt and asynchronously returns the response from the LLM.                                                    |
| METHOD      | llm.evaluatePromptStreamed(options)         | llm.StreamedResponse     | Server scripts         | Takes the ID of an existing prompt and values for variables used in the prompt and returns the streamed response from the LLM.                                                          |
| METHOD      | llm.evaluatePromptStreamed.promise(options) | Promise                  | Server scripts         | Takes the ID of an existing prompt and values for variables used in the prompt and asynchronously returns the streamed response from the LLM.                                           |
| METHOD      | llm.generateText(options)                   | llm.Response             | Server scripts         | Takes a prompt and parameters for the LLM and returns the response from the LLM.                                                                                                        |
| METHOD      | llm.generateText.promise(options)           | Promise                  | Server scripts         | Takes a prompt and parameters for the LLM and asynchronously returns the response from the LLM.                                                                                         |
| METHOD      | llm.generateTextStreamed(options)           | llm.StreamedResponse     | Server scripts         | Takes a prompt and parameters for the LLM and returns the streamed response from the LLM.                                                                                               |
| METHOD      | llm.generateTextStreamed.promise(options)   | Promise                  | Server scripts         | Takes a prompt and parameters for the LLM and asynchronously returns the streamed response from the LLM.                                                                                |
| METHOD      | llm.getRemainingFreeUsage()                 | number                   | Server scripts         | Deprecated. Calls llm.getRemainingUsage().                                                                                                                                              |
| METHOD      | llm.getRemainingFreeUsage.promise()         | Promise                  | Server scripts         | Deprecated. Calls llm.getRemainingUsage.promise().                                                                                                                                      |
| METHOD      | llm.getRemainingFreeEmbedUsage()            | number                   | Server scripts         | Deprecated. Calls llm.getRemainingUsage(). Embed usage is no longer tracked separately.                                                                                                 |
| METHOD      | llm.getRemainingFreeEmbedUsage.promise()    | Promise                  | Server scripts         | Deprecated. Calls llm.getRemainingUsage.promise(). Embed usage is no longer tracked separately.                                                                                         |
| METHOD      | llm.getRemainingUsage()                     | number                   | Server scripts         | Returns the number of AI Units remaining.                                                                                                                                               |
| METHOD      | llm.getRemainingUsage.promise()             | Promise                  | Server scripts         | Asynchronously returns the number of AI Units remaining.                                                                                                                                |
| ENUM        | llm.ChatRole                                | enum                     | Server scripts         | Holds the string values for the author (role) of a chat message. Use this enum to set the value of the options.role parameter in llm.createChatMessage(options).                        |
| ENUM        | llm.EmbedModelFamily                        | enum                     | Server scripts         | Holds the string values for the large language model to be used to generate embeddings. Use this enum to set the value of the options.embedModelFamily parameter in llm.embed(options). |
| ENUM        | llm.ModelFamily                             | enum                     | Server scripts         | Holds the string values for the large language model to be used. Use this enum to set the value of the options.model parameter in llm.generateText(options).                            |

| MEMBER TYPE | NAME                               | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                                                                                             |
|-------------|------------------------------------|--------------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ENUM        | <code>llm.SafetyMode</code>        | enum                     | Server scripts         | Holds the string values for the safety mode to be used for LLM requests. Use this enum to set the value of the <code>options.safetyMode</code> parameter in <code>llm.generateText(options)</code> and <code>llm.generateTextStreamed(options)</code> . |
| ENUM        | <code>llm.ToolParameterType</code> | enum                     | Server scripts         | Holds the data type for a tool parameter. Use this enum to set the value of the <code>options.type</code> parameter in <code>llm.createToolParameter(options)</code> .                                                                                  |
| ENUM        | <code>llm.Truncate</code>          | enum                     | Server scripts         | Holds the string values for the truncation method to use when embeddings input exceeds 512 tokens. Use this enum to set the value of the <code>options.truncate</code> parameter in <code>llm.embed(options)</code> .                                   |

ChatMessage Object Members

Return Type / Value Type

Supported Script Types

`ChatMessage.role`

The author (role) of the chat message.

`ChatMessage.text`

Text of the chat message.

| MEMBER TYPE | NAME                          | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                          |
|-------------|-------------------------------|--------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------|
| PROPERTY    | <code>ChatMessage.role</code> | string                   | Server scripts         | The author (role) of the chat message.                                                                               |
| PROPERTY    | <code>ChatMessage.text</code> | string                   | Server scripts         | Text of the chat message. This text can be either the prompt sent by the script or the response returned by the LLM. |

Citation Object Members

Return Type / Value Type

Supported Script Types

`Citation.documentIds`

The IDs of the documents where the cited text is located.

The ending position of the cited text.

The starting position of the cited text.

| MEMBER TYPE | NAME                              | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                               |
|-------------|-----------------------------------|--------------------------|------------------------|-----------------------------------------------------------|
| PROPERTY    | <code>Citation.documentIds</code> | string[]                 | Server scripts         | The IDs of the documents where the cited text is located. |
| PROPERTY    | <code>Citation.end</code>         | number                   | Server scripts         | The ending position of the cited text.                    |
| PROPERTY    | <code>Citation.start</code>       | number                   | Server scripts         | The starting position of the cited text.                  |
| PROPERTY    | <code>Citation.text</code>        | string                   | Server scripts         | The cited text from the documents.                        |

Document Object Members

Return Type / Value Type

Supported Script Types

The content of the document.

The ID of the document.

| MEMBER TYPE | NAME                       | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                  |
|-------------|----------------------------|--------------------------|------------------------|------------------------------|
| PROPERTY    | <code>Document.data</code> | string                   | Server scripts         | The content of the document. |
| PROPERTY    | <code>Document.id</code>   | string                   | Server scripts         | The ID of the document.      |

EmbedResponse Object Members

Return Type / Value Type

Supported Script Types

`EmbedResponse.embeddings`

The embeddings returned from the LLM.

`EmbedResponse.inputs`

The list of inputs used to generate the embeddings response.

| MEMBER TYPE | NAME                     | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                  |
|-------------|--------------------------|--------------------------|------------------------|--------------------------------------------------------------|
| PROPERTY    | EmbedResponse.embeddings | number[]                 | Server scripts         | The embeddings returned from the LLM.                        |
| PROPERTY    | EmbedResponse.inputs     | string[]                 | Server scripts         | The list of inputs used to generate the embeddings response. |
| PROPERTY    | EmbedResponse.model      | string                   | Server scripts         | The model used to generate the embeddings response.          |

Response Object Members

Return Type / Value Type

Supported Script Types

Response.chatHistory

llm.ChatMessage[]

List of chat messages.

Response.citations

| MEMBER TYPE | NAME                 | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                      |
|-------------|----------------------|--------------------------|------------------------|--------------------------------------------------|
| PROPERTY    | Response.chatHistory | llm.ChatMessage[]        | Server scripts         | List of chat messages.                           |
| PROPERTY    | Response.citations   | llm.Citation[]           | Server scripts         | List of citations used to generate the response. |
| PROPERTY    | Response.documents   | llm.Document[]           | Server scripts         | List of documents used to generate the response. |
| PROPERTY    | Response.model       | string                   | Server scripts         | Model used to produce the LLM response.          |
| PROPERTY    | Response.text        | string                   | Server scripts         | Text returned by the LLM.                        |
| PROPERTY    | Response.toolCalls   | llm.ToolCall[]           | Server scripts         | Tool calls requested by the LLM.                 |
| PROPERTY    | Response.usage       | llm.Usage                | Server scripts         | Token usage for a request to the LLM.            |

StreamedResponse Object Members

Return Type / Value Type

Supported Script Types

StreamedResponse.chatHistory

llm.ChatMessage[]

List of chat messages.

StreamedResponse.citations

| MEMBER TYPE | NAME                         | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                               |
|-------------|------------------------------|--------------------------|------------------------|-----------------------------------------------------------|
| PROPERTY    | StreamedResponse.chatHistory | llm.ChatMessage[]        | Server scripts         | List of chat messages.                                    |
| PROPERTY    | StreamedResponse.citations   | llm.Citation[]           | Server scripts         | List of citations used to generate the streamed response. |
| PROPERTY    | StreamedResponse.documents   | llm.Document[]           | Server scripts         | List of documents used to generate the streamed response. |
| PROPERTY    | StreamedResponse.model       | string                   | Server scripts         | Model used to produce the streamed response.              |
| PROPERTY    | StreamedResponse.text        | string                   | Server scripts         | Text returned by the LLM.                                 |
| PROPERTY    | StreamedResponse.toolCalls   | llm.ToolCall[]           | Server scripts         | Tool calls requested by the LLM.                          |

Tool Object Members

Return Type / Value Type

Supported Script Types

Tool.description

The description of the tool.

The name of the tool.

llm.ToolParameter[]

| MEMBER TYPE | NAME             | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                  |
|-------------|------------------|--------------------------|------------------------|------------------------------|
| PROPERTY    | Tool.description | string                   | Server scripts         | The description of the tool. |
| PROPERTY    | Tool.name        | string                   | Server scripts         | The name of the tool.        |
| PROPERTY    | Tool.parameters  | llm.ToolParameter[]      | Server scripts         | The parameters of the tool.  |

ToolCall Object Members

Return Type / Value Type

Supported Script Types

The name of the requested tool.

ToolCall.parameters

llm.ToolParameter[]

The parameters of the requested tool.

| MEMBER TYPE | NAME                | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                           |
|-------------|---------------------|--------------------------|------------------------|---------------------------------------|
| PROPERTY    | ToolCall.name       | string                   | Server scripts         | The name of the requested tool.       |
| PROPERTY    | ToolCall.parameters | llm.ToolParameter[]      | Server scripts         | The parameters of the requested tool. |

ToolParameter Object Members

Return Type / Value Type

Supported Script Types

ToolParameter.description

The description of the tool parameter.

ToolParameter.name

The name of the tool parameter.

| MEMBER TYPE | NAME                      | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                            |
|-------------|---------------------------|--------------------------|------------------------|----------------------------------------|
| PROPERTY    | ToolParameter.description | string                   | Server scripts         | The description of the tool parameter. |
| PROPERTY    | ToolParameter.name        | string                   | Server scripts         | The name of the tool parameter.        |
| PROPERTY    | ToolParameter.type        | string                   | Server scripts         | The type of the tool parameter.        |

ToolResult Object Members

Return Type / Value Type

Supported Script Types

The originating tool call request from the LLM.

ToolResult.outputs

The outputs from running the tool specified in the tool call request.

| MEMBER TYPE | NAME               | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                           |
|-------------|--------------------|--------------------------|------------------------|-----------------------------------------------------------------------|
| PROPERTY    | ToolResult.call    | llm.ToolCall             | Server scripts         | The originating tool call request from the LLM.                       |
| PROPERTY    | ToolResult.outputs | Object[]                 | Server scripts         | The outputs from running the tool specified in the tool call request. |

Usage Object Members

Return Type / Value Type

Supported Script Types

Usage.completionTokens

The number of tokens in the response from the LLM.

Usage.promptTokens

The number of tokens in the request to the LLM.

| MEMBER TYPE | NAME                   | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                   |
|-------------|------------------------|--------------------------|------------------------|---------------------------------------------------------------|
| PROPERTY    | Usage.completionTokens | number                   | Server scripts         | The number of tokens in the response from the LLM.            |
| PROPERTY    | Usage.promptTokens     | number                   | Server scripts         | The number of tokens in the request to the LLM.               |
| PROPERTY    | Usage.totalTokens      | number                   | Server scripts         | The total number of tokens for the entire request to the LLM. |

 **Sample 1: Send a Prompt to the LLM and Receive a Response**

```
/**
 * @NApiVersion 2.1
 */
// This example shows how to query the default LLM
require(['N/llm'],
function(llm) {
    const response = llm.generateText({
        // modelFamily is optional. When omitted, the Cohere Command A model is used.
        prompt: "Hello World!",
        modelParameters: {
            maxTokens: 1000,
            temperature: 0.2,
            topK: 3,
            topP: 0.7,
            frequencyPenalty: 0.4,
            presencePenalty: 0
        }
    });
    const responseText = response.text;
    const remainingUsage = llm.getRemainingUsage(); // View remaining AI Units
});
```

 **Sample 2: Clean Up Content for Text Area Fields After Saving a Record**

```
/**
 * @NApiVersion 2.1
 * @NScriptType UserEventScript
 */
define(['N/llm'], (llm) => {
    /**
     * @param {Object} scriptContext The updated inventory item
     * record to clean up typo errors for purchase description and
     * sales description fields. The values are set before the record
     * is submitted to be saved.
     */
    function fixTypos(scriptContext) {
        const purchaseDescription = scriptContext.newRecord.getValue({
            fieldId: 'purchasedescription'
        })
        const salesDescription = scriptContext.newRecord.getValue({
            fieldId: 'salesdescription'
        })

        const p1 = llm.generateText.promise({
            prompt: 'Please clean up typos in the following text:
                    ${purchaseDescription} and return just the corrected text.
                    Return the text as is if there's no typo
                    or you don't understand the text.'
        })
        const p2 = llm.generateText.promise({
            prompt: 'Please clean up typos in the following text:
                    ${salesDescription} and return just the corrected text.
                    Return the text as is if there's no typo
                    or you don't understand the text.'
        })

        // When both promises are resolved, set the updated values for the
        // record
        Promise.all([p1, p2]).then((results) => {
            scriptContext.newRecord.setValue({
                fieldId: 'purchasedescription',
                value: results[0].value.text
            })
            scriptContext.newRecord.setValue({
                fieldId: 'salesdescription',
                value: results[1].value.text
            })
        })
    }

    return { beforeSubmit: fixTypos }
});
```

### Sample 3: Provide an LLM-based ChatBot for NetSuite Users

```
/**
 * @NApiVersion 2.1
 * @NScriptType Suitelet
 */

define(['N/ui/serverWidget', 'N/llm'], (serverWidget, llm) => {
  /**
   * Creates NetSuite form to communicate with LLM
   */
  function onRequest (context) {
    const form = serverWidget.createForm({
      title: 'Chat Bot'
    })
    const fieldgroup = form.addFieldGroup({
      id: 'fieldgroupid',
      label: 'Chat'
    })
    fieldgroup.isSingleColumn = true
    const historySize = parseInt(
      context.request.parameters.custpage_num_chats || '0')
    const numChats = form.addField({
      id: 'custpage_num_chats',
      type: serverWidget.FieldType.INTEGER,
      container: 'fieldgroupid',
      label: 'History Size'
    })
    numChats.updateDisplayType({
      displayType: serverWidget.FieldDisplayType.HIDDEN
    })

    if (context.request.method === 'POST') {
      numChats.defaultValue = historySize + 2
      const chatHistory = []
      for (let i = historySize - 2; i >= 0; i -= 2) {
        const you = form.addField({
          id: 'custpage_hist' + (i + 2),
          type: serverWidget.FieldType.TEXTAREA,
          label: 'You',
          container: 'fieldgroupid'
        })
        const yourMessage = context.request.parameters['custpage_hist' + i]
        you.defaultValue = yourMessage
        you.updateDisplayType({
          displayType: serverWidget.FieldDisplayType.INLINE
        })

        const chatbot = form.addField({
          id: 'custpage_hist' + (i + 3),
          type: serverWidget.FieldType.TEXTAREA,
          label: 'ChatBot',
          container: 'fieldgroupid'
        })
        const chatBotMessage =
          context.request.parameters['custpage_hist' + (i + 1)]
        chatbot.defaultValue = chatBotMessage
        chatbot.updateDisplayType({
          displayType: serverWidget.FieldDisplayType.INLINE
        })
        chatHistory.push({
          role: llm.ChatRole.USER,
          text: yourMessage
        })
        chatHistory.push({
          role: llm.ChatRole.CHATBOT,
          text: chatBotMessage
        })
      }

      const prompt = context.request.parameters.custpage_text
      const promptField = form.addField({
        id: 'custpage_hist0',
        type: serverWidget.FieldType.TEXTAREA,
        label: 'You',
        container: 'fieldgroupid'
      })
      promptField.defaultValue = prompt
      promptField.updateDisplayType({
        displayType: serverWidget.FieldDisplayType.INLINE
      })
      const result = form.addField({
        id: 'custpage_hist1',
        type: serverWidget.FieldType.TEXTAREA,
        label: 'ChatBot',
        container: 'fieldgroupid'
      })
      result.defaultValue = llm.generateText({
        prompt: prompt,
        chatHistory: chatHistory
      }).text
      result.updateDisplayType({
        displayType: serverWidget.FieldDisplayType.INLINE
      })
    }
  }
})
```

```

    })
  } else {
    numChats.defaultValue = 0
  }

  form.addField({
    id: 'custpage_text',
    type: serverWidget.FieldType.TEXTAREA,
    label: 'Prompt',
    container: 'fieldgroupid'
  })

  form.addSubmitButton({
    label: 'Submit'
  })

  context.response.writePage(form)
}

return {
  onRequest: onRequest
}
})

```

#### Sample 4: Evaluate an Existing Prompt and Receive a Response

```

/**
 * @NApiVersion 2.1
 */
require(['N/llm'],
  function(llm) {
    const response = llm.evaluatePrompt({
      id: 'stdprompt_gen_purch_desc_invt_item',
      variables: {
        "form": {
          "itemid": "My Inventory Item",
          "stockdescription": "This is the stock description of the item.",
          "vendorname": "My Item Vendor Inc.",
          "isdropshipitem": "false",
          "isspecialorderitem": "true",
          "displayname": "My Amazing Inventory Item"
        },
        "text": "This is the purchase description of the item."
      }
    });
    const responseText = response.text;
    const remainingUsage = llm.getRemainingUsage(); // View remaining AI Units
  });
});

```

### Sample 5: Create a Prompt and Evaluate It

```
/**
 * @NApiVersion 2.1
 */
require(['N/record', 'N/llm'], function(record, llm) {
  const rec = record.create({
    type: "prompt"
  });

  rec.setValue({
    fieldId: "name",
    value: "Test"
  });
  rec.setValue({
    fieldId: "prompttype",
    value: "GENERIC"
  });
  rec.setValue({
    fieldId: "modelfamily",
    value: "COHERE_COMMAND"
  });
  rec.setValue({
    fieldId: "template",
    value: "${mandatoryVariable} &lt;#if optionalVariable?has_content&gt;${optionalVariable}&lt;#else&gt;World&lt;#if&gt;"
  });

  const id = rec.save();

  try {
    llm.evaluatePrompt({
      id: id
    });
  }
  catch (e) {
    if (e.name === "TEMPLATE_PROCESSING_EXCEPTION")
      log.debug("Exception", "Expected exception was thrown");
  }

  const response = llm.evaluatePrompt({
    id: id,
    variables: {
      mandatoryVariable: "Hello",
      optionalVariable: "People"
    }
  });
  if ("Hello People" === response.chatHistory[0].text)
    log.debug("Evaluation", "Correct prompt got evaluated");

  llm.evaluatePrompt.promise({
    id: id,
    variables: {
      mandatoryVariable: "Hello",
      optionalVariable: "World"
    }
  }).then(function(response) {
    if ("Hello World" === response.chatHistory[0].text)
      log.debug("Evaluation", "Correct prompt got evaluated");
    record.delete({
      type: "prompt",
      id: id
    });
    debugger;
  })
});
```

#### Sample 6: Provide Source Documents When Calling the LLM

```
/**
 * @NApiVersion 2.1
 */
require(['N/llm'], function(llm) {
  const doc1 = llm.createDocument({
    id: "doc1",
    data: "Emperor penguins are the tallest."
  });
  const doc2 = llm.createDocument({
    id: "doc2",
    data: "Emperor penguins only live in the Sahara desert."
  });

  llm.generateText({
    prompt: "Where do the tallest penguins live?",
    documents: [doc1, doc2],
    modelFamily: llm.ModelFamily.COHERE_COMMAND,
    modelParameters: {
      maxTokens: 1000,
      temperature: 0.2,
      topK: 3,
      topP: 0.7,
      frequencyPenalty: 0.4,
      presencePenalty: 0
    }
  });
});
```

#### Sample 7: Receive a Partial Response from the LLM

```
/**
 * @NApiVersion 2.1
 */
require(['N/llm'], function(llm) {
  const response = llm.generateTextStreamed({
    preamble: "You are a script writer for TV shows.",
    prompt: "Write a 300 word pitch for a TV show about tigers.",
    modelFamily: llm.ModelFamily.COHERE_COMMAND,
    modelParameters: {
      maxTokens: 1000,
      temperature: 0.8,           // High temperature values result in more varied and
                                  // creative responses
      topK: 3,
      topP: 0.7,
      frequencyPenalty: 0.4,
      presencePenalty: 0
    }
  });

  var iter = response.iterator();
  iter.each(function(token){
    log.debug("token.value: " + token.value);
    log.debug("response.text: " + response.text);
    return true;
  });
});
```

## Sample 8: Find Similar Items Using Embeddings

```
/**
 * @NApiVersion 2.1
 * @NScriptType Suitelet
 * @NModuleScope SameAccount
 */
define(['N/ui/serverWidget', 'N/query', 'N/llm'],
function(serverWidget, query, llm) {

    function cosineSimilarity(array1, array2) {
        const dotProduct = array1.reduce((sum, value, index) => sum + value
            * array2[index], 0);
        const magnitude1 = Math.sqrt(array1.reduce((sum, value) => sum +
            value * value, 0));
        const magnitude2 = Math.sqrt(array2.reduce((sum, value) => sum +
            value * value, 0));
        return dotProduct / (magnitude1 * magnitude2);
    }

    function onRequest(context) {
        if (context.request.method === 'GET') {
            var form = serverWidget.createForm({
                title: 'Item Similarity Form'
            });

            var selectField = form.addField({
                id: 'custpage_myselect',
                type: serverWidget.FieldType.SELECT,
                label: 'Select an Item to find similar items for'
            });

            var res = query.runSuiteQL("SELECT id, case when displayname
is not null then displayname else itemid end name from item
where rownum <= 96 order by id").asMappedResults();

            for (let i = 0; i < res.length; i++)
            {
                selectField.addSelectOption({
                    value: res[i]['id'],
                    text: res[i]['name']
                });
            }

            form.addSubmitButton({
                label: 'Submit'
            });

            context.response.writePage(form);
        } else {
            var selectedValue = context.request.parameters.custpage_myselect;
            var res = query.runSuiteQL("SELECT id, case when displayname
is not null then displayname else itemid end name from item
where rownum <= 96 order by id").asMappedResults();

            const inputs = [];
            let selectedName;
            let selectedIndex = 0;

            for (let i = 0; i < res.length; i++)
            {
                if (res[i]['id'] === selectedValue)
                {
                    selectedName = res[i]["name"];
                    selectedIndex = i;
                }
                inputs.push(res[i]["name"]);
            }

            var embeddingResult = llm.embed({
                embedModelFamily: llm.EmbedModelFamily.COHERE_EMBED,
                inputs: inputs
            });

            var item = embeddingResult.embeddings[selectedIndex];
            var similarityResults = [];

            for (var j = 0; j < embeddingResult.inputs.length; j++)
            {
                similarityResults.push({
                    itemName: embeddingResult.inputs[j],
                    similarity: cosineSimilarity(item, embeddingResult.embeddings[j])
                });
            }

            similarityResults.sort((a,b) => b.similarity - a.similarity);
            context.response.write(JSON.stringify(similarityResults, null, 2));
        }
    }

    return {
        onRequest: onRequest
    };
};
```

```
); }
```

## Sample 9: Use Tools to Enhance the Response from the LLM

```
/**
 * @NApiVersion 2.1
 * @NScriptType Suitelet
 */

define(['N/ui/serverWidget', 'N/llm', 'N/query'], (serverWidget, llm, query) => {
  const TOOL_HANDLERS = {
    getNumberOfCustomerPurchasesForItem: (options) => {
      const res = query.runSuiteQL({
        query: "SELECT count(*) cnt from transactionLine where entity=? and item=?",
        params: [options.userId, options.itemId]
      }).asMappedResults();
      return res[0]["cnt"];
    },

    findItemId: (options) => {
      const res = query.runSuiteQL({
        query: "SELECT id from item where itemid=?",
        params: [options.itemName]
      }).asMappedResults();
      return res.length === 0 ? -1 : res[0]["id"];
    },

    findUserId: (options) => {
      const res = query.runSuiteQL({
        query: "SELECT id from entity where fullname=?",
        params: [options.userName]
      }).asMappedResults();
      return res.length === 0 ? -1 : res[0]["id"];
    }
  };

  const TOOL_DEFINITIONS = {
    'findUserId': llm.createTool({
      name: "findUserId",
      description: "Looks up user id based on his/her name",
      parameters: [
        llm.createToolParameter({ name: "userName", description: "name of the user", type: "STRING" })
      ]
    }),
    'findItemId': llm.createTool({
      name: "findItemId",
      description: "Looks up item id based on its name",
      parameters: [
        llm.createToolParameter({ name: "itemName", description: "name of the item", type: "STRING" })
      ]
    }),
    'getNumberOfCustomerPurchasesForItem': llm.createTool({
      name: "getNumberOfCustomerPurchasesForItem",
      description: "Looks up how many times did user purchase given item, both user and item are specified by id.",
      parameters: [
        llm.createToolParameter({ name: "userId", description: "id of the user", type: "INTEGER" }),
        llm.createToolParameter({ name: "itemId", description: "id of the item", type: "INTEGER" })
      ]
    })
  };

  const createFromHeader = (context) => {
    const form = serverWidget.createForm({ title: 'Tool Use Example' });

    const fieldgroup = form.addFieldGroup({
      id: 'fieldgroupid',
      label: 'Chat'
    });
    fieldgroup.isSingleColumn = true;

    const history = form.addField({
      id: 'custpage_history',
      type: serverWidget.FieldType.LONGTEXT,
      container: 'fieldgroupid',
      label: "History"
    });
    history.updateDisplayType({ displayType: serverWidget.FieldDisplayType.HIDDEN });

    const numChats = form.addField({
      id: 'custpage_num_chats',
      type: serverWidget.FieldType.INTEGER,
      container: 'fieldgroupid',
      label: "History Size"
    });
    numChats.updateDisplayType({ displayType: serverWidget.FieldDisplayType.HIDDEN });
    return form;
  };

  const displayHistory = (context, form) => {
    const historySize = parseInt(context.request.parameters["custpage_num_chats"] || "0", 10);
    form.getField({id: 'custpage_num_chats'}).defaultValue = String(historySize + 2);
    //numChats.defaultValue = String(historySize + 2);

    // Recreate the chat history fields (in reverse order)
    for (let i = historySize - 2; i >= 0; i -= 2) {

```

```

const you = form.addField({
  id: 'custpage_hist' + (i + 2),
  type: serverWidget.FieldType.TEXTAREA,
  label: 'You',
  container: 'fieldgroupid'
});
const yourMessage = context.request.parameters["custpage_hist" + i];
you.defaultValue = yourMessage;
you.updateDisplayType({ displayType: serverWidget.FieldDisplayType.INLINE });

const chatbot = form.addField({
  id: 'custpage_hist' + (i + 3),
  type: serverWidget.FieldType.TEXTAREA,
  label: 'ChatBot',
  container: 'fieldgroupid'
});
const chatBotMessage = context.request.parameters["custpage_hist" + (i + 1)];
chatbot.defaultValue = chatBotMessage;
chatbot.updateDisplayType({ displayType: serverWidget.FieldDisplayType.INLINE });
}
}

const runToolLoop = (prompt, chatHistory) => {
  const preamble = "You are an AI assistant. Don't be chatty. If you're asking for further
    information, keep it very brief and instructive. All the input variables for tools
    have to be explicitly stated.";
  const tools = Object.values(TOOL_DEFINITIONS);

  let llmResult = llm.generateText({
    modelFamily: llm.ModelFamily.COHERE_COMMAND,
    preamble,
    tools,
    prompt,
    chatHistory: chatHistory ? JSON.parse(chatHistory) : undefined
  });

  while (llmResult.toolCalls.length > 0) {
    const toolResults = [];
    for (const call of llmResult.toolCalls) {
      const resultValue = TOOL_HANDLERS[call.name](call.parameters);
      const toolResult = llm.createToolResult({
        call,
        outputs: [{ result: resultValue }]
      });
      toolResults.push(toolResult);
    }

    llmResult = llm.generateText({
      modelFamily: llm.ModelFamily.COHERE_COMMAND,
      tools,
      toolResults,
      chatHistory: llmResult.chatHistory
    });
  }
  return llmResult;
}

const displayResult = (prompt, llmResult, form) => {
  const promptField = form.addField({
    id: 'custpage_hist0',
    type: serverWidget.FieldType.TEXTAREA,
    label: 'You',
    container: 'fieldgroupid'
  });
  promptField.defaultValue = prompt;
  promptField.updateDisplayType({ displayType: serverWidget.FieldDisplayType.INLINE });

  const resultField = form.addField({
    id: 'custpage_hist1',
    type: serverWidget.FieldType.TEXTAREA,
    label: 'ChatBot',
    container: 'fieldgroupid'
  });
  resultField.defaultValue = llmResult.text;
  resultField.updateDisplayType({ displayType: serverWidget.FieldDisplayType.INLINE });

  form.getField({id: 'custpage_history'}).defaultValue = JSON.stringify(llmResult.chatHistory);
  //history.defaultValue = JSON.stringify(llmResult.chatHistory);
}

const createFormFooter = (form) => {
  // Field for new prompt input
  form.addField({
    id: 'custpage_text',
    type: serverWidget.FieldType.TEXTAREA,
    container: 'fieldgroupid',
    label: 'Prompt'
  });

  form.addSubmitButton({ label: 'Submit' });
}

const onRequest = (context) => {
  const form = createFormHeader(context);

```

```

if (context.request.method === 'POST') {
  displayHistory(context, form);

  const prompt = context.request.parameters["custpage_text"];
  const chatHistory = context.request.parameters["custpage_history"];

  const llmResult = runToolLoop(prompt, chatHistory)

  displayResult(prompt, llmResult, form);
} else { // GET
  form.getField({id: 'custpage_num_chats'}).defaultValue = '0';
  //numChats.defaultValue = "0";
}

createFormFooter(form);
context.response.writePage(form);
};

return { onRequest };
});

```

## 26. N/log Module

Load the N/log module to access methods for logging script execution details. Module members are also supported by the global log Object.

define(['N/log'], (log) => { ... })

✓ 1 Code Sample(s)

Scripts: Client and server scripts Limitations apply to client scripts

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/log'], (log) => { // Access methods and properties on log });
```

Use the N/log module to manually access methods for logging script execution details. These methods can also be accessed using the global log object. For more information about the global log object, see log Object.

Log messages appear on the Execution Log subtab of the script deployment for deployed scripts, or on the Execution Log subtab of the SuiteScript Debugger if you are debugging a script. Log messages also appear on the Script Execution Logs page at Customization > Scripting > Script Execution Logs.

N/log Module Script Sample

### N/log Module Guidelines

The following guidelines are provided for use of the N/log module:

NetSuite governs the amount of logging that can be done in any specific 60 minute time period. A company is allowed to make up to 100,000 log object method calls across all of their scripts. Script owners are notified if NetSuite detects that one script is logging excessively and automatically adjusts the log level.

NetSuite purges logs older than 30 days.

The Server Script Log search and the Execution Log subtab of script records and script deployment records have a log storage limit of 5 million per database. Because log persistence is not guaranteed, You should use custom records if you want to store script execution logs for extended periods. The Script Execution Logs page at Customization > Scripting > Script Execution Logs does not share the same database limit.

The Execution Log subtab also lists notes returned by NetSuite such as error messages. For more information, see N/error Module.

If you deploy a client script to a form using Form.clientScriptFileId or Form.clientScriptModulePath, using the N/log module adds the logs to the deployment of the parent script. The parent script can be either a beforeLoad user event script or a SuiteScript 2.1 Suitelet Script Type.

### Using Log Levels

Use the log methods along with the Log Level field on the script deployment to specify which log messages are written to the Execution Log on the script deployment. This is useful during the debugging of a script or for providing useful execution notes for auditing or tracking purposes.

Log levels on the script deployment and N/log module methods act as a filter on the amount of information logged. The following Log Levels are supported:

Use Debug to show all messages.

This type of logging is suitable only for testing scripts. To avoid excessive logging, don't use Debug level for active scripts in production.

Use Audit to shows a record of events that have occurred during the processing of the script (for example, 'A request was made to an external site.').

Use Error to show only unexpected script errors.

| LOG LEVEL | EXAMPLE USES                                                                                                                                                                   |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DEBUG     | Use Debug to show all messages. This type of logging is suitable only for testing scripts. To avoid excessive logging, don't use Debug level for active scripts in production. |
| AUDIT     | Use Audit to shows a record of events that have occurred during the processing of the script (for example, 'A request was made to an external site.').                         |
| ERROR     | Use Error to show only unexpected script errors.                                                                                                                               |
| EMERGENCY | Use Emergency to show only the most critical errors in the script log.                                                                                                         |

| LOG LEVEL | LOG METHOD USED IN YOUR SCRIPT |                    |                    |                    |
|-----------|--------------------------------|--------------------|--------------------|--------------------|
|           | log.emergency(options)         | log.error(options) | log.audit(options) | log.debug(options) |
| EMERGENCY |                                | -                  | -                  | -                  |
| ERROR     |                                |                    | -                  | -                  |
| AUDIT     |                                |                    |                    | -                  |
| DEBUG     |                                |                    |                    |                    |

### Viewing Script Execution Logs

Log messages written in client scripts that are attached to a form are viewed in the browser debug console tab when the client script runs on the form. Log messages written in server scripts and client scripts (that are deployed at the record level) are viewed in NetSuite.

Log messages for a specific script are shown on the Execution Log of the script deployment for the script. These logs are not guaranteed to persist for 30 days and may be purged to enhance performance if volume is high.

To view script execution log details for server scripts, go to Customization > Scripting > Script Execution Logs. This list of script execution logs is an enhanced repository that stores all log details for 30 days.

On this page, you can perform the following tasks:

Search for specific logs using filter options, such as log level, execution date range, and script name.

Download the list as a CSV file or an Excel spreadsheet.

N/log Module Members

Return Type / Value Type

Supported Script Types

log.audit(options)

Client\* and server scripts

Logs an Audit message.

log.debug(options)

| MEMBER TYPE | NAME                   | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES     | DESCRIPTION                |
|-------------|------------------------|--------------------------|----------------------------|----------------------------|
| METHOD      | log.audit(options)     | void                     | Client* and server scripts | Logs an Audit message.     |
| METHOD      | log.debug(options)     | void                     | Client* and server scripts | Logs a Debug message.      |
| METHOD      | log.emergency(options) | void                     | Client* and server scripts | Logs an Emergency message. |
| METHOD      | log.error(options)     | void                     | Client* and server scripts | Logs an Error message.     |

► Official NetSuite Script Samples (1 Sample(s))

Sample 1: N/log Module Script Sample

```
/**
 * @NApiVersion 2.1
 * @NScriptType UserEventScript
 */

// This script creates each type of log message.
define(['N/log'],function(log) {
  function beforeLoad(context) {
    var myValue = 'value';

    var myObject = {
      name: 'Jane',
      id: '123'
    };

    // An audit log message
    log.audit({
      title: 'Audit Entry',
      details: myObject
    });

    // A debug log message
    log.debug({
      title: 'Debug Entry',
      details: 'Value of myValue is: ' + myValue
    });

    // An emergency log message
    log.emergency({
      title: 'Emergency Entry',
      details: 'Value of myValue is: ' + myValue
    });

    // An error log message
    log.error({
      title: 'Error Entry',
      details: 'Value of myValue is: ' + myValue
    });
  }
  return {
    beforeLoad: beforeLoad
  };
});
```

## 27. N/machineTranslation Module

Load the N/machineTranslation module to translate text into supported languages using generative AI.

```
define(['N/machineTranslation'], (machineTranslation) => { ... })
```

✓ 1 Code Sample(s)

Scripts: Server scripts SuiteScript 2.1 only

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/machineTranslation'], (machineTranslation) => { // Access methods and properties on machineTranslation });
```

The content in this help topic pertains to SuiteScript 2.1.

Use the N/machineTranslation module to translate text into supported languages using generative AI. This module uses the Oracle Cloud Infrastructure (OCI) Language service to translate text in documents you provide. For more information about this service, see Language in the OCI documentation.

You can also use the N/llm Module to translate text by providing a suitable prompt when calling generative AI methods (such as llm.generateText(options)). However, using the N/machineTranslation module provides several benefits that make it a better choice for most translation use cases:

**Reduced cost** - The N/machineTranslation module supports unlimited translation requests, but methods in the N/llm module consume AI Units. For more information about AI Units, see NetSuite AI Units and NetSuite Features and AI Units FAQ.

**No prompt engineering required** - When you use the N/machineTranslation module, you don't need to write a prompt to generate translations. The module detects the source language automatically and translates provided documents into the language you specify in your request.

**More straightforward limits** - The N/machineTranslation module limits the length of each provided document to 5,000 characters, and it also limits the total length of all provided documents to 20,000 characters. The N/llm module provides limits that are based on tokens and the context window of the LLM you use, which can be more difficult to estimate.

**More supported languages** - The N/machineTranslation module can translate text into additional languages that may not be supported by LLMs and the N/llm module.

**More reliable translations** - The N/machineTranslation module uses the OCI Language service, which is designed for specific natural language processing (NLP) tasks, including translation. It uses pretrained models that can offer more predictable and reliable translation results compared to the N/llm module. The N/llm module uses the OCI Generative AI service, which is designed for content generation and may provide more varied translation results compared to a dedicated translation service.

This module is available in NetSuite by default when the Server SuiteScript feature is enabled. For more information, see Enabling Features.

### N/machineTranslation Module Members

Return Type / Value Type

Supported Script Types

machineTranslation.Document

A document with untranslated text (to provide to machineTranslation.translate(options) for translation) or translated text (when part of a machineTranslation.Response object returned from the translation service).

machineTranslation.Error

An error returned from the translation service.

| MEMBER TYPE | NAME                                          | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                                                                                                                                                                         |
|-------------|-----------------------------------------------|--------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT      | machineTranslation.Document                   | Object                   | Server scripts         | A document with untranslated text (to provide to machineTranslation.translate(options) for translation) or translated text (when part of a machineTranslation.Response object returned from the translation service).                                                                                                               |
| OBJECT      | machineTranslation.Error                      | Object                   | Server scripts         | An error returned from the translation service.                                                                                                                                                                                                                                                                                     |
| OBJECT      | machineTranslation.Response                   | Object                   | Server scripts         | The response from the translation service, which contains machineTranslation.Document objects with translated text.                                                                                                                                                                                                                 |
| METHOD      | machineTranslation.createDocument(options)    | Object                   | Server scripts         | Creates a document to be used as source content when calling the translation service.                                                                                                                                                                                                                                               |
| METHOD      | machineTranslation.translate(options)         | Object                   | Server scripts         | Translates a set of documents into the specified language.                                                                                                                                                                                                                                                                          |
| METHOD      | machineTranslation.translate.promise(options) | Promise                  | Server scripts         | Asynchronously translates a set of documents into the specified language.                                                                                                                                                                                                                                                           |
| ENUM        | machineTranslation.Language                   | enum                     | Server scripts         | Holds string values for the source language (when creating a document) or target language (when calling machineTranslation.translate(options)). Use this enum to set the options.language parameter in machineTranslation.createDocument(options) or the options.targetLanguage parameter in machineTranslation.translate(options). |

### Document Object Members

Return Type / Value Type

Supported Script Types

The ID of the document.

Document.language

The language of the document.

The content of the document.

| MEMBER TYPE | NAME              | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                   |
|-------------|-------------------|--------------------------|------------------------|-------------------------------|
| PROPERTY    | Document.id       | string                   | Server scripts         | The ID of the document.       |
| PROPERTY    | Document.language | string                   | Server scripts         | The language of the document. |
| PROPERTY    | Document.text     | string                   | Server scripts         | The content of the document.  |

Error Object Members

Return Type / Value Type

Supported Script Types

Error.documentId

The ID of the document in which the error occurred.

The error message returned from the translation service.

| MEMBER TYPE | NAME             | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                              |
|-------------|------------------|--------------------------|------------------------|----------------------------------------------------------|
| PROPERTY    | Error.documentId | string                   | Server scripts         | The ID of the document in which the error occurred.      |
| PROPERTY    | Error.message    | string                   | Server scripts         | The error message returned from the translation service. |

Response Object Members

Return Type / Value Type

Supported Script Types

machineTranslation.Error[]

The errors returned from the translation service.

Response.results

machineTranslation.Document[]

| MEMBER TYPE | NAME             | RETURN TYPE / VALUE TYPE      | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                     |
|-------------|------------------|-------------------------------|------------------------|-----------------------------------------------------------------|
| PROPERTY    | Response.errors  | machineTranslation.Error[]    | Server scripts         | The errors returned from the translation service.               |
| PROPERTY    | Response.results | machineTranslation.Document[] | Server scripts         | The translated documents returned from the translation service. |

► Official NetSuite Script Samples (1 Sample(s))

Sample 1: N/machineTranslation Module Script Samples

```
/**
 * @NApiVersion 2.1
 */
require(['N/machineTranslation'],
  function(mt) {
    const myDocument = mt.createDocument({
      id: 'myDoc',
      text: 'Hello everyone! How are you today?',
      language: mt.Language.ENGLISH
    });

    const translationResults = mt.translate({
      documents: [myDocument],
      targetLanguage: mt.Language.CZECH
    });

    // Work with the translated document
    log.debug('Translated text', translationResults.results[0].text);
  }
);
```

## 28. N/manufacturing/productionCharges Module

Load the N/manufacturing/productionCharges module to update unit costs of manufacturing charges on Assembly Build, Work Order Close, Work Order Completion transactions.

define(['N/manufacturing/productionCharges'], (productionCharges) => { ... })

✓ 1 Code Sample(s)

Scripts: Å

Permissions: Å

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/manufacturing/productionCharges'], (productionCharges) => { // Access methods and properties on productionCharges });
```

Use the N/manufacturing/productionCharges module to update unit costs of manufacturing charges on Assembly Build, Work Order Close, Work Order Completion transactions.

The module is only usable in server-side SuiteScript 2.1 scripts.

You can use productionCharges.updateChargesToCustomUnitCost(options) method to update cost on a specific transaction line to a specified unit cost.

You can use productionCharges.updateChargesToItemPurchasePrice(options) method to update cost on a specific transaction line to a specified unit cost.

You can use productionCharges.updateAllChargesToItemPurchasePrice(options) method to update cost on all routing and non-inventory transaction lines on a specified transaction to the current price.

### Prerequisites

- Assembly Items feature must be enabled (for more information, see Enabling Assembly Items)
- Allow bulk cost updates for Production Charges preference must be enabled
- The user must have at least Edit permission for the following transactions:
  - with Manufacturing Work In Process feature DISABLED: Build Assemblies
  - with Manufacturing Work In Process feature ENABLED: Build Assemblies, Work Order Completion, Work Order Issue
- Transactions modified by this module must be:

### N/manufacturing/productionCharges Module Members

Return Type / Value Type

Supported Script Types

productionCharges.updateChargesToCustomUnitCost(options)

Updates cost on a specific transaction line to a specified unit cost.

productionCharges.updateChargesToItemPurchasePrice(options)

Updates cost on a specific transaction line according to the current purchase price of the originating item.

| MEMBER TYPE | NAME                                                           | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                      |
|-------------|----------------------------------------------------------------|--------------------------|------------------------|------------------------------------------------------------------------------------------------------------------|
| METHOD      | productionCharges.updateChargesToCustomUnitCost(options)       | void                     | Server scripts         | Updates cost on a specific transaction line to a specified unit cost.                                            |
| METHOD      | productionCharges.updateChargesToItemPurchasePrice(options)    | void                     | Server scripts         | Updates cost on a specific transaction line according to the current purchase price of the originating item.     |
| METHOD      | productionCharges.updateAllChargesToItemPurchasePrice(options) | void                     | Server scripts         | Updates cost on all routing and non-inventory transaction lines on a specified transaction to the current price. |

 Sample 1: Update Production Charges and Unit Costs on Assembly Transactions

```
/**
 * @NApiVersion 2.1
 * @NScriptType ScheduledScript
 */
define(['N/manufacturing/productionCharges', 'N/log'], (productionCharges, log) => {
  const execute = (context) => {
    try {
      // 1. Update cost on a specific transaction line to a custom unit cost
      productionCharges.updateChargesToCustomUnitCost({
        transactionId: 1045,
        line: 1,
        customUnitCost: 15.50
      });
      log.audit('Cost Update', 'Custom unit cost updated on line 1');

      // 2. Update cost on a specific line to item purchase price
      productionCharges.updateChargesToItemPurchasePrice({
        transactionId: 1045,
        line: 2
      });

      // 3. Update all routing and non-inventory charges to current item purchase price
      productionCharges.updateAllChargesToItemPurchasePrice({
        transactionId: 1045
      });
      log.audit('Bulk Update', 'All production charges updated to item purchase price');
    } catch (e) {
      log.error('Production Charges Error', e.message);
    }
  };
  return { execute };
});
```

## 29. N/pgp Module

Load the N/pgp module to send secure messages to one or multiple receivers.

`define(['N/pgp'], (pgp) => { ... })`

✓ 4 Code Sample(s)

Scripts: Server scripts SuiteScript 2.1 only

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/pgp'], (pgp) => { // Access methods and properties on pgp });
```

The content in this help topic pertains to SuiteScript 2.1.

Use the N/pgp module to enable secure messaging, file encryption, and document signing. Based on OpenPGP encryption standards.

To use the N/pgp Module, you must first generate PGP keys from GnuPG, OpenPGP, or a third party source that supports pgp key generation. The generated keys must be stored in Secrets Management to securely manage and reference the keys. To store your generated keys, go to Setup > Company > API Secrets to create a new secret key.

For more information about API Secrets in NetSuite, see Secrets Management.

If you are new to PGP, use the following resources to learn more:

RFC 4880 OpenPGP Message Format - Provides background information about the PGP standard.

GnuPG Manual - A reliable resource for practical information.

OpenPGP - Learn more about the pgp standards with documentation resources for developers.

### Limitations of N/pgp

As you are working with the N/pgp module, consider the following limitations:

You cannot generate, modify, or inspect PGP keys using the N/pgp module. You must generate keys from a third party source that supports PGP key generation.

You cannot create a message without readable PGP software.

You are limited to strings.

You are limited to data that fits into memory.

N/pgp Module Script Samples

### N/pgp Module Members

Return Type / Value Type

Supported Script Types

General configuration options that can be used for message decryption.

Cryptographic keys and its metadata.

An octet scalar that identifies a subkey.

Processed PGP data.

| MEMBER TYPE | NAME                           | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                            |
|-------------|--------------------------------|--------------------------|------------------------|------------------------------------------------------------------------|
| OBJECT      | pgp.Config                     | Object                   | Server scripts         | General configuration options that can be used for message decryption. |
| OBJECT      | pgp.Key                        | Object                   | Server scripts         | Cryptographic keys and its metadata.                                   |
| OBJECT      | pgp.KeyId                      | Object                   | Server scripts         | An octet scalar that identifies a subkey.                              |
| OBJECT      | pgp.Message                    | Object                   | Server scripts         | Processed PGP data.                                                    |
| OBJECT      | pgp.MessageData                | Object                   | Server scripts         | Message data.                                                          |
| OBJECT      | pgp.Verification               | Object                   | Server scripts         | Verification results.                                                  |
| OBJECT      | pgp.VerificationSignature      | Object                   | Server scripts         | A verification result for a single signature.                          |
| METHOD      | pgp.createConfig(options)      | pgp.Config               | Server scripts         | Creates a new configuration object.                                    |
| METHOD      | pgp.createMessageData(options) | pgp.MessageData          | Server scripts         | Creates new message data.                                              |
| METHOD      | pgp.createSigner(options)      | certificate.Signer       | Server scripts         | Creates a certificate.Signer object for signing plain strings.         |
| METHOD      | pgp.createVerification()       | pgp.Verification         | Server scripts         | Creates an empty verification object.                                  |
| METHOD      | pgp.loadKeyFromSecret(options) | pgp.Key                  | Server scripts         | Loads a key whose contents are stored securely in secret.              |
| METHOD      | pgp.parseMessage(options)      | pgp.Message              | Server scripts         | Parses a PGP message.                                                  |
| METHOD      | pgp.parseKey(options)          | pgp.Key                  | Server scripts         | Parses an existing PGP key.                                            |

| MEMBER TYPE | NAME                                  | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                                               |
|-------------|---------------------------------------|--------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ENUM        | <code>pgp.CompressionAlgorithm</code> | Enum                     | Server scripts         | Holds the values for available compression algorithms. Use this enum to set the value of the <code>options.compressionAlgorithm</code> parameter of the <code>MessageData.encrypt(options)</code> method. |
| ENUM        | <code>pgp.Format</code>               | Enum                     | Server scripts         | Literal data packet type. Use this enum to set the value for the <code>options.format</code> parameter of the <code>pgp.createMessageData(options)</code> method.                                         |

## Config Object Members

Return Type / Value Type

Supported Script Types

`Config.allowInsecureDecryptionWithSigningKeys`

Enables decryption that is not secured with signing keys.

`Config.allowMessagesWithoutIntegrityProtection`

Allows messages without integrity protection.

| MEMBER TYPE | NAME                                                        | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                               |
|-------------|-------------------------------------------------------------|--------------------------|------------------------|-----------------------------------------------------------|
| PROPERTY    | <code>Config.allowInsecureDecryptionWithSigningKeys</code>  | boolean                  | Server scripts         | Enables decryption that is not secured with signing keys. |
| PROPERTY    | <code>Config.allowMessagesWithoutIntegrityProtection</code> | boolean                  | Server scripts         | Allows messages without integrity protection.             |
| PROPERTY    | <code>Config.useRelaxedSignatureParsing</code>              | boolean                  | Server scripts         | Relaxed signature parsing for configuration objects.      |

## KeyId Object Members

Return Type / Value Type

Supported Script Types

Returns a key ID as a hexadecimal string.

| MEMBER TYPE | NAME                       | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                               |
|-------------|----------------------------|--------------------------|------------------------|-------------------------------------------|
| METHOD      | <code>KeyId.asHex()</code> | string                   | Server scripts         | Returns a key ID as a hexadecimal string. |

## Message Object Members

Return Type / Value Type

Supported Script Types

Message type that specifies how a message is processed.

`Message.asArmored()`

Converts a message to ASCII armored format.

`Message.toMessageData()`

| MEMBER TYPE | NAME                                  | RETURN TYPE / VALUE TYPE     | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                        |
|-------------|---------------------------------------|------------------------------|------------------------|----------------------------------------------------------------------------------------------------|
| PROPERTY    | <code>Message.type</code>             | boolean                      | Server Scripts         | Message type that specifies how a message is processed.                                            |
| METHOD      | <code>Message.asArmored()</code>      | string                       | Server scripts         | Converts a message to ASCII armored format.                                                        |
| METHOD      | <code>Message.toMessageData()</code>  | <code>pgp.MessageData</code> | Server scripts         | Converts a message to message data without processing. Works only if the message is not encrypted. |
| METHOD      | <code>Message.decrypt(options)</code> | <code>pgp.MessageData</code> | Server scripts         | Decrypts a message and optionally verifies the signatures.                                         |

## MessageData Object Members

Return Type / Value Type

Supported Script Types

`MessageData.filename`

The name of a file.

`MessageData.date`

The date of a message or modification date of the file.

| MEMBER TYPE | NAME                         | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                      |
|-------------|------------------------------|--------------------------|------------------------|------------------------------------------------------------------|
| PROPERTY    | MessageData.filename         | string                   | Server scripts         | The name of a file.                                              |
| PROPERTY    | MessageData.date             | Date                     | Server scripts         | The date of a message or modification date of the file.          |
| PROPERTY    | MessageData.format           | pgp.Format               | Server scripts         | Literal data packet type.                                        |
| METHOD      | MessageData.getText()        | string                   | Server scripts         | Extracts the contents of a message as text.                      |
| METHOD      | MessageData.toMessage()      | pgp.Message              | Server scripts         | Creates a message with no signature, compression, or encryption. |
| METHOD      | MessageData.encrypt(options) | pgp.Message              | Server scripts         | Creates a message that is encrypted and optionally signed.       |

## Verification Object Members

Return Type / Value Type

Supported Script Types

VerificationSignature.verified

Indicates whether verification was successful.

Verification.signatures

null | Array<VerificationSignature>;

| MEMBER TYPE | NAME                           | RETURN TYPE / VALUE TYPE             | SUPPORTED SCRIPT TYPES | DESCRIPTION                                            |
|-------------|--------------------------------|--------------------------------------|------------------------|--------------------------------------------------------|
| PROPERTY    | VerificationSignature.verified | null   boolean                       | Server scripts         | Indicates whether verification was successful.         |
| PROPERTY    | Verification.signatures        | null   Array<VerificationSignature>; | Server scripts         | A list of individual verifications, one per signature. |

## VerificationSignature Object Members

Return Type / Value Type

Supported Script Types

VerificationSignature.keyId

ID of the (sub)key that was used for signing.

VerificationSignature.dateSigned

Date when the message was signed.

| MEMBER TYPE | NAME                             | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                    |
|-------------|----------------------------------|--------------------------|------------------------|----------------------------------------------------------------|
| PROPERTY    | VerificationSignature.keyId      | pgp.KeyId                | Server Scripts         | ID of the (sub)key that was used for signing.                  |
| PROPERTY    | VerificationSignature.dateSigned | Date                     | Server Scripts         | Date when the message was signed.                              |
| PROPERTY    | VerificationSignature.verified   | boolean                  | Server scripts         | Indicates whether verification was successful for a signature. |
| PROPERTY    | VerificationSignature.problems   | string[]                 | Server scripts         | A list of problems for verification signatures.                |

 **Sample 1: Send a Message**

```
/**
 * @NApiVersion 2.1
 */
require(['N/pgp', 'N/email'], (pgp, email) => {
  const keys = {
    ours: {
      pub: pgp.loadKeyFromSecret({
        secret: { scriptId: 'custsecret_pgp_key_ours_public' }
      }),
      pri: pgp.loadKeyFromSecret({
        secret: { scriptId: 'custsecret_pgp_key_ours_private' },
        password: { scriptId: 'custsecret_pgp_key_ours_private_password' }
      })
    }
  }

  const data = pgp.createMessageData({
    content: 'Hello, world!'
  })
  const message = data.encrypt({
    encryptionKeys: keys.ours.pub,
    signingKeys: keys.ours.pri
  })
  const payload = message.asArmored()

  const senderId = -5
  const recipientId = 'notify@myCompany.com'
  email.send({
    author: senderId,
    recipients: recipientId,
    subject: 'Test PGP',
    body: 'Payload: ' + payload
  })
})
```

 **Sample 2: Receive a Message**

```
/**
 * @NApiVersion 2.1
 */
require(['N/pgp', 'N/email'], (pgp, email) => {
  const keys = {
    ours: {
      pub: pgp.loadKeyFromSecret({
        secret: { scriptId: 'custsecret_pgp_key_ours_public' }
      }),
      pri: pgp.loadKeyFromSecret({
        secret: { scriptId: 'custsecret_pgp_key_ours_private' },
        password: { scriptId: 'custsecret_pgp_key_ours_private_password' }
      })
    }
  }

  const data = pgp.createMessageData({
    content: 'Hello, world!'
  })
  const message = data.encrypt({
    encryptionKeys: keys.ours.pub,
    signingKeys: keys.ours.pri
  })
  const payload = message.asArmored()
  const parseMessage = pgp.parseMessage({
    value: payload
  })
  const msgData = parseMessage.decrypt({
    decryptionKeys: keys.ours.pri,
    verificationKeys: keys.ours.pub
  })
  msgData.getText()
})
```

### Sample 3: Send a Message to Multiple Receivers

```
/**
 * @NApiVersion 2.1
 */

require(['N/pgp'], (pgp) => {
  // public and private keys for multiple receivers
  const keys = {
    alice: {
      pub: pgp.loadKeyFromSecret({
        secret: { scriptId: 'custsecret_pgp_key_alice_public' }
      }),
      pri: pgp.loadKeyFromSecret({
        secret: { scriptId: 'custsecret_pgp_key_alice_private' },
        password: { scriptId: 'custsecret_pgp_key_bob_private_password' }
      })
    },
    bob: {
      pub: pgp.loadKeyFromSecret({
        secret: { scriptId: 'custsecret_pgp_key_bob_public' }
      }),
      pri: pgp.loadKeyFromSecret({
        secret: { scriptId: 'custsecret_pgp_key_bob_private' },
        password: { scriptId: 'custsecret_pgp_key_bob_private_password' }
      })
    },
    netsuite: {
      pub: pgp.loadKeyFromSecret({
        secret: { scriptId: 'custsecret_pgp_netsuite_pub' }
      }),
      pri: pgp.loadKeyFromSecret({
        secret: { scriptId: 'custsecret_pgp_netsuite_pri' }
      })
    },
    example: {
      pri: pgp.loadKeyFromSecret({
        secret: { scriptId: 'custsecret_pgp_example_pri' }
      })
    }
  }

  const data = pgp.createMessageData({
    content: 'Hello Alice and Bob!'
  })

  const message = data.encrypt({
    encryptionKeys: [keys.alice.pub, keys.bob.pub],
    signingKeys: [keys.netsuite.pri, keys.example.pri]
  })

  /**
   * Alice decryption.
   */
  message.decrypt({
    decryptionKeys: keys.alice.pri,
    verificationKeys: [keys.netsuite.pub]
  })

  /**
   * Bob decryption.
   */
  message.decrypt({
    decryptionKeys: [keys.bob.pri],
    verificationKeys: keys.example.pub
  })
})
```

#### Sample 4: Use a Cryptographic Key for a Signature

```
/**
 * @NApiVersion 2.1
 */
require(['N/pgp', 'N/crypto/certificate', 'N/encode'], (pgp, cryptoCertificate, encode) => {
  const keys = {
    ours: {
      pub: pgp.loadKeyFromSecret({
        secret: { scriptId: 'custsecret_pgp_key_ours_public' }
      }),
      pri: pgp.loadKeyFromSecret({
        secret: { scriptId: 'custsecret_pgp_key_ours_private' },
        password: { scriptId: 'custsecret_pgp_key_ours_private_password' }
      })
    }
  }
  const signer = pgp.createSigner({
    key: keys.ours.pri,
    algorithm: cryptoCertificate.HashAlg.SHA256
  })
  signer.update({
    input: 'Test'
  })
  const signature = signer.sign({
    outputEncoding: encode.Encoding.BASE_64_URL_SAFE
  })

  log.debug(signature)
})
```

### 30. N/piremoval Module

Load the N/piremoval module to remove personal information (PI) from system notes, workflow history, and specific field values.

`define(['N/piremoval'], (piremoval) => { ... })`

✓ 1 Code Sample(s)

Scripts: Server scripts

Permissions: Remove Personal Information Create Remove Personal Information Run

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/piremoval'], (piremoval) => { // Access methods and properties on piremoval });
```

Use the N/piremoval module to remove personal information (PI) from system notes, workflow history, and specific field values. Use the N/piremoval module to comply with privacy regulations, specifically the right to be forgotten. You can remove personal information from system notes only, or you can also remove workflow history and field values on the record. Entity records, transactions, and custom records are supported.

N/piremoval Module Script Sample

You can use the `piremoval.createTask(options)` method to create a PI removal task, or use `piremoval.loadTask(options)` to load an existing PI removal task. Both of these methods return a `piremoval.PiRemovalTask` object that represents the task. Create a `piremoval.PiRemovalTask` object for each record type that requires removal of personal information. Use the `PiRemovalTask.save()` method to save the task, then use the `PiRemovalTask.run()` method to process the task and remove the personal information.

You can use the `piremoval.getTaskStatus(options)` method to check the status of a submitted PI removal task. This method returns a `piremoval.PiRemovalTaskStatus` object that describes the current status of the removal task. The `piremoval.PiRemovalTaskStatus` object uses an iterator to provide a list of log entries in the `PiRemovalTaskStatus.logList` object.

To use the N/piremoval module, the following requirements must be met:

Remove Personal Information Create permission is required to create a PI removal task.

Remove Personal Information Run permission is required to run a PI removal task.

For more information, see Personal Information (PI) Removal.

#### N/piremoval Module Members

Return Type / Value Type

Supported Script Types

`piremoval.PiRemovalTask`

Encapsulates a personal information removal task.

Use `piremoval.createTask(options)` to create this object.

`piremoval.PiRemovalTaskLogItem`

| MEMBER TYPE | NAME                                          | RETURN TYPE / VALUE TYPE                   | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                              |
|-------------|-----------------------------------------------|--------------------------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT      | <code>piremoval.PiRemovalTask</code>          | Object                                     | Server scripts         | Encapsulates a personal information removal task. Use <code>piremoval.createTask(options)</code> to create this object.                  |
| OBJECT      | <code>piremoval.PiRemovalTaskLogItem</code>   | Object                                     | Server scripts         | Encapsulates a log item of the personal information removal task status.                                                                 |
| OBJECT      | <code>piremoval.PiRemovalTaskStatus</code>    | Object                                     | Server scripts         | Encapsulates the status of a personal information removal task. Use <code>piremoval.getTaskStatus(options)</code> to create this object. |
| METHOD      | <code>piremoval.createTask(options)</code>    | <code>piremoval.PiRemovalTask</code>       | Server scripts         | Creates a personal information removal task.                                                                                             |
| METHOD      | <code>piremoval.deleteTask(options)</code>    | void                                       | Server scripts         | Deletes a personal information removal task.                                                                                             |
| METHOD      | <code>piremoval.getTaskStatus(options)</code> | <code>piremoval.PiRemovalTaskStatus</code> | Server scripts         | Retrieves the status of a personal information removal task.                                                                             |
| METHOD      | <code>piremoval.loadTask(options)</code>      | <code>piremoval.PiRemovalTask</code>       | Server scripts         | Loads a personal information removal task.                                                                                               |

#### PiRemovalTask Object Members

The following members are available for a `piremoval.PiRemovalTask` object.

Return Type / Value Type

Supported Script Types

`PiRemovalTask.deleteTask()`

Deletes the personal information removal task.

`PiRemovalTask.run()`

| MEMBER TYPE | NAME                                    | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                    |
|-------------|-----------------------------------------|--------------------------|------------------------|------------------------------------------------|
| METHOD      | <code>PiRemovalTask.deleteTask()</code> | void                     | Server scripts         | Deletes the personal information removal task. |

| MEMBER TYPE | NAME                             | RETURN TYPE / VALUE TYPE      | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                       |
|-------------|----------------------------------|-------------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------|
| METHOD      | PiRemovalTask.run()              | void                          | Server scripts         | Runs the personal information removal task.                                                                       |
| METHOD      | PiRemovalTask.save()             | void                          | Server scripts         | Saves the personal information removal task.                                                                      |
| PROPERTY    | PiRemovalTask.fieldIds           | string[] (read-only)          | Server scripts         | Represents the field IDs that are processed by the PI removal task.                                               |
| PROPERTY    | PiRemovalTask.historyOnly        | boolean                       | Server scripts         | Indicates whether the PI removal task removes system note information only, not field values or workflow history. |
| PROPERTY    | PiRemovalTask.historyReplacement | string (read-only)            | Server scripts         | Represents the text used in system notes to replace the original values.                                          |
| PROPERTY    | PiRemovalTask.id                 | number (read-only)            | Server scripts         | Represents the ID of the personal information removal task.                                                       |
| PROPERTY    | PiRemovalTask.recordIds          | number[] (read-only)          | Server scripts         | Represents the record IDs that are processed by the PI removal task.                                              |
| PROPERTY    | PiRemovalTask.recordType         | string (read-only)            | Server scripts         | Describes the record type updated by the PI removal task.                                                         |
| PROPERTY    | PiRemovalTask.status             | piremoval.PiRemovalTaskStatus | Server scripts         | Describes the status of the submitted personal information removal task.                                          |
| PROPERTY    | PiRemovalTask.workFlowIds        | number[] (read-only)          | Server scripts         | Represents the workflow IDs whose history is processed by the PI removal task.                                    |

## PiRemovalTaskLogItem Object Members

The following members are available for a `piremoval.PiRemovalTaskLogItem` object.

Return Type/Value Type

Support Script Type

PiRemovalTaskLogItem.exception

string (read-only)

Describes the exception for the log item, including and what caused it.

| MEMBER TYPE | NAME                           | RETURN TYPE/VALUE TYPE | SUPPORT SCRIPT TYPE | DESCRIPTION                                                                                                                                        |
|-------------|--------------------------------|------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| PROPERTY    | PiRemovalTaskLogItem.exception | string (read-only)     | Server scripts      | Describes the exception for the log item, including and what caused it.                                                                            |
| PROPERTY    | PiRemovalTaskLogItem.message   | string (read-only)     | Server scripts      | Describes the message for the log item and an explanation for any errors.                                                                          |
| PROPERTY    | PiRemovalTaskLogItem.status    | string (read-only)     | Server scripts      | Describes the status of the log item. This property takes its values from <code>task.TaskStatus</code> .                                           |
| PROPERTY    | PiRemovalTaskLogItem.type      | string (read-only)     | Server scripts      | Describes the type of personal information that was removed, one of <code>FieldValue</code> , <code>SystemNote</code> , or <code>Workflow</code> . |

## PiRemovalTaskStatus Object Members

The following members are available for a `piremoval.PiRemovalTaskStatus` object.

Return Type/Value Type

Support Script Type

PiRemovalTaskStatus.logList

Represents a list of logs for the PI removal task job.

PiRemovalTaskStatus.status

| MEMBER TYPE | NAME                        | RETURN TYPE/VALUE TYPE | SUPPORT SCRIPT TYPE | DESCRIPTION                                                              |
|-------------|-----------------------------|------------------------|---------------------|--------------------------------------------------------------------------|
| PROPERTY    | PiRemovalTaskStatus.logList | list                   | Server scripts      | Represents a list of logs for the PI removal task job.                   |
| PROPERTY    | PiRemovalTaskStatus.status  | string                 | Server scripts      | Describes the status of the submitted personal information removal task. |

 Sample 1: N/piremoval Module Script Sample

```
/**
 * @NApiVersion 2.1
 */
require(['N/piremoval'], function(piremoval) {
    function removePersonalInformation() {
        var piRemovalTask = piremoval.createTask({
            recordType: 'customer',
            recordIds: [11, 19],
            fieldIds: ['comments', 'phone'],
            workflowIds: [1],
            historyOnly: false,
            historyReplacement: 'removed_value'
        });

        piRemovalTask.save();
        var taskId = piRemovalTask.id;

        var piRemovalTaskInProgress = piremoval.loadTask(taskId);
        piRemovalTaskInProgress.run();

        var status = piremoval.getTaskStatus(taskId);
    };

    removePersonalInformation();
});
```

# 31. N/plugin Module

Load the N/plugin module to load custom plug-in implementations.

define(['N/plugin'], (plugin) => { ... })

✓ 2 Code Sample(s)

Scripts: Server scripts

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/plugin'], (plugin) => { // Access methods and properties on plugin });
```

Use the N/plugin module to load custom plug-in implementations. For more information about custom plug-ins, see the following help topics:

- Custom Plug-ins
- Custom Plug-in Creation
- Custom Plug-in Development
- Custom Plug-in Implementation
- N/plugin Module Script Sample

You cannot use the SuiteScript Debugger to debug a script on demand that uses the N/plugin module. You must use deployed debugging. To use deployed debugging, you must complete the steps described in Adding a Script that Instantiates a Custom Plug-in to NetSuite. For the complete process on creating a custom plugin, see Custom Plug-in Development. For additional information about ad-hoc and deployed debugging, see SuiteScript Debugger.

## N/plugin Module Members

Return Type / Value Type

Supported Script Types

plugin.findImplementations(options)

Returns the script IDs of custom plug-in type implementations.

plugin.loadImplementation(options)

Instantiates an implementation of the custom plug-in type.

| MEMBER TYPE | NAME                                | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                    |
|-------------|-------------------------------------|--------------------------|------------------------|----------------------------------------------------------------|
| METHOD      | plugin.findImplementations(options) | string[]                 | Server scripts         | Returns the script IDs of custom plug-in type implementations. |
| METHOD      | plugin.loadImplementation(options)  | Object                   | Server scripts         | Instantiates an implementation of the custom plug-in type.     |

## ► Official NetSuite Script Samples (2 Sample(s))

### Sample 1: N/plugin Module Script Sample

```
/**
 * @NApiVersion 2.1
 * @NScriptType plugintypeimpl
 */
define(function() {
  return {
    doTheMagic: function(operand1, operand2) {
      return operand1 + operand2;
    }
  }
});
```

## Sample 2: N/plugin Module Script Sample

```
/**
 * @NApiVersion 2.1
 * @NScriptType Suitelet
 */
define(['N/plugin'], function(plugin) {
    function onRequest(context) {
        var impls = plugin.findImplementations({
            type: 'customscript_magic_plugin'
        });

        for (var i = 0; i < impls.length; i++) {
            var pl = plugin.loadImplementation({
                type: 'customscript_magic_plugin',
                implementation: impls[i]
            });
            log.debug('impl ' + impls[i] + ' result = ' + pl.doTheMagic(10, 20));
        }

        var pl = plugin.loadImplementation({
            type: 'customscript_magic_plugin'
        });
        log.debug('default impl result = ' + pl.doTheMagic(10, 20));
    }

    return {
        onRequest: onRequest
    };
});
```

# 32. N/portlet Module

Load the N/portlet module to resize or refresh a form portlet.

`define(['N/portlet'], (portlet) => { ... })`

✓ 1 Code Sample(s)

Scripts: Client scripts

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/portlet'], (portlet) => { // Access methods and properties on portlet });
```

Use the N/portlet module to resize or refresh a form portlet. For more information about portlet scripts, see SuiteScript 2.1 Portlet Script Type.

N/portlet Module Script Sample

## N/portlet Module Members

Supported Script Types

portlet.resize()

Resizes a form portlet immediately.

portlet.refresh()

Refreshes a form portlet immediately.

| MEMBER TYPE | NAME              | RETURN TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                           |
|-------------|-------------------|-------------|------------------------|---------------------------------------|
| METHOD      | portlet.resize()  | void        | Client scripts         | Resizes a form portlet immediately.   |
| METHOD      | portlet.refresh() | void        | Client scripts         | Refreshes a form portlet immediately. |

## Sample 1: N/portlet Module Script Sample

```

/**
 * @NApiVersion 2.x
 * @NScriptType Portlet
 * @NScriptPortletType form
 */

define([], function() {
    function render(context) {
        var portletObj = context.portlet;
        portletObj.title = 'Test Form Portlet';
        setComponentsForResize();
        setComponentsForRefresh();

        function setComponentsForResize() {
            var DEFAULT_HEIGHT = '50';
            var DEFAULT_WIDTH = '50';
            var inlineHTMLField = portletObj.addField({
                id: 'divfield',
                type: 'inlinehtml',
                label: 'Test inline HTML'
            });
            inlineHTMLField.defaultVal = '<div id=\''divfield_elem\'' style=\''border: 1px dotted red; height: ' + DEFAULT_HEIGHT + 'px; width: ' + DEFAULT_WIDTH + 'px;\''></div>';
            inlineHTMLField.updateLayoutType({
                layoutType: 'normal'
            });
            inlineHTMLField.updateBreakType({
                breakType: 'startcol'
            });
            var resizeHeight = portletObj.addField({
                id: 'resize_height',
                type: 'text',
                label: 'Resize Height'
            });
            resizeHeight.defaultVal = DEFAULT_HEIGHT;
            var resizeWidth = portletObj.addField({
                id: 'resize_width',
                type: 'text',
                label: 'Resize Width'
            });
            resizeWidth.defaultVal = DEFAULT_WIDTH;
            var resizeLink = portletObj.addField({
                id: 'resize_link',
                type: 'inlinehtml',
                label: 'Resize link'
            });
            resizeLink.defaultVal = '<a id=\''resize_link\'' onclick=\''require([\'SuiteScripts/portletApiTestHelper\'], function(portletApiTestHelper) {portletApiTestHelper.resizePortlet(); })\'\' href=\''#\''>Resize</a>';

            function setComponentsForRefresh() {
                var textField = portletObj.addField({
                    id: 'refresh_output',
                    type: 'text',
                    label: 'Date.now().toString()'
                });
                textField.defaultVal = Date.now().toString();
                var refreshLink = portletObj.addField({
                    id: 'refresh_link',
                    type: 'inlinehtml',
                    label: 'Refresh link'
                });
                refreshLink.defaultVal = '<a id=\''refresh_link\'' onclick=\''require([\'SuiteScripts/portletApiTestHelper\'], function(portletApiTestHelper) {portletApiTestHelper.refreshPortlet(); })\'\' href=\''#\''>Refresh</a>';
            }

            return {
                render: render
            };
        }

        // portletApiTestHelper.js
        define(['N/portlet'], function(portlet) {
            function refreshPortlet() {
                portlet.refresh();
            }

            function resizePortlet() {
                var div = document.getElementById('divfield_elem');
                var newHeight = parseInt(document.getElementById('resize_height').value);
                var newWidth = parseInt(document.getElementById('resize_width').value);
                div.style.height = newHeight + 'px';
                div.style.width = newWidth + 'px';
                portlet.resize();
            }

            return {

```

```
    refreshPortlet: refreshPortlet,  
    resizePortlet: resizePortlet  
  });  
});
```

### 33. N/query Module

Load the N/query module to create and run searches using the SuiteAnalytics Workbook query process.

define(['N/query'], (query) => { ... })

✓ 6 Code Sample(s)

Scripts: Client and server scripts

Permissions: SuiteAnalytics Workbook

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/query'], (query) => { // Access methods and properties on query });
```

Use the N/query module to create and run queries using the SuiteAnalytics Workbook query process. For more information about SuiteAnalytics Workbook, see SuiteAnalytics Workbook Overview.

- Using the query module, you can:
- Use multilevel joins to create queries using field data from multiple record types.
- Create conditions (filters) using AND, OR, and NOT logic, as well as formulas and relative dates.
- Sort query results based on the values of multiple columns.
- Load and delete existing saved queries that were created using the SuiteAnalytics Workbook interface.
- View paged query results.
- Use promises for asynchronous .
- Convert query objects to SuiteQL queries and run arbitrary SuiteQL queries.

For more information about creating scripts using the N/query module, see the following help topics:

- Scripting with the N/query Module
- Formulas in the N/query Module
- Relative Dates in the N/query Module
- SuiteQL in the N/query Module

As you use the N/query module, keep the following considerations in mind:

The N/query module lets you create and run queries using the SuiteAnalytics Workbook query process. You can load and delete existing queries, but you can't save queries with this module. To save queries, use the SuiteAnalytics Workbook interface. For details, see Navigating SuiteAnalytics Workbook.

The N/query module uses a different data source than N/search. To find record types or field IDs for filters or result columns, use the Records Catalog (not the SuiteScript Records Browser). The Records Catalog lists everything available for SuiteAnalytics Workbook and N/query. For more, see Records Catalog Overview.

- The N/query module doesn't work in unauthenticated client-side contexts.
- For details, see the SuiteAnswers article Outbound HTTPs in an unauthenticated client-side context.

#### N/query Module Members

- Return Type / Value Type
- Supported Script Types
- Client and server scripts
- The field types (query result columns) that are displayed from the query results.
- Use Query.createColumn(options) or Component.createColumn(options) to create this object.
- Client and server scripts

| MEMBER TYPE | NAME            | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------|-----------------|--------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT      | query.Column    | Object                   | Client and server scripts | The field types (query result columns) that are displayed from the query results. Use Query.createColumn(options) or Component.createColumn(options) to create this object.                                                                                                                                                                                                                                                           |
| OBJECT      | query.Component | Object                   | Client and server scripts | One component of the query definition. The query definition always contains at least one component that encapsulates the initial query type. Queries with joins contain multiple components that encapsulate the join relationships. The initial component (Query.root) is automatically created with the query definition (query.Query). Use Query.autoJoin(options) or Component.autoJoin(options) to create subsequent components. |
| OBJECT      | query.Condition | Object                   | Client and server scripts | A condition. A condition narrows the query results. Use Query.createCondition(options) or Component.createCondition(options) to create this object.                                                                                                                                                                                                                                                                                   |
| OBJECT      | query.Page      | Object                   | Client and server scripts | One page of the paged query results.                                                                                                                                                                                                                                                                                                                                                                                                  |
| OBJECT      | query.PagedData | Object                   | Client and server scripts | A set of paged query results. This object also contains information about the set of paged results it encapsulates.                                                                                                                                                                                                                                                                                                                   |

| MEMBER TYPE | NAME                                   | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                                                                                                                                                                           |
|-------------|----------------------------------------|--------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT      | query.PageRange                        | Object                   | Client and server scripts | A range of pages from the paged query results.                                                                                                                                                                                                                                        |
| OBJECT      | query.Period                           | Object                   | Client and server scripts | A period of time to use in query conditions.                                                                                                                                                                                                                                          |
| OBJECT      | query.RelativeDate                     | Object                   | Client and server scripts | A relative date to use in query conditions.                                                                                                                                                                                                                                           |
| OBJECT      | query.Result                           | Object                   | Client and server scripts | A single row of the query result set.                                                                                                                                                                                                                                                 |
| OBJECT      | query.ResultSet                        | Object                   | Client and server scripts | The set of results returned by the query.                                                                                                                                                                                                                                             |
| OBJECT      | query.Query                            | Object                   | Client and server scripts | The query definition. Use query.create(options) or query.load(options) to create this object. Important: The creation of this object is the first step in creating a query with the N/query Module.                                                                                   |
| OBJECT      | query.Sort                             | Object                   | Client and server scripts | A sort that is placed on a particular query result column. Use Query.createSort(options) or Component.createSort(options) to create this object.                                                                                                                                      |
| OBJECT      | query.SuiteQL                          | Object                   | Client and server scripts | A SuiteQL query. Use Query.toSuiteQL() to create this object.                                                                                                                                                                                                                         |
| METHOD      | query.create(options)                  | query.Query              | Client and server scripts | Creates the query definition. Important: The invocation of this method is the first step in creating a query with the N/query Module.                                                                                                                                                 |
| METHOD      | query.createPeriod(options)            | query.Period             | Client and server scripts | Creates a query.Period object that represents a period of time.                                                                                                                                                                                                                       |
| METHOD      | query.createRelativeDate(options)      | query.RelativeDate       | Client and server scripts | Creates a query.RelativeDate object that represents a date relative to the current date.                                                                                                                                                                                              |
| METHOD      | query.delete(options)                  | void                     | Client and server scripts | Deletes an existing query that was created using the SuiteAnalytics Workbook UI. The deleted query is no longer available and cannot be modified or executed.                                                                                                                         |
| METHOD      | query.listTables(options)              | <Object>                 | Client and server scripts | Lists the table view objects that are included in a workbook in SuiteAnalytics Workbook.                                                                                                                                                                                              |
| METHOD      | query.load(options)                    | query.Query              | Client and server scripts | Loads an existing query that was created using the SuiteAnalytics Workbook UI. The loaded query can be modified (for example, by setting additional property values), joined with other query types, and executed in the same way as queries created using query.create(options).     |
| METHOD      | query.load.promise(options)            | Promise Object           | Client and server scripts | Asynchronously loads an existing query that was created using the SuiteAnalytics Workbook UI.                                                                                                                                                                                         |
| METHOD      | query.runSuiteQL(options)              | query.ResultSet          | Client and server scripts | Runs an arbitrary SuiteQL query.                                                                                                                                                                                                                                                      |
| METHOD      | query.runSuiteQL.promise(options)      | Promise Object           | Client and server scripts | Asynchronously runs an arbitrary SuiteQL query.                                                                                                                                                                                                                                       |
| METHOD      | query.runSuiteQLPaged(options)         | query.PagedData          | Client and server scripts | Runs an arbitrary SuiteQL query as a paged query.                                                                                                                                                                                                                                     |
| METHOD      | query.runSuiteQLPaged.promise(options) | Promise Object           | Client and server scripts | Asynchronously runs an arbitrary query as a paged query.                                                                                                                                                                                                                              |
| ENUM        | query.Aggregate                        | enum                     | Client and server scripts | Holds the string values for aggregate functions supported with the N/query Module. This enum is used to pass the aggregate function argument to Component.createColumn(options), Component.createCondition(options), Query.createColumn(options), and Query.createCondition(options). |
| ENUM        | query.DateId                           | enum                     | Client and server scripts | Holds the string values for supported date codes in relative dates. This enum is used to pass the date ID argument to query.createRelativeDate(options).                                                                                                                              |
| ENUM        | query.FieldContext                     | enum                     | Client and server scripts | Holds the string values for the field context to use when creating a column. This enum is used to pass the context argument to Query.createColumn(options) and Component.createColumn(options).                                                                                       |
| ENUM        | query.Operator                         | enum                     | Client and server scripts | Holds the string values for operators supported with the N/query Module. This enum is used to pass the operator argument to Query.createCondition(options) and Component.createCondition(options).                                                                                    |
| ENUM        | query.PeriodAdjustment                 | enum                     | Client and server scripts | Holds the string values for adjustment types for a period. This enum is used to pass the adjustment argument to query.createPeriod(options).                                                                                                                                          |
| ENUM        | query.PeriodCode                       | enum                     | Client and server scripts | Holds the string values for period codes for a period. This enum is used to pass the code argument to query.createPeriod(options).                                                                                                                                                    |
| ENUM        | query.PeriodType                       | enum                     | Client and server scripts | Holds the string values for period types for a period. This enum is used to pass the type argument to query.createPeriod(options).                                                                                                                                                    |
| ENUM        | query.RelativeDateRange                | enum                     | Client and server scripts | Holds query.RelativeDate object values for supported date ranges in relative dates. This enum is used to pass the values argument to Query.createCondition(options) and Component.createCondition(options).                                                                           |

| MEMBER TYPE | NAME                          | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                                                                                                                                                                                                                                         |
|-------------|-------------------------------|--------------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ENUM        | <code>query.ReturnType</code> | enum                     | Client and server scripts | Holds the string values for the formula return types supported with the N/query Module. This enum is used to pass the formula return type argument to <code>Query.createColumn(options)</code> , <code>Component.createColumn(options)</code> , <code>Query.createCondition(options)</code> , and <code>Component.createCondition(options)</code> . |
| ENUM        | <code>query.SortLocale</code> | enum                     | Client and server scripts | Holds the string values for sort locales supported with the N/query Module. This enum is used to pass the sort locale argument to <code>Query.createSort(options)</code> and <code>Component.createSort(options)</code> .                                                                                                                           |
| ENUM        | <code>query.Type</code>       | enum                     | Client and server scripts | Holds the string values for supported query types used in the query definition. This enum is used to pass the initial query type argument to <code>query.create(options)</code> .                                                                                                                                                                   |

## Column Object Members

The following members are available for a `query.Column` object.

Return Type/Value Type

Supported Script Types

`Column.aggregate`

string (read-only)

Client and server scripts

| MEMBER TYPE | NAME                          | RETURN TYPE/VALUE TYPE                   | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                                                       |
|-------------|-------------------------------|------------------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PROPERTY    | <code>Column.aggregate</code> | string (read-only)                       | Client and server scripts | An aggregate function that is performed on the query result column. An aggregate function performs a calculation on the column values and returns a single value. |
| PROPERTY    | <code>Column.alias</code>     | string (read-only)                       | Client and server scripts | An alias for this column. An alias is an alternate name for a column, and the alias is used in mapped results.                                                    |
| PROPERTY    | <code>Column.component</code> | <code>query.Component</code> (read-only) | Client and server scripts | A reference to the <code>query.Component</code> object to which this query result column belongs.                                                                 |
| PROPERTY    | <code>Column.context</code>   | Object (read-only)                       | Client and server scripts | The field context for values in the query result column. The field context determines how field values are displayed in the column.                               |
| PROPERTY    | <code>Column.fieldId</code>   | string (read-only)                       | Client and server scripts | The name of the query result column. This property and the <code>Column.formula</code> property cannot be set at the same time.                                   |
| PROPERTY    | <code>Column.formula</code>   | string (read-only)                       | Client and server scripts | The formula used to create the query result column. This property and the <code>Column.fieldId</code> property cannot be set at the same time.                    |
| PROPERTY    | <code>Column.groupBy</code>   | (read-only)                              | Client and server scripts | Whether the query results are grouped by this query result column.                                                                                                |
| PROPERTY    | <code>Column.label</code>     | string (read-only)                       | Client and server scripts | The label for the column.                                                                                                                                         |
| PROPERTY    | <code>Column.type</code>      | string (read-only)                       | Client and server scripts | The return type of the formula used to create the query result column.                                                                                            |

## Component Object Members

The following members are available for a `query.Component` object.

Return Type/Value Type

Supported Script Types

`Component.autoJoin(options)`

Client and server scripts

Creates a join relationship.

| MEMBER TYPE | NAME                                            | RETURN TYPE/VALUE TYPE       | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                                                                                                                                                                                                   |
|-------------|-------------------------------------------------|------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | <code>Component.autoJoin(options)</code>        | <code>query.Component</code> | Client and server scripts | Creates a join relationship. After you create the initial query definition, use <code>Query.autoJoin(options)</code> to create your first join. Then use this method to create each subsequent join. This method selects the correct join type automatically based on the record types that are being joined. |
| METHOD      | <code>Component.createColumn(options)</code>    | <code>query.Column</code>    | Client and server scripts | Creates a query result column based on the component. Use this method to create columns based on the join relationships created with <code>Query.autoJoin(options)</code> and <code>Component.autoJoin(options)</code> .                                                                                      |
| METHOD      | <code>Component.createCondition(options)</code> | <code>query.Condition</code> | Client and server scripts | Creates a condition (filter column) based on the component. Use this method to create conditions based on the join relationships created with <code>Query.autoJoin(options)</code> and <code>Component.autoJoin(options)</code> .                                                                             |

| MEMBER TYPE | NAME                                       | RETURN TYPE/VALUE TYPE       | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------|--------------------------------------------|------------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | <code>Component.createSort(options)</code> | <code>query.Sort</code>      | Client and server scripts | Creates a sort based on the component. Use this method to create sorts based on the join relationships created with <code>Query.autoJoin(options)</code> and <code>Component.autoJoin(options)</code> .                                                                                                                                                                                                                                                                  |
| METHOD      | <code>Component.join(options)</code>       | <code>query.Component</code> | Client and server scripts | Creates a join relationship. This method is an alias to <code>Component.autoJoin(options)</code> . After you create the initial query definition, use <code>Query.autoJoin(options)</code> to create your first join. Then use this method, or <code>Component.autoJoin(options)</code> , to create each subsequent join.                                                                                                                                                |
| METHOD      | <code>Component.joinFrom(options)</code>   | <code>query.Component</code> | Client and server scripts | Creates an explicit directional join relationship from another component to this component (an inverse join). This method sets the <code>Component.source</code> property on the returned <code>query.Component</code> object. After you create the initial query definition, use this method to create explicit directional joins from other components to this component.                                                                                              |
| METHOD      | <code>Component.joinTo(options)</code>     | <code>query.Component</code> | Client and server scripts | Creates an explicit directional join relationship to another component from this component (a join). You can use this method to specify the target of the join when a field can join multiple query types. This method sets the <code>Component.target</code> property on the returned <code>query.Component</code> object. After you create the initial query definition, use this method to create explicit directional joins to other components from this component. |
| PROPERTY    | <code>Component.child</code>               | Object (read-only)           | Client and server scripts | The child components of the component. This property holds an object of key-value pairs. Each key is the name of a child component. Each value is the corresponding child <code>query.Component</code> object.                                                                                                                                                                                                                                                           |
| PROPERTY    | <code>Component.parent</code>              | string (read-only)           | Client and server scripts | The parent <code>query.Component</code> object of the component.                                                                                                                                                                                                                                                                                                                                                                                                         |
| PROPERTY    | <code>Component.source</code>              | string (read-only)           | Client and server scripts | The source query type of the component. The value of this property is set when <code>Component.joinFrom(options)</code> is called to perform an explicit directional join from another component.                                                                                                                                                                                                                                                                        |
| PROPERTY    | <code>Component.target</code>              | string (read-only)           | Client and server scripts | The target query type of the component. The value of this property is set when <code>Component.joinTo(options)</code> is called to perform an explicit directional join to another component.                                                                                                                                                                                                                                                                            |
| PROPERTY    | <code>Component.type</code>                | string (read-only)           | Client and server scripts | The query type of the component.                                                                                                                                                                                                                                                                                                                                                                                                                                         |

## Condition Object Members

The following members are available for a `query.Condition` object.

Return Type/Value Type

Supported Script Types

`Condition.aggregate`

string (read-only)

Client and server scripts

| MEMBER TYPE | NAME                             | RETURN TYPE/VALUE TYPE                                     | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                                                |
|-------------|----------------------------------|------------------------------------------------------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PROPERTY    | <code>Condition.aggregate</code> | string (read-only)                                         | Client and server scripts | An aggregate function that is performed on the condition. An aggregate function performs a calculation on the condition values and returns a single value. |
| PROPERTY    | <code>Condition.children</code>  | <code>query.Condition[]</code> (read-only)                 | Client and server scripts | An array of child conditions used to create the parent condition.                                                                                          |
| PROPERTY    | <code>Condition.component</code> | <code>query.Component</code> (read-only)                   | Client and server scripts | A reference to the <code>query.Component</code> object to which this condition belongs.                                                                    |
| PROPERTY    | <code>Condition.fieldId</code>   | string (read-only)                                         | Client and server scripts | The name of the field that is used in the condition.                                                                                                       |
| PROPERTY    | <code>Condition.formula</code>   | string (read-only)                                         | Client and server scripts | The formula used to create the condition.                                                                                                                  |
| PROPERTY    | <code>Condition.operator</code>  | string (read-only)                                         | Client and server scripts | The name of the operator used to create the condition.                                                                                                     |
| PROPERTY    | <code>Condition.type</code>      | string (read-only)                                         | Client and server scripts | The return type of the formula used to create the condition.                                                                                               |
| PROPERTY    | <code>Condition.values</code>    | string   number     &lt;string   number   &gt; (read-only) | Client and server scripts | Value or an array of values used by an operator to create the condition.                                                                                   |

## Page Object Members

The following members are available for a `query.Page` object.

Return Type/Value Type

Supported Script Types

`query.ResultSet` (read-only)

Client and server scripts

The query results contained in this page.

| MEMBER TYPE | NAME           | RETURN TYPE/VALUE TYPE      | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                |
|-------------|----------------|-----------------------------|---------------------------|------------------------------------------------------------|
| PROPERTY    | Page.data      | query.ResultSet (read-only) | Client and server scripts | The query results contained in this page.                  |
| PROPERTY    | Page.isFirst   | boolean (read-only)         | Client and server scripts | Whether this page is the first of the paged query results. |
| PROPERTY    | Page.isLast    | boolean (read-only)         | Client and server scripts | Whether this page is the last of the paged query results.  |
| PROPERTY    | Page.pagedData | query.PagedData (read-only) | Client and server scripts | The set of paged query results that this page is from.     |
| PROPERTY    | Page.pageRange | query.PageRange (read-only) | Client and server scripts | The range of query results for this page.                  |

## PagedData Object Members

The following members are available for a query.PagedData object.

Return Type/Value Type

Supported Script Types

PagedData.iterator()

Client and server scripts

Standard SuiteScript 2.0 object for iterating through results.

| MEMBER TYPE | NAME                             | RETURN TYPE/VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                           |
|-------------|----------------------------------|------------------------|---------------------------|---------------------------------------------------------------------------------------|
| METHOD      | PagedData.iterator()             | Iterator object        | Client and server scripts | Standard SuiteScript 2.0 object for iterating through results.                        |
| METHOD      | PagedData.fetch(options)         | query.Page             | Client and server scripts | Retrieves a page in the set of pages included in the PagedData object.                |
| METHOD      | PagedData.fetch.promise(options) | Promise Object         | Client and server scripts | Asynchronously retrieves a page in the set of pages included in the PagedData object. |
| PROPERTY    | PagedData.count                  | number (read-only)     | Client and server scripts | The total number of paged query results.                                              |
| PROPERTY    | PagedData.pageRanges             | query.PageRange[]      | Client and server scripts | An array of page ranges for the set of paged query results.                           |
| PROPERTY    | PagedData.pageSize               | number (read-only)     | Client and server scripts | The number of query result rows per page.                                             |

## PageRange Object Members

The following members are available for a query.PageRange object.

Return Type/Value Type

Supported Script Types

number (read-only)

Client and server scripts

The index for this page range.

| MEMBER TYPE | NAME            | RETURN TYPE/VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                         |
|-------------|-----------------|------------------------|---------------------------|-----------------------------------------------------|
| PROPERTY    | PageRange.index | number (read-only)     | Client and server scripts | The index for this page range.                      |
| PROPERTY    | PageRange.size  | number (read-only)     | Client and server scripts | The number of query result rows in this page range. |

## Period Object Members

The following members are available for a query.Period object.

Return Type/Value Type

Supported Script Types

Period.adjustment

string (read-only)

Client and server scripts

| MEMBER TYPE | NAME              | RETURN TYPE/VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                   |
|-------------|-------------------|------------------------|---------------------------|-----------------------------------------------------------------------------------------------|
| PROPERTY    | Period.adjustment | string (read-only)     | Client and server scripts | The adjustment of the period. This property uses values from the query.PeriodAdjustment enum. |
| PROPERTY    | Period.code       | string (read-only)     | Client and server scripts | The code of the period. This property uses values from the query.PeriodCode enum.             |
| PROPERTY    | Period.type       | string (read-only)     | Client and server scripts | The type of the period. This property uses values from the query.PeriodType enum.             |

## Query Object Members

The following members are available for a `query.Query` object.

Return Type/Value Type

Supported Script Types

Query.and(conditions)

Client and server scripts

Creates a new condition (a `query.Condition` object) that corresponds to a logical conjunction (AND) of the arguments passed to the method. The arguments must be one or more `query.Condition` objects.

| MEMBER TYPE | NAME                                         | RETURN TYPE/VALUE TYPE              | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------|----------------------------------------------|-------------------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | <code>Query.and(conditions)</code>           | <code>query.Condition</code>        | Client and server scripts | Creates a new condition (a <code>query.Condition</code> object) that corresponds to a logical conjunction (AND) of the arguments passed to the method. The arguments must be one or more <code>query.Condition</code> objects.                                                                                                                                                                                                                                                                         |
| METHOD      | <code>Query.autoJoin(options)</code>         | <code>query.Component</code>        | Client and server scripts | Creates a join relationship. After you create the initial query definition, use this method to create your first join or subsequent joins from the root component of the query. This method selects the correct join type automatically based on the record types that are being joined.                                                                                                                                                                                                               |
| METHOD      | <code>Query.createColumn(options)</code>     | <code>query.Column</code>           | Client and server scripts | Creates a query result column based on the <code>query.Query</code> object. Use this method to create columns on the initial query definition created with <code>query.create(options)</code> .                                                                                                                                                                                                                                                                                                        |
| METHOD      | <code>Query.createCondition(options)</code>  | <code>query.Condition</code>        | Client and server scripts | Creates a condition (filter column) based on the <code>query.Query</code> object. Use this method to create conditions on the initial query definition created with <code>query.create(options)</code> .                                                                                                                                                                                                                                                                                               |
| METHOD      | <code>Query.createSort(options)</code>       | <code>query.Sort</code>             | Client and server scripts | Creates a sort based on the <code>query.Query</code> object. The <code>query.Sort</code> object describes a sort that is placed on a particular query result column or condition.                                                                                                                                                                                                                                                                                                                      |
| METHOD      | <code>Query.join(options)</code>             | <code>query.Component</code>        | Client and server scripts | Creates a join relationship. This method is an alias to <code>Query.autoJoin(options)</code> . After you create the initial query definition, use this method, or <code>Query.autoJoin(options)</code> , to create your first join.                                                                                                                                                                                                                                                                    |
| METHOD      | <code>Query.joinFrom(options)</code>         | <code>query.Component</code>        | Client and server scripts | Creates an explicit directional join relationship from another component to the root component of the search definition (an inverse join). This method sets the <code>Component.source</code> property on the returned <code>query.Component</code> object. After you create the initial query definition, use this method to create your first join as an explicit directional join from another component to this component.                                                                         |
| METHOD      | <code>Query.joinTo(options)</code>           | <code>query.Component</code>        | Client and server scripts | Creates an explicit directional join relationship to another component from this component (a forward join). You can use this method to specify the target of the join when a field can join multiple query types. This method sets the <code>Component.target</code> property on the returned <code>query.Component</code> object. After you create the initial query definition, use this method to create your first join as an explicit directional join to another component from this component. |
| METHOD      | <code>Query.not(condition)</code>            | <code>query.Condition</code>        | Client and server scripts | Creates a new condition (a <code>query.Condition</code> object) that corresponds to a logical negation (NOT) of the argument passed to the method. The argument must be a <code>query.Condition</code> object.                                                                                                                                                                                                                                                                                         |
| METHOD      | <code>Query.or(conditions)</code>            | <code>query.Condition</code>        | Client and server scripts | Creates a new condition (a <code>query.Condition</code> object) that corresponds to a logical disjunction (OR) of the arguments passed to the method. The arguments must be one or more <code>query.Condition</code> objects.                                                                                                                                                                                                                                                                          |
| METHOD      | <code>Query.run(options)</code>              | <code>query.ResultSet</code>        | Client and server scripts | Executes the query and returns the query result set.                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| METHOD      | <code>Query.run.promise()</code>             | <code>query.ResultSet</code>        | Client and server scripts | Executes the query asynchronously and returns the query result set.                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| METHOD      | <code>Query.runPaged(options)</code>         | <code>query.PagedData</code>        | Client and server scripts | Executes the query and returns a set of paged results.                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| METHOD      | <code>Query.runPaged.promise(options)</code> | <code>query.PagedData</code>        | Client and server scripts | Executes the query asynchronously and returns a set of paged results.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| METHOD      | <code>Query.toSuiteQL()</code>               | <code>query.SuiteQL</code>          | Client and server scripts | Converts this <code>query.Query</code> object to its corresponding SuiteQL representation.                                                                                                                                                                                                                                                                                                                                                                                                             |
| PROPERTY    | <code>Query.child</code>                     | Object (read-only)                  | Client and server scripts | A reference to children of the root component of the query definition. The value of this property is an object of key-value pairs. Each key is the name of a child component. Each respective value is the corresponding <code>query.Component</code> object.                                                                                                                                                                                                                                          |
| PROPERTY    | <code>Query.columns</code>                   | <code>query.Column[]</code>         | Client and server scripts | An array of query result columns returned from the query. Before you perform the query, you must assign all created columns as values to this property.                                                                                                                                                                                                                                                                                                                                                |
| PROPERTY    | <code>Query.condition</code>                 | <code>query.Condition</code> object | Client and server scripts | The parent condition that narrows the query results. Before you perform the query, you must assign your simple or complex conditions to this property.                                                                                                                                                                                                                                                                                                                                                 |
| PROPERTY    | <code>Query.id</code>                        | number (read-only)                  | Client and server scripts | The ID of the query definition. This property has a value only for existing queries that are loaded using <code>query.load(options)</code> . If you create a query using <code>query.create(options)</code> but do not save it, this property is null.                                                                                                                                                                                                                                                 |
| PROPERTY    | <code>Query.name</code>                      | string (read-only)                  | Client and server scripts | The name of the query definition. This property has a value only for existing queries that are loaded using <code>query.load(options)</code> . If you create a query using <code>query.create(options)</code> but do not save it, this property is null.                                                                                                                                                                                                                                               |

| MEMBER TYPE | NAME       | RETURN TYPE/VALUE TYPE      | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                        |
|-------------|------------|-----------------------------|---------------------------|----------------------------------------------------|
| PROPERTY    | Query.root | query.Component (read-only) | Client and server scripts | The root component of the query definition.        |
| PROPERTY    | Query.sort | query.Column[] (read-only)  | Client and server scripts | An array of query result columns used for sorting. |
| PROPERTY    | Query.type | string (read-only)          | Client and server scripts | The query type of the initial query definition.    |

RelativeDate Object Members

The following members are available for a query.RelativeDate object.

Return Type/Value Type

Supported Script Types

RelativeDate.dateId

string (read-only)

Client and server scripts

| MEMBER TYPE | NAME                  | RETURN TYPE/VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                          |
|-------------|-----------------------|------------------------|---------------------------|------------------------------------------------------------------------------------------------------|
| PROPERTY    | RelativeDate.dateId   | string (read-only)     | Client and server scripts | The ID of the relative date.                                                                         |
| PROPERTY    | RelativeDate.end      | Object (read-only)     | Client and server scripts | The end point of the relative date.                                                                  |
| PROPERTY    | RelativeDate.interval | Object (read-only)     | Client and server scripts | The interval of the relative date (from the RelativeDate.start point to the RelativeDate.end point). |
| PROPERTY    | RelativeDate.isRange  | boolean (read-only)    | Client and server scripts | Whether the relative date represents a range of dates or a specific moment in time.                  |
| PROPERTY    | RelativeDate.start    | Object (read-only)     | Client and server scripts | The start point of the relative date.                                                                |
| PROPERTY    | RelativeDate.value    | number (read-only)     | Client and server scripts | The value of the relative date.                                                                      |

Result Object Members

The following members are available for a query.Result object.

Return Type/Value Type

Supported Script Types

Client and server scripts

Returns the query result as an object with the columns mapped to the result values.

Result.getValue(options)

| MEMBER TYPE | NAME                     | RETURN TYPE/VALUE TYPE                                            | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                         |
|-------------|--------------------------|-------------------------------------------------------------------|---------------------------|-------------------------------------------------------------------------------------|
| METHOD      | Result.asMap()           | Object                                                            | Client and server scripts | Returns the query result as an object with the columns mapped to the result values. |
| METHOD      | Result.getValue(options) | &lt;boolean   number   string   Date   null&gt; (read-only)       | Client and server scripts | Gets the value at a given index in Result.values.                                   |
| PROPERTY    | Result.values            | Array &lt;boolean   number   string   Date   null&gt; (read-only) | Client and server scripts | The result values.                                                                  |

ResultSet Object Members

The following members are available for a query.ResultSet object.

Return Type/Value Type

Supported Script Types

ResultSet.asMappedResults()

Client and server scripts

Returns a query result set as an array of mapped results.

| MEMBER TYPE | NAME                        | RETURN TYPE/VALUE TYPE     | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                    |
|-------------|-----------------------------|----------------------------|---------------------------|----------------------------------------------------------------|
| METHOD      | ResultSet.asMappedResults() | Object[]                   | Client and server scripts | Returns a query result set as an array of mapped results.      |
| METHOD      | ResultSet.iterator()        | Iterator object            | Client and server scripts | Standard SuiteScript 2.0 object for iterating through results. |
| PROPERTY    | ResultSet.columns           | query.Column[] (read-only) | Client and server scripts | An array of query result column references.                    |

| MEMBER TYPE | NAME              | RETURN TYPE/VALUE TYPE     | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                         |
|-------------|-------------------|----------------------------|---------------------------|-----------------------------------------------------|
| PROPERTY    | ResultSet.results | query.Result[] (read-only) | Client and server scripts | An array of query.Result objects.                   |
| PROPERTY    | ResultSet.types   | string[] (read-only)       | Client and server scripts | An array of the return types for ResultSet.results. |

Sort Object Members

The following members are available for a query.Sort object.

Return Type/Value Type

Supported Script Types

Client and server scripts

Whether the sort direction is ascending.

Sort.caseSensitive

| MEMBER TYPE | NAME               | RETURN TYPE/VALUE TYPE   | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                                                                                                                                                                                                                          |
|-------------|--------------------|--------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PROPERTY    | Sort.ascending     | boolean                  | Client and server scripts | Whether the sort direction is ascending.                                                                                                                                                                                                                                                                                             |
| PROPERTY    | Sort.caseSensitive | boolean                  | Client and server scripts | Whether the sort is case sensitive. If a sort is case sensitive (and the sort direction is ascending), rows with column values that start with uppercase letters are listed before rows with column values that start with lowercase letters. If a sort is not case sensitive, uppercase and lowercase letters are treated the same. |
| PROPERTY    | Sort.column        | query.Column (read-only) | Client and server scripts | The query result column that the query results are sorted by.                                                                                                                                                                                                                                                                        |
| PROPERTY    | Sort.locale        | string                   | Client and server scripts | The locale to use for the sort. A locale represents a combination of language and region, and it can affect how certain values (such as strings) are sorted.                                                                                                                                                                         |
| PROPERTY    | Sort.nullsLast     | boolean                  | Client and server scripts | Whether query results with null values are listed at the end of the query results.                                                                                                                                                                                                                                                   |

SuiteQL Object Members

The following members are available for a query.SuiteQL object.

Return Type/Value Type

Supported Script Types

Client and server scripts

Runs the SuiteQL query and returns the query results.

SuiteQL.runPaged(options)

| MEMBER TYPE | NAME                      | RETURN TYPE/VALUE TYPE           | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                          |
|-------------|---------------------------|----------------------------------|---------------------------|--------------------------------------------------------------------------------------|
| METHOD      | SuiteQL.run()             | query.ResultSet                  | Client and server scripts | Runs the SuiteQL query and returns the query results.                                |
| METHOD      | SuiteQL.runPaged(options) | query.PagedData                  | Client and server scripts | Runs the SuiteQL query as a paged query and returns the paged query results.         |
| PROPERTY    | SuiteQL.columns           | query.Column[]                   | Client and server scripts | Describes the result columns to be returned from the query.                          |
| PROPERTY    | SuiteQL.params            | <string   number   > (read-only) | Client and server scripts | Contains the parameters for the query.                                               |
| PROPERTY    | SuiteQL.query             | string (read-only)               | Client and server scripts | Holds the string representation of the query.                                        |
| PROPERTY    | SuiteQL.type              | string (read-only)               | Client and server scripts | Describes the type of the query. This property uses values from the query.Type enum. |

 Sample 1: Create a Query for Customer Records and Run It as a Non-Paged Query

```

/**
 * @NApiVersion 2.1
 */
require(['N/query'], query => {
    // Create a query definition for customer records
    let myCustomerQuery = query.create({
        type: query.Type.CUSTOMER
    });

    // Join the original query definition based on the salesrep field. In a customer
    // record, the salesrep field contains a reference to an employee record. When you
    // join based on this field, you are joining the query definition with the employee
    // query type, and you can access the fields of the joined employee record in
    // your query.
    let mySalesRepJoin = myCustomerQuery.autoJoin({
        fieldId: 'salesrep'
    });

    // Join the joined query definition based on the location field. In an employee
    // record, the location field contains a reference to a location record.
    let myLocationJoin = mySalesRepJoin.autoJoin({
        fieldId: 'location'
    });

    // Create conditions for the query
    let firstCondition = myCustomerQuery.createCondition({
        fieldId: 'id',
        operator: query.Operator.EQUAL,
        values: 107
    });
    let secondCondition = myCustomerQuery.createCondition({
        fieldId: 'id',
        operator: query.Operator.EQUAL,
        values: 2647
    });
    let thirdCondition = mySalesRepJoin.createCondition({
        fieldId: 'email',
        operator: query.Operator.START_WITH_NOT,
        values: 'example'
    });

    // Combine conditions using and() and or() operator methods. In this example,
    // the combined condition states that the id field of the customer record must
    // have a value of either 107 or 2647, and the email field of the employee
    // record (the record that is referenced in the salesrep field of the customer
    // record) must not start with 'example'.
    myCustomerQuery.condition = myCustomerQuery.and(
        thirdCondition, myCustomerQuery.or(firstCondition, secondCondition)
    );

    // Create query columns
    myCustomerQuery.columns = [
        myCustomerQuery.createColumn({
            fieldId: 'entityid'
        }),
        myCustomerQuery.createColumn({
            fieldId: 'id'
        }),
        mySalesRepJoin.createColumn({
            fieldId: 'entityid'
        }),
        mySalesRepJoin.createColumn({
            fieldId: 'email'
        }),
        mySalesRepJoin.createColumn({
            fieldId: 'hiredate'
        }),
        myLocationJoin.createColumn({
            fieldId: 'name'
        })
    ];

    // Sort the query results based on query columns
    myCustomerQuery.sort = [
        myCustomerQuery.createSort({
            column: myCustomerQuery.columns[3]
        }),
        myCustomerQuery.createSort({
            column: myCustomerQuery.columns[0],
            ascending: false
        })
    ];

    // Run the query
    let resultSet = myCustomerQuery.run();

    // Retrieve and log the results
    let results = resultSet.results;

```

```

    for (let i = results.length - 1; i >= 0; i--)
      log.debug(results[i].values);
    log.debug(resultSet.types);
  });

```

## Sample 2: Create a Query for Transaction Records and Run It as a Paged Query

```

/**
 * @NApiVersion 2.1
 */
require(['N/query'], query => {
  // Create a query definition for transaction records
  let myTransactionQuery = query.create({
    type: query.Type.TRANSACTION
  });

  // Join the original query definition based on the employee field. In a transaction
  // record, the employee field contains a reference to an employee record. When you
  // join based on this field, you are joining the query definition with the employee
  // query type, and you can access the fields of the joined employee record in
  // your query.
  let myEmployeeJoin = myTransactionQuery.autoJoin({
    fieldId: 'employee'
  });

  // Create a condition for the transaction query
  let transactionCondition = myTransactionQuery.createCondition({
    fieldId: 'isreversal',
    operator: query.Operator.IS,
    values: true
  });
  myTransactionQuery.condition = transactionCondition;

  // Create a query column
  myTransactionQuery.columns = [
    myEmployeeJoin.createColumn({
      fieldId: 'subsidiary'
    })
  ];

  // Sort the query results based on a query column
  myTransactionQuery.sort = [
    myTransactionQuery.createSort({
      column: myTransactionQuery.columns[0],
      ascending: false
    })
  ];

  // Run the query as a paged query with 10 results per page
  let results = myTransactionQuery.runPaged({
    pageSize: 10
  });

  log.debug(results.pageRanges.length);
  log.debug(results.count);

  // Retrieve the query results using an iterator
  let iterator = results.iterator();
  iterator.each(function(result) {
    let page = result.value;
    log.debug(page.pageRange.size);
    return true;
  });

  // Alternatively, retrieve the query results by looping through
  // each result
  for (let i = 0; i < results.pageRanges.length; i++) {
    let page = results.fetch(i);
    log.debug(page.pageRange.size);
  }
});

```

### Sample 3: Convert a Query to a SuiteQL and Run It

```
/**
 * @NApiVersion 2.x
 */
require(['N/query'], function(query) {
  var myCustomerQuery = query.create({
    type: query.Type.CUSTOMER
  });

  myCustomerQuery.columns = [
    myCustomerQuery.createColumn({
      fieldId: 'entityid'
    }),
    myCustomerQuery.createColumn({
      fieldId: 'email'
    })
  ];

  myCustomerQuery.condition = myCustomerQuery.createCondition({
    fieldId: 'isperson',
    operator: query.Operator.IS,
    values: [true]
  });

  var mySQLCustomerQuery = myCustomerQuery.toSuiteQL();

  var results = mySQLCustomerQuery.run();
});
```

### Sample 4: Run an Arbitrary SuiteQL Query

```
/**
 * @NApiVersion 2.x
 */
require(['N/query'], function(query) {
  var sql =
    "SELECT " +
    "  scriptDeployment.primarykey, scriptexecutioncontextmap.executioncontext " +
    " FROM " +
    "  scriptDeployment, scriptexecutioncontextmap " +
    " WHERE " +
    "  scriptexecutioncontextmap.scriptrecord = scriptDeployment.primarykey " +
    " AND " +
    "  scriptexecutioncontextmap.executioncontext IN ('WEBSTORE', 'WEBAPPLICATION')";

  var resultIterator = query.runSuiteQLPaged({
    query: sql,
    pageSize: 10
  }).iterator();

  resultIterator.each(function(page) {
    var pageIterator = page.value.data.iterator();
    pageIterator.each(function(row) {
      log.debug('ID: ' + row.value.getValue(0) + ', Context: ' + row.value.getValue(1));
      return true;
    });
    return true;
  });
});
```

#### Sample 5: Create a Query for a Custom Field

```
/*
 * @NApiVersion 2.x
 */
require(['N/query'], function(query) {
  var customFieldIdQuery = query.create({
    type: query.Type.CUSTOM_FIELD
  });
  customFieldIdQuery.columns = [
    customFieldIdQuery.createColumn({
      fieldId: 'internalid'
    })
  ];
  customFieldIdQuery.condition = customFieldIdQuery.createCondition({
    fieldId: 'scriptid',
    operator: query.Operator.IS,
    values: 'custrecord_my_custom_field'
  });

  var results = customFieldIdQuery.run().asMappedResults();
  var customFieldInternalId = results[0].internalid;
  log.debug({
    title: 'Internal ID of the custom field is ',
    details: customFieldInternalId
  });
});
```

#### Sample 6: Create a Query Using a Specific Record Field

```
/**
 * @NApiVersion 2.x
 */
require(['N/query'], function(query) {
  var mfgComponent = query.create({
    type: query.Type.MANUFACTURING_COMPONENT
  });

  var mfgOperation = mfgComponent.autoJoin({
    fieldId: 'operationdisplaytext'
  });

  mfgComponent.columns = [
    mfgComponent.createColumn({
      fieldId: 'operationdisplaytext'
    }),
    mfgComponent.createColumn({
      fieldId: 'item'
    }),
    mfgOperation.createColumn({
      fieldId: 'operationsequence'
    })
  ];

  mfgComponent.condition = mfgComponent.and(
    mfgComponent.createCondition({
      fieldId: 'transaction',
      operator: query.Operator.ANY_OF,
      values: '&lt;transactionId&gt;'
    }),
    mfgOperation.createCondition({
      fieldId: 'operationsequence',
      operator: query.Operator.EQUAL,
      values: 20
    })
  );

  var results = mfgComponent.run();
});
```

## 34. N/record Module

Load the N/record module to work with NetSuite records.

```
define(['N/record'], (record) => { ... })
```

✓ 6 Code Sample(s)

Scripts: Client and server scripts

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/record'], (record) => { // Access methods and properties on record });
```

Use the N/record module to work with NetSuite records. You can use this module to create, delete, copy, load, or make changes to a record.

For help finding a record's internal ID, see Finding Internal IDs of Records.

### N/record Module Members

Return Type / Value Type

Supported Script Types

Client and server scripts

Encapsulates a column of a sublist on a standard or custom record.

Client and server scripts

Encapsulates a body or sublist field on a standard or custom record.

| MEMBER TYPE | NAME                                              | RETURN TYPE / VALUE TYPE   | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                                     |
|-------------|---------------------------------------------------|----------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT      | <code>record.Column</code>                        | Object                     | Client and server scripts | Encapsulates a column of a sublist on a standard or custom record.                                                                              |
| OBJECT      | <code>record.Field</code>                         | Object                     | Client and server scripts | Encapsulates a body or sublist field on a standard or custom record.                                                                            |
| OBJECT      | <code>record.Macro</code>                         | Object                     | Client and server scripts | Encapsulates a NetSuite record macro.                                                                                                           |
| OBJECT      | Plain JavaScript Object                           | Object                     | Client and server scripts | A plain JavaScript object of record macros available for a record type. This object is returned by <code>Record.getMacros()</code> .            |
| OBJECT      | <code>record.Record</code>                        | Object                     | Client and server scripts | Encapsulates a NetSuite record.                                                                                                                 |
| OBJECT      | <code>record.Sublist</code>                       | Object                     | Client and server scripts | Encapsulates a sublist on a standard or custom record.                                                                                          |
| METHOD      | <code>record.attach(options)</code>               | void                       | Client and server scripts | Attaches a record to another record.                                                                                                            |
| METHOD      | <code>record.attach.promise(options)</code>       | Promise                    | Client scripts            | Attaches a record asynchronously to another record.                                                                                             |
| METHOD      | <code>record.copy(options)</code>                 | <code>record.Record</code> | Client and server scripts | Creates a new record by copying an existing record in NetSuite.                                                                                 |
| METHOD      | <code>record.copy.promise(options)</code>         | Promise                    | Client scripts            | Creates a new record asynchronously by copying an existing record in NetSuite.                                                                  |
| METHOD      | <code>record.create(options)</code>               | <code>record.Record</code> | Client and server scripts | Creates a new record.                                                                                                                           |
| METHOD      | <code>record.create.promise(options)</code>       | Promise                    | Client scripts            | Creates a new record asynchronously.                                                                                                            |
| METHOD      | <code>record.delete(options)</code>               | number                     | Client and server scripts | Deletes a record.                                                                                                                               |
| METHOD      | <code>record.delete.promise(options)</code>       | Promise                    | Client scripts            | Deletes a record asynchronously.                                                                                                                |
| METHOD      | <code>record.detach(options)</code>               | void                       | Client and server scripts | Detaches a record from another record.                                                                                                          |
| METHOD      | <code>record.detach.promise(options)</code>       | Promise                    | Client scripts            | Detaches a record from another record asynchronously.                                                                                           |
| METHOD      | <code>record.load(options)</code>                 | Object                     | Client and server scripts | Loads an existing record.                                                                                                                       |
| METHOD      | <code>record.load.promise(options)</code>         | Promise                    | Client scripts            | Loads an existing record asynchronously.                                                                                                        |
| METHOD      | <code>record.submitFields(options)</code>         | Object                     | Client and server scripts | Updates and submits one or more body fields on an existing record in NetSuite, and returns the internal ID of the parent record.                |
| METHOD      | <code>record.submitFields.promise(options)</code> | Promise                    | Client scripts            | Updates and submits one or more body fields asynchronously on an existing record in NetSuite, and returns the internal ID of the parent record. |
| METHOD      | <code>record.transform(options)</code>            | <code>record.Record</code> | Client and server scripts | Transforms a record from one type into another, using data from an existing record.                                                             |

| MEMBER TYPE   | NAME                                     | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                                                                                        |
|---------------|------------------------------------------|--------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>METHOD</b> | <b>record.transform.promise(options)</b> | Promise                  | Client scripts            | Transforms a record from one type into another asynchronously, using data from an existing record.                                                                                                 |
| <b>ENUM</b>   | <b>record.Type</b>                       | enum                     | Client and server scripts | Holds the string values for supported record types. Use this enum to set the value of the Record.type property in cases where you are working with an instance of a standard NetSuite record type. |

## Column Object Members

Return Type / Value Type

Supported Script Types

string (read-only)

Client and server scripts

Returns the internal ID of the column.

Column.isDisabled

| MEMBER TYPE     | NAME                      | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                         |
|-----------------|---------------------------|--------------------------|---------------------------|-------------------------------------------------------------------------------------|
| <b>PROPERTY</b> | <b>Column.id</b>          | string (read-only)       | Client and server scripts | Returns the internal ID of the column.                                              |
| <b>PROPERTY</b> | <b>Column.isDisabled</b>  | boolean                  | Client and server scripts | Indicates whether the column is disabled.                                           |
| <b>PROPERTY</b> | <b>Column.isDisplay</b>   | boolean (read-only)      | Client and server scripts | Indicates whether the column is displayed.                                          |
| <b>PROPERTY</b> | <b>Column.isMandatory</b> | boolean                  | Client and server scripts | Indicates whether the column is required.                                           |
| <b>PROPERTY</b> | <b>Column.isSortable</b>  | boolean (read-only)      | Client and server scripts | Indicates whether the column is sortable.                                           |
| <b>PROPERTY</b> | <b>Column.label</b>       | string (read-only)       | Client and server scripts | Returns the UI label for the column.                                                |
| <b>PROPERTY</b> | <b>Column.sublistId</b>   | string (read-only)       | Client and server scripts | Returns the internal ID of the standard or custom sublist that contains the column. |
| <b>PROPERTY</b> | <b>Column.type</b>        | string (read-only)       | Client and server scripts | Returns the column type.                                                            |

## Field Object Members

Return Type / Value Type

Supported Script Types

Field.getSelectOptions(options)

Client and server scripts

Returns an array of available options on a standard or custom select, multiselect, or radio field as key-value pairs. Only the first 1,000 available options are returned.

string (read-only)

| MEMBER TYPE     | NAME                                   | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                                                                |
|-----------------|----------------------------------------|--------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>METHOD</b>   | <b>Field.getSelectOptions(options)</b> | array                    | Client and server scripts | Returns an array of available options on a standard or custom select, multiselect, or radio field as key-value pairs. Only the first 1,000 available options are returned. |
| <b>PROPERTY</b> | <b>Field.label</b>                     | string (read-only)       | Client and server scripts | Returns the UI label for a standard or custom field body or sublist field.                                                                                                 |
| <b>PROPERTY</b> | <b>Field.id</b>                        | string (read-only)       | Client and server scripts | Returns the internal ID of a standard or custom body or sublist field.                                                                                                     |
| <b>PROPERTY</b> | <b>Field.type</b>                      | string (read-only)       | Client and server scripts | Returns the type of a body or sublist field.                                                                                                                               |
| <b>PROPERTY</b> | <b>Field.isMandatory</b>               | boolean                  | Client and server scripts | Returns true if the standard or custom field is required on the record form, or false otherwise.                                                                           |
| <b>PROPERTY</b> | <b>Field.sublistId</b>                 | string (read-only)       | Client and server scripts | Returns the ID of the sublist associated with the specified sublist field. This property is available only when working with a record in dynamic mode.                     |
| <b>PROPERTY</b> | <b>Field.isDisplay</b>                 | boolean                  | Client and server scripts | Returns true if the field is visible on the record form, or false if it is not. This property is available only when working with a record in dynamic mode.                |

## Macro Object Members

The following members are called on the record.Macro object. For information about record macros, see Overview of Record Action and Macro APIs.

Return Type / Value Type

Supported Script Types

Macro.execute(options)

Client and server scripts

Performs a macro operation and returns its result in an object.

| MEMBER TYPE | NAME                           | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                               |
|-------------|--------------------------------|--------------------------|---------------------------|---------------------------------------------------------------------------|
| METHOD      | Macro.execute(options)         | Object                   | Client and server scripts | Performs a macro operation and returns its result in an object.           |
| METHOD      | Macro.execute.promise(options) | Promise                  | Client scripts            | Performs a macro operation asynchronously.                                |
| METHOD      | Macro(options)                 | Object                   | Client and server scripts | Performs a macro operation and returns its result in an object.           |
| METHOD      | Macro.promise(options)         | Promise                  | Client scripts            | Performs a macro operation asynchronously.                                |
| PROPERTY    | Macro.id                       | string                   | Client and server scripts | The ID of the macro. For a list of macro IDs, see Supported Record Macros |
| PROPERTY    | Macro.label                    | string                   | Client and server scripts | The macro label.                                                          |
| PROPERTY    | Macro.description              | string                   | Client and server scripts | The macro description.                                                    |
| PROPERTY    | Macro.attributes               | Object                   | Client and server scripts | The macro defined attributes.                                             |

## Record Object Members

The following members are called on the record.Record object.

Return Type / Value Type

Supported Script Types

Record.cancelLine(options)

Client and server scripts

Cancels the currently selected line on a sublist.

| MEMBER TYPE | NAME                                           | RETURN TYPE / VALUE TYPE                 | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                           |
|-------------|------------------------------------------------|------------------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | Record.cancelLine(options)                     | record.Record                            | Client and server scripts | Cancels the currently selected line on a sublist.                                                                                     |
| METHOD      | Record.commitLine(options)                     | record.Record                            | Client and server scripts | Commits the currently selected line on a sublist.                                                                                     |
| METHOD      | Record.executeMacro(options)                   | Object                                   | Client and server scripts | Performs macro operation and returns its result in a plain JavaScript object.                                                         |
| METHOD      | Record.getMacros()                             | Object                                   | Client and server scripts | Provides a plain JavaScript object that contains macro objects defined for a record type, indexed by the Macro ID.                    |
| METHOD      | Record.findMatrixSublistLineWithValue(options) | number                                   | Client and server scripts | Returns the line number of the first instance where a specified value is found in a specified column of the matrix.                   |
| METHOD      | Record.findSublistLineWithValue(options)       | number                                   | Client and server scripts | Returns the line number for the first occurrence of a field value in a sublist.                                                       |
| METHOD      | Record.getCurrentMatrixSublistValue(options)   | number   Date   string   array   boolean | Client and server scripts | Gets the value for the currently selected line in the matrix.                                                                         |
| METHOD      | Record.getCurrentSublistField(options)         | record.Field                             | Client and server scripts | Returns a field object from a sublist.                                                                                                |
| METHOD      | Record.getCurrentSublistIndex(options)         | number                                   | Client and server scripts | Returns the line number of the currently selected line.                                                                               |
| METHOD      | Record.getCurrentSublistSubrecord(options)     | record.Record                            | Client and server scripts | Gets the subrecord for the associated sublist field on the current line.                                                              |
| METHOD      | Record.getCurrentSublistText(options)          | string                                   | Client and server scripts | Returns a text representation of the field value in the currently selected line.                                                      |
| METHOD      | Record.getCurrentSublistValue(options)         | number   Date   string   array   boolean | Client and server scripts | Returns the value of a sublist field on the currently selected sublist line.                                                          |
| METHOD      | Record.getField(options)                       | record.Field                             | Client and server scripts | Returns a field object from a record.                                                                                                 |
| METHOD      | Record.getFields()                             | string[]                                 | Client and server scripts | Returns the body field names (internal ids) of all the fields in the record, including machine header field and matrix header fields. |
| METHOD      | Record.getLineCount(options)                   | number                                   | Client and server scripts | Returns the number of lines in a sublist.                                                                                             |
| METHOD      | Record.getMacro(options)                       | record.Macro                             | Client and server scripts | Provides a macro to execute.                                                                                                          |

| MEMBER TYPE | NAME                                          | RETURN TYPE / VALUE TYPE                 | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                          |
|-------------|-----------------------------------------------|------------------------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | Record.getMacros()                            | Object                                   | Client and server scripts | Provides a plain JavaScript object that contains macro objects defined for a record type, indexed by the Macro ID.                   |
| METHOD      | Record.getMatrixHeaderCount(options)          | number                                   | Client and server scripts | Returns the number of columns for the specified matrix.                                                                              |
| METHOD      | Record.getMatrixHeaderField(options)          | record.Field                             | Client and server scripts | Gets the field for the specified header in the matrix.                                                                               |
| METHOD      | Record.getMatrixHeaderValue(options)          | number   Date   string   array   boolean | Client and server scripts | Gets the value for the associated header in the matrix.                                                                              |
| METHOD      | Record.getMatrixSublistField(options)         | record.Field                             | Client and server scripts | Gets the field for the specified sublist in the matrix.                                                                              |
| METHOD      | Record.getMatrixSublistValue(options)         | number   Date   string   array   boolean | Client and server scripts | Gets the value for the associated field in the matrix.                                                                               |
| METHOD      | Record.getSublist(options)                    | record.Sublist                           | Client and server scripts | Returns the specified sublist.                                                                                                       |
| METHOD      | Record.getSublists()                          | string[]                                 | Client and server scripts | Returns all the names of all the sublists.                                                                                           |
| METHOD      | Record.getSublistField(options)               | record.Field                             | Client and server scripts | Returns a field object from a sublist.                                                                                               |
| METHOD      | Record.getSublistFields(options)              | string[]                                 | Client and server scripts | Returns all the field names in a sublist.                                                                                            |
| METHOD      | Record.getSublistSubrecord(options)           | record.Record                            | Client and server scripts | Gets the subrecord associated with a sublist field. (standard mode only)                                                             |
| METHOD      | Record.getSublistText(options)                | string                                   | Client and server scripts | Returns the value of a sublist field in a text representation.                                                                       |
| METHOD      | Record.getSublistValue(options)               | number   Date   string   array   boolean | Client and server scripts | Returns the value of a sublist field.                                                                                                |
| METHOD      | Record.getSubrecord(options)                  | record.Record                            | Client and server scripts | Gets the subrecord for the associated field.                                                                                         |
| METHOD      | Record.getText(options)                       | string                                   | Client and server scripts | Returns the text representation of a field value.                                                                                    |
| METHOD      | Record.getValue(options)                      | number   Date   string   array   boolean | Client and server scripts | Returns the value of a field. Not to be used on custom password fields. Use crypto.checkPasswordField(options) instead.              |
| METHOD      | Record.hasCurrentSublistSubrecord(options)    | boolean                                  | Client and server scripts | Returns a value indicating whether the associated sublist field has a subrecord on the current line.                                 |
| METHOD      | Record.hasSublistSubrecord(options)           | boolean                                  | Client and server scripts | Returns a value indicating whether the associated sublist field contains a subrecord.                                                |
| METHOD      | Record.hasSubrecord(options)                  | boolean                                  | Client and server scripts | Returns a value indicating whether the field contains a subrecord.                                                                   |
| METHOD      | Record.insertLine(options)                    | record.Record                            | Client and server scripts | Inserts a sublist line.                                                                                                              |
| METHOD      |                                               |                                          |                           |                                                                                                                                      |
| METHOD      | Record.removeCurrentSublistSubrecord(options) | record.Record                            | Client and server scripts | Removes the subrecord for the associated sublist field on the current line.                                                          |
| METHOD      | Record.removeLine(options)                    | record.Record                            | Client and server scripts | Removes a sublist line.                                                                                                              |
| METHOD      | Record.removeSublistSubrecord(options)        | record.Record                            | Client and server scripts | Removes the subrecord for the associated sublist field. (standard mode only)                                                         |
| METHOD      | Record.removeSubrecord(options)               | record.Record                            | Client and server scripts | Removes the subrecord for the associated field.                                                                                      |
| METHOD      | Record.save(options)                          | number                                   | Client and server scripts | Submits a new record or saves edits to an existing record. This method is not available to subrecords.                               |
| METHOD      | Record.save.promise(options)                  | number                                   | Client scripts            | Submits a new record asynchronously or saves edits to an existing record asynchronously. This method is not available to subrecords. |
| METHOD      | Record.selectLine(options)                    | record.Record                            | Client and server scripts | Selects an existing line in a sublist.                                                                                               |
| METHOD      | Record.selectNewLine(options)                 | record.Record                            | Client and server scripts | Selects a new line at the end of a sublist.                                                                                          |
| METHOD      | Record.setCurrentMatrixSublistValue(options)  | record.Record                            | Client and server scripts | Sets the value for the line currently selected in the matrix.                                                                        |

| MEMBER TYPE | NAME                                   | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                           |
|-------------|----------------------------------------|--------------------------|---------------------------|---------------------------------------------------------------------------------------|
| METHOD      | Record.setCurrentSublistText(options)  | record.Record            | Client and server scripts | Sets the value for the field in the currently selected line by a text representation. |
| METHOD      | Record.setCurrentSublistValue(options) | record.Record            | Client and server scripts | Sets the value for the field in the currently selected line.                          |
| METHOD      | Record.setMatrixHeaderValue(options)   | record.Record            | Client and server scripts | Sets the value for the associated header in the matrix.                               |
| METHOD      | Record.setMatrixSublistValue(options)  | record.Record            | Client and server scripts | Sets the value for the associated field in the matrix.                                |
| METHOD      | Record.setSublistText(options)         | record.Record            | Client and server scripts | Sets the value of a sublist field by a text representation. (standard mode only)      |
| METHOD      | Record.setSublistValue(options)        | record.Record            | Client and server scripts | Sets the value of a sublist field. (standard mode only)                               |
| METHOD      | Record.setText(options)                | record.Record            | Client and server scripts | Sets the value of the field by a text representation.                                 |
| METHOD      | Record.setValue(options)               | record.Record            | Client and server scripts | Sets the value of a field.                                                            |
| PROPERTY    | Record.id                              | number (read-only)       | Client and server scripts | The internal ID of a specific record. This property is not available to subrecords.   |
| PROPERTY    | Record.isDynamic                       | boolean (read-only)      | Client and server scripts | Indicates whether the record is in dynamic or standard mode.                          |
| PROPERTY    | Record.type                            | string (read-only)       | Client and server scripts | The record type. This property is not available to subrecords.                        |

Sublist Object Members

Return Type / Value Type  
Supported Script Types  
Sublist.getColumn(options)  
Client and server scripts  
Returns a column in the sublist.  
string (read-only)

| MEMBER TYPE | NAME                       | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                    |
|-------------|----------------------------|--------------------------|---------------------------|----------------------------------------------------------------|
| METHOD      | Sublist.getColumn(options) | record.Column            | Client and server scripts | Returns a column in the sublist.                               |
| PROPERTY    | Sublist.id                 | string (read-only)       | Client and server scripts | Returns the internal ID of the sublist.                        |
| PROPERTY    | Sublist.isChanged          | boolean (read-only)      | Client and server scripts | Indicates whether the sublist has changed on the record form.  |
| PROPERTY    | Sublist.isDisplay          | boolean (read-only)      | Client and server scripts | Indicates whether the sublist is displayed on the record form. |
| PROPERTY    | Sublist.type               | string (read-only)       | Client and server scripts | Returns the sublist type.                                      |

N/record Default Values

You can specify record initialization parameters that default when creating, copying, loading, and transforming records. To enable this behavior, use the optional defaultValues parameter in the following APIs:  
record.create(options)  
record.copy(options)  
record.transform(options)  
record.load(options)  
The following table lists initialization types that are available to certain SuiteScript-supported records and the values they can contain.

| RECORD                                                             | INITIALIZATION TYPE | VALUES               |
|--------------------------------------------------------------------|---------------------|----------------------|
| ALL SUITESCRIPT-SUPPORTED RECORDS THAT SUPPORT FORM CUSTOMIZATION. | customform          | &lt;customformid>;   |
| ASSEMBLY BUILD                                                     | assemblyitem        | &lt;assemblyitemid>; |
| CASH REFUND                                                        | entity              | &lt;entityid>;       |
| CASH SALE                                                          | entity              | &lt;entityid>;       |
| CHARGE RULE                                                        | entity              | &lt;entityid>;       |
| CHECK                                                              | entity              | &lt;entityid>;       |

| RECORD                    | INITIALIZATION TYPE   | VALUES                                                               |
|---------------------------|-----------------------|----------------------------------------------------------------------|
| CREDIT MEMO               | entity                | <entityid>;                                                          |
| CUSTOMER PAYMENT          | entity                | <entityid>;                                                          |
| CUSTOMER REFUND           | entity                | <entityid>;                                                          |
| DEPOSIT                   | disablepaymentfilters | <disablepaymentfilters>;                                             |
| ESTIMATE                  | entity                | <entityid>;                                                          |
| EXPENSE REPORT            | entity                | <entityid>;                                                          |
| INVOICE                   | entity                | <entityid>;                                                          |
| ITEM RECEIPT              | entity                | <entityid>;                                                          |
| NON-INVENTORY PART        | subtype               | sale   resale   purchase                                             |
| OPPORTUNITY               | entity                | <entityid>;                                                          |
| OTHER CHARGE ITEM         | subtype               | sale   resale   purchase                                             |
| PURCHASE ORDER            | entity                | <entityid>;                                                          |
| RETURN AUTHORIZATION      | entity                | <entityid>;                                                          |
| SALES ORDER               | entity                | <entityid>;                                                          |
| SCRIPT DEPLOYMENT         | script                | <scriptid>;                                                          |
| SERVICE                   | subtype               | sale   resale   purchase                                             |
| SUBSCRIPTION CHANGE ORDER |                       | <scriptid>; See Default Values.                                      |
| TAX GROUP                 | nexuscountry          | <countrycode>; See Country Codes Used for Initialization Parameters. |
| TAX TYPE                  | country               | <countrycode>; See Country Codes Used for Initialization Parameters. |
| TOPIC                     | parenttopic           | <parenttopicid>;                                                     |
| VENDOR BILL               | entity                | <entityid>;                                                          |
| VENDOR PAYMENT            | entity                | <entityid>;                                                          |
| WORK ORDER                | assemblyitem          | <assemblyitemid>;                                                    |

### Country Codes Used for Initialization Parameters

If you are scripting the Tax Group or Tax Type records, you can initialize the record to source all values related to a specific country. In your script, use the country code for the countrycodeid value, for example:

```
record.create('taxgroup', {nexuscountry: 'AR'});
```

Country Code

- United Arab Emirates
- Antigua and Barbuda
- Bosnia and Herzegovina
- Saint Barthélemy

| COUNTRY CODE | COUNTRY NAME         |
|--------------|----------------------|
| AD           | Andorra              |
| AE           | United Arab Emirates |
| AF           | Afghanistan          |
| AG           | Antigua and Barbuda  |
| AI           | Anguilla             |
| AL           | Albania              |
| AM           | Armenia              |
| AO           | Angola               |
| AQ           | Antarctica           |
| AR           | Argentina            |
| AS           | American Samoa       |

| COUNTRY CODE | COUNTRY NAME                           |
|--------------|----------------------------------------|
| AT           | Austria                                |
| AU           | Australia                              |
| AW           | Aruba                                  |
| AX           | Åland Islands                          |
| AZ           | Azerbaijan                             |
| BA           | Bosnia and Herzegovina                 |
| BB           | Barbados                               |
| BD           | Bangladesh                             |
| BE           | Belgium                                |
| BF           | Burkina Faso                           |
| BG           | Bulgaria                               |
| BH           | Bahrain                                |
| BI           | Burundi                                |
| BJ           | Benin                                  |
| BL           | Saint Barthélemy                       |
| BM           | Bermuda                                |
| BN           | Brunei Darrussalam                     |
| BO           | Bolivia (Plurinational State of)       |
| BQ           | Bonaire, Sint Eustatius and Saba       |
| BR           | Brazil                                 |
| BS           | Bahamas                                |
| BT           | Bhutan                                 |
| BV           | Bouvet Island                          |
| BW           | Botswana                               |
| BY           | Belarus                                |
| BZ           | Belize                                 |
| CA           | Canada                                 |
| CC           | Cocos (Keeling) Islands                |
| CD           | Congo (the Democratic Republic of the) |
| CF           | Central African Republic               |
| CG           | Congo                                  |
| CH           | Switzerland                            |
| CI           | Côte d'Ivoire                          |
| CK           | Cook Islands                           |
| CL           | Chile                                  |
| CM           | Cameroon                               |
| CN           | China                                  |
| CO           | Colombia                               |
| CR           | Costa Rica                             |
| CU           | Cuba                                   |
| CV           | Cabo Verde                             |
| CW           | Curacao                                |
| CX           | Christmas Island                       |

| COUNTRY CODE | COUNTRY NAME                                 |
|--------------|----------------------------------------------|
| CY           | Cyprus                                       |
| CZ           | Czechia                                      |
| DE           | Germany                                      |
| DJ           | Djibouti                                     |
| DK           | Denmark                                      |
| DM           | Dominica                                     |
| DO           | Dominican Republic                           |
| DZ           | Algeria                                      |
| EA           | Ceuta and Melilla                            |
| EC           | Ecuador                                      |
| EE           | Estonia                                      |
| EG           | Egypt                                        |
| EH           | Western Sahara                               |
| ER           | Eritrea                                      |
| ES           | Spain                                        |
| ET           | Ethiopia                                     |
| FI           | Finland                                      |
| FJ           | Fiji                                         |
| FK           | Falkland Islands (Malvinas)                  |
| FM           | Micronesia (Federated States of)             |
| FO           | Faroe Islands                                |
| FR           | France                                       |
| GA           | Gabon                                        |
| GB           | United Kingdom                               |
| GD           | Grenada                                      |
| GE           | Georgia                                      |
| GF           | French Guiana                                |
| GG           | Guernsey                                     |
| GH           | Ghana                                        |
| GI           | Gibraltar                                    |
| GL           | Greenland                                    |
| GM           | Gambia                                       |
| GN           | Guinea                                       |
| GP           | Guadeloupe                                   |
| GQ           | Equatorial Guinea                            |
| GR           | Greece                                       |
| GS           | South Georgia and the South Sandwich Islands |
| GT           | Guatemala                                    |
| GU           | Guam                                         |
| GW           | Guinea-Bissau                                |
| GY           | Guyana                                       |
| HK           | Hong Kong                                    |
| HM           | Heard Island and McDonald Islands            |

| COUNTRY CODE | COUNTRY NAME                                |
|--------------|---------------------------------------------|
| HN           | Honduras                                    |
| HR           | Croatia                                     |
| HT           | Haiti                                       |
| HU           | Hungary                                     |
| IC           | Canary Islands                              |
| ID           | Indonesia                                   |
| IE           | Ireland                                     |
| IL           | Israel                                      |
| IM           | Isle of Man                                 |
| IN           | India                                       |
| IO           | British Indian Ocean Territory              |
| IQ           | Iraq                                        |
| IR           | Iran (Islamic Republic of)                  |
| IS           | Iceland                                     |
| IT           | Italy                                       |
| JE           | Jersey                                      |
| JM           | Jamaica                                     |
| JO           | Jordan                                      |
| JP           | Japan                                       |
| KE           | Kenya                                       |
| KG           | Kyrgyzstan                                  |
| KH           | Cambodia                                    |
| KI           | Kiribati                                    |
| KM           | Comoros                                     |
| KN           | Saint Kitts and Nevis                       |
| KP           | Korea (the Democratic People's Republic of) |
| KR           | Korea (the Republic of)                     |
| KW           | Kuwait                                      |
| KY           | Cayman Islands                              |
| KZ           | Kazakhstan                                  |
| LA           | Lao People's Democratic Republic            |
| LB           | Lebanon                                     |
| LC           | Saint Lucia                                 |
| LI           | Liechtenstein                               |
| LK           | Sri Lanka                                   |
| LR           | Liberia                                     |
| LS           | Lesotho                                     |
| LT           | Lithuania                                   |
| LU           | Luxembourg                                  |
| LV           | Latvia                                      |
| LY           | Libya                                       |
| MA           | Morocco                                     |
| MC           | Monaco                                      |

| COUNTRY CODE | COUNTRY NAME               |
|--------------|----------------------------|
| MD           | Moldova (the Republic of)  |
| ME           | Montenegro                 |
| MF           | Saint Martin (French part) |
| MG           | Madagascar                 |
| MH           | Marshall Islands           |
| MK           | North Macedonia            |
| ML           | Mali                       |
| MM           | Myanmar                    |
| MN           | Mongolia                   |
| MO           | Macao                      |
| MP           | Northern Mariana Islands   |
| MQ           | Martinique                 |
| MR           | Mauritania                 |
| MS           | Montserrat                 |
| MT           | Malta                      |
| MU           | Mauritius                  |
| MV           | Maldives                   |
| MW           | Malawi                     |
| MX           | Mexico                     |
| MY           | Malaysia                   |
| MZ           | Mozambique                 |
| NA           | Namibia                    |
| NC           | New Caledonia              |
| NE           | Niger                      |
| NF           | Norfolk Island             |
| NG           | Nigeria                    |
| NI           | Nicaragua                  |
| NL           | Netherlands                |
| NO           | Norway                     |
| NP           | Nepal                      |
| NR           | Nauru                      |
| NU           | Niue                       |
| NZ           | New Zealand                |
| OM           | Oman                       |
| PA           | Panama                     |
| PE           | Peru                       |
| PF           | French Polynesia           |
| PG           | Papua New Guinea           |
| PH           | Philippines                |
| PK           | Pakistan                   |
| PL           | Poland                     |
| PM           | Saint Pierre and Miquelon  |
| PN           | Pitcairn                   |

| COUNTRY CODE | COUNTRY NAME                                 |
|--------------|----------------------------------------------|
| PR           | Puerto Rico                                  |
| PS           | Palestine, State of                          |
| PT           | Portugal                                     |
| PW           | Palau                                        |
| PY           | Paraguay                                     |
| QA           | Qatar                                        |
| RE           | Réunion                                      |
| RO           | Romania                                      |
| RS           | Serbia                                       |
| RU           | Russian Federation                           |
| RW           | Rwanda                                       |
| SA           | Saudi Arabia                                 |
| SB           | Solomon Islands                              |
| SC           | Seychelles                                   |
| SD           | Sudan                                        |
| SE           | Sweden                                       |
| SG           | Singapore                                    |
| SH           | Saint Helena, Ascension and Tristan da Cunha |
| SI           | Slovenia                                     |
| SJ           | Svalbard and Jan Mayen                       |
| SK           | Slovakia                                     |
| SL           | Sierra Leone                                 |
| SM           | San Marino                                   |
| SN           | Senegal                                      |
| SO           | Somalia                                      |
| SR           | Suriname                                     |
| SS           | South Sudan                                  |
| ST           | Sao Tome and Principe                        |
| SV           | El Salvador                                  |
| SX           | Sint Maarten (Dutch part)                    |
| SY           | Syrian Arab Republic                         |
| SZ           | Eswatini                                     |
| TC           | Turks and Caicos Islands                     |
| TD           | Chad                                         |
| TF           | French Southern Territories                  |
| TG           | Togo                                         |
| TH           | Thailand                                     |
| TJ           | Tajikistan                                   |
| TK           | Tokelau                                      |
| TM           | Turkmenistan                                 |
| TN           | Tunisia                                      |
| TO           | Tonga                                        |
| TP           | Timor-Leste                                  |

| COUNTRY CODE | COUNTRY NAME                         |
|--------------|--------------------------------------|
| TR           | Türkiye                              |
| TT           | Trinidad and Tobago                  |
| TV           | Tuvalu                               |
| TW           | Taiwan (Province of China)           |
| TZ           | Tanzania, the United Republic of     |
| UA           | Ukraine                              |
| UG           | Uganda                               |
| UM           | United States Minor Outlying Islands |
| US           | United States                        |
| UY           | Uruguay                              |
| UZ           | Uzbekistan                           |
| VA           | Holy See                             |
| VC           | Saint Vincent and the Grenadines     |
| VE           | Venezuela (Bolivarian Republic of)   |
| VG           | Virgin Islands (British)             |
| VI           | Virgin Islands (U.S.)                |
| VN           | Viet Nam                             |
| VU           | Vanuatu                              |
| WF           | Wallis and Futuna Islands            |
| WS           | Samoa                                |
| XK           | Kosovo                               |
| YE           | Yemen                                |
| YT           | Mayotte                              |
| ZA           | South Africa                         |
| ZM           | Zambia                               |
| ZW           | Zimbabwe                             |

 Sample 1: Create and Save a Contact Record

```
/**
 * @NApiVersion 2.1
 */
require(['N/record'], record => {
  // Create an object to hold name data for the contact
  const nameData = {
    firstname: 'John',
    middlename: 'Doe',
    lastname: 'Smith'
  };

  // Create a contact record
  let objRecord = record.create({
    type: record.Type.CONTACT,
    isDynamic: true
  });

  // Set the values of the subsidiary, firstname, middlename,
  // and lastname properties
  objRecord.setValue({
    fieldId: 'subsidiary',
    value: '1'
  });
  for (let key in nameData) {
    if (nameData.hasOwnProperty(key)) {
      objRecord.setValue({
        fieldId: key,
        value: nameData[key]
      });
    }
  }

  // Save the record
  let recordId = objRecord.save({
    enableSourcing: false,
    ignoreMandatoryFields: false
  });
});
```

### Sample 2: Create and Save a Contact Record Asynchronously Using Promise Methods

```
/**
 * @NApiVersion 2.1
 * @NScriptType ClientScript
 */
require(['N/record'], record => {
  // Create an object to hold name data for the contact
  const nameData = {
    firstname: 'John',
    middlename: 'Doe',
    lastname: 'Smith'
  };

  // Create a contact record using the promise method
  let createRecordPromise = record.create.promise({
    type: record.Type.CONTACT,
    isDynamic: true
  });

  // When the promise is fulfilled, set the values of the subsidiary,
  // firstname, middlename, and lastname properties, and save the
  // record
  createRecordPromise.then(objRecord => {
    log.debug('Start evaluating promise content...');
    objRecord.setValue({
      fieldId: 'subsidiary',
      value: '1'
    });
    for (let key in nameData) {
      if (nameData.hasOwnProperty(key)) {
        objRecord.setValue({
          fieldId: key,
          value: nameData[key]
        });
      }
    }
    let recordId = objRecord.save({
      enableSourcing: false,
      ignoreMandatoryFields: false
    });
  }, function(e) {
    log.error('Unable to create contact', e.name);
  });
});
```

### Sample 3: Create Multiple Sales Records Using a Scheduled Script

```
/**
 * @NApiVersion 2.1
 * @NScriptType ScheduledScript
 */

// This script creates multiple sales records and logs the record creation progress.
define(['N/runtime', 'N/record'], function(runtime, record) {
  return {
    execute: function(context) {
      var script = runtime.getCurrentScript();
      for (x = 0; x < 500; x++) {
        var rec = record.create({
          type: record.Type.SALES_ORDER
        });
        script.percentComplete = (x * 100)/500;
        log.debug({
          title: 'New Sales Orders',
          details: 'Record creation progress: ' + script.percentComplete + '%'
        });
      }
    }
  };
});
```

#### Sample 4: Access Sublists and a Subrecord from a Record

```
/**
 * @NApiVersion 2.x
 */
require(['N/record'], function(record) {
  function createPurchaseOrder() {
    var rec = record.create({
      type: 'purchaseorder',
      isDynamic: true
    });
    rec.setValue({
      fieldId: 'entity',
      value: 52
    });
    rec.setValue({
      fieldId: 'location',
      value: 2
    });
    rec.selectNewLine({
      sublistId: 'item'
    });
    rec.setCurrentSublistValue({
      sublistId: 'item',
      fieldId: 'item',
      value: 190
    });
    rec.setCurrentSublistValue({
      sublistId: 'item',
      fieldId: 'quantity',
      value: 2
    });
    subrecordInvDetail = rec.getCurrentSublistSubrecord({
      sublistId: 'item',
      fieldId: 'inventorydetail'
    });
    subrecordInvDetail.selectNewLine({
      sublistId: 'inventoryassignment'
    });
    subrecordInvDetail.setCurrentSublistValue({
      sublistId: 'inventoryassignment',
      fieldId: 'receiptinventorynumber',
      value: 'myinventoryNumber'
    });
    subrecordInvDetail.commitLine({
      sublistId: 'inventoryassignment'
    });
    subrecordInvDetail.selectLine({
      sublistId: 'inventoryassignment',
      line: 0
    });
    var myInventoryNumber = subrecordInvDetail.getCurrentSublistValue({
      sublistId: 'inventoryassignment',
      fieldId: 'receiptinventorynumber'
    });
    rec.commitLine({
      sublistId: 'item'
    });
    var recordId = rec.save();
  }
  createPurchaseOrder();
});
```

#### Sample 5: Access Sublists and a Subrecord from a Record Asynchronously Using Promise Methods

```
/**
 * @NApiVersion 2.x
 */
require(['N/record'], function(record) {
  function createPurchaseOrder() {
    var createRecordPromise = record.create.promise({
      type: 'purchaseorder',
      isDynamic: true
    });
    createRecordPromise.then(function(rec) {
      rec.setValue({
        fieldId: 'entity',
        value: 52
      });
      rec.setValue({
        fieldId: 'location',
        value: 2
      });
      rec.selectNewLine({
        sublistId: 'item'
      });
      rec.setCurrentSublistValue({
        sublistId: 'item',
        fieldId: 'item',
        value: 190
      });
      rec.setCurrentSublistValue({
        sublistId: 'item',
        fieldId: 'quantity',
        value: 2
      });
      subrecordInvDetail = rec.getCurrentSublistSubrecord({
        sublistId: 'item',
        fieldId: 'inventorydetail'
      });
      subrecordInvDetail.selectNewLine({
        sublistId: 'inventoryassignment'
      });
      subrecordInvDetail.setCurrentSublistValue({
        sublistId: 'inventoryassignment',
        fieldId: 'receiptinventorynumber',
        value: 'myinventoryNumber'
      });
      subrecordInvDetail.commitLine({
        sublistId: 'inventoryassignment'
      });
      subrecordInvDetail.selectLine({
        sublistId: 'inventoryassignment',
        line: 0
      });
      var myInventoryNumber = subrecordInvDetail.getCurrentSublistValue({
        sublistId: 'inventoryassignment',
        fieldId: 'receiptinventorynumber'
      });
      rec.commitLine({
        sublistId: 'item'
      });
      var recordId = rec.save();
    }, function(err) {
      log.error('Unable to create purchase order!', err.name);
    });
  }
  createPurchaseOrder();
});
```

## Sample 6: Call a Macro on a Sales Order Record

```
/**
 * @NApiVersion 2.x
 */
require(['N/record'],
function(record) {

    var recordObj = record.create({
        type: record.Type.SALES_ORDER,
        isDynamic: true
    });

    var ENTITY_VALUE = 1;
    var ITEM_VALUE = 1;
    recordObj.setValue({
        fieldId: 'entity',
        value: ENTITY_VALUE
    });
    recordObj.selectNewLine({
        sublistId: 'item'
    });
    recordObj.setCurrentSublistValue({
        sublistId: 'item',
        fieldId: 'item',
        value: ITEM_VALUE
    });
    recordObj.setCurrentSublistValue({
        sublistId: 'item',
        fieldId: 'quantity',
        value: 1
    });
    recordObj.commitLine({
        sublistId: 'item'
    });

    var totalBeforeTax = recordObj.getValue({fieldId: 'total'});

    // get macros available on the record
    var macros = recordObj.getMacros();

    // execute the macro
    if ('calculateTax' in macros)
    {
        macros.calculateTax(); // For promise version use: macros.calculateTax.promise()
    }
    // Alternative (direct) macro execution
    // var calculateTax = recordObj.getMacro({id: 'calculateTax'});
    // calculateTax(); // For promise version use: calculateTax.promise()
    var totalAfterTax = recordObj.getValue({fieldId: 'total'});

    var recordId = recordObj.save({
        enableSourcing: false,
        ignoreMandatoryFields: false
    });
});
```

### 35. N/recordContext Module

Load the N/recordContext module to get the context type of a record.

`define(['N/recordContext'], (recordContext) => { ... })`

✓ 1 Code Sample(s)

Scripts: Client and server scripts

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/recordContext'], (recordContext) => { // Access methods and properties on recordContext });
```

Use the N/recordContext module to get all the available context types of the record, such as localization. The localization context type indicates which country a script is using for execution. You can also use the N/recordContext module to create conditional statements within a script so that the script behaves differently based on the context.

For more information about localization context, see Localization Context.

N/recordContext Module Sample

#### N/recordContext Module Members

Return Type / Value Type

Supported Script Types

recordContext.RecordContext

Object (read-only)

Client and server scripts

Contains key-value pairs that represent context types and their values.

| MEMBER TYPE | NAME                              | RETURN TYPE / VALUE TYPE    | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                                  |
|-------------|-----------------------------------|-----------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT      | recordContext.RecordContext       | Object (read-only)          | Client and server scripts | Contains key-value pairs that represent context types and their values.                                                                      |
| METHOD      | recordContext.getContext(options) | recordContext.RecordContext | Client and server scripts | Returns the record context object for a record.                                                                                              |
| ENUM        | recordContext.ContextType         | enum (read-only)            | Client and server scripts | Holds the values for the context type. Used to set the value for the contextTypes parameter of the recordContext.getContext(options) method. |

 Sample 1: N/recordContext Module Sample

```
/**
 *@NApiVersion 2.x
 */
// This example calls the getContext() API with options recordType, recordId and record.
require(['N/record', 'N/recordContext'],
function(record, recordContext) {
    // Create record
    var employee = record.create({
        type : record.Type.EMPLOYEE,
        isDynamic: true
    })
    // Setup CA subsidiary
    employee.setValue('subsidiary', 2);
    employee.setValue('entityid', 'test_emp_' + Date.now())
    employeeId = employee.save()

    // getContext() with options recordType, recordId and contextTypes
    var employeeContext = recordContext.getContext({
        recordType : record.Type.EMPLOYEE,
        recordId: employeeId,
        contextTypes: [recordContext.ContextType.LOCALIZATION]
    })
    log.debug(employeeContext); // expected log will list {"localization":["CA"]}

    // Change employee subsidiary to AU
    employee.setValue('subsidiary', 3);

    // getContext() with options record and contextTypes
    var employeeContext = recordContext.getContext({
        record : employee,
        contextTypes: [recordContext.ContextType.LOCALIZATION]
    })
    log.debug(employeeContext); // expected log will list {"localization":["AU"]}

    // Delete record
    record.delete({
        type : record.Type.EMPLOYEE,
        id: 1
    })
});
```

## 36. N/redirect Module

Load the N/redirect module to redirect users to a URL, a Suitelet, a record, a task link, a saved search, or an unsaved search.

define(['N/redirect'], (redirect) => { ... })

✓ 1 Code Sample(s)

Scripts: Server scripts

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/redirect'], (redirect) => { // Access methods and properties on redirect });
```

Use the N/redirect module to customize navigation within NetSuite by setting up a redirect URL that resolves to a NetSuite resource or external URL. You can redirect users to one of the following:

- URL
- Suitelet
- Record
- Task link
- Saved search
- Unsaved search
- N/redirect Module Script Sample

Suitelets, beforeLoad user events, and synchronous afterSubmit user events are supported. This module does not support beforeSubmit or asynchronous afterSubmit user events. This module is only supported when triggered from the UI. Backend contexts such as CSV Import and Scheduled Scripts are not supported.

### N/redirect Module Members

- Supported Script Types
- redirect.redirect(options)  
Suitelets, beforeLoad user events, and synchronous afterSubmit user events  
Redirects to the URL of a Suitelet that is available externally (available without login).
- redirect.toRecord(options)  
Suitelets, beforeLoad user events, and synchronous afterSubmit user events

| MEMBER TYPE | NAME                                  | RETURN TYPE | SUPPORTED SCRIPT TYPES                                                     | DESCRIPTION                                                                                |
|-------------|---------------------------------------|-------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| METHOD      | redirect.redirect(options)            | void        | Suitelets, beforeLoad user events, and synchronous afterSubmit user events | Redirects to the URL of a Suitelet that is available externally (available without login). |
| METHOD      | redirect.toRecord(options)            | void        | Suitelets, beforeLoad user events, and synchronous afterSubmit user events | Redirects to a NetSuite record.                                                            |
| METHOD      | redirect.toRecordTransform(options)   | void        | workflow action scripts                                                    | Redirects to a standard or custom transaction instance.                                    |
| METHOD      | redirect.toSavedSearch(options)       | void        | afterSubmit user events                                                    | Redirects to a saved search.                                                               |
| METHOD      | redirect.toSavedSearchResult(options) | void        | afterSubmit user events                                                    | Redirects to a saved search result.                                                        |
| METHOD      | redirect.toSearch(options)            | void        | afterSubmit user events                                                    | Redirects to search.                                                                       |
| METHOD      | redirect.toSearchResult(options)      | void        | afterSubmit user events                                                    | Redirects to search results.                                                               |
| METHOD      | redirect.toSuitelet(options)          | void        | Suitelets, beforeLoad user events, and synchronous afterSubmit user events | Redirects to a Suitelet.                                                                   |
| METHOD      | redirect.toTaskLink(options)          | void        | Suitelets, beforeLoad user events, and synchronous afterSubmit user events | Redirects to a tasklink.                                                                   |

 Sample 1: N/redirect Module Script Sample

```
/**
 * @NApiVersion 2.x
 */
require(['N/record', 'N/redirect'], function(record, redirect) {
  function redirectToTaskRecord() {
    var taskTitle = 'New Opportunity';
    var taskRecord = record.create({
      type: record.Type.TASK
    });
    taskRecord.setValue('title', taskTitle);

    var taskRecordId = taskRecord.save();

    redirect.toRecord({
      type: record.Type.TASK,
      id: taskRecordId
    });
  }

  redirectToTaskRecord();
});
```

## 37. N/render Module

Load the N/render module to create forms or email from templates and to print to PDF or HTML.

```
define(['N/render'], (render) => { ... })
```

✓ 4 Code Sample(s)

Scripts: Server scripts

Permissions: Advanced PDF/HTML Templates Custom PDF Layouts

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/render'], (render) => { // Access methods and properties on render });
```

Use the N/render module for printing, PDF creation, form creation from templates, and email creation from templates.

Direct manipulation of the print URL is not supported.

### N/render Module Members

Return Type / Value Type

Supported Script Types

render.EmailMergeResult

Encapsulates an email merge result.

render.TemplateRenderer

Encapsulates a template object that produces HTML and PDF printed forms utilizing advanced PDF/HTML template capabilities.

| MEMBER TYPE | NAME                          | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                                                      |
|-------------|-------------------------------|--------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT      | render.EmailMergeResult       | Object                   | Server scripts         | Encapsulates an email merge result.                                                                                                                                                                              |
| OBJECT      | render.TemplateRenderer       | Object                   | Server scripts         | Encapsulates a template object that produces HTML and PDF printed forms utilizing advanced PDF/HTML template capabilities.                                                                                       |
| METHOD      | render.bom(options)           | file.File                | Server scripts         | Creates a PDF or HTML file object containing a bill of materials.                                                                                                                                                |
| METHOD      | render.create()               | render.TemplateRenderer  | Server scripts         | Creates a render.TemplateRenderer object.                                                                                                                                                                        |
| METHOD      | render.mergeEmail(options)    | render.EmailMergeResult  | Server scripts         | Creates a render.EmailMergeResult object.                                                                                                                                                                        |
| METHOD      | render.packingSlip(options)   | file.File                | Server scripts         | Creates a PDF or HTML file object containing a packing slip.                                                                                                                                                     |
| METHOD      | render.pickingTicket(options) | file.File                | Server scripts         | Creates a PDF or HTML file object containing a picking ticket.                                                                                                                                                   |
| METHOD      | render.statement(options)     | file.File                | Server scripts         | Creates a PDF or HTML file object containing a statement.                                                                                                                                                        |
| METHOD      | render.transaction(options)   | file.File                | Server scripts         | Creates a PDF or HTML file object containing a transaction.                                                                                                                                                      |
| METHOD      | render.xmlToPdf(options)      | file.File                | Server scripts         | Passes XML to the BFO tag library (which is stored by NetSuite), and returns a PDF file.                                                                                                                         |
| ENUM        | render.DataSource             | enum                     | Server scripts         | Holds the string values for supported data source types. Use this enum to set the options.format parameter of the TemplateRenderer.addCustomDataSource(options) method.                                          |
| ENUM        | render.PrintMode              | enum                     | Server scripts         | Holds the string values for supported print output types. Use this enum to set the options.printMode parameter of the render.bom(options), render.pickingTicket(options), and render.statement(options) methods. |

### EmailMergeResult Object Members

Return Type / Value Type

Supported Script Types

EmailMergeResult.body

string (read-only)

The body of the email distribution in string format.

EmailMergeResult.subject

| MEMBER TYPE | NAME                     | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                             |
|-------------|--------------------------|--------------------------|------------------------|---------------------------------------------------------|
| PROPERTY    | EmailMergeResult.body    | string (read-only)       | Server scripts         | The body of the email distribution in string format.    |
| PROPERTY    | EmailMergeResult.subject | string (read-only)       | Server scripts         | The subject of the email distribution in string format. |

### TemplateRenderer Object Members

Return Type / Value Type

## Supported Script Types

TemplateRenderer.addCustomDataSource(options)

Adds an XML file or JSON object to an advanced template as a custom data source.

TemplateRenderer.addQuery(options)

Uses Query as the renderer's data source.

| MEMBER TYPE | NAME                                            | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                      |
|-------------|-------------------------------------------------|--------------------------|------------------------|----------------------------------------------------------------------------------|
| METHOD      | TemplateRenderer.addCustomDataSource(options)   | void                     | Server scripts         | Adds an XML file or JSON object to an advanced template as a custom data source. |
| METHOD      | TemplateRenderer.addQuery(options)              | void                     | Server scripts         | Uses Query as the renderer's data source.                                        |
| METHOD      | TemplateRenderer.addRecord(options)             | void                     | Server scripts         | Binds a record to a template variable.                                           |
| METHOD      | TemplateRenderer.addSearchResults(options)      | void                     | Server scripts         | Binds a search result to a template variable.                                    |
| METHOD      | TemplateRenderer.renderAsPdf()                  | Object                   | Server scripts         | Uses an advanced template to produce a PDF printed form.                         |
| METHOD      | TemplateRenderer.renderPdfToResponse(options)   | void                     | Server scripts         | Renders PDF template content as a server response.                               |
| METHOD      | TemplateRenderer.renderAsString()               | string                   | Server scripts         | Returns template content in string form.                                         |
| METHOD      | TemplateRenderer.setTemplateById(options)       | void                     | Server scripts         | Sets the template using the internal ID.                                         |
| METHOD      | TemplateRenderer.setTemplateByScriptId(options) | void                     | Server scripts         | Sets the template using the script ID.                                           |
| METHOD      | TemplateRenderer.renderToResponse(options)      | void                     | Server scripts         | Renders HTML template content as a server response.                              |
| PROPERTY    | TemplateRenderer.templateContent                | string                   | Server scripts         | Content of the template.                                                         |

## ► Official NetSuite Script Samples (4 Sample(s))

### Sample 1: Generate a PDF File from a Raw XML String

```
/**
 * @NApiVersion 2.x
 */

require(['N/render'],
function(render) {
    function generatePdfFileFromRawXml() {
        var xmlStr = "<?xml version='1.0'?">\n" +
            "<!DOCTYPE pdf PUBLIC \"-//big.faceless.org//report\" \"report-1.1.dtd\">\n" +
            "<pdf>\n<body font-size='18'>\nHello World!\n</body>\n</pdf>";
        var pdfFile = render.xmlToPdf({
            xmlString: xmlStr
        });
        generatePdfFileFromRawXml();
    }
});
```

### Sample 2: Render a Transaction Record Into an HTML Page

```
/**
 * @NApiVersion 2.x
 */

require(['N/render'],
function(render) {
    function renderTransactionToHtml() {
        var transactionFile = render.transaction({
            entityId: 23,
            printMode: render.PrintMode.HTML
        });
        renderTransactionToHtml();
    }
});
```

### Sample 3: Render an Invoice Into a PDF File Using an XML Template

```
/**
 * @NApiVersion 2.x
 */

// This sample shows how to render an invoice into a PDF file using an XML template in the file cabinet.
// Note that this example requires the Advanced PDF/HTML Templates feature.
require(['N/render', 'N/file', 'N/record'],
function(render, file, record) {
    function renderRecordToPdfWithTemplate() {
        var xmlTemplateFile = file.load('Templates/PDF Templates/invoicePDFTemplate.xml');
        var renderer = render.create();
        renderer.templateContent = xmlTemplateFile.getContents();
        renderer.addRecord('record', record.load({
            type: record.Type.INVOICE,
            id: 37
        }));
        var invoicePdf = renderer.renderAsPdf();
    }
    renderRecordToPdfWithTemplate();
});
```

### Sample 4: Render Search Results Into a PDF File

```
/**
 * @NApiVersion 2.1
 * @NScriptType Suitelet
 */
// This sample shows how to render search results into a PDF file.
define(['N/render', 'N/search'], function(render, search) {
    function onRequest(options) {
        var request = options.request;
        var response = options.response;

        var xmlStr = '<?xml version="1.0" encoding="UTF-8">\n' +
            '<!DOCTYPE pdf PUBLIC "-//big.faceless.org//report" "report-1.1.dtd">\n' +
            '<pdf lang="ru=RU" xml:lang="ru-RU">\n' + '<head>\n' +
            '<link name="russianfont" type="font" subtype="opentype" ' +
            'src="NetSuiteFonts/verdana.ttf" " + "src-bold="NetSuiteFonts/verdanab.ttf"' +
            'src-italic="NetSuiteFonts/verdanai.ttf" " + "src-bolditalic="NetSuiteFonts/verdanabi.ttf"' +
            'bytes="2"/>\n' + '</head>\n' +
            '<body font-family="russianfont" font-size="18">\n????? </body>\n' + '</pdf>';

        var rs = search.create({
            type: search.Type.TRANSACTION,
            columns: ['trandate', 'amount', 'entity'],
            filters: []
        }).run();

        var results = rs.getRange(0, 1000);
        var renderer = render.create();
        renderer.templateContent = xmlStr;
        renderer.addSearchResults({
            templateName: 'results',
            searchResult: results
        });

        var newfile = renderer.renderAsPdf();
        response.writeFile(newfile, false);
    }

    return {
        onRequest: onRequest
    };
});
```

## 38. N/runtime Module

Load the N/runtime module to access runtime settings for company, script, session, system, user, or version.

```
define(['N/runtime'], (runtime) => { ... })
```

✓ 2 Code Sample(s)

Scripts: Client and server scripts

Permissions: Å

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/runtime'], (runtime) => { // Access methods and properties on runtime });
```

Use the N/runtime module to view runtime settings for the script, the session, or the user. You can also use this module to set a session key and to see whether a particular feature is enabled in your account.

### N/runtime Module Members

Return Type / Value Type

Supported Script Types

Client and server scripts

Encapsulates the runtime settings of the currently executing script.

Client and server scripts

Encapsulates the user session for the currently executing script.

| MEMBER TYPE | NAME                               | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                              |
|-------------|------------------------------------|--------------------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT      | runtime.Script                     | Object                   | Client and server scripts | Encapsulates the runtime settings of the currently executing script.                                                                     |
| OBJECT      | runtime.Session                    | Object                   | Client and server scripts | Encapsulates the user session for the currently executing script.                                                                        |
| OBJECT      | runtime.User                       | Object                   | Client and server scripts | Encapsulates the properties and preferences of the user currently executing the script.                                                  |
| METHOD      | runtime.getCurrentScript()         | runtime.Script           | Client and server scripts | Returns a runtime.Script object that represents the currently executing script.                                                          |
| METHOD      | runtime.getCurrentSession()        | runtime.Session          | Client and server scripts | Returns a runtime.Session object that represents the user session for the currently executing script.                                    |
| METHOD      | runtime.getCurrentUser()           | runtime.User             | Client and server scripts | Returns a runtime.User object that represents the properties and preferences of the user currently executing the script.                 |
| METHOD      | runtime.isFeatureInEffect(options) | boolean                  | Client and server scripts | Indicates whether a particular feature is enabled in a NetSuite account. These are the features that appear on the Enable Features page. |
| METHOD      | runtime.isNextActive()             | boolean                  | Client and server scripts | Returns whether NetSuite Next is active for the user running the script.                                                                 |
| PROPERTY    | runtime.accountId                  | string (read-only)       | Client and server scripts | The account ID for the current user.                                                                                                     |
| PROPERTY    | runtime.country                    | string (read-only)       | Client and server scripts | The country for the current company.                                                                                                     |
| PROPERTY    | runtime.envType                    | string (read-only)       | Client and server scripts | The current environment in which the script is executing. This property uses values from the runtime.EnvType enum.                       |
| PROPERTY    | runtime.executionContext           | string (read-only)       | Client and server scripts | The trigger of the current script. This property uses values from the runtime.ContextType enum.                                          |
| PROPERTY    | runtime.processorCount             | number (read-only)       | Client and server scripts | The number of processors available to the current account.                                                                               |
| PROPERTY    | runtime.queueCount                 | number (read-only)       | Client and server scripts | The number of scheduled script queues available to the current account.                                                                  |
| PROPERTY    | runtime.version                    | string (read-only)       | Client and server scripts | The version of NetSuite that the method is called in. For example, this property in an account running NetSuite 2023.2 is 2023.2.        |
| ENUM        | runtime.ContextType                | enum                     | Client and server scripts | Holds the context values for script triggers. This is the type for the runtime.executionContext property.                                |
| ENUM        | runtime.EnvType                    | enum                     | Client and server scripts | Holds all possible environment types that the current script can execute in. This is the type for the runtime.envType property.          |
| ENUM        | runtime.Permission                 | enum                     | Client and server scripts | Holds the user permission level for a specific permission ID. This is the type returned by the User.getPermission(options) method.       |

### Script Object Members

Return Type / Value Type

## Supported Script Types

Script.getParameter(options)

number | Date | string | boolean

Client and server scripts

Returns the value of a script parameter for the currently executing script.

| MEMBER TYPE | NAME                         | RETURN TYPE / VALUE TYPE         | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                                           |
|-------------|------------------------------|----------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | Script.getParameter(options) | number   Date   string   boolean | Client and server scripts | Returns the value of a script parameter for the currently executing script.                                                                           |
| METHOD      | Script.getRemainingUsage()   | number                           | Client and server scripts | Returns the number of units remaining (per governance limitations) for the currently executing script.                                                |
| PROPERTY    | Script.apiVersion            | string (read-only)               | Client and server scripts | The current script's runtime version.                                                                                                                 |
| PROPERTY    | Script.bundleIds             | Array (read-only)                | Client and server scripts | An array of bundle IDs for the bundles that include the currently executing script.                                                                   |
| PROPERTY    | Script.deploymentId          | string (read-only)               | Server scripts            | The deployment ID for the script deployment of the currently executing script.                                                                        |
| PROPERTY    | Script.id                    | string (read-only)               | Client and server scripts | The script ID for the currently executing script.                                                                                                     |
| PROPERTY    | Script.logLevel              | string (read-only)               | Server scripts            | The script logging level for the currently executing script.                                                                                          |
| PROPERTY    | Script.percentComplete       | number                           | Client and server scripts | The percent complete for the current scheduled script execution. This value will appear in the % Complete column on the Scheduled Script Status page. |

## Session Object Members

Return Type / Value Type

Supported Script Types

Session.get(options)

Returns the user-defined session object value associated with the session object key. Both the session object value and associated key are defined using Session.set(options).

Session.set(options)

Sets a key and value for a user-defined session object.

| MEMBER TYPE | NAME                 | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                    |
|-------------|----------------------|--------------------------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | Session.get(options) | string   null            | Server scripts         | Returns the user-defined session object value associated with the session object key. Both the session object value and associated key are defined using Session.set(options). |
| METHOD      | Session.set(options) | void                     | Server scripts         | Sets a key and value for a user-defined session object.                                                                                                                        |

## User Object Members

Return Type / Value Type

Supported Script Types

User.getPermission(options)

Client and server scripts

Returns a runtime.Permission user permission level for the specified permission.

User.getPreference(options)

| MEMBER TYPE | NAME                        | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                                                                                                                             |
|-------------|-----------------------------|--------------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | User.getPermission(options) | string                   | Client and server scripts | Returns a runtime.Permission user permission level for the specified permission.                                                                                                                                                        |
| METHOD      | User.getPreference(options) | string                   | Client and server scripts | Returns the value of a NetSuite preference. Currently only General Preferences and Accounting Preferences are exposed in SuiteScript. For more information about these preferences, see General Preferences and Accounting Preferences. |
| PROPERTY    | User.contact                | number (read-only)       | Client and server scripts | The internal ID of the currently logged-in contact.                                                                                                                                                                                     |
| PROPERTY    | User.department             | number (read-only)       | Client and server scripts | The internal ID of the department for the current user.                                                                                                                                                                                 |
| PROPERTY    | User.email                  | string (read-only)       | Client and server scripts | The email address of the current user.                                                                                                                                                                                                  |
| PROPERTY    | User.id                     | number (read-only)       | Client and server scripts | The internal ID of the current user.                                                                                                                                                                                                    |

| MEMBER TYPE | NAME            | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                |
|-------------|-----------------|--------------------------|---------------------------|----------------------------------------------------------------------------|
| PROPERTY    | User.location   | number (read-only)       | Client and server scripts | The internal ID of the location of the current user.                       |
| PROPERTY    | User.name       | string (read-only)       | Client and server scripts | The name of the current user.                                              |
| PROPERTY    | User.role       | number (read-only)       | Client and server scripts | The internal ID of the role for the current user.                          |
| PROPERTY    | User.roleCenter | string (read-only)       | Client and server scripts | The string value of the center type, or role center, for the current user. |
| PROPERTY    | User.roleId     | string (read-only)       | Client and server scripts | The custom scriptId of the role for the current user.                      |
| PROPERTY    | User.subsidiary | number (read-only)       | Client and server scripts | The internal ID of the subsidiary for the current user.                    |

## ► Official NetSuite Script Samples (2 Sample(s))

### Sample 1: Return User and Session Information

```

/**
 * @NApiVersion 2.1
 * @NScriptType Suitelet
 */

// This script writes user and session information for the currently executing script to the response.
define(['N/runtime'], function(runtime) {
    function onRequest(context) {
        var remainingUsage = runtime.getCurrentScript().getRemainingUsage();
        var userRole = runtime.getCurrentUser().role;
        var currentSession = runtime.getCurrentSession();

        // Set the current sessions's scope
        currentSession.set({
            name: 'scope',
            value: 'global'
        });

        var sessionScope = runtime.getCurrentSession().get({
            name: 'scope'
        });

        log.debug('Remaining Usage:', remainingUsage);
        log.debug('Role:', userRole);
        log.debug('Session Scope:', sessionScope);

        context.response.write('Executing under role: ' + userRole
            + '. Session scope: ' + sessionScope + '.');
    }
    return {
        onRequest: onRequest
    };
});

```

### Sample 2: Create Multiple Sales Records Using a Scheduled Script

```

/**
 * @NApiVersion 2.1
 * @NScriptType ScheduledScript
 */

// This script creates multiple sales records and logs the record creation progress.
define(['N/runtime', 'N/record'], function(runtime, record) {
    return {
        execute: function(context) {
            var script = runtime.getCurrentScript();
            for (x = 0; x < 500; x++) {
                var rec = record.create({
                    type: record.Type.SALES_ORDER
                });
                script.percentComplete = (x * 100)/500;
                log.debug({
                    title: 'New Sales Orders',
                    details: 'Record creation progress: ' + script.percentComplete + '%'
                });
            }
        }
    };
});

```

### 39. N/scriptTypes/restlet Module

Load the N/scriptTypes/restlet module to create custom handling for your RESTlet script.

`define(['N/scriptTypes/restlet'], (restlet) => { ... })`

✓ 1 Code Sample(s)

Scripts: RESTlet scripts

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/scriptTypes/restlet'], (restlet) => { // Access methods and properties on restlet });
```

Use the N/scriptTypes/restlet module to create custom HTTP responses for your RESTlet script.

This module is available only to the RESTlet script type.

N/scriptTypes/restlet Module Script Sample

#### N/scriptTypes/restlet Module Members

Return Type / Value Type

Supported Script Types

restlet.Response

Object (read-only)

An HTTP response of a RESTlet script.

restlet.createResponse(options)

| MEMBER TYPE | NAME                            | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                          |
|-------------|---------------------------------|--------------------------|------------------------|------------------------------------------------------|
| OBJECT      | restlet.Response                | Object (read-only)       | RESTlet script         | An HTTP response of a RESTlet script.                |
| METHOD      | restlet.createResponse(options) | restlet.Response         | RESTlet script         | Creates a custom HTTP response for a RESTlet script. |

#### Response Object Members

The following members are called on the restlet.Response object.

Return Type / Value Type

Supported Script Types

Response.content

string (read-only)

The content of the RESTlet HTTP response.

| MEMBER TYPE | NAME                 | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                           |
|-------------|----------------------|--------------------------|------------------------|-------------------------------------------------------|
| PROPERTY    | Response.content     | string (read-only)       | RESTlet script         | The content of the RESTlet HTTP response.             |
| PROPERTY    | Response.contentType | string (read-only)       | RESTlet script         | The Content-Type header of the RESTlet HTTP response. |

## Sample 1: N/scriptTypes/restlet Module Script Sample

```
/**
 * @NApiVersion 2.1
 * @NScriptType RESTlet
 */
define(['N/scriptTypes/restlet', 'N/query'],
function(restlet, query) {

    // RESTlet entry point
    const get = function (requestParams) {
        // Create Employee query
        const qObj = query.create({
            type: query.Type.EMPLOYEE
        });
        // Set query condition for the Supervisor field on the employee record, it should not be empty
        qObj.condition = qObj.createCondition({
            fieldId: 'supervisor',
            operator: query.Operator.EMPTY_NOT
        });
        // Add supervisor value in the query results
        qObj.columns = [qObj.createColumn({
            fieldId: 'supervisor'
        })]

        const returnArray = [];

        // Retrieve the results
        const results = qObj.run().results;
        for (var i = results.length - 1; i >= 0; i--) {
            const obj = results[i];
            returnArray.push(obj.values[0]);
        }

        // Create a RESTlet custom response to set the Content-Type header to JSON object
        const response = restlet.createResponse({
            content: JSON.stringify(returnArray),
            contentType: 'application/json'
        });

        log.debug('response', JSON.stringify(response))

        return response;
    }

    return { get };
});
```

## 40. N/search Module

Load the N/search module to create and run on-demand or saved searches and analyze and iterate through the search results. You can search for a single record by keywords, create saved searches, search for duplicate records, or return a set of records that match filters you define.

```
define(['N/search'], (search) => { ... })
```

✓ 9 Code Sample(s)

Scripts: Client and server scripts

Permissions: Perform Search Persist Search Publish Search

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/search'], (search) => { // Access methods and properties on search });
```

Use the N/search module to create and run on-demand or saved searches, then analyze and work through the results. With this module, you can do things like:

Search for a single record using keywords

Create and save searches

Load and run saved searches

Search for duplicate records

Return a set of records that match your defined filter criteria

You can also paginate results and add navigation to jump between pages. This makes it great for handling large result sets.

The N/search module doesn't work in unauthenticated client-side contexts.

For details, see the SuiteAnswers Outbound HTTPs in an unauthenticated client-side context.

### N/search Module Members

Return Type / Value Type

Supported Script Types

Client and server scripts

Encapsulates a single search column in a search.Search object. Use the methods and properties available to the Column object to get or set Column properties.

Client and server scripts

Encapsulates a search filter used in a search. Use the properties for the Filter object to get and set the filter properties.

| MEMBER TYPE | NAME                           | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                                                   |
|-------------|--------------------------------|--------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT      | search.Column                  | Object                   | Client and server scripts | Encapsulates a single search column in a search.Search object. Use the methods and properties available to the Column object to get or set Column properties. |
| OBJECT      | search.Filter                  | Object                   | Client and server scripts | Encapsulates a search filter used in a search. Use the properties for the Filter object to get and set the filter properties.                                 |
| OBJECT      | search.Page                    | Object                   | Client and server scripts | Encapsulates a set of search results for a single search page.                                                                                                |
| OBJECT      | search.PagedData               | Object                   | Client and server scripts | Holds metadata about a paginated query.                                                                                                                       |
| OBJECT      | search.PageRange               | Object                   | Client and server scripts | Defines the page range to bound the result set for a paginated query.                                                                                         |
| OBJECT      | search.Result                  | Object                   | Client and server scripts | Encapsulate a single search result row. Use the methods and properties for the Result object to get the column values for the result row.                     |
| OBJECT      | search.ResultSet               | Object                   | Client and server scripts | Encapsulates a set of search results returned by Search.run().                                                                                                |
| OBJECT      | search.Search                  | Object                   | Client and server scripts | Encapsulates a NetSuite search. Use the methods available to the Search object to create a search, run a search, or save a search.                            |
| OBJECT      | search.Setting                 | Object                   | Client and server scripts | Encapsulates a search setting. Search settings let you specify search parameters that are typically available only in the UI.                                 |
| METHOD      | search.create(options)         | search.Search            | Client and server scripts | Creates a new search and returns it as a search.Search object.                                                                                                |
| METHOD      | search.create.promise(options) | search.Search            | Client and server scripts | Creates a new search asynchronously and returns it as a search.Search object.                                                                                 |
| METHOD      | search.createColumn(options)   | search.Column            | Client and server scripts | Creates a new search column as a search.Column object.                                                                                                        |
| METHOD      | search.createFilter(options)   | search.Filter            | Client and server scripts | Creates a new search filter as a search.Filter object.                                                                                                        |

| MEMBER TYPE | NAME                                              | RETURN TYPE / VALUE TYPE     | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                                                                                                 |
|-------------|---------------------------------------------------|------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | <code>search.createSetting(options)</code>        | <code>search.Setting</code>  | Client and server scripts | Creates a new search setting and returns it as a <code>search.Setting</code> object.                                                                                                                        |
| METHOD      | <code>search.delete(options)</code>               | <code>void</code>            | Client and server scripts | Deletes an existing saved search asynchronously and returns it as a <code>search.Search</code> object.                                                                                                      |
| METHOD      | <code>search.delete.promise(options)</code>       | <code>void</code>            | Client and server scripts | Deletes an existing saved search and returns it as a <code>search.Search</code> object.                                                                                                                     |
| METHOD      | <code>search.duplicates(options)</code>           | <code>search.Result[]</code> | Client and server scripts | Performs a search for duplicate records based on the duplicate detection configuration for the account. Returns an array of <code>search.Result</code> objects.                                             |
| METHOD      | <code>search.duplicates.promise(options)</code>   | <code>search.Result[]</code> | Client and server scripts | Performs a search for duplicate records asynchronously based on the duplicate detection configuration for the account. Returns an array of <code>search.Result</code> objects.                              |
| METHOD      | <code>search.global(options)</code>               | <code>search.Result[]</code> | Client and server scripts | Performs a global search against a single keyword or multiple keywords.                                                                                                                                     |
| METHOD      | <code>search.global.promise(options)</code>       | <code>search.Result[]</code> | Client and server scripts | Performs a global search asynchronously against a single keyword or multiple keywords.                                                                                                                      |
| METHOD      | <code>search.load(options)</code>                 | <code>search.Search</code>   | Client and server scripts | Loads an existing saved search and returns it as a <code>search.Search</code> object.                                                                                                                       |
| METHOD      | <code>search.load.promise(options)</code>         | <code>search.Search</code>   | Client and server scripts | Loads an existing saved search asynchronously and returns it as a <code>search.Search</code> object.                                                                                                        |
| METHOD      | <code>search.lookupFields(options)</code>         | <code>Object   array</code>  | Client and server scripts | Performs a search for one or more body fields on a record. Returns select fields as an object with value and text properties. Returns multiselect fields as an object with value:text pairs.                |
| METHOD      | <code>search.lookupFields.promise(options)</code> | <code>Object   array</code>  | Client and server scripts | Performs a search asynchronously for one or more body fields on a record. Returns select fields as an object with value and text properties. Returns multiselect fields as an object with value:text pairs. |
| ENUM        | <code>search.Operator</code>                      | <code>enum</code>            | Client and server scripts | Holds the values for search operators to use with the <code>search.Filter</code> .                                                                                                                          |
| ENUM        | <code>search.Sort</code>                          | <code>enum</code>            | Client and server scripts | Holds the values for supported sorting directions used with <code>search.createColumn(options)</code> .                                                                                                     |
| ENUM        | <code>search.Summary</code>                       | <code>enum</code>            | Client and server scripts | Holds the values for summary types used by the <code>Column.summary</code> object.                                                                                                                          |
| ENUM        | <code>search.Type</code>                          | <code>enum</code>            | Client and server scripts | Holds the string values for search types supported in the N/search Module. Use this enum to set the value for the <code>options.type</code> parameter of the <code>search.create(options)</code> method.    |

## Column Object Members

The following members are available for a `search.Column` object.

Return Type / Value Type

Supported Script Types

`Column.setWhenOrderedBy(options)`

Client and server scripts

Returns the search column for which the minimal or maximal value should be found when returning the `search.Column` value.

| MEMBER TYPE | NAME                                          | RETURN TYPE / VALUE TYPE        | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                           |
|-------------|-----------------------------------------------|---------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | <code>Column.setWhenOrderedBy(options)</code> | <code>search.Column</code>      | Client and server scripts | Returns the search column for which the minimal or maximal value should be found when returning the <code>search.Column</code> value. |
| PROPERTY    | <code>Column.formula</code>                   | <code>string</code>             | Client and server scripts | Formula used for a search column as a string.                                                                                         |
| PROPERTY    | <code>Column.function</code>                  | <code>string</code>             | Client and server scripts | Special function used in the search column as a string.                                                                               |
| PROPERTY    | <code>Column.join</code>                      | <code>string (read-only)</code> | Client and server scripts | Join ID for a search column as a string.                                                                                              |
| PROPERTY    | <code>Column.label</code>                     | <code>string</code>             | Client and server scripts | Label used for the search column. You can only get or set custom labels with this property.                                           |
| PROPERTY    | <code>Column.name</code>                      | <code>string (read-only)</code> | Client and server scripts | Name of a search column as a string.                                                                                                  |
| PROPERTY    | <code>Column.summary</code>                   | <code>string (read-only)</code> | Client and server scripts | Returns the summary type for a search column.                                                                                         |

## Filter Object Members

The following members are available for a `search.Filter` object.

Return Type / Value Type

Supported Script Types

Client and server scripts

Formula used by the search filter.

string (read-only)

| MEMBER TYPE | NAME                   | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                              |
|-------------|------------------------|--------------------------|---------------------------|------------------------------------------|
| PROPERTY    | <b>Filter.formula</b>  | string                   | Client and server scripts | Formula used by the search filter.       |
| PROPERTY    | <b>Filter.join</b>     | string (read-only)       | Client and server scripts | Join ID for the search filter.           |
| PROPERTY    | <b>Filter.name</b>     | string (read-only)       | Client and server scripts | Name or internal ID of the search field. |
| PROPERTY    | <b>Filter.operator</b> | string (read-only)       | Client and server scripts | Operator used for the search filter.     |
| PROPERTY    | <b>Filter.summary</b>  | search.Summary           | Client and server scripts | Summary type for the search filter.      |

## Page Object Members

The following members are available for a search.Page object.

Return Type / Value Type

Supported Script Types

Client and server scripts

Gets the next segment of data from a paginated search

Page.next.promise()

| MEMBER TYPE | NAME                       | RETURN TYPE / VALUE TYPE     | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                              |
|-------------|----------------------------|------------------------------|---------------------------|--------------------------------------------------------------------------|
| METHOD      | <b>Page.next()</b>         | void                         | Client and server scripts | Gets the next segment of data from a paginated search                    |
| METHOD      | <b>Page.next.promise()</b> | void                         | Client scripts            | Asynchronously gets the next segment of data from a paginated search     |
| METHOD      | <b>Page.prev()</b>         | void                         | Client and server scripts | Gets the previous segment of data from a paginated search                |
| METHOD      | <b>Page.prev.promise()</b> | void                         | Client scripts            | Asynchronously gets the previous segment of data from a paginated search |
| PROPERTY    | <b>Page.data</b>           | search.Result[]              | Client and server scripts | The results from a paginated search.                                     |
| PROPERTY    | <b>Page.isFirst</b>        | boolean (read-only)          | Client and server scripts | Indicates whether a page is the first page of data for a result set      |
| PROPERTY    | <b>Page.isLast</b>         | boolean (read-only)          | Client and server scripts | Indicates whether a page is the last page of data for a result set.      |
| PROPERTY    | <b>Page.pagedData</b>      | search.PagedData (read-only) | Client and server scripts | The PagedData Object used to fetch this Page Object.                     |
| PROPERTY    | <b>Page.pageRange</b>      | search.PageRange (read-only) | Client and server scripts | The PageRange Object used to fetch this Page Object.                     |

## PagedData Object Members

The following members are available for a search.PagedData object.

Return Type / Value Type

Supported Script Types

PagedData.fetch(options)

Client and server scripts

Retrieves the data within the specified page range.

| MEMBER TYPE | NAME                              | RETURN TYPE / VALUE TYPE       | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                |
|-------------|-----------------------------------|--------------------------------|---------------------------|--------------------------------------------------------------------------------------------|
| METHOD      | <b>PagedData.fetch(options)</b>   | search.Page                    | Client and server scripts | Retrieves the data within the specified page range.                                        |
| METHOD      | <b>PagedData.fetch.promise()</b>  | search.Page                    | Client scripts            | Asynchronously retrieves the data within the specified page range.                         |
| PROPERTY    | <b>PagedData.count</b>            | number (read-only)             | Client and server scripts | The total number of results when Search.runPaged(options) was executed.                    |
| PROPERTY    | <b>PagedData.pageRanges</b>       | search.PageRange[] (read-only) | Client and server scripts | The collection of PageRange objects that divide the entire result set into smaller groups. |
| PROPERTY    | <b>PagedData.pageSize</b>         | number (read-only)             | Client and server scripts | The maximum number of entries per page                                                     |
| PROPERTY    | <b>PagedData.searchDefinition</b> | search.Search (read-only)      | Client and server scripts | The search criteria used when Search.runPaged(options) was executed.                       |

## PageRange Object Members

The following members are available for a search.PageRange object.

Return Type / Value Type

Supported Script Types

PageRange.compoundLabel

string (read-only)

Client and server scripts

| MEMBER TYPE | NAME                    | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                      |
|-------------|-------------------------|--------------------------|---------------------------|------------------------------------------------------------------|
| PROPERTY    | PageRange.compoundLabel | string (read-only)       | Client and server scripts | Human-readable label with beginning and ending range identifiers |
| PROPERTY    | PageRange.index         | number (read-only)       | Client and server scripts | The index of this page range.                                    |

## Result Object Members

The following members are available for a search.Result object.

Return Type / Value Type

Supported Script Types

Result.getText(column)

Client and server scripts

The text value for a search.Column if it is a stored select field.

| MEMBER TYPE | NAME                     | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                                                                        |
|-------------|--------------------------|--------------------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | Result.getText(column)   | string                   | Client and server scripts | The text value for a search.Column if it is a stored select field.                                                                                                                 |
| METHOD      | Result.getText(options)  | string                   | Client and server scripts | The UI display name, or text value, for a search result column. This method is supported only for non-stored select, image, and document fields.                                   |
| METHOD      | Result.getValue(options) | string                   | Client and server scripts | Used on formula fields and non-formula (standard) fields to get the value of a specified search return column.                                                                     |
| METHOD      | Result.getValue(column)  | string                   | Client and server scripts | Used on formula and non-formula (standard) fields. Returns the string value of a specified search result column. For convenience, this method takes a single search.Column Object. |
| PROPERTY    | Result.columns           | search.Column[]          | Client and server scripts | Array of search.Column objects that encapsulate the columns returned in the search result row.                                                                                     |
| PROPERTY    | Result.id                | string (read-only)       | Client and server scripts | The internal ID for the record returned in a search result row.                                                                                                                    |
| PROPERTY    | Result.recordType        | string (read-only)       | Client and server scripts | The type of record returned in a search result row.                                                                                                                                |

## ResultSet Object Members

The following members are available for a search.ResultSet object.

Return Type / Value Type

Supported Script Types

ResultSet.each(callback)

Client and server scripts

Use a developer-defined function to invoke on each row in the search results, up to 4000 results at a time.

| MEMBER TYPE | NAME                                | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                |
|-------------|-------------------------------------|--------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------|
| METHOD      | ResultSet.each(callback)            | void                     | Client and server scripts | Use a developer-defined function to invoke on each row in the search results, up to 4000 results at a time.                |
| METHOD      | ResultSet.each.promise(callback)    | void                     | Client scripts            | Asynchronously use a developer-defined function to invoke on each row in the search results, up to 4000 results at a time. |
| METHOD      | ResultSet.getRange(options)         | search.Result[]          | Client and server scripts | Retrieve a slice of the search result as an array of search.Result objects.                                                |
| METHOD      | ResultSet.getRange.promise(options) | search.Result[]          | Client scripts            | Asynchronously retrieve a slice of the search result as an array of search.Result objects.                                 |
| PROPERTY    | ResultSet.columns                   | search.Column[]          | Client and server scripts | An array of search.Column objects that represent the columns returned in the search results.                               |

## Search Object Members

The following members are available for a search.Search object.

Return Type / Value Type

Supported Script Types

search.ResultSet

Client and server scripts

Runs an on demand search created with search.create(options) or a search loaded with search.load(options), returning the results as a search.ResultSet.

| MEMBER TYPE | NAME                             | RETURN TYPE / VALUE TYPE    | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                                             |
|-------------|----------------------------------|-----------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | Search.run()                     | search.ResultSet            | Client and server scripts | Runs an on demand search created with search.create(options) or a search loaded with search.load(options), returning the results as a search.ResultSet. |
| METHOD      | Search.runPaged(options)         | search.PagedData            | Client and server scripts | Runs the current search and returns a search.PagedData Object.                                                                                          |
| METHOD      | Search.runPaged.promise(options) | search.PagedData            | Client and server scripts | Asynchronously runs the current search and returns a search.PagedData Object.                                                                           |
| METHOD      | Search.save()                    | number                      | Client and server scripts | Saves a search created by search.create(options) or loaded with search.load(options). Returns the internal ID of the saved search.                      |
| METHOD      | Search.save.promise()            | number                      | Client and server scripts | Asynchronously saves a search created by search.create(options) or loaded with search.load(options). Returns the internal ID of the saved search.       |
| PROPERTY    | Search.columns                   | search.Column[]   string[]  | Client and server scripts | Columns to return for this search as an array of search.Column objects or a string array of column names.                                               |
| PROPERTY    | Search.filterExpression          | Object[]                    | Client and server scripts | Search filter expression for the search as an array of expression objects.                                                                              |
| PROPERTY    | Search.filters                   | search.Filter[]             | Client and server scripts | Filters for the search as an array of search.Filter objects.                                                                                            |
| PROPERTY    | Search.id                        | string                      | Client and server scripts | Script ID for a saved search, starting with customsearch.                                                                                               |
| PROPERTY    | Search.isPublic                  | boolean                     | Client and server scripts | Value is true if the search is public, or false if it is not.                                                                                           |
| PROPERTY    | Search.packageId                 | string                      | Client and server scripts | The application ID for the search.                                                                                                                      |
| PROPERTY    | Search.searchId                  | number                      | Client and server scripts | Internal ID of a search.                                                                                                                                |
| PROPERTY    | Search.searchType                | string                      | Client and server scripts | Search type on which a search is based.                                                                                                                 |
| PROPERTY    | Search.settings                  | search.Setting[]   string[] | Client and server scripts | Search settings for this search as an array of search.Setting objects or a string array of column names.                                                |
| PROPERTY    | Search.title                     | string                      | Client and server scripts | Title for a saved search. Use this property to set the title for a search before you save it for the first time.                                        |

## Setting Object Members

The following members are available for a search.Setting object.

Return Type / Value Type

Supported Script Types

string (read-only)

Client and server scripts

The name of the search parameter.

| MEMBER TYPE | NAME          | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                        |
|-------------|---------------|--------------------------|---------------------------|------------------------------------|
| PROPERTY    | Setting.name  | string (read-only)       | Client and server scripts | The name of the search parameter.  |
| PROPERTY    | Setting.value | string (read-only)       | Client and server scripts | The value of the search parameter. |

 **Sample 1: Search for Customer Records and Log First 50 Results**

```
/**
 * @NApiVersion 2.1
 */

require(['/N/search'], function(search) {
    var mySearch = search.create({
        type: search.Type.CUSTOMER,
        columns: ['entityid', 'firstname', 'lastname', 'salesrep'],
        filters: ['entityid', 'contains', 'Adam']
    });

    var myResultSet = mySearch.run();

    var resultRange = myResultSet.getRange({
        start: 0,
        end: 50
    });

    for (var i = 0; i < resultRange.length; i++) {
        log.debug(resultRange[i]);
    }
});
```

 **Sample 2: Search for Sales Order Records**

```
/**
 * @NApiVersion 2.x
 */

require(['/N/search'], function(search) {
    function createSearch() {
        var mySalesOrderSearch = search.create({
            type: search.Type.SALES_ORDER,
            title: 'My SalesOrder Search',
            id: 'customsearch_my_so_search',
            columns: ['entity', 'subsidiary', 'name', 'currency'],
            filters: [
                ['mainline', 'is', 'T'],
                'and', ['subsidiary.name', 'contains', 'CAD']
            ]
        });

        mySalesOrderSearch.save();
    }

    createSearch();
});
```

 **Sample 3: Load a Search for Sales Order Records and Use a Callback Function to Process Results**

```
/**
 * @NApiVersion 2.x
 */

require(['/N/search'], function(search) {
    function loadAndRunSearch() {
        var mySearch = search.load({
            id: 'customsearch_my_so_search'
        });

        mySearch.run().each(function(result) {
            var entity = result.getValue({
                name: 'entity'
            });
            var subsidiary = result.getValue({
                name: 'subsidiary'
            });

            return true;
        });
    }

    loadAndRunSearch();
});
```

#### Sample 4: Load a Search for Sales Order Records and Return the First 100 Search Results

```
/**
 * @NApiVersion 2.x
 */

require(['N/search'], function(search) {
    function runSearchAndFetchResult() {
        var mySearch = search.load({
            id: 'customsearch_my_so_search'
        });

        var searchResult = mySearch.run().getRange({
            start: 0,
            end: 100
        });
        for (var i = 0; i < searchResult.length; i++) {
            var entity = searchResult[i].getValue({
                name: 'entity'
            });
            var subsidiary = searchResult[i].getValue({
                name: 'subsidiary'
            });
        }

        runSearchAndFetchResult();
    }
});
```

#### Sample 5: Load and Run a Paginated Search and Process the Results

```
/**
 * @NApiVersion 2.x
 */

require(['N/search'], function(search) {
    function loadAndRunSearch() {
        var mySearch = search.load({
            id: 'customsearch_my_so_search'
        });

        var myPagedData = mySearch.runPaged();
        myPagedData.pageRanges.forEach(function(pageRange){
            var myPage = myPagedData.fetch({index: pageRange.index});
            myPage.data.forEach(function(result){
                var entity = result.getValue({
                    name: 'entity'
                });
                var subsidiary = result.getValue({
                    name: 'subsidiary'
                });
            });
        });

        loadAndRunSearch();
    }
});
```

#### Sample 6: Create a Search for a Custom Record Type

```
/**
 * @NApiVersion 2.x
 */

require(['N/search'], function(search) {
    var myCustomRecordSearch = search.create({
        type: search.Type.CUSTOM_RECORD + '6',
        title: 'My Search Title',
        columns: ['custrecord1']
    }).run().each(function(result) {
        // Process each result
        return true;
    });
});
```

#### Sample 7: Search for Items in a Custom List

```
/**
 * @NApiVersion 2.x
 */
require(['N/search'], function(search) {
    var internalId = -1;
    var myCustomListSearch = search.create({
        type: 'customlist_mylist',
        columns: [
            { name: 'internalId' },
            { name: 'abbreviation' }
        ]
    });

    myCustomListSearch.filters = [
        search.createFilter({
            name: 'formulatext',
            formula: '{abbreviation}',
            operator: search.Operator.IS,
            values: abbreviation
        })
    ];

    var resultSet = myCustomListSearch.run();
    var results = resultSet.getRange({
        start: 0,
        end: 1
    });
    for(var i in results) {
        // log.debug('Found custom list record', results[i]);
        internalId = results[i].getValue({
            name: 'internalId'
        });
    };
});
```

#### Sample 8: Delete a Saved Search

```
/**
 * @NApiVersion 2.x
 */
require(['N/search'], function(search) {
    function deleteSearch() {
        search.delete({
            id: 'customsearch_my_so_search'
        });
    }

    deleteSearch();
});
```

#### Sample 9: Search Using a Specific Record Field

```
/**
 * @NApiVersion 2.x
 */
require(['N/search'], function(search) {
    var mySearch = search.create({
        type: search.Type.TRANSACTION,
        columns: ['item', 'operationdisplaytext'],
        filters: [
            ['internalid', search.Operator.ANYOF, '&lt;transactionId&gt;'],
            'and',
            ['operationdisplaytext', search.Operator.EQUALTO, 20]]
    });

    var myResultSet = mySearch.run();
    var resultRange = myResultSet.getRange({
        start: 0,
        end: 20
    });
});
```

# 41. N/sftp Module

Load the N/sftp module to connect to a remote FTP server using SFTP and transfer files.

`define(['N/sftp'], (sftp) => { ... })`

✓ 3 Code Sample(s)

Scripts: Server scripts

Permissions: Key Access (needed when public key authentication is used)

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/sftp'], (sftp) => { // Access methods and properties on sftp });
```

Use the N/sftp module to manage folders and upload or download files from external SSH file transfer (SFTP) servers. You can perform the following SFTP functions using the N/sftp module:

SFTP servers can be hosted by your organization or by a third party. NetSuite does not provide SFTP server functionality. All SFTP transfers to or from NetSuite must originate from SuiteScript. It is not possible for external clients to initiate file transfers using SFTP.

Use SSH keys to establish an SFTP connection. By using the keys, you can manage files and directories by using the SFTP protocol. For more information, see SSH Keys for SFTP. For more information about working with keys in SuiteScript, see the N/keyControl Module, see N/keyControl Module.

All paths, directories, and filenames that contain wildcards such as ? and \* must have those characters escaped, unless these characters are specifically intended to work as wildcards.

To use an external server to initiate a NetSuite file transfer that doesn't use SFTP, you can use RESTlets or SOAP web services. In SuiteScript, RESTlets can respond to requests containing file data and save them in the File Cabinet. RESTlets can also respond to requests for file data by loading the contents from the File Cabinet and returning them in the response. Note that binary file content must be received or sent as Base64 encoded Strings. See SuiteScript 2.1 RESTlet Script Type for more information.

In SOAP web services, applications can invoke CRUD operations on the file record to populate or change the contents of the File Cabinet. See SuiteTalk SOAP Web Services Platform Guide and File for more information.

## N/sftp Module Members

Return Type / Value Type

Supported Script Types

Represents a connection to the account on the remote FTP server.

sftp.createConnection(options)

Establishes a connection to a remote FTP server.

sftp.MAX\_CONNECT\_TIMEOUT

| MEMBER TYPE | NAME                           | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                    |
|-------------|--------------------------------|--------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT      | sftp.Connection                | Object                   | Server scripts         | Represents a connection to the account on the remote FTP server.                                                                                               |
| METHOD      | sftp.createConnection(options) | sftp.Connection          | Server scripts         | Establishes a connection to a remote FTP server.                                                                                                               |
| ENUM        | sftp.MAX_CONNECT_TIMEOUT       | enum                     | Server scripts         | Holds the values for maximum connection timeout.                                                                                                               |
| ENUM        | sftp.MIN_CONNECT_TIMEOUT       | enum                     | Server scripts         | Holds the values for minimum connection timeout.                                                                                                               |
| ENUM        | sftp.MAX_PORT_NUMBER           | enum                     | Server scripts         | Holds the values for the maximum port number.                                                                                                                  |
| ENUM        | sftp.MIN_PORT_NUMBER           | enum                     | Server scripts         | Holds the values for the minimum port number.                                                                                                                  |
| ENUM        | sftp.DEFAULT_PORT_NUMBER       | enum                     | Server scripts         | Holds the values for the default port number.                                                                                                                  |
| ENUM        | sftp.Sort                      | enum                     | Server scripts         | Holds the values to be used to sort the listed directory. Use this enum to set the value of the options.sort parameter of the Connection.list(options) method. |

## Connection Object Members

The following members are called on the sftp.Connection object.

Return Type / Value Type

Supported Script Types

Connection.download(options)

Downloads a file from the remote FTP server.

Connection.upload(options)

| MEMBER TYPE | NAME                         | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                  |
|-------------|------------------------------|--------------------------|------------------------|----------------------------------------------|
| METHOD      | Connection.download(options) | file.File                | Server scripts         | Downloads a file from the remote FTP server. |

| MEMBER TYPE | NAME                                | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                             |
|-------------|-------------------------------------|--------------------------|------------------------|---------------------------------------------------------|
| METHOD      | Connection.upload(options)          | void                     | Server scripts         | Uploads a file to the remote FTP server.                |
| METHOD      | Connection.makeDirectory(options)   | string                   | Server scripts         | Creates an empty directory.                             |
| METHOD      | Connection.removeDirectory(options) | void                     | Server scripts         | Removes an empty directory.                             |
| METHOD      | Connection.removeFile(options)      | void                     | Server scripts         | Removes a file in a directory.                          |
| METHOD      | Connection.move(options)            | void                     | Server scripts         | Moves a file or directory from one location to another. |
| METHOD      | Connection.list(options)            | Array<Object>;           | Server scripts         | Lists the remote directory.                             |
| ENUM        | Connection.MAX_FILE_SIZE            | enum                     | Server scripts         | Holds the values for the maximum file size.             |
| ENUM        | Connection.MAX_TRANSFER_TIMEOUT     | enum                     | Server scripts         | Holds the values for the maximum transfer timeout.      |

► Official NetSuite Script Samples (3 Sample(s))

 Sample 1: Upload and Download a File

```
/**
 * @NApiVersion 2.1
 */
require(['N/sftp', 'N/file'], (sftp, file) => {
  const myPwdGuid = "B34672495064525E5D65032D63B52301";
  const myHostKey = "AAA1234567890Q=";

  // Establish a connection to a remote FTP server
  let connection = sftp.createConnection({
    username: 'myuser',
    passwordGuid: myPwdGuid,
    url: 'host.somewhere.com',
    directory: 'myuser/wheres/my/file',
    hostKey: myHostKey
  });

  // Create a file to upload using the N/file module
  let myFileToUpload = file.create({
    name: 'originalname.js',
    fileType: file.Type.PLAINTEXT,
    contents: 'I am a test file.'
  });

  // Upload the file to the remote server
  connection.upload({
    directory: 'relative/path/to/remote/dir',
    filename: 'newFileNameOnServer.js',
    file: myFileToUpload,
    replaceExisting: true
  });

  // Download the file from the remote server
  let downloadedFile = connection.download({
    directory: 'relative/path/to/file',
    filename: 'downloadMe.js'
  });
});
```

## Sample 2: Manage Files and Directories

```
/**
 * @NApiVersion 2.1
 */
require(['N/sftp', 'N/file'], (sftp, file) => {
  // Establish a connection
  log.debug('Establishing SFTP connection...');
  let connection = sftp.createConnection({
    username: 'sftpuser',
    keyId: 'custkeysftp_nft_demo_key',
    url: 'myurl',
    port: 22,
    directory: 'inbound',
    hostKey: 'myhostkey'
  });
  log.debug('Connection established!');

  // List the directory and log the number of elements there
  let list = connection.list({
    path: 'yyy/test'
  });
  log.debug('Items in directory "test" at the beginning: ' + list.length);

  // Generate the test file
  log.debug('Generating test file...');
  let myFileToUpload = file.create({
    name: 'asdf.txt',
    fileType: file.Type.PLAINTEXT,
    contents: 'I am a test file.'
  });
  log.debug('Test file generated, uploading to "test" directory...');

  // Upload the test file
  connection.upload({
    directory: 'yyy/test',
    filename: 'af.txt',
    file: myFileToUpload,
    replaceExisting: true
  });
  log.debug('Upload complete!');

  // List the directory to confirm there is one more file than before
  list = connection.list({
    path: 'yyy/test'
  });
  log.debug('Items in directory "test" after the upload: ' + list.length);

  // Create a new directory
  log.debug('Creating directory "test2"...');
  try {
    connection.makeDirectory({
      path: 'yyy/test2'
    });
    log.debug('Directory created.');
```

```

}

// The directory is not empty so delete the file first
log.debug('Removing test file from "test2" directory...');
connection.removeFile({
  path: 'yyy/test2/af.txt'
});
log.debug('Test file removed!');

list = connection.list({
  path: 'yyy/test2'
});
log.debug('Items in directory "test2": ' + list.length);

// Try to remove the directory again
log.debug('Removing directory "test2" ...');
try {
  connection.removeDirectory({
    path: 'yyy/test2'
  });
  log.debug('Directory removed!');
} catch (e) {
  log.debug('Directory not removed!');
  log.error(e.message);
}

// Try to list the removed directory
log.debug('Trying to list directory "test2" ...');
try {
  list = connection.list({
    path: 'yyy/test2'
  });
} catch (e) {
  log.error(e.message);
}
});

```

### Sample 3: Set Conditional Default Settings Using N/ftp Enums

```
/**
 * @NApiVersion 2.1
 * @NScriptType UserEventScript
 * @NModuleScope SameAccount
 */

define(['N/file', 'N/sftp', 'N/error'],
function(file, sftp, error) {
    return {
        beforeLoad: function(){
            var portNumber = -1;
            var connectTimeout = -1;
            var transferTimeout = -1;
            //these variables can be taken as parameters of the script instead

            if (portNumber < sftp.MIN_PORT_NUMBER || portNumber > sftp.MAX_PORT_NUMBER)
                portNumber = sftp.DEFAULT_PORT_NUMBER;
            if (connectTimeout < sftp.MIN_CONNECT_TIMEOUT)
                connectTimeout = sftp.MIN_CONNECT_TIMEOUT;
            else if (connectTimeout > sftp.MAX_CONNECT_TIMEOUT)
                connectTimeout = sftp.MAX_CONNECT_TIMEOUT;

            var connection = sftp.createConnection({
                username: 'sftpuser',
                keyId: 'custkey1',
                url: '192.168.0.100',
                port: portNumber,
                directory: 'inbound',
                timeout: connectTimeout,
                hostKey: "AAAAB3NzaC1yc2EAAAADAQABAAQDMiFkH2vTxditye8nem7+LS3x7dTQR/A67KdsR/5C2WUcDipBzYhHbnG6Am12Nd2tLM01LnaBZA6/8P4Y9x/sGTxtsdE/MzeGDUBn
6HBlQvgIrhX62wgoKQG+P2LEA01+Vz8y3/HB1NmD7Fc62cJ9Mu88YA6jwJOIPZeHYNVyIm90rYvYzYyV3hH0x7SXyvGnijJQF4G8C4c8u/UVpF/sE16xKZtLy2Rx0aDL2FsDrtpyPmM602/R6ISbsmgab3M
zzAEIu+zLDMdIBJn3cDhNt1F7Rar6Tu0u18KCKk8GPxbnxDu64sCN0nXPYkDXSMUbm/ocRjY6tqdZUMmeTf3"
            });

            // can also be a big file (created for example by async search)
            var myFileToUpload = file.create({
                name: 'originalname.txt',
                fileType: file.Type.PLAINTEXT,
                contents: 'I am a test file.'
            });

            if (myFileToUpload.size > connection.MAX_FILE_SIZE)
                throw error.create({name:"FILE_IS_TOO_BIG", message:"The file you are trying to upload is too big"});

            var minTransferTimeout = 10;
            if (transferTimeout > connection.MAX_TRANSFER_TIMEOUT)
                transferTimeout = connection.MAX_TRANSFER_TIMEOUT;
            else if (transferTimeout < minTransferTimeout)
                transferTimeout = minTransferTimeout;

            connection.upload({
                directory: 'files',
                filename: 'test.txt',
                file: myFileToUpload,
                replaceExisting: true,
                timeout: transferTimeout
            });
        }
    };
});
```

## 42. N/sso Module

Note: As of NetSuite 2025.1, the support ended for the SuiteSignOn feature, which means the N/sso module is also no longer supported. Load the N/sso module to generate outbound single sign-on (SuiteSignOn) tokens.

`define(['N/sso'], (sso) => { ... })`

✓ 1 Code Sample(s)

Scripts: Client and server scripts

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/sso'], (sso) => { // Access methods and properties on sso });
```

As of NetSuite 2025.1, support ended for the SuiteSignOn feature, which includes the N/sso module. If you still use SuiteSignOn or the N/sso module in your customizations, you must transition to a supported alternative, such as SuiteScript OAuth 2.0 or OpenID Connect (OIDC) via N/auth and N/https modules.

### Deprecation & Migration Notice

Status: Deprecated as of NetSuite release 2025.1.

Alternative: Use N/auth and N/https modules for OAuth 2.0 and OpenID Connect authentication workflows.

Legacy Capabilities: Previously provided single sign-on authentication token generation.

### ► Official NetSuite Script Samples (1 Sample(s))

#### Sample 1: Migration from N/sso to OAuth 2.0 / OIDC Authentication

```
/**
 * @NApiVersion 2.1
 * @NScriptType UserEventScript
 *
 * Note: N/sso is deprecated as of NetSuite 2025.1.
 * This sample demonstrates the official supported migration to OAuth 2.0 / OIDC using N/https and N/auth.
 */
define(['N/https', 'N/auth', 'N/log'], (https, auth, log) => {
  const beforeSubmit = (context) => {
    // Modern replacement: Authenticate using OAuth 2.0 client credentials or token exchange
    try {
      const tokenResponse = https.post({
        url: 'https://oauth2.provider.com/token',
        headers: {
          'Content-Type': 'application/x-www-form-urlencoded'
        },
        body: {
          grant_type: 'client_credentials',
          client_id: 'YOUR_CLIENT_ID',
          client_secret: 'YOUR_CLIENT_SECRET'
        }
      });

      const tokenData = JSON.parse(tokenResponse.body);
      log.audit('OAuth 2.0 Auth', 'Obtained token: ' + tokenData.access_token);
    } catch (e) {
      log.error('Authentication Error', e);
    }
  };
  return { beforeSubmit };
});
```

### 43. N/suiteAppInfo Module

Load the N/suiteAppInfo module when you want to access information related to SuiteApps and Bundles.

`define(['N/suiteAppInfo'], (suiteAppInfo) => { ... })`

✓ 1 Code Sample(s)

Scripts: Client and server scripts

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/suiteAppInfo'], (suiteAppInfo) => { // Access methods and properties on suiteAppInfo });
```

Use the N/suiteAppInfo module to access information related to SuiteApps and Bundles. This module is available for all script types.

N/suiteAppInfo Module Script Sample

#### N/suiteAppInfo Module Members

Return Type / Value Type

Supported Script Types

suiteAppInfo.isBundleInstalled(options)

Client and server scripts

Returns true if the specific bundle is installed.

suiteAppInfo.isSuiteAppInstalled(options)

| MEMBER TYPE | NAME                                                 | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                        |
|-------------|------------------------------------------------------|--------------------------|---------------------------|----------------------------------------------------------------------------------------------------|
| METHOD      | suiteAppInfo.isBundleInstalled(options)              | boolean                  | Client and server scripts | Returns true if the specific bundle is installed.                                                  |
| METHOD      | suiteAppInfo.isSuiteAppInstalled(options)            | boolean                  | Client and server scripts | Returns true if the specified SDF SuiteApp is installed.                                           |
| METHOD      | suiteAppInfo.listBundlesContainingScripts(options)   | Object                   | Client and server scripts | Returns the IDs for bundles that contain the specified script, for each script specified.          |
| METHOD      | suiteAppInfo.listInstalledBundles()                  | Object[]                 | Client and server scripts | Returns a list of bundles that are installed.                                                      |
| METHOD      | suiteAppInfo.listInstalledSuiteApps()                | Object[]                 | Client and server scripts | Returns a list of SDF SuiteApps that are installed.                                                |
| METHOD      | suiteAppInfo.listSuiteAppsContainingScripts(options) | Object                   | Client and server scripts | Returns the ID for the SDF SuiteApp that contains the specified script, for each script specified. |

 Sample 1: N/suiteAppInfo Module Script Sample

```
/*
 * @NApiVersion 2.x
 *
 */

// This script sample uses each method available in the N/suiteAppInfo module to retrieve information about installed bundles and SuiteApps installed in the
current account.

require(['N/suiteAppInfo'], function (suiteAppInfo){
    var isBundleInstalled = suiteAppInfo.isBundleInstalled({
        bundleId: '1234'
    });

    var isSuiteAppInstalled = suiteAppInfo.isSuiteAppInstalled({
        suiteAppId: '789'
    });

    var allMyBundlesInstalled = suiteAppInfo.listInstalledBundles();

    var allMySuiteAppsInstalled = suiteAppInfo.listInstalledSuiteApps();

    var myScripts = ["scriptA", "scriptB", "scriptC"];
    var scriptInBundles = suiteAppInfo.listBundlesContainingScripts({
        scriptIds: myScripts
    });

    var scriptInSuiteApps = suiteAppInfo.listSuiteAppsContainingScripts({
        scriptsIds: myScripts
    });
})
```

## 44. N/task Module

Load the N/task module to create tasks and place them in the internal NetSuite scheduling or task queue. Use this module to schedule scripts, run Map/Reduce scripts, import CSV files, merge duplicate records, and execute asynchronous workflows.

`define(['N/task'], (task) => { ... })`

✓ 5 Code Sample(s)

Scripts: Server scripts

Permissions: Tasks

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/task'], (task) => { // Access methods and properties on task });
```

Use the N/task module to create tasks and place them in the internal NetSuite scheduling or task queue. You can use this module to create tasks for the following:

- To submit a scheduled script
- To run a map/reduce script
- To import CSV files
- To merge duplicate records
- To execute asynchronous searches, asynchronous document capture tasks, constructed queries, SuiteQL queries, and workflows

Each task is a specific task type (task.TaskType) and each task type has its own corresponding object types. Use the methods available to each object type to configure, submit, and monitor the tasks.

Regardless of task type, tasks are always triggered asynchronously.

### N/task Module Members

Return Type / Value Type

Supported Script Types

task.CsvImportTask

The properties of a CSV import task.

Use the methods and properties for this object to submit a CSV import task into the task queue and asynchronously import record data into NetSuite.

task.CsvImportTaskStatus

| MEMBER TYPE | NAME                               | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                              |
|-------------|------------------------------------|--------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT      | task.CsvImportTask                 | Object                   | Server scripts         | The properties of a CSV import task. Use the methods and properties for this object to submit a CSV import task into the task queue and asynchronously import record data into NetSuite. |
| OBJECT      | task.CsvImportTaskStatus           | Object                   | Server scripts         | The status of a CSV import task placed into the NetSuite scheduling queue.                                                                                                               |
| OBJECT      | task.DocumentCaptureTask           | Object                   | Server scripts         | The properties of a document capture task. Use the methods and properties of this object to submit a document capture task into the NetSuite task queue.                                 |
| OBJECT      | task.DocumentCaptureTaskStatus     | Object                   | Server scripts         | The status of a document capture task placed into the NetSuite task queue.                                                                                                               |
| OBJECT      | task.EntityDeduplicationTask       | Object                   | Server scripts         | All the properties of a merge duplicate records task request. Use the methods and properties of this object to submit a merge duplicate record job task into the NetSuite task queue.    |
| OBJECT      | task.EntityDeduplicationTaskStatus | Object                   | Server scripts         | The status of a merge duplicate record task placed into the NetSuite task queue.                                                                                                         |
| OBJECT      | task.MapReduceScriptTask           | Object                   | Server scripts         | A map/reduce script deployment.                                                                                                                                                          |
| OBJECT      | task.MapReduceScriptTaskStatus     | Object                   | Server scripts         | The status of a map/reduce script deployment that has been submitted for processing.                                                                                                     |
| OBJECT      | task.QueryTask                     | Object                   | Server scripts         | The properties of a query task. Use the methods and properties of this object to submit a query task into the NetSuite task queue.                                                       |
| OBJECT      | task.QueryTaskStatus               | Object                   | Server scripts         | The status of a query task placed into the NetSuite task queue.                                                                                                                          |
| OBJECT      | task.RecordActionTask              | Object                   | Server scripts         | The properties of a record action task. Use this object to place a record action task into the NetSuite scheduling queue.                                                                |
| OBJECT      | task.RecordActionTaskStatus        | Object                   | Server scripts         | The status of a record action task in the NetSuite scheduling queue.                                                                                                                     |

| MEMBER TYPE | NAME                                        | RETURN TYPE / VALUE TYPE                                                                                                                                                                                                                                                                                             | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                                                                                    |
|-------------|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT      | <code>task.ScheduledScriptTask</code>       | Object                                                                                                                                                                                                                                                                                                               | Server scripts         | All the properties of a scheduled script task in SuiteScript. Use this object to place a scheduled script deployment into the NetSuite scheduling queue.                                                                                       |
| OBJECT      | <code>task.ScheduledScriptTaskStatus</code> | Object                                                                                                                                                                                                                                                                                                               | Server scripts         | The status of a scheduled script placed into the NetSuite scheduling queue.                                                                                                                                                                    |
| OBJECT      | <code>task.SearchTask</code>                | Object                                                                                                                                                                                                                                                                                                               | Server scripts         | The properties required to initiate an asynchronous search.                                                                                                                                                                                    |
| OBJECT      | <code>task.SearchTaskStatus</code>          | Object                                                                                                                                                                                                                                                                                                               | Server scripts         | The status of an asynchronous search initiation task that is placed into the NetSuite task queue.                                                                                                                                              |
| OBJECT      | <code>task.SuiteQLTask</code>               | Object                                                                                                                                                                                                                                                                                                               | Server scripts         | The properties of a SuiteQL task. Use the methods and properties of this object to submit a query task into the NetSuite task queue.                                                                                                           |
| OBJECT      | <code>task.SuiteQLTaskStatus</code>         | Object                                                                                                                                                                                                                                                                                                               | Server scripts         | The status of a SuiteQL task placed into the NetSuite task queue.                                                                                                                                                                              |
| OBJECT      | <code>task.WorkflowTriggerTask</code>       | Object                                                                                                                                                                                                                                                                                                               | Server scripts         | All the properties required to asynchronously initiate a workflow. Use <code>WorkflowTriggerTask</code> to create a task that initiates an instance of a specific workflow.                                                                    |
| OBJECT      | <code>task.WorkflowTriggerTaskStatus</code> | Object                                                                                                                                                                                                                                                                                                               | Server scripts         | The status of an asynchronous workflow initiation task placed into the NetSuite task queue.                                                                                                                                                    |
| METHOD      | <code>task.checkStatus(options)</code>      | <code>task.CsvImportTaskStatus   task.DocumentCaptureTaskStatus   task.EntityDeduplicationTaskStatus   task.MapReduceScriptTaskStatus   task.QueryTaskStatus   task.RecordActionTaskStatus   task.ScheduledScriptTaskStatus   task.SearchTaskStatus   task.SuiteQLTaskStatus   task.WorkflowTriggerTaskStatus</code> | Server scripts         | Returns a task status object associated with a specific task ID.                                                                                                                                                                               |
| METHOD      | <code>task.create(options)</code>           | <code>task.CsvImportTask   task.DocumentCaptureTask   task.EntityDeduplicationTask   task.MapReduceScriptTask   task.QueryTask   task.RecordActionTask   task.ScheduledScriptTask   task.SearchTask   task.SuiteQLTask   task.WorkflowTriggerTask</code>                                                             | Server scripts         | Creates an object for a specific task type and returns the task object.                                                                                                                                                                        |
| ENUM        | <code>task.ActionCondition</code>           | enum                                                                                                                                                                                                                                                                                                                 | Server scripts         | Holds the string values for the possible record action conditions. This enum is returned by <code>RecordActionTask.condition</code> .                                                                                                          |
| ENUM        | <code>task.DedupeEntityType</code>          | enum                                                                                                                                                                                                                                                                                                                 | Server scripts         | Holds the string values for entity types for which you can merge duplicate records with <code>task.EntityDeduplicationTask</code> .                                                                                                            |
| ENUM        | <code>task.DedupeMode</code>                | enum                                                                                                                                                                                                                                                                                                                 | Server scripts         | Holds the string values for available deduplication modes when merging duplicate records with <code>task.EntityDeduplicationTask</code> . Use this enum to set the <code>EntityDeduplicationTask.entityType</code> .                           |
| ENUM        | <code>task.MapReduceStage</code>            | enum                                                                                                                                                                                                                                                                                                                 | Server scripts         | Holds the string values for the stages of a map/reduce script deployment, which is encapsulated by the <code>task.MapReduceScriptTask</code> object. This enum is returned by <code>MapReduceScriptTaskStatus.stage</code> .                   |
| ENUM        | <code>task.MasterSelectionMode</code>       | enum                                                                                                                                                                                                                                                                                                                 | Server scripts         | Holds the string values for supported master selection modes when merging duplicate records with <code>task.EntityDeduplicationTask</code> . Use this enum to set the <code>EntityDeduplicationTask.masterSelectionMode</code> property.       |
| ENUM        | <code>task.TaskStatus</code>                | enum                                                                                                                                                                                                                                                                                                                 | Server scripts         | Holds the string values for the possible status of tasks created and submitted with the N/task Module.                                                                                                                                         |
| ENUM        | <code>task.TaskType</code>                  | enum                                                                                                                                                                                                                                                                                                                 | Server scripts         | Holds the string values for the types of task objects you can create using <code>task.create(options)</code> . Use this enum to set the value for the <code>options.taskType</code> parameter of the <code>task.create(options)</code> method. |

## CsvImportTask Object Members

The following members are available for a `task.CsvImportTask` object.

Return Type / Value Type

Supported Script Types

`CsvImportTask.submit()`

Directs NetSuite to place a CSV import task into the NetSuite task queue and returns a unique ID for the task.

You can use this method only in bundle installation scripts, scheduled scripts, and RESTlets.

| MEMBER TYPE     | NAME                                   | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                                                  |
|-----------------|----------------------------------------|--------------------------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>METHOD</b>   | <code>CsvImportTask.submit()</code>    | string                   | Server scripts         | Directs NetSuite to place a CSV import task into the NetSuite task queue and returns a unique ID for the task. You can use this method only in bundle installation scripts, scheduled scripts, and RESTlets. |
| <b>PROPERTY</b> | <code>CsvImportTask.id</code>          | string                   | Server scripts         | The ID of the task.                                                                                                                                                                                          |
| <b>PROPERTY</b> | <code>CsvImportTask.importFile</code>  | file.File   string       | Server scripts         | CSV file to import. Use a file.File object or a string that represents the CSV text to be imported.                                                                                                          |
| <b>PROPERTY</b> | <code>CsvImportTask.linkedFiles</code> | Object                   | Server scripts         | A map of key-value pairs that sets the data to be imported in a linked file for a multi-file import job, by referencing a file in the File Cabinet or the raw CSV data to import.                            |
| <b>PROPERTY</b> | <code>CsvImportTask.mappingId</code>   | number   string          | Server scripts         | Script ID or internal ID of the saved import map that you created when you ran the Import Assistant.                                                                                                         |
| <b>PROPERTY</b> | <code>CsvImportTask.name</code>        | string                   | Server scripts         | Name for the CSV import task.                                                                                                                                                                                |
| <b>PROPERTY</b> | <code>CsvImportTask.queueId</code>     | number                   | Server scripts         | Overrides the Queue Number property under Advanced Options on the Import Options page of the Import Assistant.                                                                                               |

## CsvImportTaskStatus Object Members

The following members are available for a task.CsvImportTaskStatus object.

Return Type / Value Type

Supported Script Types

CsvImportTaskStatus.status

string (read-only)

Status for a CSV import task. Returns a task.TaskStatus enum value.

| MEMBER TYPE     | NAME                                    | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                         |
|-----------------|-----------------------------------------|--------------------------|------------------------|---------------------------------------------------------------------|
| <b>PROPERTY</b> | <code>CsvImportTaskStatus.status</code> | string (read-only)       | Server scripts         | Status for a CSV import task. Returns a task.TaskStatus enum value. |
| <b>PROPERTY</b> | <code>CsvImportTaskStatus.taskId</code> | string (read-only)       | Server scripts         | The task ID associated with the specified task.                     |

## DocumentCaptureTask Object Members

The following members are available for a task.DocumentCaptureTask object.

Return Type / Value Type

Supported Script Types

DocumentCaptureTask.addInboundDependency(options)

Adds a scheduled script task to the document capture task as a dependent task.

DocumentCaptureTask.submit()

| MEMBER TYPE     | NAME                                                           | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                            |
|-----------------|----------------------------------------------------------------|--------------------------|------------------------|--------------------------------------------------------------------------------------------------------|
| <b>METHOD</b>   | <code>DocumentCaptureTask.addInboundDependency(options)</code> | void                     | Server scripts         | Adds a scheduled script task to the document capture task as a dependent task.                         |
| <b>METHOD</b>   | <code>DocumentCaptureTask.submit()</code>                      | string                   | Server scripts         | Submits the document capture task for asynchronous processing and returns the task ID.                 |
| <b>PROPERTY</b> | <code>DocumentCaptureTask.documentType</code>                  | string                   | Server scripts         | The document type.                                                                                     |
| <b>PROPERTY</b> | <code>DocumentCaptureTask.features</code>                      | string[]                 | Server scripts         | The features to extract from the document (such as fields, tables, or text).                           |
| <b>PROPERTY</b> | <code>DocumentCaptureTask.id</code>                            | string                   | Server scripts         | The ID of the task.                                                                                    |
| <b>PROPERTY</b> | <code>DocumentCaptureTask.inboundDependencies</code>           | Object[]                 | Server scripts         | Key-value pairs that contain information about the dependent tasks added to the document capture task. |
| <b>PROPERTY</b> | <code>DocumentCaptureTask.inputFile</code>                     | file.File                | Server scripts         | The document to extract content from.                                                                  |
| <b>PROPERTY</b> | <code>DocumentCaptureTask.language</code>                      | string                   | Server scripts         | The language of the document.                                                                          |
| <b>PROPERTY</b> | <code>DocumentCaptureTask.outputFilePath</code>                | string                   | Server scripts         | The path of the JSON file to export document capture results to.                                       |
| <b>PROPERTY</b> | <code>DocumentCaptureTask.ociConfig</code>                     | Object                   | Server scripts         | Important: This object is no longer supported. Any values specified in this object are ignored.        |

## DocumentCaptureTaskStatus Object Members

The following members are available for a task.DocumentCaptureTaskStatus object.

Return Type / Value Type

Supported Script Types

DocumentCaptureTaskStatus.status

string (read-only)

The status for a document capture task (as a task.TaskStatus enum value).

| MEMBER TYPE | NAME                             | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                               |
|-------------|----------------------------------|--------------------------|------------------------|---------------------------------------------------------------------------|
| PROPERTY    | DocumentCaptureTaskStatus.status | string (read-only)       | Server scripts         | The status for a document capture task (as a task.TaskStatus enum value). |
| PROPERTY    | DocumentCaptureTaskStatus.taskId | string (read-only)       | Server scripts         | The task ID associated with the specified task.                           |

## EntityDeduplicationTask Object Members

The following members are available for a task.EntityDeduplicationTask object.

Return Type / Value Type

Supported Script Types

EntityDeduplicationTask.submit()

Directs NetSuite to place the merge duplicate records task into the NetSuite task queue and returns a unique ID for the task.

EntityDeduplicationTask.dedupeMode

| MEMBER TYPE | NAME                                        | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                            |
|-------------|---------------------------------------------|--------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | EntityDeduplicationTask.submit()            | string                   | Server scripts         | Directs NetSuite to place the merge duplicate records task into the NetSuite task queue and returns a unique ID for the task.                                          |
| PROPERTY    | EntityDeduplicationTask.dedupeMode          | string                   | Server scripts         | The mode in which to merge or delete duplicate records. Use values from the task.DedupeMode enum.                                                                      |
| PROPERTY    | EntityDeduplicationTask.entityType          | string                   | Server scripts         | The type of entity on which you want to merge duplicate records. Use a task.DedupeEntityType enum to set the value.                                                    |
| PROPERTY    | EntityDeduplicationTask.id                  | string                   | Server scripts         | The ID of the task.                                                                                                                                                    |
| PROPERTY    | EntityDeduplicationTask.masterRecordId      | number                   | Server scripts         | Master record ID. When you merge duplicate records, you can delete all duplicates for a record or merge information from the duplicate records into the master record. |
| PROPERTY    | EntityDeduplicationTask.masterSelectionMode | string                   | Server scripts         | Master selection mode. Use values from the task.MasterSelectionMode enum.                                                                                              |
| PROPERTY    | EntityDeduplicationTask.recordIds           | number[]                 | Server scripts         | Number array of record internal IDs to perform the merge or delete operation on.                                                                                       |

## EntityDeduplicationTaskStatus Object Members

The following members are available for a task.EntityDeduplicationTaskStatus object.

Return Type / Value Type

Supported Script Types

EntityDeduplicationTaskStatus.status

string (read-only)

Status for a merge duplicate record task. Returns a task.TaskStatus enum value.

| MEMBER TYPE | NAME                                 | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                     |
|-------------|--------------------------------------|--------------------------|------------------------|---------------------------------------------------------------------------------|
| PROPERTY    | EntityDeduplicationTaskStatus.status | string (read-only)       | Server scripts         | Status for a merge duplicate record task. Returns a task.TaskStatus enum value. |
| PROPERTY    | EntityDeduplicationTaskStatus.taskId | string (read-only)       | Server scripts         | The task ID associated with the specified task.                                 |

## MapReduceScriptTask Object Members

The following members are available for a task.MapReduceScriptTask object.

Return Type / Value Type

Supported Script Types

MapReduceScriptTask.submit()

Submits a map/reduce script deployment for processing.

MapReduceScriptTask.deploymentId

| MEMBER TYPE | NAME                             | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                |
|-------------|----------------------------------|--------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------|
| METHOD      | MapReduceScriptTask.submit()     | string                   | Server scripts         | Submits a map/reduce script deployment for processing.                                                                     |
| PROPERTY    | MapReduceScriptTask.deploymentId | string                   | Server scripts         | Script ID (as a string), for the script deployment record for a map/reduce script.                                         |
| PROPERTY    | MapReduceScriptTask.id           | string                   | Server scripts         | The ID of the task.                                                                                                        |
| PROPERTY    | MapReduceScriptTask.params       | Object                   | Server scripts         | Object that represents key-value pairs that override static script parameter field values on the script deployment record. |
| PROPERTY    | MapReduceScriptTask.scriptId     | number   string          | Server scripts         | Internal ID (as a number), or script ID (as a string), for the map/reduce script record.                                   |

## MapReduceScriptTaskStatus Object Members

The following members are available for a task.MapReduceScriptTaskStatus object.

Return Type / Value Type

Supported Script Types

MapReduceScriptTaskStatus.getCurrentTotalSize()

Returns the total size in bytes of all stored work in progress by a task.MapReduceScriptTask.

MapReduceScriptTaskStatus.getPendingMapCount()

| MEMBER TYPE | NAME                                               | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                            |
|-------------|----------------------------------------------------|--------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | MapReduceScriptTaskStatus.getCurrentTotalSize()    | number                   | Server scripts         | Returns the total size in bytes of all stored work in progress by a task.MapReduceScriptTask.                                          |
| METHOD      | MapReduceScriptTaskStatus.getPendingMapCount()     | number                   | Server scripts         | Returns the total number of records or rows not yet processed by the map stage of a task.MapReduceScriptTask.                          |
| METHOD      | MapReduceScriptTaskStatus.getPendingMapSize()      | number                   | Server scripts         | Returns the total number of bytes not yet processed by the map stage, as a component of total size, of a task.MapReduceScriptTask.     |
| METHOD      | MapReduceScriptTaskStatus.getPendingOutputCount()  | number                   | Server scripts         | Returns the total number of records or rows not yet processed by a task.MapReduceScriptTask.                                           |
| METHOD      | MapReduceScriptTaskStatus.getPendingOutputSize()   | number                   | Server scripts         | Returns the total size in bytes of all key-value pairs written as output, as a component of total size, by a task.MapReduceScriptTask. |
| METHOD      | MapReduceScriptTaskStatus.getPendingReduceCount()  | number                   | Server scripts         | Returns the total number of records or rows not yet processed by the reduce stage of a task.MapReduceScriptTask.                       |
| METHOD      | MapReduceScriptTaskStatus.getPendingReduceSize()   | number                   | Server scripts         | Returns the total number of bytes not yet processed by the reduce stage, as a component of total size, of a task.MapReduceScriptTask.  |
| METHOD      | MapReduceScriptTaskStatus.getPercentageCompleted() | number                   | Server scripts         | Returns the current percentage complete for the current stage of a task.MapReduceScriptTask.                                           |
| METHOD      | MapReduceScriptTaskStatus.getTotalMapCount()       | number                   | Server scripts         | Returns the total number of records or rows passed as input to the map stage of a task.MapReduceScriptTask.                            |
| METHOD      | MapReduceScriptTaskStatus.getTotalOutputCount()    | number                   | Server scripts         | Returns the total number of records or rows passed as inputs to the output phase of a task.MapReduceScriptTask.                        |
| METHOD      | MapReduceScriptTaskStatus.getTotalReduceCount()    | number                   | Server scripts         | Returns the total number of record or row inputs to the reduce stage of a task.MapReduceScriptTask.                                    |
| PROPERTY    | MapReduceScriptTaskStatus.deploymentId             | string (read-only)       | Server scripts         | Script ID for a script deployment record associated with a specific task.MapReduceScriptTask.                                          |
| PROPERTY    | MapReduceScriptTaskStatus.scriptId                 | number (read-only)       | Server scripts         | Internal ID for a map/reduce script record associated with a specific task.MapReduceScriptTask.                                        |
| PROPERTY    | MapReduceScriptTaskStatus.stage                    | string (read-only)       | Server scripts         | The current stage of a map/reduce script deployment that is being processed. See task.MapReduceStage for supported values.             |
| PROPERTY    | MapReduceScriptTaskStatus.status                   | string (read-only)       | Server scripts         | Status for a map/reduce script task. Returns a task.TaskStatus enum value.                                                             |
| PROPERTY    | MapReduceScriptTaskStatus.taskId                   | string (read-only)       | Server scripts         | The task ID associated with the specified task.                                                                                        |

## QueryTask Object Members

The following members are available for a task.QueryTask object.

Return Type / Value Type

Supported Script Types

QueryTask.addInboundDependency(options)

Adds a scheduled script task or map/reduce script task to the query task as a dependent task.

QueryTask.submit()

| MEMBER TYPE | NAME                                    | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                        |
|-------------|-----------------------------------------|--------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | QueryTask.addInboundDependency(options) | void                     | Server scripts         | Adds a scheduled script task or map/reduce script task to the query task as a dependent task.                                      |
| METHOD      | QueryTask.submit()                      | string                   | Server scripts         | Submits the query task for asynchronous processing and returns the task ID.                                                        |
| PROPERTY    | QueryTask.fileId                        | number                   | Server scripts         | Internal ID of the CSV file to export query results to. This property is mutually exclusive with the QueryTask.filePath parameter. |
| PROPERTY    | QueryTask.filePath                      | string                   | Server scripts         | Path of the CSV file to export query results to. This property is mutually exclusive with the QueryTask.fileId property.           |
| PROPERTY    | QueryTask.id                            | string                   | Server scripts         | The ID of the task.                                                                                                                |
| PROPERTY    | QueryTask.inboundDependencies           | Object[]                 | Server scripts         | Key-value pairs that contain information about the dependent tasks added to the query task.                                        |
| PROPERTY    | QueryTask.query                         | string                   | Server scripts         | Query definition for the query task.                                                                                               |

## QueryTaskStatus Object Members

The following members are available for a task.QueryTaskStatus object.

Return Type / Value Type

Supported Script Types

QueryTaskStatus.fileId

number (read-only)

Internal ID of the CSV file that query results are exported to.

| MEMBER TYPE | NAME                   | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                     |
|-------------|------------------------|--------------------------|------------------------|-----------------------------------------------------------------|
| PROPERTY    | QueryTaskStatus.fileId | number (read-only)       | Server scripts         | Internal ID of the CSV file that query results are exported to. |
| PROPERTY    | QueryTaskStatus.query  | query.Query (read-only)  | Server scripts         | Query definition for the submitted query task.                  |
| PROPERTY    | QueryTaskStatus.status | string (read-only)       | Server scripts         | Status of the submitted query task.                             |
| PROPERTY    | QueryTaskStatus.taskId | string (read-only)       | Server scripts         | ID of the submitted query task.                                 |

## RecordActionTask Object Members

The following members are available for a task.RecordActionTaskStatus object.

Return Type / Value Type

Supported Script Types

RecordActionTask.submit()

Submits a record action task for processing and returns its task ID.

RecordActionTask.action

| MEMBER TYPE | NAME                           | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                                                                   |
|-------------|--------------------------------|--------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | RecordActionTask.submit()      | string                   | Server scripts         | Submits a record action task for processing and returns its task ID.                                                                                                                                                          |
| PROPERTY    | RecordActionTask.action        | string                   | Server scripts         | The ID of the action to be invoked.                                                                                                                                                                                           |
| PROPERTY    | RecordActionTask.condition     | Object                   | Server scripts         | The condition used to select record IDs of records for which the action is to be executed. Only the action.ALL_QUALIFIED_INSTANCES constant is currently supported.                                                           |
| PROPERTY    | RecordActionTask.id            | string                   | Server scripts         | The ID of the task.                                                                                                                                                                                                           |
| PROPERTY    | RecordActionTask.paramCallback | Object                   | Server scripts         | Function that takes record ID and returns the parameter object for the specified record ID.                                                                                                                                   |
| PROPERTY    | RecordActionTask.params        | Object[]                 | Server scripts         | An array of parameter objects. Each object corresponds to one record ID of the record for which the action is to be executed. The object has the following form: {recordId: 1, someParam: 'example1', otherParam: 'example2'} |
| PROPERTY    | RecordActionTask.recordType    | string                   | Server scripts         | The record type on which the action is to be performed. For a list of record types, see record.Type.                                                                                                                          |

## RecordActionTaskStatus Object Members

The following members are available for a task.RecordActionTaskStatus object.

Return Type / Value Type

Supported Script Types

RecordActionTaskStatus.complete

number (read-only)

The number of record action tasks with a completed status.

| MEMBER TYPE | NAME                             | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                                                                        |
|-------------|----------------------------------|--------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PROPERTY    | RecordActionTaskStatus.complete  | number (read-only)       | Server scripts         | The number of record action tasks with a completed status.                                                                                                                                                                         |
| PROPERTY    | RecordActionTaskStatus.errors    | Object (read-only)       | Server scripts         | The error details of failed action executions. The value of the property is the record instance ID and the corresponding error details. The error details are returned in an unnamed object with two properties: code and message. |
| PROPERTY    | RecordActionTaskStatus.failed    | number (read-only)       | Server scripts         | The number of record action tasks with a failed status.                                                                                                                                                                            |
| PROPERTY    | RecordActionTaskStatus.pending   | number (read-only)       | Server scripts         | The number of record action tasks with a pending status.                                                                                                                                                                           |
| PROPERTY    | RecordActionTaskStatus.results   | Object (read-only)       | Server scripts         | The results of successfully executed record action tasks. The value of the property is the task instance ID and the corresponding action result.                                                                                   |
| PROPERTY    | RecordActionTaskStatus.status    | string (read-only)       | Server scripts         | Represents the record action task status. Returns a value from the task.TaskStatus enum.                                                                                                                                           |
| PROPERTY    | RecordActionTaskStatus.succeeded | number (read-only)       | Server scripts         | The number of record action tasks with a succeeded status.                                                                                                                                                                         |
| PROPERTY    | RecordActionTaskStatus.taskId    | string (read-only)       | Server scripts         | The task ID associated with the specified task.                                                                                                                                                                                    |

## ScheduledScriptTask Object Members

The following members are available for a task.ScheduledScriptTask object.

Return Type / Value Type

Supported Script Types

ScheduledScriptTask.submit()

Directs NetSuite to place a scheduled script deployment into the NetSuite scheduling queue and returns a unique ID for the task.

ScheduledScriptTask.deploymentsId

| MEMBER TYPE | NAME                              | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                      |
|-------------|-----------------------------------|--------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | ScheduledScriptTask.submit()      | string                   | Server scripts         | Directs NetSuite to place a scheduled script deployment into the NetSuite scheduling queue and returns a unique ID for the task. |
| PROPERTY    | ScheduledScriptTask.deploymentsId | string                   | Server scripts         | Script ID (as a string), for the script deployment record associated with a task.ScheduledScriptTask object.                     |
| PROPERTY    | ScheduledScriptTask.id            | string                   | Server scripts         | The ID of the task.                                                                                                              |
| PROPERTY    | ScheduledScriptTask.params        | Object                   | Server scripts         | Object with key-value pairs that override the static script parameter field values on the script deployment.                     |
| PROPERTY    | ScheduledScriptTask.scriptId      | number   string          | Server scripts         | Internal ID (as a number), or script ID (as a string) for the script record associated with a task.ScheduledScriptTask object.   |

## ScheduledScriptTaskStatus Object Members

The following members are available for a task.ScheduledScriptTaskStatus object.

Return Type / Value Type

Supported Script Types

ScheduledScriptTaskStatus.deploymentsId

string (read-only)

Script ID for a script deployment record associated with a specific task.ScheduledScriptTask object.

| MEMBER TYPE | NAME                                    | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                          |
|-------------|-----------------------------------------|--------------------------|------------------------|------------------------------------------------------------------------------------------------------|
| PROPERTY    | ScheduledScriptTaskStatus.deploymentsId | string (read-only)       | Server scripts         | Script ID for a script deployment record associated with a specific task.ScheduledScriptTask object. |
| PROPERTY    | ScheduledScriptTaskStatus.scriptId      | number (read-only)       | Server scripts         | Internal ID for a script record associated with a specific task.ScheduledScriptTask object.          |
| PROPERTY    | ScheduledScriptTaskStatus.status        | string (read-only)       | Server scripts         | Status for a scheduled script task. Returns a task.TaskStatus enum value.                            |

| MEMBER TYPE | NAME                                          | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                     |
|-------------|-----------------------------------------------|--------------------------|------------------------|-------------------------------------------------|
| PROPERTY    | <code>ScheduledScriptTaskStatus.taskId</code> | string (read-only)       | Server scripts         | The task ID associated with the specified task. |

## SearchTask Object Members

The following members are available for a task.SearchTask object.

Return Type / Value Type

Supported Script Types

SearchTask.addInboundDependency()

Adds a scheduled script task or map/reduce script task to the search task as a dependent script. Dependent scripts are processed automatically when the search task is complete. For more information, see SuiteCloud Processors.

SearchTask.submit()

| MEMBER TYPE | NAME                                           | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                                                                       |
|-------------|------------------------------------------------|--------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | <code>SearchTask.addInboundDependency()</code> | void                     | Server scripts         | Adds a scheduled script task or map/reduce script task to the search task as a dependent script. Dependent scripts are processed automatically when the search task is complete. For more information, see SuiteCloud Processors. |
| METHOD      | <code>SearchTask.submit()</code>               | string                   | Server scripts         | Places the asynchronous search initiation task into the SuiteScript task queue, and returns a unique ID for the task.                                                                                                             |
| PROPERTY    | <code>SearchTask.fileId</code>                 | number                   | Server scripts         | ID of the CSV file to export search results into.                                                                                                                                                                                 |
| PROPERTY    | <code>SearchTask.filePath</code>               | string                   | Server scripts         | Path of the CSV file to export search results into.                                                                                                                                                                               |
| PROPERTY    | <code>SearchTask.id</code>                     | string                   | Server scripts         | The ID of the task.                                                                                                                                                                                                               |
| PROPERTY    | <code>SearchTask.inboundDependencies</code>    | Object[] (read-only)     | Server scripts         | Key-value pairs to describe the dependent scripts added to the search task.                                                                                                                                                       |
| PROPERTY    | <code>SearchTask.savedSearchId</code>          | number                   | Server scripts         | ID of the saved search to be executed during the task.                                                                                                                                                                            |

## SearchTaskStatus Object Members

The following members are available for a task.SearchTaskStatus object.

Return Type / Value Type

Supported Script Types

SearchTaskStatus.fileId

number (read-only)

ID of the CSV file into which search results are exported.

| MEMBER TYPE | NAME                                        | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                              |
|-------------|---------------------------------------------|--------------------------|------------------------|--------------------------------------------------------------------------------------------------------------------------|
| PROPERTY    | <code>SearchTaskStatus.fileId</code>        | number (read-only)       | Server scripts         | ID of the CSV file into which search results are exported.                                                               |
| PROPERTY    | <code>SearchTaskStatus.savedSearchId</code> | number (read-only)       | Server scripts         | ID of the saved search executed during the task.                                                                         |
| PROPERTY    | <code>SearchTaskStatus.status</code>        | string (read-only)       | Server scripts         | Status of an asynchronous search task placed in the NetSuite task queue. Returns one of the task.TaskStatus enum values. |
| PROPERTY    | <code>SearchTaskStatus.taskId</code>        | string (read-only)       | Server scripts         | ID of the asynchronous task.                                                                                             |

## SuiteQLTask Object Members

The following members are available for a task.SuiteQLTask object.

Return Type / Value Type

Supported Script Types

SuiteQLTask.addInboundDependency(options)

Adds a scheduled script task or map/reduce script task to the SuiteQL task as a dependent task.

SuiteQLTask.submit()

| MEMBER TYPE | NAME                                                   | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                     |
|-------------|--------------------------------------------------------|--------------------------|------------------------|-------------------------------------------------------------------------------------------------|
| METHOD      | <code>SuiteQLTask.addInboundDependency(options)</code> | void                     | Server scripts         | Adds a scheduled script task or map/reduce script task to the SuiteQL task as a dependent task. |
| METHOD      | <code>SuiteQLTask.submit()</code>                      | string                   | Server scripts         | Submits the SuiteQL task for asynchronous processing and returns the task ID.                   |

| MEMBER TYPE | NAME                            | RETURN TYPE / VALUE TYPE          | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                  |
|-------------|---------------------------------|-----------------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| PROPERTY    | SuiteQLTask.fileId              | number                            | Server scripts         | Internal ID of the CSV file to export SuiteQL query results to. This property is mutually exclusive with the SuiteQLTask.filePath parameter. |
| PROPERTY    | SuiteQLTask.filePath            | string                            | Server scripts         | Path of the CSV file to export SuiteQL query results to. This property is mutually exclusive with the SuiteQLTask.field property.            |
| PROPERTY    | SuiteQLTask.id                  | string                            | Server scripts         | The ID of the task.                                                                                                                          |
| PROPERTY    | SuiteQLTask.inboundDependencies | Object[]                          | Server scripts         | Key-value pairs that contain information about the dependent tasks added to the SuiteQL task.                                                |
| PROPERTY    | SuiteQLTask.params              | Array<string   boolean   number>; | Server scripts         | Parameters for the SuiteQL query.                                                                                                            |
| PROPERTY    | SuiteQLTask.query               | string                            | Server scripts         | SuiteQL query definition for the SuiteQL task.                                                                                               |

## SuiteQLTaskStatus Object Members

The following members are available for a task.SuiteQLTaskStatus object.

Return Type / Value Type

Supported Script Types

SuiteQLTaskStatus.field

number (read-only)

Internal ID of the CSV file that SuiteQL query results are exported to.

| MEMBER TYPE | NAME                     | RETURN TYPE / VALUE TYPE                      | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                             |
|-------------|--------------------------|-----------------------------------------------|------------------------|-------------------------------------------------------------------------|
| PROPERTY    | SuiteQLTaskStatus.fileId | number (read-only)                            | Server scripts         | Internal ID of the CSV file that SuiteQL query results are exported to. |
| PROPERTY    | SuiteQLTaskStatus.params | Array<string   boolean   number>; (read-only) | Server scripts         | Parameters for the SuiteQL query.                                       |
| PROPERTY    | SuiteQLTaskStatus.query  | string (read-only)                            | Server scripts         | SuiteQL query definition for the SuiteQL task.                          |
| PROPERTY    | SuiteQLTaskStatus.status | string (read-only)                            | Server scripts         | Status of the SuiteQL task.                                             |
| PROPERTY    | SuiteQLTaskStatus.taskId | string (read-only)                            | Server scripts         | ID of the submitted SuiteQL task.                                       |

## WorkflowTriggerTask Object Members

The following members are available for a task.WorkflowTriggerTask object.

Return Type / Value Type

Supported Script Types

WorkflowTriggerTask.submit()

Directs NetSuite to place the asynchronous workflow initiation task into the NetSuite scheduling queue and returns a unique ID for the task.

WorkflowTriggerTask.id

| MEMBER TYPE | NAME                           | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                  |
|-------------|--------------------------------|--------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | WorkflowTriggerTask.submit()   | string                   | Server scripts         | Directs NetSuite to place the asynchronous workflow initiation task into the NetSuite scheduling queue and returns a unique ID for the task. |
| PROPERTY    | WorkflowTriggerTask.id         | string                   | Server scripts         | The ID of the task.                                                                                                                          |
| PROPERTY    | WorkflowTriggerTask.params     | Object                   | Server scripts         | Object that contains key-value pairs to set default values on fields specific to the workflow.                                               |
| PROPERTY    | WorkflowTriggerTask.recordId   | number                   | Server scripts         | Internal ID of the workflow definition base record. For example, 55 or 124.                                                                  |
| PROPERTY    | WorkflowTriggerTask.recordType | string                   | Server scripts         | Record type of the workflow base record. For example, customer, salesorder, or lead.                                                         |
| PROPERTY    | WorkflowTriggerTask.workflowId | number   string          | Server scripts         | Internal ID (as a number), or script ID (as a string), for the workflow definition.                                                          |

## WorkflowTriggerTaskStatus Object Members

The following members are available for a task.WorkflowTriggerTaskStatus object.

Return Type / Value Type

Supported Script Types

WorkflowTriggerTaskStatus.status

string (read-only)

Status for a asynchronous workflow placed in the NetSuite task queue. Returns a value from the task.TaskStatus enum.

| MEMBER TYPE | NAME                             | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                          |
|-------------|----------------------------------|--------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------|
| PROPERTY    | WorkflowTriggerTaskStatus.status | string (read-only)       | Server scripts         | Status for a asynchronous workflow placed in the NetSuite task queue. Returns a value from the task.TaskStatus enum. |
| PROPERTY    | WorkflowTriggerTaskStatus.taskId | string (read-only)       | Server scripts         | The task ID associated with the specified task.                                                                      |

► Official NetSuite Script Samples (5 Sample(s))

 Sample 1: Create and Submit a Map/Reduce Script Task

```
/**
 * @NApiVersion 2.1
 */
require(['N/task', 'N/runtime', 'N/email'], (task, runtime, email) => {
  // Store the script ID of the script to submit
  //
  // Update the following statement so it uses the script ID
  // of the map/reduce script record you want to submit
  const mapReduceScriptId = 'customscript_test_mapreduce_script';

  // Create a map/reduce task
  //
  // Update the deploymentId parameter to use the script ID of
  // the deployment record for your map/reduce script
  let mrTask = task.create({
    taskType: task.TaskType.MAP_REDUCE,
    scriptId: mapReduceScriptId,
    deploymentId: 'customdeploy_test_mapreduce_script'
  });

  // Submit the map/reduce task
  let mrTaskId = mrTask.submit();

  // Check the status of the task, and send an email if the
  // task has a status of FAILED
  //
  // Update the authorId value with the internal ID of the user
  // who is the email sender. Update the recipientEmail value
  // with the email address of the recipient.
  let taskStatus = task.checkStatus(mrTaskId);
  if (taskStatus.status === 'FAILED') {
    const authorId = -5;
    const recipientEmail = 'notify@myCompany.com';
    email.send({
      author: authorId,
      recipients: recipientEmail,
      subject: 'Failure executing map/reduce job!',
      body: 'Map reduce task: ' + mapReduceScriptId + ' has failed.'
    });
  }
});
```

### Sample 2: Create and Submit an Asynchronous Search Task and Export the Results into a CSV File

```
/**
 * @NApiVersion 2.1
 */
require(['N/task'], task => {
  // Do one of the following:
  //
  // - Create a saved search and capture its ID. To do this, you can use
  //   the following code snippet (replacing the type, id, filters, and
  //   columns values as appropriate):
  //
  //   let mySearch = search.create({
  //     type: search.Type.SALES_ORDER,
  //     id: 'customsearch_my_search',
  //     filters: [...],
  //     columns: [...]
  //   });
  //   mySearch.save();
  //   let savedSearchId = mySearch.searchId;
  //
  // - Use the ID of an existing saved search. This is the approach that
  //   this script sample uses. Update the following statement with the
  //   internal ID of the search you want to use.
  const savedSearchId = 669;

  // Create the search task
  let myTask = task.create({
    taskType: task.TaskType.SEARCH
  });
  myTask.savedSearchId = savedSearchId;

  // Specify the ID of the file that search results will be exported into
  //
  // Update the following statement so it uses the internal ID of the file
  // you want to use
  myTask.fileId = 448;

  // Submit the search task
  let myTaskId = myTask.submit();

  // Retrieve the status of the search task
  let taskStatus = task.checkStatus({
    taskId: myTaskId
  });

  // Optionally, add logic that executes when the task is complete
  if (taskStatus.status === task.TaskStatus.COMPLETE) {
    // Add any code that is appropriate. For example, if this script created
    // a saved search, you may want to delete it.
  }
});
```

### Sample 3: Create and Submit a Task with Dependent Scripts

```
/**
 * @NApiVersion 2.1
 */
require(['N/task'], (task) => {
  // Specify a file for the search results
  const asyncSearchResultFile = 'SuiteScripts/ExportFile.csv';

  // Create a map/reduce script task
  const mapReduceScript = task.create({
    taskType: task.TaskType.MAP_REDUCE
  });
  mapReduceScript.scriptId = 'customscript_mr_create_submit';
  mapReduceScript.deployedId = 'customdeploy_mr_create_submit';
  mapReduceScript.params = {
    'custscript_mr_as_srch_res' : asyncSearchResultFile
  };

  // Create the search task
  const asyncTask = task.create({
    taskType: task.TaskType.SEARCH
  });
  asyncTask.savedSearchId = 'customsearch35';
  asyncTask.filePath = asyncSearchResultFile;

  // Add dependent scripts to the search task before it is submitted
  asyncTask.addInboundDependency(mapReduceScript);

  // Submit the search task
  const asyncTaskId = asyncTask.submit();
});
```

#### Sample 4: Create and Submit a Task with Dependent Scripts

```
/**
 * @NApiVersion 2.1
 * @NScriptType MapReduceScript
 * @NModuleScope SameAccount
 */

define(['N/runtime', 'N/file', 'N/log', 'N/email'], (runtime, file, log, email) => {
  // Load the search results file, count the number of letters in the file, and
  // store this count in another file

  function getInputData() {
    // Retrieve the ID of the search results file
    //
    // Update the completionScriptParameterName value to use the script
    // ID of the original search task
    const fileId = runtime.getCurrentScript().getParameter({
      name: 'custscript_mr_new_submit_task_field'
    });

    if (!fileId) {
      log.error({
        details: 'fileId is not valid. Please check the script parameter stored in the completionScriptParameterName variable in getInputData().'
      });
    }

    return {
      type: 'file',
      id: fileId
    };
  }

  function map(context) {
    log.debug('map context: ', context);
    context.write({
      key: context.key,
      value: 1
    });
  }

  function reduce(context) {
    log.debug('reduce context: ', context);
    log.debug('context.values.length', context.values.length);

    // Count the number of rows
    let rowCount = 0;
    rowCount = context.values.length - 1 // Subtracting 1 to remove header row from the count.
    log.debug('rowCount: ', rowCount);

    // Send an email to the user who ran the script, and attach the
    // CSV file with the search results
    const completionScriptParameterName = 'custscript_mr_new_submit_task_field';
    const resFileId = runtime.getCurrentScript().getParameter({
      name: completionScriptParameterName
    });
    const fileObj = file.load({
      id: resFileId
    });
    const userId = runtime.getCurrentUser().id;
    email.send({
      author: userId,
      recipients: userId,
      subject: 'Search completed',
      body: 'CSV file attached, ' + rowCount + ' record(s) found.',
      attachments: [fileObj]
    });
  }

  function summarize(summary) {
    const type = summary.toString();
    log.audit({ title: type + ' Usage Consumed ', details: summary.usage });
    log.audit({ title: type + ' Concurrency Number ', details: summary.concurrency });
    log.audit({ title: type + ' Number of Yields ', details: summary.yields });
  }

  return {
    getInputData: getInputData,
    map: map,
    reduce: reduce,
    summarize: summarize
  };
});
```

#### Sample 5: Submit a Record Action Task and Check Status

```
/**
 * @NApiVersion 2.x
 */
require(['N/task'], function(task) {
    var recordActionTask = task.create({
        taskType: task.TaskType.RECORD_ACTION
    });
    recordActionTask.recordType = 'timebill';
    recordActionTask.action = 'approve';
    recordActionTask.params = [
        {recordId: 1, note: 'This is a note for 1'},
        {recordId: 5, note: 'This is a note for 5'},
        {recordId: 23, note: 'This is a note for 23'}
    ];

    var handle = recordActionTask.submit();

    var res = task.checkStatus({
        taskId: handle
    }); // Returns a RecordActionTaskStatus object
    log.debug('Initial status: ' + res.status);
});
```

## 45. N/task/accounting/recognition Module

Load the task/accounting/recognition module to merge revenue arrangements or revenue elements. This module lets you combine revenue arrangements or revenue elements from multiple sources to represent a single contract for revenue allocation and recognition.

```
define(['N/task/accounting/recognition'], (recognition) => { ... })
```

[✓ 3 Code Sample\(s\)](#)[Scripts: Server scripts](#)

Permissions: (Transactions) Revenue Arrangement

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/task/accounting/recognition'], (recognition) => { // Access methods and properties on recognition });
```

Use the N/task/accounting/recognition module to merge revenue arrangements or revenue elements. A revenue arrangement is a transaction that records the details of a sale for the purposes of revenue allocation and recognition. The N/task/accounting/recognition module lets you combine revenue arrangements or revenue elements from multiple sources to represent a single contract obligation for revenue allocation and recognition.

You can use the recognition.create(options) method to create a merge task that combines entire revenue arrangements or individual revenue elements. This method returns a recognition.MergeArrangementsTask object (when merging revenue arrangements) or recognition.MergeElementsTask object (when merging revenue elements). After you obtain one of these objects, you can set its properties, such as the list of arrangements or elements to merge, the date on the merged revenue arrangement, whether to prospectively merge arrangements, and so on. You can use these properties to specify the same input data that you can specify when you merge revenue arrangements using the NetSuite UI. After you set its properties, you can submit the task for processing. Merge tasks are processed asynchronously.

You cannot merge more than 10,000 revenue elements at one time. An error is thrown in the script if you attempt to merge more than 10,000 revenue elements. For UI related limitations, see Limitations for Creating Transactions.

You can use the recognition.checkStatus(options) method to check the status of a submitted merge task. This method returns a recognition.MergeArrangementsTaskStatus object that describes the current status of the merge task (pending, processing, complete, or failed). This object represents the current status for either a recognition.MergeArrangementsTask or a recognition.MergeElementsTask. If the task completes successfully, this object includes the ID of the merged revenue arrangement record that was created. If the task fails, this object includes an error message that describes the failure.

To merge revenue arrangements or revenue elements using the N/task/accounting/recognition module, the following requirements must be met:

The Advanced Revenue Management feature must be enabled in your account. For more information, see Enabling the Advanced Revenue Management (Essentials) Feature.

Your role must have the (Transactions) Revenue Arrangement permission assigned at a level of Create or higher. For more information, see NetSuite Permissions Overview.

For more information about revenue arrangements, see the following help topics:

Revenue Arrangement Management - This topic describes revenue arrangements in general.

Combination and Modification of Performance Obligations - This topic describes the different types of merge results (combined revenue arrangements and prospective change orders).

Revenue Arrangement - This topic describes the revenue arrangement record type, including scripting considerations, supported script types, and sublist fields.

### N/task/accounting/recognition Module Members

Return Type / Value Type

Supported Script Types

recognition.MergeArrangementsTask

Encapsulates a task to merge all of the revenue elements from a specified list of revenue arrangements.

Use recognition.create(options) to create this object.

recognition.MergeArrangementsTaskStatus

| MEMBER TYPE | NAME                                    | RETURN TYPE / VALUE TYPE                                          | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                             |
|-------------|-----------------------------------------|-------------------------------------------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT      | recognition.MergeArrangementsTask       | Object                                                            | Server scripts         | Encapsulates a task to merge all of the revenue elements from a specified list of revenue arrangements. Use recognition.create(options) to create this object.                          |
| OBJECT      | recognition.MergeArrangementsTaskStatus | Object                                                            | Server scripts         | Encapsulates the current status of a submitted merge task. Use recognition.checkStatus(options) to create this object.                                                                  |
| OBJECT      | recognition.MergeElementsTask           | Object                                                            | Server scripts         | Encapsulates a task to merge all of the specified revenue elements. Use recognition.create(options) to create this object.                                                              |
| METHOD      | recognition.checkStatus(options)        | recognition.MergeArrangementsTaskStatus                           | Server scripts         | Checks the status of a submitted merge task.                                                                                                                                            |
| METHOD      | recognition.create(options)             | recognition.MergeArrangementsTask   recognition.MergeElementsTask | Server scripts         | Creates a merge task that combines entire revenue arrangements or individual revenue elements. Use values in the recognition.TaskType enum to specify the type of merge task to create. |

| MEMBER TYPE | NAME                   | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                    |
|-------------|------------------------|--------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ENUM        | recognition.TaskStatus | enum                     | Server scripts         | Holds the string values for supported merge task statuses. This enum is used to represent the task status in a recognition.MergeArrangementsTaskStatus object. |
| ENUM        | recognition.TaskType   | enum                     | Server scripts         | Holds the string values for supported merge task types. This enum is used to pass the task type argument to recognition.create(options).                       |

## MergeArrangementsTask Object Members

The following members are available for a recognition.MergeArrangementsTask object.

Return Type / Value Type

Supported Script Types

MergeArrangementsTask.submit()

number (read-only)

Submits the merge task for processing.

| MEMBER TYPE | NAME                                                            | RETURN TYPE / VALUE TYPE           | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------|-----------------------------------------------------------------|------------------------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | MergeArrangementsTask.submit()                                  | number (read-only)                 | Server scripts         | Submits the merge task for processing. This method returns a task ID that uniquely identifies the merge task.                                                                                                                                                                                                                                                                                     |
| PROPERTY    | MergeArrangementsTask.arrangements                              | Array<number   string> (read-only) | Server scripts         | Holds an array of internal IDs of the revenue arrangement records to merge.                                                                                                                                                                                                                                                                                                                       |
| PROPERTY    | MergeArrangementsTask.contractAcquisitionDeferredExpenseAccount | number   string (read-only)        | Server scripts         | References the contract acquisition deferred expense account for the new revenue arrangement. This property is valid only if the accounting preference Enable Advanced Cost Amortization is enabled. For more information, see Advanced Cost Amortization. The default value is the account specified by the accounting preference Contract Acquisition Deferred Expense Account in your account. |
| PROPERTY    | MergeArrangementsTask.contractAcquisitionExpenseAccount         | number   string (read-only)        | Server scripts         | References the contract acquisition expense account for the new revenue arrangement. This property is valid only if the accounting preference Enable Advanced Cost Amortization is enabled. For more information, see Advanced Cost Amortization. The default value is the account specified by the accounting preference Contract Acquisition Expense Account in your account.                   |
| PROPERTY    | MergeArrangementsTask.contractCostAccrualDate                   | JavaScript Date (read-only)        | Server scripts         | Describes the contract cost accrual date to use for the new revenue arrangement. This property is valid only if the accounting preference Enable Advanced Cost Amortization is enabled. For more information, see Advanced Cost Amortization. The default value is today's date.                                                                                                                  |
| PROPERTY    | MergeArrangementsTask.mergeResidualRevenueAmounts               | boolean (read-only)                | Server scripts         | Indicates whether the revenue arrangements are merged prospectively. For more information about prospective merges, see Prospective Merges. The default value is false.                                                                                                                                                                                                                           |
| PROPERTY    | MergeArrangementsTask.recalculateResidualFairValue              | boolean (read-only)                | Server scripts         | Indicates whether to recalculate the fair value on residual elements when revenue arrangements are prospectively merged. For more information about prospective merges, see Prospective Merges. This property can be set to true only if the MergeArrangementsTask.mergeResidualRevenueAmounts property is also set to true. The default value is false.                                          |
| PROPERTY    | MergeArrangementsTask.revenueArrangementDate                    | JavaScript Date (read-only)        | Server scripts         | Describes the date of the new revenue arrangement. The default value is today's date.                                                                                                                                                                                                                                                                                                             |

## MergeArrangementsTaskStatus Object Members

The following members are available for a recognition.MergeArrangementsTaskStatus object.

Return Type / Value Type

Supported Script Types

MergeArrangementsTaskStatus.errorMessage

string (read-only)

Holds an error message that describes the failure of the merge task. This property is valid only if the value of the status property is TaskStatus.FAILED.

| MEMBER TYPE | NAME                                             | RETURN TYPE / VALUE TYPE    | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                                  |
|-------------|--------------------------------------------------|-----------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PROPERTY    | MergeArrangementsTaskStatus.errorMessage         | string (read-only)          | Server scripts         | Holds an error message that describes the failure of the merge task. This property is valid only if the value of the status property is TaskStatus.FAILED.                                   |
| PROPERTY    | MergeArrangementsTaskStatus.inputArrangements    | number[] (read-only)        | Server scripts         | Holds an array of internal IDs of the revenue arrangement records to merge. This property is valid only if the merge task was created using a task type of TaskType.MERGE_ARRANGEMENTS_TASK. |
| PROPERTY    | MergeArrangementsTaskStatus.inputElements        | number[] (read-only)        | Server scripts         | Holds an array of internal IDs of the revenue elements to merge. This property is valid only if the merge task was created using a task type of TaskType.MERGE_ELEMENTS_TASK.                |
| PROPERTY    | MergeArrangementsTaskStatus.resultingArrangement | number   string (read-only) | Server scripts         | References the internal ID of the new revenue arrangement that was created. This property is valid only if the value of the status property is TaskStatus.COMPLETE.                          |
| PROPERTY    | MergeArrangementsTaskStatus.status               | string (read-only)          | Server scripts         | Represents the current status of the merge task. This property uses values in the recognition.TaskStatus enum.                                                                               |
| PROPERTY    | MergeArrangementsTaskStatus.submissionId         | number   string (read-only) | Server scripts         | References the submission ID of the merge arrangements bulk process.                                                                                                                         |
| PROPERTY    | MergeArrangementsTaskStatus.taskId               | number   string (read-only) | Server scripts         | Holds the task ID of the merge task. The task ID is assigned to the merge task when you call recognition.create(options).                                                                    |

## MergeElementsTask Object Members

The following members are available for a recognition.MergeElementsTask object.

Return Type / Value Type

Supported Script Types

MergeElementsTask.submit()

number (read-only)

Submits the merge task for processing.

| MEMBER TYPE | NAME                                                        | RETURN TYPE / VALUE TYPE           | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------|-------------------------------------------------------------|------------------------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | MergeElementsTask.submit()                                  | number (read-only)                 | Server scripts         | Submits the merge task for processing. This method returns a task ID that uniquely identifies the merge task.                                                                                                                                                                                                                                                                                     |
| PROPERTY    | MergeElementsTask.contractAcquisitionDeferredExpenseAccount | number   string (read-only)        | Server scripts         | References the contract acquisition deferred expense account for the new revenue arrangement. This property is valid only if the accounting preference Enable Advanced Cost Amortization is enabled. For more information, see Advanced Cost Amortization. The default value is the account specified by the accounting preference Contract Acquisition Deferred Expense Account in your account. |
| PROPERTY    | MergeElementsTask.contractAcquisitionExpenseAccount         | number   string (read-only)        | Server scripts         | References the contract acquisition expense account for the new revenue arrangement. This property is valid only if the accounting preference Enable Advanced Cost Amortization is enabled. For more information, see Advanced Cost Amortization. The default value is the account specified by the accounting preference Contract Acquisition Expense Account in your account.                   |
| PROPERTY    | MergeElementsTask.contractCostAccrualDate                   | JavaScript Date (read-only)        | Server scripts         | Describes the contract cost accrual date to use for the new revenue arrangement. This property is valid only if the accounting preference Enable Advanced Cost Amortization is enabled. For more information, see Advanced Cost Amortization. The default value is today's date.                                                                                                                  |
| PROPERTY    | MergeElementsTask.elements                                  | Array<number   string> (read-only) | Server scripts         | Holds an array of internal IDs of the revenue element records to merge.                                                                                                                                                                                                                                                                                                                           |
| PROPERTY    | MergeElementsTask.revenueArrangementDate                    | JavaScript Date (read-only)        | Server scripts         | Describes the date of the new revenue arrangement. The default value is today's date.                                                                                                                                                                                                                                                                                                             |

 **Sample 1: Merge Revenue Elements Using Internal IDs**

```
/**
 * @NApiVersion 2.x
 */

require(['N/task/accounting/recognition'], function(recognition){
  var elementsList = [];
  elementsList.push(401);
  elementsList.push(402);

  var recognitionTask = recognition.create({
    taskType: recognition.TaskType.MERGE_ELEMENTS_TASK
  });
  recognitionTask.elements = elementsList;
  var taskStatusId = recognitionTask.submit();

  var mergeTaskState = recognition.checkStatus({
    taskId: taskStatusId
  });

  log.debug('Submission ID = ' + mergeTaskState.submissionId);
  log.debug('Resulting Arrangement ID = ' + mergeTaskState.resultingArrangement);
  log.debug('status = ' + mergeTaskState.status);
});
```

 **Sample 2: Merge Revenue Arrangements Using a Saved Search**

```
/**
 * @NApiVersion 2.x
 */

require(['N/task/accounting/recognition', 'N/search'], function(recognition, search){
  var mySearch = search.load({
    id: 'customsearch22'
  });

  var elementsList = [];
  mySearch.run().each(function(result) {
    var id = result.getValue({
      name: 'internalid'
    });
    elementsList.push(id);
  });

  var recognitionTask = recognition.create({
    taskType: recognition.TaskType.MERGE_ARRANGEMENTS_TASK
  });

  recognitionTask.arrangements = elementsList;
  recognitionTask.revenueArrangementDate = new Date(2019, 2, 10);

  var taskStatusId = recognitionTask.submit();
  log.debug('taskId = ' + taskStatusId);

  var mergeTaskState = recognition.checkStatus({
    taskId: taskStatusId
  });

  log.debug('Submission ID = ' + mergeTaskState.submissionId);
  log.debug('Resulting Arrangement ID = ' + mergeTaskState.resultingArrangement);
  log.debug('status = ' + mergeTaskState.status);
  log.debug('Error message = ' + mergeTaskState.errorMessage);
});
```

### Sample 3: Merge Revenue Arrangements Using an Ad-Hoc Search

```
/**
 * @NApiVersion 2.x
 */
require(['N/task/accounting/recognition', 'N/search'], function(recognition, search) {
    var elementsList = [];
    var rs = search.create({
        type: 'revenueelement',
        columns: [
            'internalid'
        ]
    }).run();

    var results = rs.getRange(0, 50);
    for (var i = 0; i < results.length; i++) {
        var id = result.getValue('elementsList');
        elementsList.push(id);
    }

    var t = recognition.create({
        taskType: recognition.TaskType.MERGE_ELEMENTS_TASK
    });
    t.elements = elementsList;
    t.revenueArrangementDate = new Date(2019, 1, 1);

    var taskId = t.submit();
    log.debug('Initial status: ' + res.status);
});
```

# 46. N/transaction Module

Load the N/transaction module to void transactions.

define(['N/transaction'], (transaction) => { ... })

✓ 1 Code Sample(s)

Scripts: Client and server scripts

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/transaction'], (transaction) => { // Access methods and properties on transaction });
```

Use the N/transaction module to void transactions.

When you void a transaction, the total and all the line items for the transaction are set to zero. The transaction is not removed from the system. NetSuite supports two types of voids: direct voids and voids by reversing journal. For additional information, see Voiding, Deleting, or Closing Transactions.

The type of void performed with your script depends on the targeted account's preference settings:

If the Using Reversing Journals preference is disabled, a direct void is performed.

If the Using Reversing Journals preference is enabled, a void by reversing journal is performed.

After you successfully void a transaction, you can no longer make changes to the transaction that impact the general ledger.

N/transaction Module Script Sample

## N/transaction Module Members

Return Type / Value Type

Supported Script Types

transaction.void(options)

Client and server scripts

voids a transaction record.

transaction.void.promise(options)

| MEMBER TYPE | NAME                              | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                                   |
|-------------|-----------------------------------|--------------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | transaction.void(options)         | number                   | Client and server scripts | Voids a transaction record.                                                                                                                   |
| METHOD      | transaction.void.promise(options) | number                   | Client and server scripts | Voids a transaction record asynchronously.                                                                                                    |
| ENUM        | transaction.Type                  | enum                     | Client and server scripts | Holds the string values for supported record types. This enum is used for the options.type parameter of the transaction.void(options) method. |

## Sample 1: N/transaction Module Script Sample

```

/**
 * @NApiVersion 2.x
 */
require(['N/transaction', 'N/config', 'N/record'], function(transaction, config, record) {
    function voidSalesOrder() {
        var accountingConfig = config.load({
            type: config.Type.ACCOUNTING_PREFERENCES
        });
        accountingConfig.setValue({
            fieldId: 'REVERSALVOIDING',
            value: false
        });

        accountingConfig.save();

        var salesOrderObj = record.create({
            type: 'salesorder',
            isDynamic: false
        });
        salesOrderObj.setValue({
            fieldId: 'entity',
            value: 107
        });
        salesOrderObj.setSublistValue({
            sublistId: 'item',
            fieldId: 'item',
            value: 233,
            line: 0
        });
        salesOrderObj.setSublistValue({
            sublistId: 'item',
            fieldId: 'amount',
            value: 1,
            line: 0
        });

        var salesOrderId = salesOrderObj.save();

        var voidSalesOrderId = transaction.void({
            type: record.Type.SALES_ORDER,
            id: salesOrderId
        });

        var salesOrder = record.load({
            type: 'salesorder',
            id: voidSalesOrderId
        });

        // The value of the memo field is 'VOID'
        var memo = salesOrder.getValue({
            fieldId: 'memo'
        });
    }

    voidSalesOrder();
});

```

## 47. N/translation Module

Load the N/translation module to load NetSuite Translation Collections in SuiteScript.

define(['N/translation'], (translation) => { ... })

✓ 5 Code Sample(s)

Scripts: Client and server scripts

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/translation'], (translation) => { // Access methods and properties on translation });
```

Use the N/translation module to allow SuiteScript developers to interact with NetSuite Translation Collections programmatically. For more information about Translation Collections, see Translation Collections Overview.

You can watch a video that demonstrates how to use the N/translation module to work with Translation Collections.

The N/translation module provides read-only access. If you want to create or modify Translation Collections, you can do so in the NetSuite UI at Customization > Translations > Manage Translations

For more information about managing Translation Collections in the UI, see Manage Translations.

A Translation Collection is encapsulated in the translation.Handle object. The translation.Handle object is a hierarchical object, which means that each node in the object is either another translation.Handle object or a translation.Translator function. Translator functions combine strings with parameters. When you create a Translation Collection in the NetSuite UI, you can include parameter placeholders in your translation strings. The translator function injects the specified parameter values into the placeholders in the returned translation string.

In your scripts, use translation.get(options) to get a translation.Translator function you can use to obtain specific translated strings in a collection. Consider the following code sample:

// key HELLO\_1 = 'Hello, {1}' message: translation.get({ collection: 'custcollection\_my\_strings', key: 'HELLO\_1' })({ params: ['NetSuite'] }) In this sample, if the string value of the HELLO\_1 key is "Hello, {1}", the translation.Translator function combines the string with the params parameter value and returns "Hello, NetSuite". You can also use translation.load(options) to load translation strings from one or more Translation Collections. For information about the way strings are added to and formatted in collections, see Working with Translation Collection Strings.

You can load collections in different language locales by using the locales parameter of translation.load(options). You can also use translation.selectLocale(options) to create a translation.Handle object in a specific locale from an existing translation.Handle object.

### N/translation Module Members

Return Type / Value Type

Supported Script Type

translation.Handle

Client and server scripts

Encapsulates a Translation Collection for a locale.

translation.Translator

| MEMBER TYPE | NAME                              | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPE     | DESCRIPTION                                                                                                                                                           |
|-------------|-----------------------------------|--------------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT      | translation.Handle                | Object                   | Client and server scripts | Encapsulates a Translation Collection for a locale.                                                                                                                   |
| OBJECT      | translation.Translator            | Object / Function        | Client and server scripts | Represents a translator function that returns translated strings. The translated strings include variables that are passed as parameters to the translator function.  |
| METHOD      | translation.get(options)          | translation.Translator   | Client and server scripts | Creates a translator function for a key in the specified Translation Collection and locale.                                                                           |
| METHOD      | translation.load(options)         | translation.Handle       | Client and server scripts | Creates a translation.Handle object with translations for the specified Translation Collections and locales.                                                          |
| METHOD      | translation.selectLocale(options) | translation.Handle       | Client and server scripts | Creates a translation.Handle object in the specified locale from an existing translation.Handle object.                                                               |
| ENUM        | translation.Locale                | enum                     | Client and server scripts | Holds the supported locales for Translation Collections. Use this enum to pass the locale argument to translation.get(options) and translation.selectLocale(options). |

#### Sample 1: Access Translation Strings

```
/**
 * @NApiVersion 2.x
 */

require(['N/ui/message', 'N/translation'],
function(message, translation) {

    // Create a message with translated strings
    var myMsg = message.create({
        title: translation.get({
            collection: 'custcollection_my_strings',
            key: 'MY_TITLE'
        })(),
        message: translation.get({
            collection: 'custcollection_my_strings',
            key: 'MY_MESSAGE'
        })(),
        type: message.Type.CONFIRMATION
    });

    // Show the message for 5 seconds
    myMsg.show({
        duration: 5000
    });

});
```

#### Sample 2: Access Translation Strings Using a Non-Default Locale

```
/**
 * @NApiVersion 2.x
 */

require(['N/ui/message', 'N/translation'],
function(message, translation) {

    // Create a message with translated strings
    var myMsg = message.create({
        title: translation.get({
            collection: 'custcollection_my_strings',
            key: 'MY_TITLE',
            locale: translation.Locale.COMPANY_DEFAULT
        })(),
        message: translation.get({
            collection: 'custcollection_my_strings',
            key: 'MY_MESSAGE',
            locale: translation.Locale.COMPANY_DEFAULT
        })(),
        type: message.Type.CONFIRMATION
    });

    // Show the message for 5 seconds
    myMsg.show({
        duration: 5000
    });

});
```

### Sample 3: Access Parameterized Translation Strings

```
/**
 * @NApiVersion 2.x
 */

require(['N/ui/message', 'N/translation'],
function(message, translation) {

    // Create a message with translated strings
    var myMsg = message.create({
        title: translation.get({
            collection: 'custcollection_my_strings',
            key: 'MY_TITLE'
        })(),
        message: translation.get({
            collection: 'custcollection_my_strings',
            key: 'HELLO_1'
        })({
            params: ['NetSuite']
        }),
        type: message.Type.CONFIRMATION
    });

    // Show the message for 5 seconds
    myMsg.show({
        duration: 5000
    });
});
```

### Sample 4: Load Specific Translation Strings from a Collection

```
/**
 * @NApiVersion 2.x
 */

require(['N/ui/message', 'N/translation'],
function(message, translation) {

    // Load translation strings by key
    var localizedStrings = translation.load({
        collections: [{
            alias: 'myCollection',
            collection: 'custcollection_my_strings',
            keys: ['MY_TITLE', 'MY_MESSAGE']
        }]
    });

    // Create a message with translated strings
    var myMsg = message.create({
        title: localizedStrings.myCollection.MY_TITLE(),
        message: localizedStrings.myCollection.MY_MESSAGE(),
        type: message.Type.CONFIRMATION
    });

    // Show the message for 5 seconds
    myMsg.show({
        duration: 5000
    });
});
```

### Sample 5: Load Translation Strings By Key from Multiple Translation Collections

```
/**
 * @NApiVersion 2.x
 */
require(['N/ui/message', 'N/translation'],
function(message, translation) {

    // Load two Translation Collections
    var localizedStrings = translation.load({
        collections: [{
            alias: 'myCollection',
            collection: 'custcollection_my_strings',
            keys: ['MY_TITLE']
        }, {
            alias: 'myOtherCollection',
            collection: 'custcollection_other_strings',
            keys: ['MY_OTHER_MESSAGE']
        }]
    });

    // Create a message with translated strings
    var myMsg = message.create({
        title: localizedStrings.myCollection.MY_TITLE(),
        message: localizedStrings.myOtherCollection.MY_OTHER_MESSAGE(),
        type: message.Type.CONFIRMATION
    });

    // Show the message for 5 seconds
    myMsg.show({
        duration: 5000
    });
});
```

# 48. N/ui/dialog Module

Load the N/ui/dialog module to create a modal dialog that is displayed until a button on the dialog is pressed.

define(['N/ui/dialog'], (dialog) => { ... })

✓ 4 Code Sample(s)

Scripts: Client scripts

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/ui/dialog'], (dialog) => { // Access methods and properties on dialog });
```

Use the N/ui/dialog module to create a modal dialog that persists until a button on the dialog is pressed.

## N/ui/dialog Module Members

Property Type / Method Return Type

Supported Script Types

dialog.alert(options)

Creates an Alert dialog with an OK button.

dialog.confirm(options)

Creates a Confirm dialog with OK and Cancel buttons.

| MEMBER TYPE | NAME                    | PROPERTY TYPE / METHOD RETURN TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                          |
|-------------|-------------------------|------------------------------------|------------------------|------------------------------------------------------|
| METHOD      | dialog.alert(options)   | Promise                            | Client scripts         | Creates an Alert dialog with an OK button.           |
| METHOD      | dialog.confirm(options) | Promise                            | Client scripts         | Creates a Confirm dialog with OK and Cancel buttons. |
| METHOD      | dialog.create(options)  | Promise                            | Client scripts         | Creates a dialog with specified buttons.             |

### ► Official NetSuite Script Samples (4 Sample(s))

#### Sample 1: Create an Alert Dialog

```
/**
 * @NApiVersion 2.1
 * @NScriptType ClientScript
 */
define(['N/ui/dialog'], (dialog) => {
  function pageInit() {
    let options = {
      title: 'I am an Alert',
      message: 'Click OK to continue.'
    };

    function success(result) {
      console.log('Success with value ' + result);
    }

    function failure(reason) {
      console.log('Failure: ' + reason);
    }

    dialog.alert(options).then(success).catch(failure);
  }

  return {
    pageInit: pageInit
  };
});
```

### Sample 2: Create a Confirmation Dialog

```
/**
 * @NApiVersion 2.1
 * @NScriptType ClientScript
 */
define(['N/ui/dialog'], (dialog) => {
  function pageInit() {
    let options = {
      title: 'I am a Confirmation',
      message: 'Press OK or Cancel'
    };

    function success(result) {
      console.log('Success with value ' + result);
    }

    function failure(reason) {
      console.log('Failure: ' + reason);
    }

    dialog.confirm(options).then(success).catch(failure);
  }

  return {
    pageInit: pageInit
  };
});
```

### Sample 3: Create a Dialog with Buttons

```
/**
 * @NApiVersion 2.1
 * @NScriptType ClientScript
 */
define(['N/ui/dialog'], (dialog) => {
  function pageInit() {
    let button1 = {
      label: 'I am A',
      value: 1
    };
    let button2 = {
      label: 'I am B',
      value: 2
    };
    let button3 = {
      label: 'I am C',
      value: 3
    };
    let options = {
      title: 'Alphabet Test',
      message: 'Which One?',
      buttons: [button1, button2, button3]
    };

    function success(result) {
      console.log('Success with value ' + result);
    }

    function failure(reason) {
      console.log('Failure: ' + reason);
    }

    dialog.create(options).then(success).catch(failure);
  }

  return {
    pageInit: pageInit
  };
});
```

#### Sample 4: Create a Dialog that Includes a Default Button

```
/**
 * @NApiVersion 2.1
 * @NScriptType ClientScript
 */
define(['N/ui/dialog'], (dialog) => {
  function pageInit() {
    let options = {
      title: 'I am a Dialog with the default button',
      message: 'Click a button to continue.',
    };

    function success(result) {
      console.log('Success with value ' + result);
    }

    function failure(reason) {
      console.log('Failure: ' + reason);
    }

    dialog.create(options).then(success).catch(failure);
  }

  return {
    pageInit: pageInit
  };
});
```

# 49. N/ui/message Module

Load the N/message module to display a message at the top of the screen under the menu bar.

`define(['N/ui/message'], (message) => { ... })`

✓ 1 Code Sample(s)

Scripts: Client scripts

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/ui/message'], (message) => { // Access methods and properties on message });
```

Use the N/ui/message module to display a message at the top of the screen under the menu bar.

N/ui/message Module Script Sample

## N/ui/message Members

Return Type / Value Type

Supported Script Types

Encapsulates the Message object that gets created when calling the message.create(options) method.

message.create(options)

Creates a message that can be displayed or hidden near the top of the page.

Indicates the type of message to create and display, which specifies the background color of the message and other message indicators.

| MEMBER TYPE | NAME                    | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                                                                                |
|-------------|-------------------------|--------------------------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT      | message.Message         | void                     | Client scripts         | Encapsulates the Message object that gets created when calling the message.create(options) method.                                                                                                                                         |
| METHOD      | message.create(options) | message.Message          | Client scripts         | Creates a message that can be displayed or hidden near the top of the page.                                                                                                                                                                |
| ENUM        | message.Type            | enum                     | Client scripts         | Indicates the type of message to create and display, which specifies the background color of the message and other message indicators. Use this enum to set the value of the options.type parameter of the message.create(options) method. |

## Message Object Members

Return Type / Value Type

Supported Script Types

Hides the message.

Message.show(options)

Shows the message.

| MEMBER TYPE | NAME                  | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION        |
|-------------|-----------------------|--------------------------|------------------------|--------------------|
| METHOD      | Message.hide()        | void                     | Client scripts         | Hides the message. |
| METHOD      | Message.show(options) | void                     | Client scripts         | Shows the message. |

 Sample 1: N/ui/message Module Script Sample

```
/**
 * @NApiVersion 2.1
 */
define(['N/ui/message'], (message) => {
  let myMsg = message.create({
    title: 'My Title',
    message: 'My Message',
    type: message.Type.CONFIRMATION
  });
  myMsg.show({
    duration: 5000 // will disappear after 5s
  });

  let myMsg2 = message.create({
    title: 'My Title 2',
    message: 'My Message 2',
    type: message.Type.INFORMATION
  });
  myMsg2.show();
  setTimeout(myMsg2.hide, 15000); // will disappear after 15s

  let myMsg3 = message.create({
    title: 'My Title 3',
    message: 'My Message 3',
    type: message.Type.WARNING,
    duration: 20000
  });
  myMsg3.show(); // will disappear after 20s

  let myMsg4 = message.create({
    title: 'My Title 4',
    message: 'My Message 4',
    type: message.Type.ERROR
  });
  myMsg4.show(); // will stay up until hide is called.
});
```

## 50. N/ui/serverWidget Module

Load the N/serverWidget module to work with the user interface within NetSuite.

```
define(['N/ui/serverWidget'], (serverWidget) => { ... })
```

✓ 10 Code Sample(s)

Scripts: Server scripts

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/ui/serverWidget'], (serverWidget) => { // Access methods and properties on serverWidget });
```

Use the N/ui/serverWidget module to work with the user interface within NetSuite. You can use Suitelets to build custom pages and wizards that have a NetSuite look-and-feel. You can also create various components of the NetSuite UI (for example, forms, fields, sublists, tabs).

When you add a UI object to an existing NetSuite page, to minimize the occurrence of field/object name conflicts, the internal ID that references the object must be prefixed with custpage.

### N/ui/serverWidget Module Members

Return Type / Value Type

Supported Script Types

serverWidget.Assistant

A scriptable, multi-step NetSuite assistant.

serverWidget.AssistantStep

A step within a custom NetSuite assistant.

| MEMBER TYPE | NAME                                  | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES               | DESCRIPTION                                                                                                                                           |
|-------------|---------------------------------------|--------------------------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT      | serverWidget.Assistant                | Object                   | Suitelets                            | A scriptable, multi-step NetSuite assistant.                                                                                                          |
| OBJECT      | serverWidget.AssistantStep            | Object                   | Suitelets                            | A step within a custom NetSuite assistant.                                                                                                            |
| OBJECT      | serverWidget.Button                   | Object                   | Suitelets and beforeLoad user events | A button that appears in a UI object.                                                                                                                 |
| OBJECT      | serverWidget.Field                    | Object                   | Suitelets and beforeLoad user events | A NetSuite field.                                                                                                                                     |
| OBJECT      | serverWidget.FieldGroup               | Object                   | Suitelets and beforeLoad user events | A field group.                                                                                                                                        |
| OBJECT      | serverWidget.Form                     | Object                   | Suitelets and beforeLoad user events | A NetSuite form.                                                                                                                                      |
| OBJECT      | serverWidget.List                     | Object                   | Suitelets                            | A list.                                                                                                                                               |
| OBJECT      | serverWidget.ListColumn               | Object                   | Suitelets                            | A list column.                                                                                                                                        |
| OBJECT      | serverWidget.Sublist                  | Object                   | Suitelets and beforeLoad user events | A NetSuite sublist.                                                                                                                                   |
| OBJECT      | serverWidget.Tab                      | Object                   | Suitelets and beforeLoad user events | A NetSuite tab and subtabs.                                                                                                                           |
| METHOD      | serverWidget.createAssistant(options) | serverWidget.Assistant   | Suitelets                            | Creates and returns a new assistant object.                                                                                                           |
| METHOD      | serverWidget.createForm(options)      | serverWidget.Form        | Suitelets                            | Creates and returns a new form object.                                                                                                                |
| METHOD      | serverWidget.createList(options)      | serverWidget.List        | Suitelets                            | Creates a List object (specifying the title, and whether to hide the navigation bar).                                                                 |
| ENUM        | serverWidget.AssistantSubmitAction    | string (read-only)       | Suitelets                            | Holds the string values for submit actions performed by the user. This enum is used to set the value of the Assistant.getLastAction().                |
| ENUM        | serverWidget.FieldBreakType           | string (read-only)       | Suitelets and beforeLoad user events | Holds the string values for supported field break types. This enum is used to set the value of the Field.updateBreakType(options) property.           |
| ENUM        | serverWidget.FieldDisplayType         | string (read-only)       | Suitelets and beforeLoad user events | Holds the string values for supported field display types. This enum is used to set the value of the Field.updateDisplayType(options) property.       |
| ENUM        | serverWidget.FieldLayoutType          | string (read-only)       | Suitelets and beforeLoad user events | Holds the string values for the supported types of field layouts. This enum is used to set the value of the Field.updateLayoutType(options) property. |

| MEMBER TYPE | NAME                             | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES               | DESCRIPTION                                                                                                                                                    |
|-------------|----------------------------------|--------------------------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ENUM        | serverWidget.FieldType           | string (read-only)       | Suitelets and beforeLoad user events | Holds the values for supported field types. This enum is used to set the value of the type parameter when Form.addField(options) is called.                    |
| ENUM        | serverWidget.FormPageLinkType    | string (read-only)       | Suitelets and beforeLoad user events | Holds the string values for supported page link types on a form. This enum is used to set the value of the type parameter for Form.addPageLink(options).       |
| ENUM        | serverWidget.LayoutJustification | string (read-only)       | Suitelets and beforeLoad user events | Holds the string values for supported justification layouts. This enum is used to set the value of the align parameter when List.addColumn(options) is called. |
| ENUM        | serverWidget.ListStyle           | string (read-only)       | Suitelets                            | Holds the string values for supported list styles. This enum is used to set the value of the List.style property.                                              |
| ENUM        | serverWidget.SublistDisplayType  | string (read-only)       | Suitelets and beforeLoad user events | Holds the string values for supported sublist display types. This enum is used to set the value of the Sublist.displayType property.                           |
| ENUM        | serverWidget.SublistType         | string (read-only)       | Suitelets and beforeLoad user events | Holds the string values for valid sublist types. This enum is used to define the type parameter when Form.addSublist(options) is called.                       |

## Assistant Object Members

The following members are called on the serverWidget.Assistant object.

Return Type / Value Type

Supported Script Types

Assistant.addField(options)

serverWidget.Field

Adds a field to an assistant.

| MEMBER TYPE | NAME                                          | RETURN TYPE / VALUE TYPE     | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                       |
|-------------|-----------------------------------------------|------------------------------|------------------------|-----------------------------------------------------------------------------------|
| METHOD      | Assistant.addField(options)                   | serverWidget.Field           | Suitelets              | Adds a field to an assistant.                                                     |
| METHOD      | Assistant.addFieldGroup(options)              | serverWidget.FieldGroup      | Suitelets              | Adds a field group to an assistant.                                               |
| METHOD      | Assistant.addStep(options)                    | serverWidget.AssistantStep   | Suitelets              | Adds a step to an assistant.                                                      |
| METHOD      | Assistant.addSublist(options)                 | serverWidget.Sublist         | Suitelets              | Adds a sublist to an assistant.                                                   |
| METHOD      | Assistant.getField(options)                   | serverWidget.Field           | Suitelets              | Gets a field object.                                                              |
| METHOD      | Assistant.getFieldGroup(options)              | serverWidget.FieldGroup      | Suitelets              | Gets a field group object.                                                        |
| METHOD      | Assistant.getFieldGroupIds()                  | string[]                     | Suitelets              | Gets all the field group IDs in an assistant.                                     |
| METHOD      | Assistant.getFieldIds()                       | string[]                     | Suitelets              | Gets all the field IDs in an assistant.                                           |
| METHOD      | Assistant.getFieldIdsByFieldGroup(fieldGroup) | string[]                     | Suitelets              | Gets all field IDs in the assistant field group.                                  |
| METHOD      | Assistant.getLastAction()                     | string                       | Suitelets              | Gets the last action submitted by the user.                                       |
| METHOD      | Assistant.getLastStep()                       | serverWidget.AssistantStep   | Suitelets              | Gets the step that the last submitted action came from.                           |
| METHOD      | Assistant.getNextStep()                       | serverWidget.AssistantStep   | Suitelets              | Gets the next step prompted by the assistant.                                     |
| METHOD      | Assistant.getStep(options)                    | serverWidget.AssistantStep   | Suitelets              | Returns a step in an assistant.                                                   |
| METHOD      | Assistant.getStepCount()                      | number                       | Suitelets              | Gets the total count of steps in the assistant.                                   |
| METHOD      | Assistant.getSteps()                          | serverWidget.AssistantStep[] | Suitelets              | Gets all the steps in the assistant.                                              |
| METHOD      | Assistant.getSublist(options)                 | serverWidget.Sublist         | Suitelets              | Get a Sublist object from its ID.                                                 |
| METHOD      | Assistant.getSublistIds()                     | string[]                     | Suitelets              | Gets all the sublist IDs in an assistant.                                         |
| METHOD      | Assistant.hasErrorHtml()                      | Boolean                      | Suitelets              | Indicates whether the assistant has an error message to display.                  |
| METHOD      | Assistant.isFinished()                        | Boolean                      | Suitelets              | Indicates the status of the assistant. If set to true, the assistant is finished. |
| METHOD      | Assistant.sendRedirect(options)               | void                         | Suitelets              | Manages redirects in an assistant.                                                |
| METHOD      | Assistant.setSplash(options)                  | void                         | Suitelets              | Define a splash message.                                                          |
| METHOD      | Assistant.updateDefaultValues(values)         | void                         | Suitelets              | Sets the default values of an array of fields that are specific to the assistant. |

| MEMBER TYPE | NAME                             | RETURN TYPE / VALUE TYPE               | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                               |
|-------------|----------------------------------|----------------------------------------|------------------------|---------------------------------------------------------------------------|
| PROPERTY    | Assistant.clientScriptFileId     | number                                 | Suitelets              | The File Cabinet ID of client script file to be used in this assistant.   |
| PROPERTY    | Assistant.clientScriptModulePath | string                                 | Suitelets              | The relative path to the client script file to be used in this assistant. |
| PROPERTY    | Assistant.currentStep            | serverWidget.AssistantStep (read-only) | Suitelets              | The current step.                                                         |
| PROPERTY    | Assistant.errorHtml              | string                                 | Suitelets              | The error message text.                                                   |
| PROPERTY    | Assistant.finishedHtml           | string                                 | Suitelets              | The text displayed after an assistant is finished.                        |
| PROPERTY    | Assistant.hideAddToShortcutsLink | Boolean                                | Suitelets              | Whether the Add to Shortcuts Link is displayed in the UI.                 |
| PROPERTY    | Assistant.hideStepNumber         | Boolean                                | Suitelets              | Whether the current and total step numbers are displayed in the UI.       |
| PROPERTY    | Assistant.isNotOrdered           | Boolean                                | Suitelets              | Whether assistant steps are ordered or unordered.                         |
| PROPERTY    | Assistant.title                  | string                                 | Suitelets              | The title of the assistant.                                               |

## AssistantStep Object Members

The following members are called on the serverWidget.AssistantStep object.

Return Type / Value Type

Supported Script Types

AssistantStep.getFieldIds()

Gets all the field IDs in an assistant step.

AssistantStep.getLineCount(options)

| MEMBER TYPE | NAME                                   | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                      |
|-------------|----------------------------------------|--------------------------|------------------------|------------------------------------------------------------------|
| METHOD      | AssistantStep.getFieldIds()            | string[]                 | Suitelets              | Gets all the field IDs in an assistant step.                     |
| METHOD      | AssistantStep.getLineCount(options)    | number                   | Suitelets              | Gets the number of lines previously entered by a user in a step. |
| METHOD      | AssistantStep.getLineCount(options)    | string[]                 | Suitelets              | Gets all the field IDs in a list.                                |
| METHOD      | AssistantStep.getSublistValue(options) | string                   | Suitelets              | Gets the current value of a sublist field (line item) in a step. |
| METHOD      | AssistantStep.getSubmittedSublistIds() | string[]                 | Suitelets              | Gets the IDs for all the sublist fields (line items) in a step.  |
| METHOD      | AssistantStep.getValue(options)        | string   string[]        | Suitelets              | Gets the current value of a field.                               |
| PROPERTY    | AssistantStep.helpText                 | string                   | Suitelets              | The help text for a step.                                        |
| PROPERTY    | AssistantStep.id                       | string (read-only)       | Suitelets              | The internal ID of the step.                                     |
| PROPERTY    | AssistantStep.label                    | string                   | Suitelets              | The label for a step.                                            |
| PROPERTY    | AssistantStep.stepNumber               | number                   | Suitelets              | Indicates where this step appears sequentially in an assistant.  |

## Button Object Members

The following members are called on the serverWidget.Button object.

Supported Script Types

Button.isDisabled

Suitelets and beforeLoad user events

Determines whether a button is dimmed.

Suitelets and beforeLoad user events

| MEMBER TYPE | NAME              | PROPERTY TYPE | SUPPORTED SCRIPT TYPES               | DESCRIPTION                                        |
|-------------|-------------------|---------------|--------------------------------------|----------------------------------------------------|
| PROPERTY    | Button.isDisabled | Boolean       | Suitelets and beforeLoad user events | Determines whether a button is dimmed.             |
| PROPERTY    | Button.isHidden   | Boolean       | Suitelets and beforeLoad user events | Determines whether the button is hidden in the UI. |
| PROPERTY    | Button.label      | string        | Suitelets and beforeLoad user events | The label for the button.                          |

Field Object Members

The following members are called on the serverWidget.Field object.

Return Type / Value Type

Supported Script Types

Field.addSelectOption(options)

Suitelets and beforeLoad user events

Adds a select option to a dropdown list for a selectable field.

| MEMBER TYPE | NAME                             | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES               | DESCRIPTION                                                                              |
|-------------|----------------------------------|--------------------------|--------------------------------------|------------------------------------------------------------------------------------------|
| METHOD      | Field.addSelectOption(options)   | void                     | Suitelets and beforeLoad user events | Adds a select option to a dropdown list for a selectable field.                          |
| METHOD      | Field.getSelectOptions(options)  | Object[]                 | Suitelets and beforeLoad user events | Returns the internal ID and label of the options for a select field as name/value pairs. |
| METHOD      | Field.setHelpText(options)       | serverWidget.Field       | Suitelets and beforeLoad user events | Sets the help text that appears in the field help popup.                                 |
| METHOD      | Field.updateBreakType(options)   | serverWidget.Field       | Suitelets and beforeLoad user events | Updates the break type used to add a break in flow layout for the field.                 |
| METHOD      | Field.updateDisplaySize(options) | serverWidget.Field       | Suitelets and beforeLoad user events | Updates the height and width for the field.                                              |
| METHOD      | Field.updateDisplayType(options) | serverWidget.Field       | Suitelets and beforeLoad user events | Updates the type of display for the field.                                               |
| METHOD      | Field.updateLayoutType(options)  | serverWidget.Field       | Suitelets and beforeLoad user events | Updates the layout type for the field.                                                   |
| PROPERTY    | Field.alias                      | string                   | Suitelets and beforeLoad user events | The alias used to set the field value.                                                   |
| PROPERTY    | Field.defaultValue               | string                   | Suitelets and beforeLoad user events | The default value for the field.                                                         |
| PROPERTY    | Field.helpText                   | string (read-only)       | Suitelets and beforeLoad user events | The help text for the field.                                                             |
| PROPERTY    | Field.id                         | string (read-only)       | Suitelets and beforeLoad user events | The internal ID for the field.                                                           |
| PROPERTY    | Field.isMandatory                | Boolean                  | Suitelets and beforeLoad user events | Whether the field is required.                                                           |
| PROPERTY    | Field.label                      | string                   | Suitelets and beforeLoad user events | The label for the field.                                                                 |
| PROPERTY    | Field.linkText                   | string                   | Suitelets and beforeLoad user events | The text displayed for a link in place of the URL.                                       |
| PROPERTY    | Field.maxLength                  | number                   | Suitelets and beforeLoad user events | The maximum length, in characters, for the field.                                        |
| PROPERTY    | Field.padding                    | number                   | Suitelets and beforeLoad user events | The number of empty vertical character spaces above the field.                           |
| PROPERTY    | Field.richTextHeight             | number                   | Suitelets and beforeLoad user events | The height of a rich text field, in pixels.                                              |
| PROPERTY    | Field.richTextWidth              | number                   | Suitelets and beforeLoad user events | The width of a rich text field, in pixels.                                               |
| PROPERTY    | Field.type                       | string (read-only)       | Suitelets and beforeLoad user events | The type of field.                                                                       |

FieldGroup Object Members

The following members are called on the serverWidget.FieldGroup object.

Supported Script Types

FieldGroup.isBorderHidden

Suitelets and beforeLoad user events

Whether a border appears around the field group.

FieldGroup.isCollapsible

| MEMBER TYPE | NAME                      | PROPERTY TYPE | SUPPORTED SCRIPT TYPES               | DESCRIPTION                                                                     |
|-------------|---------------------------|---------------|--------------------------------------|---------------------------------------------------------------------------------|
| PROPERTY    | FieldGroup.isBorderHidden | Boolean       | Suitelets and beforeLoad user events | Whether a border appears around the field group.                                |
| PROPERTY    | FieldGroup.isCollapsible  | Boolean       | Suitelets and beforeLoad user events | Whether the field group is collapsible.                                         |
| PROPERTY    | FieldGroup.isCollapsed    | Boolean       | Suitelets and beforeLoad user events | Whether the field group is initially collapsed or expanded in the default view. |

| MEMBER TYPE | NAME                      | PROPERTY TYPE | SUPPORTED SCRIPT TYPES               | DESCRIPTION                                              |
|-------------|---------------------------|---------------|--------------------------------------|----------------------------------------------------------|
| PROPERTY    | FieldGroup.isSingleColumn | Boolean       | Suitelets and beforeLoad user events | Whether the field group is displayed in a single column. |
| PROPERTY    | FieldGroup.label          | string        | Suitelets and beforeLoad user events | The label for the field group.                           |

## Form Object Members

The following members are called on the serverWidget.Form object.

Return Type / Value Type

Supported Script Types

Form.addButton(options)

serverWidget.Button

Suitelets and beforeLoad user events

| MEMBER TYPE | NAME                              | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES               | DESCRIPTION                                                                                                                         |
|-------------|-----------------------------------|--------------------------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | Form.addButton(options)           | serverWidget.Button      | Suitelets and beforeLoad user events | Adds a button to the form.                                                                                                          |
| METHOD      | Form.addCredentialField(options)  | serverWidget.Field       | Suitelets and beforeLoad user events | Adds a field that store credentials in NetSuite for invoking services provided by third parties.                                    |
| METHOD      | Form.addField(options)            | serverWidget.Field       | Suitelets and beforeLoad user events | Adds a field to the form.                                                                                                           |
| METHOD      | Form.addFieldGroup(options)       | serverWidget.FieldGroup  | Suitelets and beforeLoad user events | Adds a group of fields to the form.                                                                                                 |
| METHOD      | Form.addPageInitMessage(options)  | void                     | Suitelets and beforeLoad user events | Shows a message on a form in view mode. You can use this method to show a message on a form based on its user event script context. |
| METHOD      | Form.addPageLink(options)         | void                     | Suitelets and beforeLoad user events | Adds a link to a form.                                                                                                              |
| METHOD      | Form.addResetButton(options)      | serverWidget.Button      | Suitelets and beforeLoad user events | Adds a Reset button to a form that clears the values of all fields.                                                                 |
| METHOD      | Form.addSecretKeyField(options)   | serverWidget.Field       | Suitelets and beforeLoad user events | Add a secret key field to the form.                                                                                                 |
| METHOD      | Form.addSublist(options)          | serverWidget.Sublist     | Suitelets and beforeLoad user events | Adds a sublist to the form.                                                                                                         |
| METHOD      | Form.addSubmitButton(options)     | serverWidget.Button      | Suitelets and beforeLoad user events | Adds a submit button to a form that saves user inputs.                                                                              |
| METHOD      | Form.addSubtab(options)           | serverWidget.Tab         | Suitelets and beforeLoad user events | Adds a subtab to a form.                                                                                                            |
| METHOD      | Form.addTab(options)              | serverWidget.Tab         | Suitelets and beforeLoad user events | Adds a tab to a form.                                                                                                               |
| METHOD      | Form.getButton(options)           | serverWidget.Button      | Suitelets and beforeLoad user events | Returns a button by internal ID.                                                                                                    |
| METHOD      | Form.getField(options)            | serverWidget.Field       | Suitelets and beforeLoad user events | Returns a field by internal ID.                                                                                                     |
| METHOD      | Form.getSublist(options)          | serverWidget.Sublist     | Suitelets and beforeLoad user events | Returns a sublist by internal ID.                                                                                                   |
| METHOD      | Form.getSubtab(options)           | serverWidget.Tab         | Suitelets and beforeLoad user events | Returns a subtab by internal ID.                                                                                                    |
| METHOD      | Form.getTab(options)              | serverWidget.Tab         | Suitelets and beforeLoad user events | Returns a tab object from its internal ID.                                                                                          |
| METHOD      | Form.getTabs()                    | string[]                 | Suitelets and beforeLoad user events | Returns the internal IDs of all tabs.                                                                                               |
| METHOD      | Form.insertField(options)         | void                     | Suitelets and beforeLoad user events | Inserts a field before another field within a form.                                                                                 |
| METHOD      | Form.insertSublist(options)       | void                     | Suitelets and beforeLoad user events | Inserts a sublist before another sublist on a form.                                                                                 |
| METHOD      | Form.insertSubtab(options)        | void                     | Suitelets and beforeLoad user events | Inserts a subtab before another subtab on a form.                                                                                   |
| METHOD      | Form.insertTab(options)           | void                     | Suitelets and beforeLoad user events | Inserts a tab before another tab on a form.                                                                                         |
| METHOD      | Form.removeButton(options)        | void                     | Suitelets and beforeLoad user events | Removes a button from a form.                                                                                                       |
| METHOD      | Form.updateDefaultValues(options) | void                     | Suitelets and beforeLoad user events | Sets the default values of many fields on a form.                                                                                   |

| MEMBER TYPE | NAME                               | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES               | DESCRIPTION                                                          |
|-------------|------------------------------------|--------------------------|--------------------------------------|----------------------------------------------------------------------|
| PROPERTY    | <b>Form.clientScriptFileId</b>     | number                   | Suitelets and beforeLoad user events | The File Cabinet ID of client script file to be used in this form.   |
| PROPERTY    | <b>Form.clientScriptModulePath</b> | string                   | Suitelets and beforeLoad user events | The relative path to the client script file to be used in this form. |
| PROPERTY    | <b>Form.title</b>                  | string                   | Suitelets and beforeLoad user events | The title used for the form.                                         |

## List Object Members

The following members are called on the `serverWidget.List` object.

Return Type / Value Type

Supported Script Types

`List.addButton(options)`

`serverWidget.Button`

Adds a button to a list.

| MEMBER TYPE | NAME                               | RETURN TYPE / VALUE TYPE             | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                     |
|-------------|------------------------------------|--------------------------------------|------------------------|---------------------------------------------------------------------------------|
| METHOD      | <b>List.addButton(options)</b>     | <code>serverWidget.Button</code>     | Suitelets              | Adds a button to a list.                                                        |
| METHOD      | <b>List.addColumn(options)</b>     | <code>serverWidget.ListColumn</code> | Suitelets              | Adds a column to a list.                                                        |
| METHOD      | <b>List.addEditColumn(options)</b> | <code>serverWidget.ListColumn</code> | Suitelets              | Adds a column containing Edit or Edit/View links to a Suitelet or Portlet list. |
| METHOD      | <b>List.addPageLink(options)</b>   | <code>serverWidget.List</code>       | Suitelets              | Adds a link to a list.                                                          |
| METHOD      | <b>List.addRow(options)</b>        | <code>serverWidget.List</code>       | Suitelets              | Adds a single row to a list.                                                    |
| METHOD      | <b>List.addRows(options)</b>       | <code>serverWidget.List</code>       | Suitelets              | Adds multiple rows to a list.                                                   |
| PROPERTY    | <b>List.clientScriptFileId</b>     | number                               | Suitelets              | The File Cabinet ID of client script file to be used in this list.              |
| PROPERTY    | <b>List.clientScriptModulePath</b> | string                               | Suitelets              | The relative path to the client script file to be used in this list.            |
| PROPERTY    | <b>List.style</b>                  | string                               | Suitelets              | The display style for this list.                                                |
| PROPERTY    | <b>List.title</b>                  | string                               | Suitelets              | The List title.                                                                 |

## ListColumn Object Members

The following members are called on the `serverWidget.ListColumn` object.

Return Type / Value Type

Supported Script Types

`ListColumn.addParamToURL(options)`

`serverWidget.ListColumn`

Adds a URL parameter (optionally defined per row) to the list column's URL.

| MEMBER TYPE | NAME                                     | RETURN TYPE / VALUE TYPE             | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                 |
|-------------|------------------------------------------|--------------------------------------|------------------------|-----------------------------------------------------------------------------|
| METHOD      | <b>ListColumn.addParamToURL(options)</b> | <code>serverWidget.ListColumn</code> | Suitelets              | Adds a URL parameter (optionally defined per row) to the list column's URL. |
| METHOD      | <b>ListColumn.setURL(options)</b>        | <code>serverWidget.ListColumn</code> | Suitelets              | Sets the base URL for the list column.                                      |
| PROPERTY    | <b>ListColumn.label</b>                  | string                               | Suitelets              | The label of this list column.                                              |

## Sublist Object Members

The following members are called on the `serverWidget.Sublist` object.

Return Type / Value Type

Supported Script Types

`Sublist.addButton(options)`

`serverWidget.Button`

Suitelets and beforeLoad user events

| MEMBER TYPE | NAME                                    | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES               | DESCRIPTION                                                                                                                         |
|-------------|-----------------------------------------|--------------------------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | Sublist.addButton(options)              | serverWidget.Button      | Suitelets and beforeLoad user events | Adds a button to a sublist.                                                                                                         |
| METHOD      | Sublist.addField(options)               | serverWidget.Field       | Suitelets and beforeLoad user events | Add a field to a sublist.                                                                                                           |
| METHOD      | Sublist.addMarkAllButtons()             | serverWidget.Button[]    | Suitelets and beforeLoad user events | Adds a Mark All or Unmark All button.                                                                                               |
| METHOD      | Sublist.addRefreshButton()              | serverWidget.Button      | Suitelets and beforeLoad user events | Adds a Reset button.                                                                                                                |
| METHOD      | Sublist.getField(options)               | serverWidget.Field       | Suitelets and beforeLoad user events | Returns a Field object on a specified sublist.                                                                                      |
| METHOD      | Sublist.getSublistValue(options)        | string                   | Suitelets and beforeLoad user events | Gets a field value on a sublist.                                                                                                    |
| METHOD      | Sublist.insertField(options)            | void                     | Suitelets and beforeLoad user events | Inserts a field before another field on a sublist.                                                                                  |
| METHOD      | Sublist.setSublistValue(options)        | void                     | Suitelets and beforeLoad user events | Sets the value of a sublist field.                                                                                                  |
| METHOD      | Sublist.updateTotallingFieldId(options) | serverWidget.Sublist     | Suitelets and beforeLoad user events | Updates the ID of a field designated as a totalling column, which is used to calculate and display a running total for the sublist. |
| METHOD      | Sublist.updateUniqueFieldId(options)    | serverWidget.Sublist     | Suitelets and beforeLoad user events | Updates a field ID that is to have unique values across the rows in the sublist.                                                    |
| PROPERTY    | Sublist.displayType                     | string                   | Suitelets and beforeLoad user events | The display style for a sublist.                                                                                                    |
| PROPERTY    | Sublist.helpText                        | string                   | Suitelets and beforeLoad user events | The inline help text for a sublist.                                                                                                 |
| PROPERTY    | Sublist.label                           | string                   | Suitelets and beforeLoad user events | The label for a sublist.                                                                                                            |
| PROPERTY    | Sublist.lineCount                       | number (read-only)       | Suitelets and beforeLoad user events | The number of line items in a sublist.                                                                                              |

## Tab Object Members

The following members are called on the serverWidget.Tab object.

Supported Script Types

Suitelets and beforeLoad user events

The inline help text for a tab or subtab.

Suitelets and beforeLoad user events

The label for a tab or subtab.

| MEMBER TYPE | NAME         | VALUE TYPE | SUPPORTED SCRIPT TYPES               | DESCRIPTION                               |
|-------------|--------------|------------|--------------------------------------|-------------------------------------------|
| PROPERTY    | Tab.helpText | string     | Suitelets and beforeLoad user events | The inline help text for a tab or subtab. |
| PROPERTY    | Tab.label    | string     | Suitelets and beforeLoad user events | The label for a tab or subtab.            |

 **Sample 1: Create a Form with a Credential Field**

```
/**
 * @NApiVersion 2.1
 * @NScriptType Suitelet
 */

// This script creates a form with a credential field.
define(['N/ui/serverWidget', 'N/https', 'N/url'], (serverWidget, https, url) => {
  function onRequest(context) {
    if (context.request.method === 'GET') {
      const form = serverWidget.createForm({
        title: 'Password Form'
      });

      const credField = form.addCredentialField({
        id: 'password',
        label: 'Password',
        restrictToDomains: ['<accountID>.app.netsuite.com'],
        restrictToCurrentUser: false,
        restrictToScriptIds: 'customscript_my_script'
      });

      credField.maxLength = 32;

      form.addSubmitButton();

      context.response.writePage({
        pageObject: form
      });
    }
    else {
      // Request to an existing Suitelet with credentials
      let passwordGuid = context.request.parameters.password;

      // Replace SCRIPTID and DEPLOYMENTID with the internal ID of the suitelet script and deployment in your account
      let baseUrl = url.resolveScript({
        scriptId: SCRIPTID,
        deploymentId: DEPLOYMENTID,
        returnExternalURL: true
      });

      let authUrl = baseUrl + '&pwd=' + passwordGuid + ' ';

      let secureStringUrl = https.createSecureString({
        input: authUrl
      });

      let headers = ({
        'pwd': passwordGuid
      });

      let response = https.post({
        credentials: [passwordGuid],
        url: secureStringUrl,
        body: {authorization: ' ' + passwordGuid + ' ', data: 'anything can be here'},
        headers: headers
      });
    }
  }
  return {
    onRequest: onRequest
  };
});
```

 **Sample 2: Complete Custom List Page Sample Script**

```
/**
 * @NApiVersion 2.1
 * @NScriptType Suitelet
 */
```

 **Sample 3: Complete Custom List Page Sample Script**

```
/**
 * @NApiVersion 2.1
 * @NScriptType Suitelet
 */
define(['N/ui/serverWidget'],
  function(serverWidget) {
    ;
  });
```

#### Sample 4: Complete Custom List Page Sample Script

```
/**
 *@ApiVersion 2.1
 *@NScriptType Suitelet
 */
define(['/N/ui/serverWidget'],
function(serverWidget) {
    function onRequest(context){
        if(context.request.method === 'GET'){
            // Section One - List - See 'Steps for Creating a List', Step Five
            // Section Two - Columns - See 'Steps for Creating a List', Step Seven
            context.response.writePage(list);
        }else{
        }
    }
    return {
        onRequest: onRequest
    }
});
```

#### Sample 5: Complete Custom List Page Sample Script

```
var list = serverWidget.createList({
    title: 'Purchase History'
});
list.style = serverWidget.ListStyle.REPORT;
```

#### Sample 6: Complete Custom List Page Sample Script

```
list.addButton({
    id : 'buttonid',
    label : 'Test',
    functionName: '' // the function called when the button is pressed
});
```

#### Sample 7: Complete Custom List Page Sample Script

```
var datecol = list.addColumn({
    id : 'column1',
    type : serverWidget.FieldType.DATE,
    label : 'Date',
    align : serverWidget.LayoutJustification.RIGHT
});

list.addColumn({
    id : 'column2',
    type : serverWidget.FieldType.TEXT,
    label : 'Product',
    align : serverWidget.LayoutJustification.RIGHT
});

list.addColumn({
    id : 'column3',
    type : serverWidget.FieldType.INTEGER,
    label : 'Quantity',
    align : serverWidget.LayoutJustification.RIGHT
});

list.addColumn({
    id : 'column4',
    type : serverWidget.FieldType.CURRENCY,
    label : 'Unit Cost',
    align : serverWidget.LayoutJustification.RIGHT
});
```

#### Sample 8: Complete Custom List Page Sample Script

```
list.addRow({
    row : { column1 : '02/12/2018', column2 : 'Widget', column3: '4', column4:'4.50' }
});
```

### Sample 9: Complete Custom List Page Sample Script

```
list.addRows({
  rows :[{column1 : '02/12/2018', column2 : 'Widget', column3: '4', column4:'4.50'},
    {column1 : '02/14/2018', column2 : 'Sprocket', column3: '6', column4:'11.50'},
    {column1 : '02/16/2018', column2 : 'Gizmo', column3: '9', column4:'1.25'}]
});
```

### Sample 10: Complete Custom List Page Sample Script

```
/**
 * @NApiVersion 2.1
 * @NScriptType Suitelet
 */
define(['N/ui/serverWidget'], function(serverWidget) {
  function onRequest(context){
    if(context.request.method === 'GET'){
      // Section One - List - See 'Steps for Creating a Custom List Page', Step Five
      var list = serverWidget.createList({
        title: 'Purchase History'
      });

      list.style = serverWidget.ListStyle.REPORT;

      list.addButton({
        id: 'buttonid',
        label: 'Test',
        functionName: '' // the function called when the button is pressed
      });

      // Section Two - Columns - See 'Steps for Creating a Custom List Page', Step Seven
      var datecol = list.addColumn({
        id: 'column1',
        type: serverWidget.FieldType.DATE,
        label: 'Date',
        align: serverWidget.LayoutJustification.RIGHT
      });

      list.addColumn({
        id: 'column2',
        type: serverWidget.FieldType.TEXT,
        label: 'Product',
        align: serverWidget.LayoutJustification.RIGHT
      });

      list.addColumn({
        id: 'column3',
        type: serverWidget.FieldType.INTEGER,
        label: 'Quantity',
        align: serverWidget.LayoutJustification.RIGHT
      });

      list.addColumn({
        id: 'column4',
        type: serverWidget.FieldType.CURRENCY,
        label: 'Unit Cost',
        align: serverWidget.LayoutJustification.RIGHT
      });

      list.addRows({
        rows: [{column1: '05/30/2018', column2: 'Widget', column3: '4', column4: '4.50'},
          {column1: '05/30/2018', column2: 'Sprocket', column3: '6', column4: '11.50'},
          {column1: '05/30/2018', column2: 'Gizmo', column3: '9', column4: '1.25'}]
      });
      context.response.writePage(list);
    }
  }

  return {
    onRequest: onRequest
  }
});
```

# 51. N/url Module

Load the N/url module to determine URL navigation paths within NetSuite or format URL strings.

define(['N/url'], (url) => { ... })

✓ 4 Code Sample(s)

Scripts: Client and server scripts

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/url'], (url) => { // Access methods and properties on url });
```

Use the N/url module to determine URL navigation paths within NetSuite and format URL strings.

## N/url Module Members

Return Type / Value Type

Supported Script Types

url.format(options)

Client and server scripts

Converts (serializes) URL query parameters into a string.

url.resolveDomain(options)

| MEMBER TYPE | NAME                         | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                                                         |
|-------------|------------------------------|--------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | url.format(options)          | string                   | Client and server scripts | Converts (serializes) URL query parameters into a string.                                                                                                           |
| METHOD      | url.resolveDomain(options)   | string                   | Client and server scripts | Returns a domain name for a NetSuite account.                                                                                                                       |
| METHOD      | url.resolveRecord(options)   | string                   | Client and server scripts | Returns an internal URL to a NetSuite record.                                                                                                                       |
| METHOD      | url.resolveScript(options)   | string                   | Client and server scripts | Returns an external or internal URL to a script.                                                                                                                    |
| METHOD      | url.resolveTaskLink(options) | string                   | Client and server scripts | Returns an internal URL for a tasklink.                                                                                                                             |
| ENUM        | url.HostType                 | enum                     | Client and server scripts | Holds the string values that describe a category of domain name. Use this enum to set the value of the hostType parameter of the url.resolveDomain(options) method. |

## ► Official NetSuite Script Samples (4 Sample(s))

### Sample 1: Retrieve the Relative URL of a Record

```
/**
 * @NApiVersion 2.x
 */

// This script retrieves the relative URL of a record.
require(['N/url'], function(url) {
  var output = url.resolveRecord({
    recordType: 'salesorder',
    recordId: 6,
    isEditMode: true
  });
});
```

#### Sample 2: Generate an Absolute URL to a Specific Resource

```
/**
 * @NApiVersion 2.x
 */

// This script generates an absolute URL to a specific resource.
require(['N/url', 'N/record'], function(url, record) {
  function resolveRecordUrl() {
    var scheme = 'https://';
    var host = url.resolveDomain({
      hostType: url.HostType.APPLICATION
    });
    var relativePath = url.resolveRecord({
      recordType: record.Type.SALES_ORDER,
      recordId: 6,
      isEditMode: true
    });
    var myURL = scheme + host + relativePath;
  }
  resolveRecordUrl();
});
```

#### Sample 3: Retrieve the Domain for Calling a RESTlet

```
/**
 * @NApiVersion 2.x
 */

// This script retrieves the domain for calling a RESTlet.
require(['N/url'], function(url) {
  function resolveDomainUrl() {
    var sCompId = 'MSTRWLF';
    var output = url.resolveDomain({
      hostType: url.HostType.RESTLET,
      accountId: sCompId
    });
  }
  resolveDomainUrl();
});
```

#### Sample 4: Create a URL and Send a Secure HTTPS Post Request to the URL

```
/**
 * @NApiVersion 2.x
 */

// This script creates a URL, sends a secure HTTPS POST request to that URL, and logs the server's response.
require(['N/url', 'N/https'], function(url, https) {
  var script = 'customscript1';
  var deployment = 'customdeploy1';
  var parameters = '';
  try {
    var suiteletURL = url.resolveScript({
      scriptId: script,
      deploymentId: deployment
    });
    var response = https.post({
      url: suiteletURL,
      body: parameters
    });
    log.debug(response.body.toString());
  }
  catch(e) {
    log.error(e.toString());
  }
});
```

## 52. N/util Module

Load the N/util module to manually access util methods. Module members are also supported by the global util Object.

`define(['N/util'], (util) => { ... })`

✓ 1 Code Sample(s)

Scripts: Client and server scripts

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/util'], (util) => { // Access methods and properties on util });
```

Use the N/util module to manually access methods that verify object and primitive types in a SuiteScript 2.x script. These methods can also be accessed using the global util object. For more information about the global util object, see SuiteScript 2.1 Global Objects.

### N/util Module Script Sample

Use the N/util module to access methods that verify object and primitive types in a SuiteScript 2.x script.

Each method (for example, util.isArray(obj)) returns a boolean value, based on evaluation of the obj parameter.

If you need to identify a type specific to SuiteScript 2.x, use the toString() global method.

### N/util Module Members

Return Type / Value Type  
Supported Script Types  
util.each(iterable, callback)  
Client and server scripts  
Iterates over each member in an Object or Array.  
util.extend(receiver, contributor)

| MEMBER TYPE | NAME                               | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                        |
|-------------|------------------------------------|--------------------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | util.each(iterable, callback)      | Object or Array          | Client and server scripts | Iterates over each member in an Object or Array.                                                                                   |
| METHOD      | util.extend(receiver, contributor) | Object                   | Client and server scripts | Copies the properties in a source object to a destination object.                                                                  |
| METHOD      | util.isArray(obj)                  | boolean                  | Client and server scripts | Returns true if the obj parameter is a JavaScript Array object and false otherwise.                                                |
| METHOD      | util.isAsyncFunction(obj)          | boolean                  | Client and server scripts | Returns true if the obj parameter is JavaScript Async Function and false otherwise.                                                |
| METHOD      | util.isBoolean(obj)                | boolean                  | Client and server scripts | Returns true if the obj parameter is a JavaScript Boolean and false otherwise.                                                     |
| METHOD      | util.isDate(obj)                   | boolean                  | Client and server scripts | Returns true if the obj parameter is a JavaScript Date object and false otherwise.                                                 |
| METHOD      | util.isFunction(obj)               | boolean                  | Client and server scripts | Returns true if the obj parameter is a JavaScript Function or Async Function and false otherwise.                                  |
| METHOD      | util.isNumber(obj)                 | boolean                  | Client and server scripts | Returns true if the obj parameter is a JavaScript Number object or a value that evaluates to a Number object, and false otherwise. |
| METHOD      | util.isObject(obj)                 | boolean                  | Client and server scripts | Returns true if the obj parameter is a plain JavaScript object (new Object() or {} for example), and false otherwise.              |
| METHOD      | util.isRegExp(obj)                 | boolean                  | Client and server scripts | Returns true if the obj parameter is a JavaScript RegExp object or a value that evaluates to a RegExp object, and false otherwise. |
| METHOD      | util.isString(obj)                 | boolean                  | Client and server scripts | Returns true if the obj parameter is a JavaScript String object or a value that evaluates to a String object, and false otherwise. |

 Sample 1: N/util Module Script Sample

```
/**
 * @NApiVersion 2.x
 */
require(['N/record'], function(record){
    // Create a sales order
    var rec = record.create({
        type: 'salesorder',
        isDynamic: true
    });
    rec.setValue({
        fieldId: 'entity',
        value: 107
    });

    // Set up an object containing an item's internal ID and the corresponding quantity
    var itemList = {
        39: 5,
        38: 1
    }

    // Iterate through the object and set the key-value pairs on the record
    util.each(itemList, function(quantity, itemId){    // (5, 39) and (1, 38)
        rec.selectNewLine('item');
        rec.setCurrentSublistValue('item', 'item', itemId);
        rec.setCurrentSublistValue('item', 'quantity', quantity);
        rec.commitLine('item');
    });

    var id = rec.save();
});
```

## 53. N/workbook Module

Load the N/workbook module to create and manage workbooks in SuiteAnalytics Workbook. Use this module with the N/dataset module to manage all aspects of your datasets and workbooks in SuiteAnalytics Workbook.

define(['N/workbook'], (workbook) => { ... })

✓ 2 Code Sample(s)

Scripts: Server scripts

Permissions: SuiteAnalytics Workbook

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/workbook'], (workbook) => { // Access methods and properties on workbook });
```

Use the N/workbook module to create new workbooks, load existing ones, or list all available workbooks.

A workbook can contain:

- Pivots
- Tables with columns, filters, and field contexts
- Charts with axes and legends
- Selectors
- Sections
- Data dimensions
- Sorts
- Conditional and limiting filters
- Expressions
- Data measures and calculated measures

Workbooks help you analyze your dataset query results using different components, like table views. Every workbook is based on a dataset. For more, see the N/dataset Module. To learn more about SuiteAnalytics workbooks and datasets, check out:

- SuiteAnalytics Workbook Overview
- Getting Started with SuiteAnalytics Workbook
- Custom Workbooks and Datasets
- The N/workbook module doesn't work in unauthenticated client-side contexts.
- For details, see the SuiteAnswers Outbound HTTPs in an unauthenticated client-side context.

### N/workbook Module Members

- Return Type / Value Type
- Supported Script Types
- workbook.CalculatedMeasure
- A calculated measure.
- workbook.Category
- A chart category.

| MEMBER TYPE | NAME                           | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                               |
|-------------|--------------------------------|--------------------------|------------------------|---------------------------------------------------------------------------|
| OBJECT      | workbook.Aspect                | Object                   | Server scripts         | An aspect.                                                                |
| OBJECT      | workbook.CalculatedMeasure     | Object                   | Server scripts         | A calculated measure.                                                     |
| OBJECT      | workbook.Category              | Object                   | Server scripts         | A chart category.                                                         |
| OBJECT      | workbook.Chart                 | Object                   | Server scripts         | A chart.                                                                  |
| OBJECT      | workbook.ChartAxis             | Object                   | Server scripts         | A chart axis object which is used when you create a category or a legend. |
| OBJECT      | workbook.ChildNodesSelector    | Object                   | Server scripts         | A selector for child nodes.                                               |
| OBJECT      | workbook.Color                 | Object                   | Server scripts         | A color.                                                                  |
| OBJECT      | workbook.ConditionalFilter     | Object                   | Server scripts         | A conditional filter.                                                     |
| OBJECT      | workbook.ConditionalFormat     | Object                   | Server scripts         | A conditional format.                                                     |
| OBJECT      | workbook.ConditionalFormatRule | Object                   | Server scripts         | A conditional format rule.                                                |

| MEMBER TYPE | NAME                                   | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                                      |
|-------------|----------------------------------------|--------------------------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT      | workbook.Currency                      | Object                   | Server scripts         | A currency amount and currency type.                                                                                                                                                             |
| OBJECT      | workbook.DataDimension                 | Object                   | Server scripts         | A data dimension.                                                                                                                                                                                |
| OBJECT      | workbook.DataDimensionItem             | Object                   | Server scripts         | A data dimension item.                                                                                                                                                                           |
| OBJECT      | workbook.DataDimensionItemValue        | Object                   | Server scripts         | The value of a data dimension item.                                                                                                                                                              |
| OBJECT      | workbook.DataDimensionValue            | Object                   | Server scripts         | The value of a data dimension.                                                                                                                                                                   |
| OBJECT      | workbook.DataMeasure                   | Object                   | Server scripts         | A data measure.                                                                                                                                                                                  |
| OBJECT      | workbook.DescendantorSelfNodesSelector | Object                   | Server scripts         | A selector for descendant or self nodes.                                                                                                                                                         |
| OBJECT      | workbook.DimensionSelector             | Object                   | Server scripts         | A dimension selector.                                                                                                                                                                            |
| OBJECT      | workbook.Duration                      | Object                   | Server scripts         | A duration.                                                                                                                                                                                      |
| OBJECT      | workbook.Expression                    | Object                   | Server scripts         | An expression.                                                                                                                                                                                   |
| OBJECT      | workbook.FieldContext                  | Object                   | Server scripts         | A field context.                                                                                                                                                                                 |
| OBJECT      | workbook.FontSize                      | Object                   | Server scripts         | A font size.                                                                                                                                                                                     |
| OBJECT      | workbook.Legend                        | Object                   | Server scripts         | A chart legend.                                                                                                                                                                                  |
| OBJECT      | workbook.LimitingFilter                | Object                   | Server scripts         | A limiting filter.                                                                                                                                                                               |
| OBJECT      | workbook.MeasureSelector               | Object                   | Server scripts         | A measure selector.                                                                                                                                                                              |
| OBJECT      | workbook.MeasureValue                  | Object                   | Server scripts         | A measure value.                                                                                                                                                                                 |
| OBJECT      | workbook.MeasureValueSelector          | Object                   | Server scripts         | A measure value selector.                                                                                                                                                                        |
| OBJECT      | workbook.PathSelector                  | Object                   | Server scripts         | A path selector.                                                                                                                                                                                 |
| OBJECT      | workbook.Pivot                         | Object                   | Server scripts         | A pivot definition. A pivot is a workbook component that enables you to pivot your dataset query results by defining measures and dimensions, so that you can analyze different subsets of data. |
| OBJECT      | workbook.PivotAxis                     | Object                   | Server scripts         | A pivot axis.                                                                                                                                                                                    |
| OBJECT      | workbook.PivotIntersection             | Object                   | Server scripts         | A pivot intersection.                                                                                                                                                                            |
| OBJECT      | workbook.PositionPercent               | Object                   | Server scripts         | A position defined by percentages of the x and y axes.                                                                                                                                           |
| OBJECT      | workbook.PositionUnits                 | Object                   | Server scripts         | A position defined by units.                                                                                                                                                                     |
| OBJECT      | workbook.PositionValues                | Object                   | Server scripts         | A position defined by horizontal and vertical position values.                                                                                                                                   |
| OBJECT      | workbook.Range                         | Object                   | Server scripts         | A date or date-time range.                                                                                                                                                                       |
| OBJECT      | workbook.Record                        | Object                   | Server scripts         | A record.                                                                                                                                                                                        |
| OBJECT      | workbook.RecordKey                     | Object                   | Server scripts         | A record key.                                                                                                                                                                                    |
| OBJECT      | workbook.ReportStyle                   | Object                   | Server scripts         | A report style.                                                                                                                                                                                  |
| OBJECT      | workbook.ReportStyleRule               | Object                   | Server scripts         | A report style rule.                                                                                                                                                                             |
| OBJECT      | workbook.Section                       | Object                   | Server scripts         | A workbook section.                                                                                                                                                                              |
| OBJECT      | workbook.SectionValue                  | Object                   | Server scripts         | A section value.                                                                                                                                                                                 |
| OBJECT      | workbook.Series                        | Object                   | Server scripts         | A series in a workbook. A series is used when you create a chart definition.                                                                                                                     |
| OBJECT      | workbook.Sort                          | Object                   | Server scripts         | A sort.                                                                                                                                                                                          |
| OBJECT      | workbook.SortByDataDimensionItem       | Object                   | Server scripts         | A sort based on a data dimension item.                                                                                                                                                           |
| OBJECT      | workbook.SortByMeasure                 | Object                   | Server scripts         | A sort based on a measure.                                                                                                                                                                       |
| OBJECT      | workbook.SortDefinition                | Object                   | Server scripts         | A sort definition.                                                                                                                                                                               |
| OBJECT      | workbook.Style                         | Object                   | Server scripts         | A style.                                                                                                                                                                                         |
| OBJECT      | workbook.Table                         | Object                   | Server scripts         | A table.                                                                                                                                                                                         |
| OBJECT      | workbook.TableColumn                   | Object                   | Server scripts         | A table column.                                                                                                                                                                                  |
| OBJECT      | workbook.TableColumnCondition          | Object                   | Server scripts         | Condition for a table view column.                                                                                                                                                               |

| MEMBER TYPE | NAME                                                       | RETURN TYPE / VALUE TYPE                    | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                                                                                                     |
|-------------|------------------------------------------------------------|---------------------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT      | <code>workbook.TableColumnFilter</code>                    | Object                                      | Server scripts         | A table column filter.                                                                                                                                                                                                                                          |
| OBJECT      | <code>workbook.Workbook</code>                             | Object                                      | Server scripts         | A workbook. Workbooks are where you analyze the results of your dataset queries using different components, such as table views and pivots. All workbooks are based on a dataset, and a single dataset can be used as the basis for multiple workbooks.         |
| METHOD      | <code>workbook.create(options)</code>                      | <code>workbook.Workbook</code>              | Server scripts         | Creates a workbook. Workbooks are where you analyze the results of your dataset queries using different components, such as table views and pivots. All workbooks are based on a dataset, and a single dataset can be used as the basis for multiple workbooks. |
| METHOD      | <code>workbook.createAspect(options)</code>                | <code>workbook.Aspect</code>                | server scripts         | Creates an aspect for a chart series. An aspect includes a measure and an aspect type.                                                                                                                                                                          |
| METHOD      | <code>workbook.createCalculatedMeasure(options)</code>     | <code>workbook.CalculatedMeasure</code>     | Server scripts         | Creates a calculated measure.                                                                                                                                                                                                                                   |
| METHOD      | <code>workbook.createCategory(options)</code>              | <code>workbook.Category</code>              | Server scripts         | Creates a chart category, which includes an axis, a data root, and a sort definition. A chart category is used in a <code>workbook.Chart</code> .                                                                                                               |
| METHOD      | <code>workbook.createChart(options)</code>                 | <code>workbook.Chart</code>                 | Server scripts         | Creates a chart.                                                                                                                                                                                                                                                |
| METHOD      | <code>workbook.createChartAxis(options)</code>             | <code>workbook.ChartAxis</code>             | Server scripts         | Creates an X-axis or a Y-axis for the chart.                                                                                                                                                                                                                    |
| METHOD      | <code>workbook.createColor(options)</code>                 | <code>workbook.Color</code>                 | Server scripts         | Creates a color.                                                                                                                                                                                                                                                |
| METHOD      | <code>workbook.createComplexRecordKey</code>               | <code>workbook.RecordKey</code>             | Server scripts         | Creates a complex <code>RecordKey</code> object from another object.                                                                                                                                                                                            |
| METHOD      | <code>workbook.createConditionalFilter(options)</code>     | <code>workbook.ConditionalFilter</code>     | Server scripts         | Creates a conditional filter.                                                                                                                                                                                                                                   |
| METHOD      | <code>workbook.createConditionalFormat(options)</code>     | <code>workbook.ConditionalFormat</code>     | Server scripts         | Creates a conditional format.                                                                                                                                                                                                                                   |
| METHOD      | <code>workbook.createConditionalFormatRule(options)</code> | <code>workbook.ConditionalFormatRule</code> | Server scripts         | Creates a conditional format rule.                                                                                                                                                                                                                              |
| METHOD      | <code>workbook.createConstant(options)</code>              | <code>workbook.Expression</code>            | Server scripts         | Creates a constant expression.                                                                                                                                                                                                                                  |
| METHOD      | <code>workbook.createCurrency(options)</code>              | <code>workbook.Currency</code>              | Server scripts         | Creates a currency.                                                                                                                                                                                                                                             |
| METHOD      | <code>workbook.createDataDimension(options)</code>         | <code>workbook.DataDimension</code>         | Server scripts         | Creates a data dimension.                                                                                                                                                                                                                                       |
| METHOD      | <code>workbook.createDataDimensionItem(options)</code>     | <code>workbook.DataDimensionItem</code>     | Server scripts         | Creates a data dimension item.                                                                                                                                                                                                                                  |
| METHOD      | <code>workbook.createDataMeasure(options)</code>           | <code>workbook.DataMeasure</code>           | Server scripts         | Creates a data measure.                                                                                                                                                                                                                                         |
| METHOD      | <code>workbook.createDimensionSelector(options)</code>     | <code>workbook.DimensionSelector</code>     | Server scripts         | Creates a dimension selector.                                                                                                                                                                                                                                   |
| METHOD      | <code>workbook.createDuration(options)</code>              | <code>workbook.Duration</code>              | Server scripts         | Creates a duration.                                                                                                                                                                                                                                             |
| METHOD      | <code>workbook.createExpression(options)</code>            | <code>workbook.Expression</code>            | Server scripts         | Creates an expression.                                                                                                                                                                                                                                          |
| METHOD      | <code>workbook.createFieldContext(options)</code>          | <code>workbook.FieldContext</code>          | Server scripts         | Creates a field context for table column.                                                                                                                                                                                                                       |
| METHOD      | <code>workbook.createFontSize(options)</code>              | <code>workbook.FontSize</code>              | Server scripts         | Creates a font size.                                                                                                                                                                                                                                            |
| METHOD      | <code>workbook.createLegend(options)</code>                | <code>workbook.Legend</code>                | Server scripts         | Creates a chart legend.                                                                                                                                                                                                                                         |
| METHOD      | <code>workbook.createLimitingFilter(options)</code>        | <code>workbook.LimitingFilter</code>        | Server scripts         | Creates a limiting filter.                                                                                                                                                                                                                                      |
| METHOD      | <code>workbook.createMeasureSelector(options)</code>       | <code>workbook.MeasureSelector</code>       | Server scripts         | Creates a measure selector.                                                                                                                                                                                                                                     |
| METHOD      | <code>workbook.createMeasureValueSelector(options)</code>  | <code>workbook.MeasureValueSelector</code>  | Server scripts         | Creates a measure value selector.                                                                                                                                                                                                                               |
| METHOD      | <code>workbook.createPathSelector(options)</code>          | <code>workbook.PathSelector</code>          | Server scripts         | Creates a path selector.                                                                                                                                                                                                                                        |
| METHOD      | <code>workbook.createPivot(options)</code>                 | <code>workbook.Pivot</code>                 | Server scripts         | Creates a pivot definition. A pivot is a workbook component that enables you to pivot your dataset query results by defining measures and dimensions, so that you can analyze different subsets of data.                                                        |
| METHOD      | <code>workbook.createPivotAxis(options)</code>             | <code>workbook.PivotAxis</code>             | Server scripts         | Creates a pivot axis, which includes a data root and a sort definition.                                                                                                                                                                                         |
| METHOD      | <code>workbook.createPositionPercent(options)</code>       | <code>workbook.PositionPercent</code>       | Server scripts         | Creates a position defined by percentages of the x and y axes.                                                                                                                                                                                                  |
| METHOD      | <code>workbook.createPositionUnits(options)</code>         | <code>workbook.PositionUnits</code>         | Server scripts         | Creates a position defined by units.                                                                                                                                                                                                                            |
| METHOD      | <code>workbook.createPositionValues(options)</code>        | <code>workbook.PositionValues</code>        | Server scripts         | Creates a position defined by horizontal and vertical position values.                                                                                                                                                                                          |
| METHOD      | <code>workbook.createRange(options)</code>                 | <code>workbook.Range</code>                 | Server scripts         | Creates a range.                                                                                                                                                                                                                                                |

| MEMBER TYPE | NAME                                                         | RETURN TYPE / VALUE TYPE                      | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                                                                           |
|-------------|--------------------------------------------------------------|-----------------------------------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | <code>workbook.createReportStyle(options)</code>             | <code>workbook.ReportStyle</code>             | Server scripts         | Creates a report style.                                                                                                                                                                                                               |
| METHOD      | <code>workbook.createReportStyleRule(options)</code>         | <code>workbook.ReportStyleRule</code>         | Server scripts         | Creates a report style rule.                                                                                                                                                                                                          |
| METHOD      | <code>workbook.createSection(options)</code>                 | <code>workbook.Section</code>                 | Server scripts         | Creates a section.                                                                                                                                                                                                                    |
| METHOD      | <code>workbook.createSeries(options)</code>                  | <code>workbook.Series</code>                  | Server scripts         | Creates a chart series, which is a set of aspects.                                                                                                                                                                                    |
| METHOD      | <code>workbook.createSimpleRecordKey</code>                  | <code>workbook.RecordKey</code>               | Server scripts         | Creates a record key.                                                                                                                                                                                                                 |
| METHOD      | <code>workbook.createSort(options)</code>                    | <code>workbook.Sort</code>                    | Server scripts         | Creates a sort.                                                                                                                                                                                                                       |
| METHOD      | <code>workbook.createSortByDataDimensionItem(options)</code> | <code>workbook.SortByDataDimensionItem</code> | Server scripts         | Creates a sort based on a data dimension item.                                                                                                                                                                                        |
| METHOD      | <code>workbook.createSortByMeasure(options)</code>           | <code>workbook.SortByMeasure</code>           | Server scripts         | Creates a sort based on a measure.                                                                                                                                                                                                    |
| METHOD      | <code>workbook.createSortDefinition(options)</code>          | <code>workbook.SortDefinition</code>          | Server scripts         | Creates a sort definition.                                                                                                                                                                                                            |
| METHOD      | <code>workbook.createStyle(options)</code>                   | <code>workbook.Style</code>                   | Server scripts         | Creates a style.                                                                                                                                                                                                                      |
| METHOD      | <code>workbook.createTable(options)</code>                   | <code>workbook.Table</code>                   | Server scripts         | Creates a table view.                                                                                                                                                                                                                 |
| METHOD      | <code>workbook.createTableColumn(options)</code>             | <code>workbook.TableColumn</code>             | Server scripts         | Creates a table column.                                                                                                                                                                                                               |
| METHOD      | <code>workbook.createTableColumnCondition(options)</code>    | <code>workbook.TableColumnCondition</code>    | Server scripts         | Creates a table column condition.                                                                                                                                                                                                     |
| METHOD      | <code>workbook.createTableColumnFilter(options)</code>       | <code>workbook.TableColumnFilter</code>       | Server scripts         | Creates a table filter.                                                                                                                                                                                                               |
| METHOD      | <code>workbook.createTranslation(options)</code>             | <code>workbook.Expression</code>              | Server scripts         | Creates a translation (a translation expression).                                                                                                                                                                                     |
| METHOD      | <code>workbook.list()</code>                                 | <code>Object[]</code>                         | Server scripts         | Lists all existing workbooks.                                                                                                                                                                                                         |
| METHOD      | <code>workbook.listPaged(options)</code>                     | <code>PagedInfoData</code>                    | Server scripts         | Retrieves a set of pages with metadata about workbooks.                                                                                                                                                                               |
| METHOD      | <code>workbook.loadWorkbook(options)</code>                  | <code>workbook.Workbook</code>                | Server scripts         | Loads an existing workbook.                                                                                                                                                                                                           |
| ENUM        | <code>workbook.Aggregation</code>                            | enum                                          | Server scripts         | Holds string values for aggregation types. Used to set the value of the <code>options.aggregation</code> parameter of the <code>workbook.createDataMeasure(options)</code> method.                                                    |
| ENUM        | <code>workbook.AspectType</code>                             | enum                                          | Server scripts         | Holds string values for aspect types. Used to set the <code>options.type</code> parameter of the <code>workbook.createAspect(options)</code> method.                                                                                  |
| ENUM        | <code>workbook.ChartType</code>                              | enum                                          | Server scripts         | Holds string values for chart types. Used to pass the type value to <code>workbook.createChart(options)</code>                                                                                                                        |
| ENUM        | <code>workbook.Color</code>                                  | enum                                          | Server scripts         | Holds string values for colors. Used to set the <code>options.backgroundColor</code> , <code>options.color</code> , and <code>options.textDecorationColor</code> parameters of the <code>workbook.createStyle(options)</code> method. |
| ENUM        | <code>workbook.ConstantType</code>                           | enum                                          | Server scripts         | Holds string values for constant types. Used to set the value of the <code>options.type</code> parameter of the <code>workbook.createConstant(options)</code> method.                                                                 |
| ENUM        | <code>workbook.DateTimeHierarchy</code>                      | enum                                          | Server scripts         | Holds string values for workbook date-time hierarchy types.                                                                                                                                                                           |
| ENUM        | <code>workbook.DateTimeProperty</code>                       | enum                                          | Server scripts         | Holds string values for workbook date-time property types. Used to set the value of the <code>DATE_TIME_PROPERTY</code> in <code>workbook.ExpressionType</code> .                                                                     |
| ENUM        | <code>workbook.ExpressionType</code>                         | enum                                          | Server scripts         | Holds string values for workbook expression types. Use these values for the <code>options.functionId</code> parameter when creating an expression using <code>workbook.createExpression(options)</code>                               |
| ENUM        | <code>workbook.FontSize</code>                               | enum                                          | Server scripts         | Holds string values for font sizes. Used to set the value for the <code>options.fontSize</code> parameter of the <code>workbook.createStyle(options)</code> method.                                                                   |
| ENUM        | <code>workbook.FontStyle</code>                              | enum                                          | Server scripts         | Holds string values for font sizes. Used to set the value for the <code>options.fontSize</code> parameter of the <code>workbook.createStyle(options)</code> method.                                                                   |
| ENUM        | <code>workbook.FontWeight</code>                             | enum                                          | Server scripts         | Holds string values for font weights. Used to set the value for the <code>options.fontWeight</code> parameter of the <code>workbook.createStyle(options)</code> method.                                                               |
| ENUM        | <code>workbook.Image</code>                                  | enum                                          | Server scripts         | Holds string values for images that you can use in workbooks. Used as a value for the <code>Style.backgroundImage</code> property.                                                                                                    |
| ENUM        | <code>workbook.Position</code>                               | enum                                          | Server scripts         | Holds string values for positions. Used to set the value for the <code>PositionValues.horizontal</code> and <code>PositionValues.vertical</code> properties.                                                                          |

| MEMBER TYPE | NAME                         | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                                              |
|-------------|------------------------------|--------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ENUM        | workbook.Stacking            | enum                     | Server scripts         | Holds stacking types. Used to pass the stacking value to workbook.createChart(options).                                                                                                                  |
| ENUM        | workbook.TemporalUnit        | enum                     | Server scripts         | Holds string values for temporal units, such as hours or minutes. Used to set the value of the options.start and options.end parameters of the workbook.createDuration(options) method.                  |
| ENUM        | workbook.TextAlign           | enum                     | Server scripts         | Holds string values for text alignments. Used to set the value for the options.textAlign parameter of the workbook.createStyle(options) method.                                                          |
| ENUM        | workbook.TextDecorationLine  | enum                     | Server scripts         | Holds string values for text decoration line types, such as underline and strikethrough. Used to set the value for the options.textDecorationLine parameter of the workbook.createStyle(options) method. |
| ENUM        | workbook.TextDecorationStyle | enum                     | Server scripts         | Holds string values for text decoration line styles, such as solid and dashed. Used to set the value for the options.textDecoractionStyle parameter of the workbook.createStyle(options) method.         |
| ENUM        | workbook.TotalLine           | enum                     | Server scripts         | Holds string values for predefined total line formats. Used to pass the totalLine value to workbook.createDataDimension(options) and to workbook.createSection(options).                                 |
| ENUM        | workbook.Unit                | enum                     | Server scripts         | Holds string values for units of measurement. Used to set the options.unit parameter in the workbook.createPositionUnits(options) method.                                                                |

Aspect Object Members

The following members are available for a workbook.Aspect object.

Return Type / Value Type

Supported Script Types

workbook.CalculatedMeasure | workbook.DataMeasure

The measure of the aspect.

The type of the aspect.

| MEMBER TYPE | NAME           | RETURN TYPE / VALUE TYPE                          | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                       |
|-------------|----------------|---------------------------------------------------|------------------------|-------------------------------------------------------------------|
| PROPERTY    | Aspect.measure | workbook.CalculatedMeasure   workbook.DataMeasure | Server scripts         | The measure of the aspect.                                        |
| PROPERTY    | Aspect.type    | string                                            | Server scripts         | The type of the aspect. Set this value using workbook.AspectType. |

CalculatedMeasure Object Members

The following members are available for a workbook.CalculatedMeasure object.

Return Type / Value Type

Supported Script Types

CalculatedMeasure.expression

workbook.Expression

The expression for the calculated measure.

| MEMBER TYPE | NAME                         | RETURN TYPE / VALUE TYPE     | SUPPORTED SCRIPT TYPES | DESCRIPTION                                |
|-------------|------------------------------|------------------------------|------------------------|--------------------------------------------|
| PROPERTY    | CalculatedMeasure.expression | workbook.Expression          | Server scripts         | The expression for the calculated measure. |
| PROPERTY    | CalculatedMeasure.label      | string   workbook.Expression | Server scripts         | The label of the calculated measure.       |

Category Object Members

The following members are available for a object.

Return Type / Value Type

Supported Script Types

workbook.ChartAxis

The axis of the category.

workbook.DataDimension | workbook.Section

| MEMBER TYPE | NAME                     | RETURN TYPE / VALUE TYPE                  | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                         |
|-------------|--------------------------|-------------------------------------------|------------------------|---------------------------------------------------------------------|
| PROPERTY    | Category.axis            | workbook.ChartAxis                        | Server scripts         | The axis of the category.                                           |
| PROPERTY    | Category.root            | workbook.DataDimension   workbook.Section | Server scripts         | The section or data dimension (that is, the fields for the x-axis). |
| PROPERTY    | Category.sortDefinitions | workbook.SortDefinition[]                 | Server scripts         | The sort definitions of the category.                               |

### ChartAxis Object Members

The following members are available for a workbook.ChartAxis object.

Return Type / Value Type

Supported Script Types

The title of the chart axis.

| MEMBER TYPE | NAME            | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                  |
|-------------|-----------------|--------------------------|------------------------|------------------------------|
| PROPERTY    | ChartAxis.title | string                   | Server scripts         | The title of the chart axis. |

### Chart Object Members

The following members are available for a workbook.Chart object.

Return Type / Value Type

Supported Script Types

Chart.aggregationFilters

Array<workbook.LimitingFilter | workbook.ConditionalFilter>;

Limiting and conditional filters for the chart.

| MEMBER TYPE | NAME                     | RETURN TYPE / VALUE TYPE                                     | SUPPORTED SCRIPT TYPES | DESCRIPTION                                     |
|-------------|--------------------------|--------------------------------------------------------------|------------------------|-------------------------------------------------|
| PROPERTY    | Chart.aggregationFilters | Array<workbook.LimitingFilter   workbook.ConditionalFilter>; | Server scripts         | Limiting and conditional filters for the chart. |
| PROPERTY    | Chart.category           | string                                                       | Server scripts         | The category of the chart.                      |
| PROPERTY    | Chart.dataset            | dataset.Dataset                                              | Server scripts         | The underlying dataset for the chart.           |
| PROPERTY    | Chart.filterExpressions  | workbook.Expression[]                                        | Server scripts         | The filter expressions of the chart.            |
| PROPERTY    | Chart.id                 | string                                                       | Server scripts         | The ID of the chart.                            |
| PROPERTY    | Chart.legend             | workbook.Legend                                              | Server scripts         | The legend of the chart.                        |
| PROPERTY    | Chart.name               | string                                                       | Server scripts         | The name of chart.                              |
| PROPERTY    | Chart.series             | workbook.Series                                              | Server scripts         | The series of the chart.                        |
| PROPERTY    | Chart.stacking           | string                                                       | Server scripts         | The stacking type for the chart.                |
| PROPERTY    | Chart.subTitle           | string                                                       | Server scripts         | The subtitle of the chart.                      |
| PROPERTY    | Chart.title              | string                                                       | Server scripts         | The title of chart.                             |
| PROPERTY    | Chart.type               | workbook.ChartType                                           | Server scripts         | The type of the chart.                          |

### Color Object Members

The following members are available for a workbook.Color object.

Return Type / Value Type

Supported Script Types

The opacity, or transparency, of the color.

The blue portion of the color.

The green portion of the color.

| MEMBER TYPE | NAME        | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                 |
|-------------|-------------|--------------------------|------------------------|---------------------------------------------|
| PROPERTY    | Color.alpha | number                   | Server scripts         | The opacity, or transparency, of the color. |
| PROPERTY    | Color.blue  | number                   | Server scripts         | The blue portion of the color.              |
| PROPERTY    | Color.green | number                   | Server scripts         | The green portion of the color.             |

| MEMBER TYPE | NAME      | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                   |
|-------------|-----------|--------------------------|------------------------|-------------------------------|
| PROPERTY    | Color.red | number                   | Server scripts         | The red portion of the color. |

ConditionalFilter Object Members

The following members are available for a workbook.ConditionalFilter object.

Return Type / Value Type

Supported Script Types

ConditionalFilter.columnSelector

workbook.DescendantorSelfNodesSelector | workbook.PathSelector | workbook.DimensionSelector | workbook.ChildNodesSelector

The column selector.

| MEMBER TYPE | NAME                                    | RETURN TYPE / VALUE TYPE                                                                                                  | SUPPORTED SCRIPT TYPES | DESCRIPTION                                            |
|-------------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------|------------------------|--------------------------------------------------------|
| PROPERTY    | ConditionalFilter.columnSelector        | workbook.DescendantorSelfNodesSelector   workbook.PathSelector   workbook.DimensionSelector   workbook.ChildNodesSelector | Server scripts         | The column selector.                                   |
| PROPERTY    | ConditionalFilter.filteredNodesSelector | workbook.PathSelector   workbook.DimensionSelector                                                                        | Server scripts         | The selected filters.                                  |
| PROPERTY    | ConditionalFilter.measure               | workbook.CalculatedMeasure   workbook.DataMeasure                                                                         | Server scripts         | The measure of the filter.                             |
| PROPERTY    | ConditionalFilter.otherAxisSelector     | workbook.PathSelector   workbook.DimensionSelector                                                                        | Server scripts         | The filter selector for the other axis.                |
| PROPERTY    | ConditionalFilter.predicate             | workbook.Expression                                                                                                       | Server scripts         | The predicate which indicates if the condition is met. |
| PROPERTY    | ConditionalFilter.row                   | boolean                                                                                                                   | Server scripts         | The row axis indicator.                                |
| PROPERTY    | ConditionalFilter.rowSelector           | workbook.DescendantorSelfNodesSelector   workbook.PathSelector   workbook.DimensionSelector   workbook.ChildNodesSelector | Server scripts         | The row selector.                                      |

ConditionalFormat Object Members

The following members are available for a workbook.ConditionalFormat object.

Return Type / Value Type

Supported Script Types

ConditionalFormat.rules

workbook.ConditionalFormatRule[]

The conditional formatting rules that are included in the conditional format.

| MEMBER TYPE | NAME                    | RETURN TYPE / VALUE TYPE         | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                   |
|-------------|-------------------------|----------------------------------|------------------------|-------------------------------------------------------------------------------|
| PROPERTY    | ConditionalFormat.rules | workbook.ConditionalFormatRule[] | Server scripts         | The conditional formatting rules that are included in the conditional format. |

ConditionalFormatRule Object Members

The following members are available for a workbook.ConditionalFormatRule object.

Return Type / Value Type

Supported Script Types

ConditionalFormatRule.filter

workbook.TableColumnFilter

The filter that determines which rows or cells to apply the conditional format to.

| MEMBER TYPE | NAME                         | RETURN TYPE / VALUE TYPE   | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                        |
|-------------|------------------------------|----------------------------|------------------------|------------------------------------------------------------------------------------|
| PROPERTY    | ConditionalFormatRule.filter | workbook.TableColumnFilter | Server scripts         | The filter that determines which rows or cells to apply the conditional format to. |
| PROPERTY    | ConditionalFormatRule.style  | workbook.Style             | Server scripts         | The style to apply as the conditional format.                                      |

Currency Object Members

The following members are available for a workbook.Currency object.

Return Type / Value Type

Supported Script Types

The amount of the currency.

The ID of the currency (for example, USD, EUR, GBP, and so on).

| MEMBER TYPE | NAME            | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                     |
|-------------|-----------------|--------------------------|------------------------|-----------------------------------------------------------------|
| PROPERTY    | Currency.amount | number                   | Server scripts         | The amount of the currency.                                     |
| PROPERTY    | Currency.id     | string                   | Server scripts         | The ID of the currency (for example, USD, EUR, GBP, and so on). |

DataDimension Object Members

The following members are available for a workbook.DataDimension object.

Return Type / Value Type

Supported Script Types

DataDimension.children

Array<workbook.DataDimension | workbook.Section>;

The children of the data dimension.

| MEMBER TYPE | NAME                    | RETURN TYPE / VALUE TYPE                          | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                        |
|-------------|-------------------------|---------------------------------------------------|------------------------|------------------------------------------------------------------------------------|
| PROPERTY    | DataDimension.children  | Array<workbook.DataDimension   workbook.Section>; | Server scripts         | The children of the data dimension.                                                |
| PROPERTY    | DataDimension.items     | workbook.DataDimensionItem[]                      | Server scripts         | The items of the data dimension.                                                   |
| PROPERTY    | DataDimension.totalLine | string                                            | Server scripts         | The formatting option for the total line. Set this value using workbook.TotalLine. |

DataDimensionItem Object Members

The following members are available for a workbook.DataDimensionItem object.

Return Type / Value Type

Supported Script Types

DataDimensionItem.expression

workbook.Expression

The expression of data dimension item.

| MEMBER TYPE | NAME                         | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                            |
|-------------|------------------------------|--------------------------|------------------------|----------------------------------------|
| PROPERTY    | DataDimensionItem.expression | workbook.Expression      | Server scripts         | The expression of data dimension item. |
| PROPERTY    | DataDimensionItem.label      | string                   | Server scripts         | The label of the data dimension item.  |

DataDimensionItemValue Object Members

The following members are available for a workbook.DataDimensionItemValue object.

Return Type / Value Type

Supported Script Types

DataDimensionItemValue.item

workbook.DataDimensionItem

The data dimension item.

| MEMBER TYPE | NAME                         | RETURN TYPE / VALUE TYPE                                                                             | SUPPORTED SCRIPT TYPES | DESCRIPTION                           |
|-------------|------------------------------|------------------------------------------------------------------------------------------------------|------------------------|---------------------------------------|
| PROPERTY    | DataDimensionItemValue.item  | workbook.DataDimensionItem                                                                           | Server scripts         | The data dimension item.              |
| PROPERTY    | DataDimensionItemValue.value | string   number   boolean   workbook.Record   workbook.Currency   workbook.Range   workbook.Duration | Server scripts         | The value of the data dimension item. |

DataDimensionValue Object Members

The following members are available for a workbook.DataDimensionValue object.

Return Type / Value Type

Supported Script Types

DataDimensionValue.dataDimension

workbook.DataDimension

The data dimension.

| MEMBER TYPE | NAME                             | RETURN TYPE / VALUE TYPE          | SUPPORTED SCRIPT TYPES | DESCRIPTION                             |
|-------------|----------------------------------|-----------------------------------|------------------------|-----------------------------------------|
| PROPERTY    | DataDimensionValue.dataDimension | workbook.DataDimension            | Server scripts         | The data dimension.                     |
| PROPERTY    | DataDimensionValue.itemValues    | workbook.DataDimensionItemValue[] | Server scripts         | The item values for the data dimension. |

DataMeasure Object Members

The following members are available for a workbook.DataMeasure object.

Return Type / Value Type

Supported Script Types

DataMeasure.aggregation

The aggregation of the data measure.

DataMeasure.expression

| MEMBER TYPE | NAME                    | RETURN TYPE / VALUE TYPE     | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                       |
|-------------|-------------------------|------------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------|
| PROPERTY    | DataMeasure.aggregation | string                       | Server scripts         | The aggregation of the data measure.                                                                              |
| PROPERTY    | DataMeasure.expression  | workbook.Expression          | Server scripts         | The expression for the data measure. This property is used if the data measure is a single-expression measure.    |
| PROPERTY    | DataMeasure.expressions | workbook.Expression[]        | Server scripts         | The expressions for the data measure. This property is used if the data measure is a multiple-expression measure. |
| PROPERTY    | DataMeasure.label       | string   workbook.Expression | Server scripts         | The label of the data measure.                                                                                    |

DimensionSelector Object Members

The following members are available for a workbook.DimensionSelector object.

Return Type / Value Type

Supported Script Types

DimensionSelector.dimension

workbook.DataDimension | workbook.Section

The dimension of the dimension selector.

| MEMBER TYPE | NAME                        | RETURN TYPE / VALUE TYPE                  | SUPPORTED SCRIPT TYPES | DESCRIPTION                              |
|-------------|-----------------------------|-------------------------------------------|------------------------|------------------------------------------|
| PROPERTY    | DimensionSelector.dimension | workbook.DataDimension   workbook.Section | Server scripts         | The dimension of the dimension selector. |

Duration Object Members

The following members are available for a workbook.Duration object.

Return Type / Value Type

Supported Script Types

The amount of the duration.

The units of the duration.

| MEMBER TYPE | NAME            | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                 |
|-------------|-----------------|--------------------------|------------------------|-----------------------------|
| PROPERTY    | Duration.amount | number                   | Server scripts         | The amount of the duration. |
| PROPERTY    | Duration.units  | Object                   | Server scripts         | The units of the duration.  |

Expression Object Members

The following members are available for a workbook.Expression object.

Return Type / Value Type

Supported Script Types

Expression.functionId

The ID of the function used in the expression.

Expression.parameters

| MEMBER TYPE | NAME                  | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                    |
|-------------|-----------------------|--------------------------|------------------------|------------------------------------------------|
| PROPERTY    | Expression.functionId | string                   | Server scripts         | The ID of the function used in the expression. |

| MEMBER TYPE | NAME                  | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                       |
|-------------|-----------------------|--------------------------|------------------------|-----------------------------------|
| PROPERTY    | Expression.parameters | Object                   | Server scripts         | The parameters of the expression. |

FieldContext Object Members

The following members are available for a workbook.FieldContext object.

Return Type / Value Type

Supported Script Types

FieldContext.name

The name of the field context (for example, DISPLAY or CONSOLIDATED)

FieldContext.parameters

| MEMBER TYPE | NAME                    | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                          |
|-------------|-------------------------|--------------------------|------------------------|----------------------------------------------------------------------|
| PROPERTY    | FieldContext.name       | string                   | Server scripts         | The name of the field context (for example, DISPLAY or CONSOLIDATED) |
| PROPERTY    | FieldContext.parameters | Object                   | Server scripts         | The parameters of the field context.                                 |

FontSize Object Members

The following members are available for a workbook.FontSize object.

Return Type / Value Type

Supported Script Types

The numeric size of the font size.

The unit of the font size.

| MEMBER TYPE | NAME          | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                        |
|-------------|---------------|--------------------------|------------------------|------------------------------------|
| PROPERTY    | FontSize.size | number                   | Server scripts         | The numeric size of the font size. |
| PROPERTY    | FontSize.unit | string                   | Server scripts         | The unit of the font size.         |

Legend Object Members

The following members are available for a object.

Return Type / Value Type

Supported Script Types

workbook.ChartAxis[]

The axes of the legend.

workbook.DataDimension | workbook.Section

| MEMBER TYPE | NAME                   | RETURN TYPE / VALUE TYPE                  | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                          |
|-------------|------------------------|-------------------------------------------|------------------------|----------------------------------------------------------------------|
| PROPERTY    | Legend.axes            | workbook.ChartAxis[]                      | Server scripts         | The axes of the legend.                                              |
| PROPERTY    | Legend.root            | workbook.DataDimension   workbook.Section | Server scripts         | The section or data dimension (that is., the fields for the y-axis). |
| PROPERTY    | Legend.sortDefinitions | workbook.SortDefinition[]                 | Server scripts         | The sort definitions of the legend.                                  |

LimitingFilter Object Members

The following members are available for a workbook.LimitingFilter object.

Return Type / Value Type

Supported Script Types

LimitingFilter.filteredNodesSelector

workbook.PathSelector | workbook.DimensionSelector

The selected filter.

| MEMBER TYPE | NAME                                 | RETURN TYPE / VALUE TYPE                           | SUPPORTED SCRIPT TYPES | DESCRIPTION                      |
|-------------|--------------------------------------|----------------------------------------------------|------------------------|----------------------------------|
| PROPERTY    | LimitingFilter.filteredNodesSelector | workbook.PathSelector   workbook.DimensionSelector | Server scripts         | The selected filter.             |
| PROPERTY    | LimitingFilter.limit                 | number                                             | Server scripts         | The limit number for the filter. |

| MEMBER TYPE | NAME                   | RETURN TYPE / VALUE TYPE                                          | SUPPORTED SCRIPT TYPES | DESCRIPTION                            |
|-------------|------------------------|-------------------------------------------------------------------|------------------------|----------------------------------------|
| PROPERTY    | LimitingFilter.row     | boolean                                                           | Server scripts         | The row axis indicator for the filter. |
| PROPERTY    | LimitingFilter.sortBys | Array<workbook.SortByDataDimensionItem   workbook.SortByMeasure>; | Server scripts         | The ordering elements of the filter.   |

### MeasureSelector Object Members

The following members are available for a workbook.MeasureSelector object.

Return Type / Value Type

Supported Script Types

MeasureSelector.measures

workbook.CalculatedMeasure[] | workbook.DataMeasure

The measures for the measure selector.

| MEMBER TYPE | NAME                     | RETURN TYPE / VALUE TYPE                            | SUPPORTED SCRIPT TYPES | DESCRIPTION                            |
|-------------|--------------------------|-----------------------------------------------------|------------------------|----------------------------------------|
| PROPERTY    | MeasureSelector.measures | workbook.CalculatedMeasure[]   workbook.DataMeasure | Server scripts         | The measures for the measure selector. |

### MeasureValue Object Members

The following members are available for a workbook.MeasureValue object.

Return Type / Value Type

Supported Script Types

MeasureValue.measure

workbook.MeasureValue

The measure to use for the measure value.

| MEMBER TYPE | NAME                 | RETURN TYPE / VALUE TYPE                                                                             | SUPPORTED SCRIPT TYPES | DESCRIPTION                               |
|-------------|----------------------|------------------------------------------------------------------------------------------------------|------------------------|-------------------------------------------|
| PROPERTY    | MeasureValue.measure | workbook.MeasureValue                                                                                | Server scripts         | The measure to use for the measure value. |
| PROPERTY    | MeasureValue.value   | string   number   boolean   workbook.Record   workbook.Currency   workbook.Range   workbook.Duration | Server scripts         | The value to use for the measure value.   |

### MeasureValueSelector Object Members

The following members are available for a workbook.MeasureValueSelector object.

Return Type / Value Type

Supported Script Types

MeasureValueSelector.columnSelector

workbook.DimensionSelector | workbook.PathSelector | workbook.DescendantorSelfNodesSelector

The column selector.

| MEMBER TYPE | NAME                                 | RETURN TYPE / VALUE TYPE                                                                    | SUPPORTED SCRIPT TYPES | DESCRIPTION            |
|-------------|--------------------------------------|---------------------------------------------------------------------------------------------|------------------------|------------------------|
| PROPERTY    | MeasureValueSelector.columnSelector  | workbook.DimensionSelector   workbook.PathSelector   workbook.DescendantorSelfNodesSelector | Server scripts         | The column selector.   |
| PROPERTY    | MeasureValueSelector.measureSelector | workbook.MeasureSelector[]                                                                  | Server scripts         | The measure selectors. |
| PROPERTY    | MeasureValueSelector.rowSelector     | workbook.DimensionSelector   workbook.PathSelector   workbook.DescendantorSelfNodesSelector | Server scripts         | The row selector.      |

### PathSelector Object Members

The following members are available for a workbook.PathSelector object.

Return Type / Value Type

Supported Script Types

PathSelector.elements

workbook.DimensionSelector

The elements denoting 'xpath' of the selector.

| MEMBER TYPE | NAME                  | RETURN TYPE / VALUE TYPE   | SUPPORTED SCRIPT TYPES | DESCRIPTION                                    |
|-------------|-----------------------|----------------------------|------------------------|------------------------------------------------|
| PROPERTY    | PathSelector.elements | workbook.DimensionSelector | Server scripts         | The elements denoting 'xpath' of the selector. |

PivotAxis Object Members

The following members are available for a workbook.PivotAxis object.

Return Type / Value Type

Supported Script Types

workbook.DataDimension | workbook.Section

The data for the pivot axis.

PivotAxis.sortDefinitions

| MEMBER TYPE | NAME                      | RETURN TYPE / VALUE TYPE                  | SUPPORTED SCRIPT TYPES | DESCRIPTION                             |
|-------------|---------------------------|-------------------------------------------|------------------------|-----------------------------------------|
| PROPERTY    | PivotAxis.root            | workbook.DataDimension   workbook.Section | Server scripts         | The data for the pivot axis.            |
| PROPERTY    | PivotAxis.sortDefinitions | workbook.SortDefinition[]                 | Server scripts         | The sort definitions of the pivot axis. |

Pivot Object Members

The following members are available for a workbook.Pivot object.

Return Type / Value Type

Supported Script Types

Pivot.aggregationFilters

Array<workbook.ConditionalFilter | workbook.LimitingFilter>;

The limiting and conditional filters of the pivot definition.

| MEMBER TYPE | NAME                     | RETURN TYPE / VALUE TYPE                                     | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                   |
|-------------|--------------------------|--------------------------------------------------------------|------------------------|---------------------------------------------------------------|
| PROPERTY    | Pivot.aggregationFilters | Array<workbook.ConditionalFilter   workbook.LimitingFilter>; | Server scripts         | The limiting and conditional filters of the pivot definition. |
| PROPERTY    | Pivot.columnAxis         | workbook.PivotAxis                                           | Server scripts         | The column axis of the pivot definition.                      |
| PROPERTY    | Pivot.dataset            | dataset.Dataset                                              | Server scripts         | The underlying dataset of the pivot definition.               |
| PROPERTY    | Pivot.datasetLink        | datasetLink.DatasetLink                                      | Server scripts         | Underlying dataset for the pivot.                             |
| PROPERTY    | Pivot.filterExpressions  | workbook.Expression                                          | Server scripts         | The filter expressions of the pivot definition.               |
| PROPERTY    | Pivot.id                 | string                                                       | Server scripts         | The ID of the pivot definition.                               |
| PROPERTY    | Pivot.name               | string                                                       | Server scripts         | The name of the pivot definition.                             |
| PROPERTY    | Pivot.portletName        | string   workbook.Expression                                 | Server scripts         | The name of the portlet for the pivot.                        |
| PROPERTY    | Pivot.reportStyles       | workbook.ReportStyle[]                                       | Server scripts         | Report styles for the pivot.                                  |
| PROPERTY    | Pivot.rowAxis            | workbook.PivotAxis                                           | Server scripts         | The row axis of the pivot definition.                         |

PivotIntersection Object Members

The following members are available for a workbook.PivotIntersection object.

Return Type / Value Type

Supported Script Types

PivotIntersection.column

workbook.DataDimensionValue | workbook.SectionValue

The column dimension value.

| MEMBER TYPE | NAME                            | RETURN TYPE / VALUE TYPE                            | SUPPORTED SCRIPT TYPES | DESCRIPTION                                   |
|-------------|---------------------------------|-----------------------------------------------------|------------------------|-----------------------------------------------|
| PROPERTY    | PivotIntersection.column        | workbook.DataDimensionValue   workbook.SectionValue | Server scripts         | The column dimension value.                   |
| PROPERTY    | PivotIntersection.measureValues | workbook.MeasureValue[]                             | Server scripts         | The measure values in the pivot intersection. |
| PROPERTY    | PivotIntersection.row           | workbook.DataDimensionValue   workbook.SectionValue | Server scripts         | The row dimension value.                      |

### PositionPercent Object Members

The following members are available for a workbook.PositionPercent object.

Return Type / Value Type

Supported Script Types

PositionPercent.percentX

The percentage of the x dimension.

PositionPercent.percentY

| MEMBER TYPE | NAME                     | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                        |
|-------------|--------------------------|--------------------------|------------------------|------------------------------------|
| PROPERTY    | PositionPercent.percentX | number                   | Server scripts         | The percentage of the x dimension. |
| PROPERTY    | PositionPercent.percentY | number                   | Server scripts         | The percentage of the y dimension. |

### PositionUnits Object Members

The following members are available for a workbook.PositionUnits object.

Return Type / Value Type

Supported Script Types

PositionUnits.unit

The units for the position.

The x value of the position.

| MEMBER TYPE | NAME               | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                  |
|-------------|--------------------|--------------------------|------------------------|------------------------------|
| PROPERTY    | PositionUnits.unit | string                   | Server scripts         | The units for the position.  |
| PROPERTY    | PositionUnits.x    | number                   | Server scripts         | The x value of the position. |
| PROPERTY    | PositionUnits.y    | number                   | Server scripts         | The y value of the position. |

### PositionValues Object Members

The following members are available for a workbook.PositionValues object.

Return Type / Value Type

Supported Script Types

PositionValues.horizontal

The horizontal value of the position.

PositionValues.vertical

| MEMBER TYPE | NAME                      | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                           |
|-------------|---------------------------|--------------------------|------------------------|---------------------------------------|
| PROPERTY    | PositionValues.horizontal | string                   | Server scripts         | The horizontal value of the position. |
| PROPERTY    | PositionValues.vertical   | string                   | Server scripts         | The vertical value of the position.   |

### Range Object Members

The following members are available for a workbook.Range object.

Return Type / Value Type

Supported Script Types

The end date or date-time of the range.

The start date or date-time of the range.

| MEMBER TYPE | NAME        | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                               |
|-------------|-------------|--------------------------|------------------------|-------------------------------------------|
| PROPERTY    | Range.end   | string                   | Server scripts         | The end date or date-time of the range.   |
| PROPERTY    | Range.start | string                   | Server scripts         | The start date or date-time of the range. |

### Record Object Members

The following members are available for a workbook.Record object.

Return Type / Value Type

Supported Script Types

The name of the record type for the record.

Record.primaryKey

The primary key of the record.

| MEMBER TYPE | NAME              | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                 |
|-------------|-------------------|--------------------------|------------------------|---------------------------------------------|
| PROPERTY    | Record.name       | string                   | Server scripts         | The name of the record type for the record. |
| PROPERTY    | Record.primaryKey | number                   | Server scripts         | The primary key of the record.              |
| PROPERTY    | Record.properties | Object                   | Server scripts         | The properties of the record.               |

RecordKey Object Members

The following members are available for a workbook.RecordKey object.

Return Type / Value Type

Supported Script Types

RecordKey.properties

The properties of the record key.

| MEMBER TYPE | NAME                 | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                       |
|-------------|----------------------|--------------------------|------------------------|-----------------------------------|
| PROPERTY    | RecordKey.properties | Object                   | Server scripts         | The properties of the record key. |

ReportStyle Object Members

The following members are available for a workbook.ReportStyle object.

Return Type / Value Type

Supported Script Types

ReportStyle.rules

workbook.ReportStyleRule[]

The formatting rules for the report style.

| MEMBER TYPE | NAME                  | RETURN TYPE / VALUE TYPE        | SUPPORTED SCRIPT TYPES | DESCRIPTION                                |
|-------------|-----------------------|---------------------------------|------------------------|--------------------------------------------|
| PROPERTY    | ReportStyle.rules     | workbook.ReportStyleRule[]      | Server scripts         | The formatting rules for the report style. |
| PROPERTY    | ReportStyle.selectors | workbook.MeasureValueSelector[] | Server scripts         | The selectors for the report style.        |

ReportStyleRule Object Members

The following members are available for a workbook.ReportStyleRule object.

Return Type / Value Type

Supported Script Types

ReportStyleRule.expression

workbook.Expression

A boolean expression indicating whether the style should be applied.

| MEMBER TYPE | NAME                       | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                          |
|-------------|----------------------------|--------------------------|------------------------|----------------------------------------------------------------------|
| PROPERTY    | ReportStyleRule.expression | workbook.Expression      | Server scripts         | A boolean expression indicating whether the style should be applied. |
| PROPERTY    | ReportStyleRule.style      | workbook.Style           | Server scripts         | The style to be applied.                                             |

Section Object Members

The following members are available for a workbook.Section object.

Return Type / Value Type

Supported Script Types

Section.children

Array<workbook.CalculatedMeasure | workbook.DataMeasure | workbook.DataDimension | workbook.DataDimensionItem>;

The children of the section.

| MEMBER TYPE | NAME             | RETURN TYPE / VALUE TYPE                                                                                        | SUPPORTED SCRIPT TYPES | DESCRIPTION                  |
|-------------|------------------|-----------------------------------------------------------------------------------------------------------------|------------------------|------------------------------|
| PROPERTY    | Section.children | Array<workbook.CalculatedMeasure   workbook.DataMeasure   workbook.DataDimension   workbook.DataDimensionItem>; | Server scripts         | The children of the section. |

| MEMBER TYPE | NAME              | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                        |
|-------------|-------------------|--------------------------|------------------------|------------------------------------------------------------------------------------|
| PROPERTY    | Section.totalLine | string                   | Server scripts         | The formatting option for the total line. Set this value using workbook.TotalLine. |

### SectionValue Object Members

The following members are available for a workbook.SectionValue object.

Return Type / Value Type

Supported Script Types

SectionValue.section

workbook.Section

The section of the section value.

| MEMBER TYPE | NAME                 | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                       |
|-------------|----------------------|--------------------------|------------------------|-----------------------------------|
| PROPERTY    | SectionValue.section | workbook.Section         | Server scripts         | The section of the section value. |

### Series Object Members

The following members are available for a workbook.Series object.

Return Type / Value Type

Supported Script Types

The aspects for the series.

| MEMBER TYPE | NAME           | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                 |
|-------------|----------------|--------------------------|------------------------|-----------------------------|
| PROPERTY    | Series.aspects | string                   | Server scripts         | The aspects for the series. |

### Sort Object Members

The following members are available for a workbook.Sort object.

Return Type / Value Type

Supported Script Types

When set to true, indicates the sort is in ascending order.

Sort.caseSensitive

When set to true, indicates the sort is case sensitive.

| MEMBER TYPE | NAME               | RETURN TYPE / VALUE TYPE     | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                         |
|-------------|--------------------|------------------------------|------------------------|---------------------------------------------------------------------|
| PROPERTY    | Sort.ascending     | boolean                      | Server scripts         | When set to true, indicates the sort is in ascending order.         |
| PROPERTY    | Sort.caseSensitive | boolean                      | Server scripts         | When set to true, indicates the sort is case sensitive.             |
| PROPERTY    | Sort.locale        | query.SortLocale (read-only) | Server scripts         | The locale of the sort.                                             |
| PROPERTY    | Sort.nullsLast     | boolean                      | Server scripts         | When set to true, indicates that nulls are placed last in the sort. |
| PROPERTY    | Sort.order         | number                       | Server scripts         | Sort order indicator.                                               |

### SortDefinition Object Members

The following members are available for a workbook.SortDefinition object.

Return Type / Value Type

Supported Script Types

SortDefinition.selector

workbook.DimensionSelector | workbook.PathSelector

The selector for the sort definition.

| MEMBER TYPE | NAME                    | RETURN TYPE / VALUE TYPE                                          | SUPPORTED SCRIPT TYPES | DESCRIPTION                             |
|-------------|-------------------------|-------------------------------------------------------------------|------------------------|-----------------------------------------|
| PROPERTY    | SortDefinition.selector | workbook.DimensionSelector   workbook.PathSelector                | Server scripts         | The selector for the sort definition.   |
| PROPERTY    | SortDefinition.sortBys  | Array<workbook.SortByDataDimensionItem   workbook.SortByMeasure>; | Server scripts         | The sort order for the sort definition. |

SortByDataDimensionItem Object Members

The following members are available for a workbook.SortByDataDimensionItem object.

Return Type / Value Type

Supported Script Types

SortByDataDimensionItem.item

workbook.DataDimensionItem

The data dimension item to use for the sort.

| MEMBER TYPE | NAME                         | RETURN TYPE / VALUE TYPE   | SUPPORTED SCRIPT TYPES | DESCRIPTION                                  |
|-------------|------------------------------|----------------------------|------------------------|----------------------------------------------|
| PROPERTY    | SortByDataDimensionItem.item | workbook.DataDimensionItem | Server scripts         | The data dimension item to use for the sort. |
| PROPERTY    | SortByDataDimensionItem.sort | workbook.Sort              | Server scripts         | The sort to use.                             |

SortByMeasure Object Members

The following members are available for a workbook.SortByMeasure object.

Return Type / Value Type

Supported Script Types

SortByMeasure.measure

workbook.CalculatedMeasure | workbook.DataMeasure

The measure for the sort.

| MEMBER TYPE | NAME                            | RETURN TYPE / VALUE TYPE                                                                    | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                      |
|-------------|---------------------------------|---------------------------------------------------------------------------------------------|------------------------|----------------------------------------------------------------------------------|
| PROPERTY    | SortByMeasure.measure           | workbook.CalculatedMeasure   workbook.DataMeasure                                           | Server scripts         | The measure for the sort.                                                        |
| PROPERTY    | SortByMeasure.otherAxisSelector | workbook.DescendantorSelfNodesSelector   workbook.PathSelector   workbook.DimensionSelector | Server scripts         | The selector for the axis that is not defined in the associated sort definition. |
| PROPERTY    | SortByMeasure.sort              | workbook.Sort                                                                               | Server scripts         | The sort to use.                                                                 |

Style Object Members

The following members are available for a workbook.Style object.

Return Type / Value Type

Supported Script Types

Style.backgroundColor

string | workbook.Color

The background color of the style.

| MEMBER TYPE | NAME                      | RETURN TYPE / VALUE TYPE                                                    | SUPPORTED SCRIPT TYPES | DESCRIPTION                             |
|-------------|---------------------------|-----------------------------------------------------------------------------|------------------------|-----------------------------------------|
| PROPERTY    | Style.backgroundColor     | string   workbook.Color                                                     | Server scripts         | The background color of the style.      |
| PROPERTY    | Style.backgroundImage     | string                                                                      | Server scripts         | The background image of the style.      |
| PROPERTY    | Style.backgroundPosition  | workbook.PositionPercent   workbook.PositionUnits   workbook.PositionValues | Server scripts         | The background position of the style.   |
| PROPERTY    | Style.color               | string   workbook.Color                                                     | Server scripts         | The color of the style.                 |
| PROPERTY    | Style.fontSize            | string   workbook.FontSize                                                  | Server scripts         | The font size of the style.             |
| PROPERTY    | Style.fontStyle           | string                                                                      | Server scripts         | The font style of the style.            |
| PROPERTY    | Style.fontWeight          | string                                                                      | Server scripts         | The font weight of the style.           |
| PROPERTY    | Style.textAlign           | string                                                                      | Server scripts         | The text alignment of the style.        |
| PROPERTY    | Style.textDecorationColor | string   workbook.Color                                                     | Server scripts         | The text decoration color of the style. |
| PROPERTY    | Style.textDecorationLine  | string                                                                      | Server scripts         | The text decoration line of the style.  |
| PROPERTY    | Style.textDecorationStyle | string                                                                      | Server scripts         | The text decoration style of the style. |

Table Object Members

The following members are available for a workbook.Table object.

Return Type / Value Type

Supported Script Types

workbook.TableColumn

The columns in the table view.

The dataset for the table view.

| MEMBER TYPE | NAME          | RETURN TYPE / VALUE TYPE     | SUPPORTED SCRIPT TYPES | DESCRIPTION                     |
|-------------|---------------|------------------------------|------------------------|---------------------------------|
| PROPERTY    | Table.columns | workbook.TableColumn         | Server scripts         | The columns in the table view.  |
| PROPERTY    | Table.dataset | dataset.Dataset              | Server scripts         | The dataset for the table view. |
| PROPERTY    | Table.id      | string                       | Server scripts         | The ID of the table view.       |
| PROPERTY    | Table.name    | string   workbook.Expression | Server scripts         | The label of the table view.    |

TableColumn Object Members

The following members are available for a workbook.TableColumn object.

Return Type / Value Type

Supported Script Types

TableColumn.alias

The alias for the table column.

TableColumn.datasetColumnAlias

| MEMBER TYPE | NAME                           | RETURN TYPE / VALUE TYPE   | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                              |
|-------------|--------------------------------|----------------------------|------------------------|--------------------------------------------------------------------------|
| PROPERTY    | TableColumn.alias              | string                     | Server scripts         | The alias for the table column.                                          |
| PROPERTY    | TableColumn.datasetColumnAlias | string                     | Server scripts         | The alias of the dataset column from which the table column was created. |
| PROPERTY    | TableColumn.fieldContext       | workbook.FieldContext      | Server scripts         | The field context for the field used in the table column.                |
| PROPERTY    | TableColumn.filters            | workbook.TableColumnFilter | Server scripts         | The filters for the table column.                                        |
| PROPERTY    | TableColumn.label              | string                     | Server scripts         | The label for the table column.                                          |
| PROPERTY    | TableColumn.sort               | workbook.Sort              | Server scripts         | The sort for the table column.                                           |
| PROPERTY    | TableColumn.width              | number                     | Server scripts         | The desired width of the table column when displayed in the UI.          |

TableColumnCondition Object Members

The following members are available for a workbook.TableColumnCondition object.

Return Type / Value Type

Supported Script Types

TableColumnCondition.filters

workbook.TableColumnFilter[]

The filters for the condition

| MEMBER TYPE | NAME                          | RETURN TYPE / VALUE TYPE     | SUPPORTED SCRIPT TYPES | DESCRIPTION                     |
|-------------|-------------------------------|------------------------------|------------------------|---------------------------------|
| PROPERTY    | TableColumnCondition.filters  | workbook.TableColumnFilter[] | Server scripts         | The filters for the condition   |
| PROPERTY    | TableColumnCondition.operator | string                       | Server scripts         | The operator for the condition. |

TableColumnFilter Object Members

The following members are available for a workbook.TableColumnFilter object.

Return Type / Value Type

Supported Script Types

TableColumnFilter.operator

The operator for the table column filter.

TableColumnFilter.values

| MEMBER TYPE | NAME                       | RETURN TYPE / VALUE TYPE                                 | SUPPORTED SCRIPT TYPES | DESCRIPTION                               |
|-------------|----------------------------|----------------------------------------------------------|------------------------|-------------------------------------------|
| PROPERTY    | TableColumnFilter.operator | string                                                   | Server scripts         | The operator for the table column filter. |
| PROPERTY    | TableColumnFilter.values   | Array<null   Object   boolean   number   string   Date>; | Server scripts         | The values for the table column filter.   |

Workbook Object Members

The following members are available for a workbook.Workbook object.

Return Type / Value Type

Supported Script Types

Workbook.runPivot(options)

workbook.PivotIntersection[]

Runs a pivot in the workbook and returns the results as a set of row-column intersections.

| MEMBER TYPE | NAME                       | RETURN TYPE / VALUE TYPE     | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                |
|-------------|----------------------------|------------------------------|------------------------|--------------------------------------------------------------------------------------------|
| METHOD      | Workbook.runPivot(options) | workbook.PivotIntersection[] | Server scripts         | Runs a pivot in the workbook and returns the results as a set of row-column intersections. |
| PROPERTY    | Workbook.description       | string                       | Server scripts         | The description of the workbook.                                                           |
| PROPERTY    | Workbook.id                | string                       | Server scripts         | The ID of the workbook.                                                                    |
| PROPERTY    | Workbook.name              | string                       | Server scripts         | The name of the workbook.                                                                  |
| PROPERTY    | Workbook.pivots            | workbook.Pivot[]             | Server scripts         | The pivots in the workbook.                                                                |
| PROPERTY    | Workbook.tables            | workbook.Table[]             | Server scripts         | The tables in the workbook.                                                                |

 **Sample 1: Create Datasets, Dataset Links, and a Workbook with a Pivot and Run the Workbook**

```

/**
 * @NApiVersion 2.x
 */
// The following script creates a simple workbook that contains a simple table
require(['N/workbook', 'N/dataset', 'N/datasetLink'], function(nWorkbook, nDataset, datasetLink) {
  var join = nDataset.createJoin({
    fieldId: "budgetmachine"
  });

  var period = nDataset.createColumn({
    join: join,
    fieldId: "period",
    alias: "budgetmachineperiod",
    label: "Accounting Period"
  });

  var department = nDataset.createColumn({
    fieldId: "department",
    alias: "department",
    label: "Department"
  });

  var total = nDataset.createColumn({
    fieldId: "total",
    alias: "total",
    label: "Amount (Total)"
  });

  var budget = nDataset.create({
    type: 'budgets',
    columns: [period, department, total]
  });

  var postingperiod = nDataset.createColumn({
    fieldId: "postingperiod",
    alias: "postingperiod",
    label: "Posting Period"
  });

  var amount = nDataset.createColumn({
    fieldId: "amount",
    alias: "amount",
    label: "Amount"
  });

  var sales = nDataset.create({
    type: 'salesinvoiced',
    columns: [postingperiod, department, amount],
  });

  var budgetmachineperiod = budget.getExpressionFromColumn({
    alias: "budgetmachineperiod"
  });
  var postingperiodExpression = sales.getExpressionFromColumn({
    alias: "postingperiod"
  });
  var link = datasetLink.create({
    datasets: [budget, sales],
    expressions: [[budgetmachineperiod, postingperiodExpression]],
    id: "link"
  });

  var postingPeriodItem = nWorkbook.createDataDimensionItem({
    expression: postingperiodExpression
  });
  var postingPeriodDimension = nWorkbook.createDataDimension({
    items: [postingPeriodItem]
  });
  var rowSection = nWorkbook.createSection({
    children: [postingPeriodDimension]
  });

  var departmentItem = nWorkbook.createDataDimensionItem({
    expression: budget.getExpressionFromColumn({
      alias: "department"
    })
  });
  var departmentDimension = nWorkbook.createDataDimension({
    items: [departmentItem]
  });

  var sumTotal = nWorkbook.createDataMeasure({
    label: 'Sum Total',
    expression: budget.getExpressionFromColumn({
      alias: 'total'
    }),
    aggregation: 'SUM'
  });

```

```

});

var sumAmountNet = nWorkbook.createDataMeasure({
  label: 'Sum Amount',
  expression: sales.getExpressionFromColumn({
    alias: 'amount'
  }),
  aggregation: 'SUM'
});

var columnSection = nWorkbook.createSection({
  children: [departmentDimension, sumTotal, sumAmountNet]
});

var pivot = nWorkbook.createPivot({
  id: "pivot",
  rowAxis: nWorkbook.createPivotAxis({
    root: rowSection
  }),
  columnAxis: nWorkbook.createPivotAxis({
    root: columnSection
  }),
  name: "Pivot",
  datasetLink: link
});

var wb = nWorkbook.create({
  pivots: [pivot]
});

wb.runPivot.promise("pivot").then(function(intersections){
  for (var i in intersections)
  {
    var intersection = intersections[i];
    if (intersection.row.itemValues) //skip header
    {
      console.log("Period: " + intersection.row.itemValues[0].value.name);
      console.log(intersection.column.section.children[1].label + ":");
      console.log(intersection.measureValues[0] ? intersection.measureValues[0].value.amount : 0);
      console.log(intersection.column.section.children[2].label + ":");
      console.log(intersection.measureValues[1] ? intersection.measureValues[1].value.amount : 0);
    }
  }
});

```

## Sample 2: Create a Comprehensive Workbook

```
/**
 * @NApiVersion 2.x
 */
// The following script creates a workbook that includes a chart, a table, and a pivot. This sample uses a dataset that is not included in your account, so you
// will need to change the dataset id to a valid value from your account. This dataset needs to include columns for 'id', 'name', 'date', and 'total'.

require(['N/workbook', 'N/dataset'], function(workbook, dataset){
  var myDataset = dataset.load({
    id: 'dataset_7'
  });

  var theIDExpression = myDataset.getExpressionFromColumn({
    alias: 'id'
  });
  var sort = workbook.createSort({
    ascending: false
  });
  var columnID = workbook.createTableColumn({
    datasetColumnAlias: 'id'
  });
  var columnName = workbook.createTableColumn({
    datasetColumnAlias: 'name'
  });
  var columnDate = workbook.createTableColumn({
    datasetColumnAlias: 'date'
  });
  var columnTotal = myBasicWorkbook.createTableColumn({
    datasetColumnAlias: 'total'
  });
  var tableView = workbook.createTableDefinition({
    id: 'view',
    name: 'View',
    dataset: myDataset,
    columns: [columnID, columnName, columnDate, columnTotal]
  });
  var theDateExpression = dataset.getExpressionFromColumn({
    alias: 'date'
  });
  var rowItem = workbook.createDataDimensionitem({
    label: 'A',
    expression: theDateExpression
  });
  var rowDataDimension = workbook.createDataDimension({
    items: [rowItem]
  });
  var rowSection = workbook.createSection({
    children: [rowDataDimension]
  });
  var theTotalExpression = dataset.getExpressionFromColumn({
    alias: 'total'
  });
  var columnItem = workbook.createDataDimensionItem({
    label: 'B',
    expression: theTotalExpression
  });
  var columnDataDimension = workbook.createDataDimension({
    items: [columnItem]
  });
  var columnMeasure = workbook.createMeasure({
    label: 'M',
    expression: theIDExpression,
    aggregation: workbook.Aggregation.MAX
  });
  var columnSection = workbook.createSection({
    children: [columnDataDimension, columnMeasure]
  });
  var constExpr = workbook.createConstant({
    constant: 1
  });
  var anyOfExpr = workbook.createExpression({
    functionId: workbook.ExpressionType.AND,
    parameters: {
      expression: theIDExpression,
      set: [constExpr]
    }
  });
  var notExpr = workbook.createExpression({
    functionId: workbook.ExpressionType.NOT,
    parameters: {
      a: anyOfExpr
    }
  });
  var allSubNodesSelector = workbook.createAllSubNodesSelector();
  var rowItemSelector = workbook.createDimensionSelector({
    dimension: rowDataDimension
  });
  var columnItemSelector = workbook.createDimensionSelector({
    dimension: columnDataDimension
  });
  var rowSelector = workbook.createPathSelector({
    elements: [allSubNodesSelector, rowItemSelector]
  });
});
```

```

var columnSelector = workbook.createPathSelector({
  elements: [allSubNodesSelector, columnItemSelector]
});
var rowSort = workbook.createDimensionSort({
  item: rowItem,
  sort: sort
});
var columnSort = workbook.createMeasureSort({
  measure: columnMeasure,
  sort: sort,
  otherAxisSelector: allSubNodesSelector
});
var rowSortDefinition = workbook.createSortDefinition({
  sortBys: [rowSort],
  selector: rowSelector
});
var columnSortDefinition = workbook.createSortDefinition({
  sortBys: [columnSort],
  selector: columnSelector
});
var rowAxis = workbook.createPivotAxis({
  root: rowSection,
  sortDefinitions: [rowSortDefinition]
});
var columnAxis = workbook.createPivotAxis({
  root: columnSection,
  sortDefinitions: [columnSortDefinition]
});
var limitingFilter = workbook.createLimitingFilter({
  row: true,
  filteredNodesSelector: rowSelector,
  limit: 1,
  sortBys: [rowSort]
});
var conditionalFilter = workbook.createConditionalFilter({
  row: false,
  filteredNodesSelector: rowSelector,
  otherAxisSelector: columnSelector,
  measure: columnMeasure,
  predicate: notExpr
});
var pivot = workbook.createPivotDefinition({
  id: 'pivot',
  name: 'Pivot',
  dataset: myDataset,
  rowAxis: rowAxis,
  columnAxis: columnAxis,
  filterExpressions: [notExpr],
  aggregationFilters: [limitingFilter, conditionalFilter]
});
var firstAxis = workbook.createChartAxis({
  title: 'First axis'
});
var secondAxis = workbook.createChartAxis({
  title: 'Second axis'
});
var category = workbook.createCategory({
  axis: firstAxis,
  root: rowSection
});
var legend = workbook.createLegend({
  axes: [secondAxis],
  root: columnSection
});
var aspect = workbook.createAspect({
  measure: columnMeasure
});
var series = workbook.createSeries({
  aspects: [aspect]
});
var chart = workbook.createChartDefinition({
  id: 'chart',
  name: 'Chart',
  type: workbook.ChartType.AREA,
  dataset: myDataset,
  category: category,
  legend: legend,
  series: [series]
});
var myNewWorkbook = workbook.create({
  description: 'My new updated workbook',
  name: 'Workbook Updated',
  tableDefinitions: [tableView],
  pivotDefinitions: [pivot],
  chartDefinitions: [chart]
});
var workbookList = workbook.list();

log.debug({
  title: "MyNewWorkbook",
  details: myNewWorkbook
});
});

```

# 54. N/workflow Module

Load the N/workflow module to initiate new workflow instances or trigger existing workflow instances.

define(['N/workflow'], (workflow) => { ... })

✓ 1 Code Sample(s)

Scripts: Server scripts

Permissions: Workflow

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/workflow'], (workflow) => { // Access methods and properties on workflow });
```

Use the N/workflow module to initiate new workflow instances or trigger existing workflow instances.

N/workflow Module Script Sample

## N/workflow Module Members

Return Type / Value Type

Supported Script Types

workflow.initiate(options)

Initiates a workflow on-demand. This method is the programmatic equivalent of the Initiate Workflow Action action in SuiteFlow.

Returns the internal ID of the workflow instance used to track the workflow against the record.

workflow.trigger(options)

| MEMBER TYPE | NAME                       | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES | DESCRIPTION                                                                                                                                                                                                                                                                               |
|-------------|----------------------------|--------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | workflow.initiate(options) | number                   | Server scripts         | Initiates a workflow on-demand. This method is the programmatic equivalent of the Initiate Workflow Action action in SuiteFlow. Returns the internal ID of the workflow instance used to track the workflow against the record.                                                           |
| METHOD      | workflow.trigger(options)  | number                   | Server scripts         | Triggers a workflow on a record. The actions and transitions of the workflow are evaluated for the record in the workflow instance, based on the current state for the workflow instance. Returns the internal ID of the workflow instance used to track the workflow against the record. |

### ► Official NetSuite Script Samples (1 Sample(s))

#### Sample 1: N/workflow Module Script Sample

```
/**
 * @NApiVersion 2.1
 */
require(['N/workflow', 'N/search', 'N/error', 'N/record'],
function(workflow, search, error, record) {
    function initiateWorkflow() {
        var workflowInstanceId = workflow.initiate({
            recordType: 'customer',
            recordId: 24,
            workflowId: 'customworkflow_myWorkFlow'
        });
        var customerRecord = record.load({
            type: record.Type.CUSTOMER,
            id: 24
        });
    }
    initiateWorkflow();
});
```

# 55. N/xml Module

Load the N/xml module to validate, parse, read, and modify XML documents.

define(['N/xml'], (xml) => { ... })

✓ 1 Code Sample(s)

Scripts: Client and server scripts

Permissions: -

```
/** * @NApiVersion 2.1 * @NScriptType UserEventScript */ define(['N/xml'], (xml) => { // Access methods and properties on xml });
```

Use the N/xml module to validate, parse, read, and modify XML documents.

## N/xml Module Members

Return Type / Value Type

Supported Script Types

Client and server scripts

Represents an attribute node of an xml.Element object.

Client and server scripts

Represents an entire XML document. The XML DOM presents a document as a hierarchy of node objects. Use the methods and properties available to the xml.Document object to manipulate the XML document and the nodes in the document tree.

| MEMBER TYPE | NAME                  | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                                                                                                                               |
|-------------|-----------------------|--------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT      | xml.Attr              | Object                   | Client and server scripts | Represents an attribute node of an xml.Element object.                                                                                                                                                                                    |
| OBJECT      | xml.Document          | Object                   | Client and server scripts | Represents an entire XML document. The XML DOM presents a document as a hierarchy of node objects. Use the methods and properties available to the xml.Document object to manipulate the XML document and the nodes in the document tree. |
| OBJECT      | xml.Element           | Object                   | Client and server scripts | Represents an element in an XML document. Elements may contain attributes, other elements, or text. If an element contains text, the text is represented in a text node of type TEXT_NODE.                                                |
| OBJECT      | xml.Node              | Object                   | Client and server scripts | Represents a generic XML node in an XML document. A node can be a Document, Element, or Attribute.                                                                                                                                        |
| OBJECT      | xml.Parser            | Object                   | Client and server scripts | Encapsulates the functionality used by NetSuite to parse XML.                                                                                                                                                                             |
| OBJECT      | xml.XPath             | Object                   | Client and server scripts | Encapsulates the functionality used by NetSuite to run XPath expressions. XPath is a standard for enumerating paths in an XML document collection.                                                                                        |
| METHOD      | xml.escape(options)   | string                   | Client and server scripts | Prepares a string for use in XML by escaping XML markup, such as angle brackets, quotation marks, and ampersands.                                                                                                                         |
| METHOD      | xml.validate(options) | void                     | Server scripts            | Validates an XML document against an XML Schema (XSD).                                                                                                                                                                                    |
| ENUM        | xml.NodeType          | string (read-only)       | Client and server scripts | Holds the string values for the supported node types. Use this enum to set the Node.nodeType property.                                                                                                                                    |

## Attr Object Members

The following members are called on the xml.Attr object.

Return Type / Value Type

Supported Script Types

Attr.ownerElement

xml.Element (read-only)

Client and server scripts

| MEMBER TYPE | NAME              | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                                        |
|-------------|-------------------|--------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| PROPERTY    | Attr.ownerElement | xml.Element (read-only)  | Client and server scripts | The xml.Element object that is the parent of the xml.Attr object.                                                                                  |
| PROPERTY    | Attr.name         | string (read-only)       | Client and server scripts | The name of an attribute.                                                                                                                          |
| PROPERTY    | Attr.specified    | boolean                  | Client and server scripts | Returns true if the attribute value is set in the parsed XML document, and false if it is a default value in a DTD or Schema.                      |
| PROPERTY    | Attr.value        | string                   | Client and server scripts | Value of an attribute. The value of the attribute is returned as a string. Character and general entity references are replaced with their values. |

## Document Object Members

In addition to the Document object members, Document objects inherit the members of the Node object. The methods and properties associated with a Node object can be used as members of a Document object. For more information, see Node Object Members.

The following members are called on the `xml.Document` object.

Return Type / Value Type

Supported Script Types

`Document.adoptNode(options)`

Client and server scripts

| MEMBER TYPE | NAME                                                       | RETURN TYPE / VALUE TYPE             | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                                         |
|-------------|------------------------------------------------------------|--------------------------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | <code>Document.adoptNode(options)</code>                   | <code>xml.Node</code>                | Client and server scripts | Attempts to adopt a node from another document to this document.                                                                                    |
| METHOD      | <code>Document.createAttribute(options)</code>             | <code>xml.Attr</code>                | Client and server scripts | Creates an attribute node of type <code>ATTRIBUTE_NODE</code> with the optional specified value.                                                    |
| METHOD      | <code>Document.createAttributeNS(options)</code>           | <code>xml.Attr</code>                | Client and server scripts | Creates an attribute node of type <code>ATTRIBUTE_NODE</code> , with the specified namespace value and optional specified value.                    |
| METHOD      | <code>Document.createCDATASection(options)</code>          | <code>xml.Node</code>                | Client and server scripts | Creates a CDATA section node of type <code>DOCUMENT_FRAGMENT_NODE</code> with the specified data.                                                   |
| METHOD      | <code>Document.createComment(options)</code>               | <code>xml.Node</code>                | Client and server scripts | Creates a Comment node of type <code>COMMENT_NODE</code> with the specified string.                                                                 |
| METHOD      | <code>Document.createDocumentFragment()</code>             | <code>xml.Node</code>                | Client and server scripts | Creates a node of type <code>DOCUMENT_FRAGMENT_NODE</code> .                                                                                        |
| METHOD      | <code>Document.createElement(options)</code>               | <code>xml.Element</code>             | Client and server scripts | Creates a new node of type <code>ELEMENT_NODE</code> with the specified name.                                                                       |
| METHOD      | <code>Document.createElementNS(options)</code>             | <code>xml.Element</code>             | Client and server scripts | Creates a new node of type <code>ELEMENT_NODE</code> with the specified namespace URI and name.                                                     |
| METHOD      | <code>Document.createProcessingInstruction(options)</code> | <code>xml.Node</code>                | Client and server scripts | Creates a new node of type <code>PROCESSING_INSTRUCTION_NODE</code> with the specified target and data.                                             |
| METHOD      | <code>Document.createTextNode(options)</code>              | <code>xml.Node</code>                | Client and server scripts | Creates a new node of type <code>TEXT_NODE</code> .                                                                                                 |
| METHOD      | <code>Document.getElementById(options)</code>              | <code>xml.Element</code>             | Client and server scripts | Returns the element that has an ID attribute with the specified value as an <code>xml.Element</code> object.                                        |
| METHOD      | <code>Document.getElementsByTagName(options)</code>        | <code>xml.Element[]</code>           | Client and server scripts | Returns an array of <code>xml.Element</code> objects with a specific tag name, in the order in which they appear in the XML document.               |
| METHOD      | <code>Document.getElementsByTagNameNS(options)</code>      | <code>xml.Element[]</code>           | Client and server scripts | Returns an array of <code>xml.Element</code> objects with a specific tag name and namespace, in the order in which they appear in the XML document. |
| METHOD      | <code>Document.importNode(options)</code>                  | <code>xml.Node</code>                | Client and server scripts | Imports a node from another document to this document. Creates a new copy of the source node.                                                       |
| PROPERTY    | <code>Document.doctype</code>                              | Object (read-only)                   | Client and server scripts | Returns a node of type <code>DOCUMENT_TYPE_NODE</code> that represents the doctype of the XML document.                                             |
| PROPERTY    | <code>Document.documentElement</code>                      | <code>xml.Element</code> (read-only) | Client and server scripts | Root node of the XML document.                                                                                                                      |
| PROPERTY    | <code>Document.documentURI</code>                          | string (read-only)                   | Client and server scripts | Location of the document or null if undefined.                                                                                                      |
| PROPERTY    | <code>Document.inputEncoding</code>                        | string (read-only)                   | Client and server scripts | Encoding used for an XML document at the time the document was parsed.                                                                              |
| PROPERTY    | <code>Document.xmlEncoding</code>                          | string (read-only)                   | Client and server scripts | Part of the XML declaration, the XML encoding of the XML document.                                                                                  |
| PROPERTY    | <code>Document.xmlStandalone</code>                        | boolean                              | Client and server scripts | Part of the XML declaration, returns true if the current XML document is standalone or returns false if it is not.                                  |
| PROPERTY    | <code>Document.xmlVersion</code>                           | string                               | Client and server scripts | Part of the XML declaration, the version number of the XML document.                                                                                |

## Element Object Members

In addition to the Element object members, Element objects inherit the members of the Node object. The methods and properties associated with a Node object can be used as members of a Element object. For more information, see Node Object Members.

The following members are called on the `xml.Element` object.

Return Type / Value Type

Supported Script Types

`Element.getAttribute(options)`

Client and server scripts

| MEMBER TYPE | NAME                                                 | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                                                                                                             |
|-------------|------------------------------------------------------|--------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | <code>Element.getAttribute(options)</code>           | string                   | Client and server scripts | Returns the value of the specified attribute.                                                                                                                                                                           |
| METHOD      | <code>Element.getAttributeNode(options)</code>       | xml.Attr                 | Client and server scripts | Retrieves an attribute node by name.                                                                                                                                                                                    |
| METHOD      | <code>Element.getAttributeNodeNS(options)</code>     | string                   | Client and server scripts | Returns an attribute node with the specified namespace URI and local name.                                                                                                                                              |
| METHOD      | <code>Element.getAttributeNS(options)</code>         | xml.Attr                 | Client and server scripts | Returns an attribute value with the specified namespace URI and local name.                                                                                                                                             |
| METHOD      | <code>Element.getElementsByTagName(options)</code>   | xml.Element[]            | Client and server scripts | Returns an array of descendant xml.Element objects with a specific tag name, in the order in which they appear in the XML document.                                                                                     |
| METHOD      | <code>Element.getElementsByTagNameNS(options)</code> | xml.Element[]            | Client and server scripts | Returns an array of descendant xml.Element objects with a specific tag name and namespace, in the order in which they appear in the XML document.                                                                       |
| METHOD      | <code>Element.hasAttribute(options)</code>           | boolean                  | Client and server scripts | Returns true if the current element has an attribute with the specified name or if that attribute has a default value. Otherwise, returns false.                                                                        |
| METHOD      | <code>Element.hasAttributeNS(options)</code>         | boolean                  | Client and server scripts | Returns true if the current element has an attribute with the specified local name and namespace or if that attribute has a default value. Otherwise, returns false.                                                    |
| METHOD      | <code>Element.removeAttribute(options)</code>        | void                     | Client and server scripts | Removes the attribute with the specified name.                                                                                                                                                                          |
| METHOD      | <code>Element.removeAttributeNode(options)</code>    | xml.Attr                 | Client and server scripts | Removes the attribute specified as a xml.Attr object.                                                                                                                                                                   |
| METHOD      | <code>Element.removeAttributeNS(options)</code>      | void                     | Client and server scripts | Removes the attribute with the specified namespace URI and local name.                                                                                                                                                  |
| METHOD      | <code>Element.setAttribute(options)</code>           | void                     | Client and server scripts | Adds a new attribute with the specified name. If an attribute with that name is already present in the element, its value is changed to the value specified in method argument.                                         |
| METHOD      | <code>Element.setAttributeNode(options)</code>       | xml.Attr                 | Client and server scripts | Adds the specified attribute node. If an attribute with the same name is already present in the element, it is replaced by the new one.                                                                                 |
| METHOD      | <code>Element.setAttributeNodeNS(options)</code>     | xml.Attr                 | Client and server scripts | Adds the specified attribute node. If an attribute with the same local name and namespace URI is already present in the element, it is replaced by the new one.                                                         |
| METHOD      | <code>Element.setAttributeNS(options)</code>         | void                     | Client and server scripts | Adds a new attribute with the specified name and namespace URI. If an attribute with the same name and namespace URI is already present in the element, its value is changed to the value specified in method argument. |
| PROPERTY    | <code>Element.tagName</code>                         | string (read-only)       | Client and server scripts | The tag name of this xml.Element object.                                                                                                                                                                                |

## Node Object Members

The following members are called on the xml.Node object.

Return Type / Value Type

Supported Script Types

Node.appendChild(options)

Client and server scripts

Appends a node after the last child node of a specific element node. Returns the new child node.

| MEMBER TYPE | NAME                                               | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                                                                                                       |
|-------------|----------------------------------------------------|--------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| METHOD      | <code>Node.appendChild(options)</code>             | xml.Node                 | Client and server scripts | Appends a node after the last child node of a specific element node. Returns the new child node.                                                                                                  |
| METHOD      | <code>Node.cloneNode(options)</code>               | xml.Node                 | Client and server scripts | Creates a copy of a node. Returns the copied node.                                                                                                                                                |
| METHOD      | <code>Node.compareDocumentPosition(options)</code> | number                   | Client and server scripts | Returns a number that reflects where two nodes are located, compared to each other.                                                                                                               |
| METHOD      | <code>Node.hasAttributes()</code>                  | boolean                  | Client and server scripts | Returns true if the current node has any attributes. Note that only element nodes can have attributes.                                                                                            |
| METHOD      | <code>Node.hasChildNodes()</code>                  | boolean                  | Client and server scripts | Returns true if the current node has child nodes or returns false if the current node does not have child nodes.                                                                                  |
| METHOD      | <code>Node.insertBefore(options)</code>            | xml.Node                 | Client and server scripts | Inserts a new child node before an existing child node for the current node.                                                                                                                      |
| METHOD      | <code>Node.isDefaultNamespace(options)</code>      | boolean                  | Client and server scripts | Returns true if the specified namespace uniform resource identifier (URI) is the default namespace for the current node or returns false if the specified namespace is not the default namespace. |

| MEMBER TYPE | NAME                                          | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                                                                              |
|-------------|-----------------------------------------------|--------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------|
| METHOD      | <code>Node.isEqualNode(options)</code>        | boolean                  | Client and server scripts | Returns true if two nodes are equal or returns false if two nodes are not equal.                                         |
| METHOD      | <code>Node.isSameNode(options)</code>         | boolean                  | Client and server scripts | Returns true if two nodes reference the same object or returns false if two nodes do not reference the same object.      |
| METHOD      | <code>Node.lookupNamespaceURI(options)</code> | string                   | Client and server scripts | Returns the namespace uniform resource identifier (URI) that matches the specified namespace prefix.                     |
| METHOD      | <code>Node.lookupPrefix(options)</code>       | string                   | Client and server scripts | Returns the namespace prefix associated with the specified namespace uniform resource identifier (URI).                  |
| METHOD      | <code>Node.normalize()</code>                 | void                     | Client and server scripts | Puts all text nodes underneath a node, including attribute nodes, into a normal form.                                    |
| METHOD      | <code>Node.removeChild(options)</code>        | xml.Node                 | Client and server scripts | Removes the specified child node. Returns the removed child node.                                                        |
| METHOD      | <code>Node.replaceChild(options)</code>       | xml.Node                 | Client and server scripts | Replaces a specific child node with another child node in a list of child nodes.                                         |
| PROPERTY    | <code>Node.attributes</code>                  | Object (read-only)       | Client and server scripts | Key-value pairs for all attributes for an xml.Element node. Returns null for all other node types.                       |
| PROPERTY    | <code>Node.baseURI</code>                     | string (read-only)       | Client and server scripts | Absolute base uniform resource identifier (URI) of a node or null if the URI cannot be determined.                       |
| PROPERTY    | <code>Node.childNodes</code>                  | xml.Node[] (read-only)   | Client and server scripts | Array of all child nodes of a node or an empty array if there are no child nodes.                                        |
| PROPERTY    | <code>Node.firstChild</code>                  | xml.Node (read-only)     | Client and server scripts | First child node for a specific node or null if there are no child nodes.                                                |
| PROPERTY    | <code>Node.lastChild</code>                   | xml.Node (read-only)     | Client and server scripts | Last child node for a specific node or null if there is no last child node.                                              |
| PROPERTY    | <code>Node.localName</code>                   | string (read-only)       | Client and server scripts | The local part of the qualified name of a node.                                                                          |
| PROPERTY    | <code>Node.namespaceURI</code>                | string (read-only)       | Client and server scripts | The namespace uniform resource identifier (URI) of a node or null if there is no namespace URI for the node.             |
| PROPERTY    | <code>Node.nextSibling</code>                 | xml.Node (read-only)     | Client and server scripts | The next node in a node list or null if the current node is the last node.                                               |
| PROPERTY    | <code>Node.nodeName</code>                    | string (read-only)       | Client and server scripts | Name of a node, depending on the type. For example, for a node of type xml.Element, the name is the name of the element. |
| PROPERTY    | <code>Node.nodeType</code>                    | string                   | Client and server scripts | The type of node defined as a value from the xml.NodeType enum.                                                          |
| PROPERTY    | <code>Node.nodeValue</code>                   | string                   | Client and server scripts | The value of a node, depending on its type.                                                                              |
| PROPERTY    | <code>Node.ownerDocument</code>               | xml.Document (read-only) | Client and server scripts | The root element for a node as a xml.Document object.                                                                    |
| PROPERTY    | <code>Node.parentNode</code>                  | xml.Node (read-only)     | Client and server scripts | The parent node of a node.                                                                                               |
| PROPERTY    | <code>Node.prefix</code>                      | string                   | Client and server scripts | The namespace prefix of the node, or null if the node does not have a namespace.                                         |
| PROPERTY    | <code>Node.previousSibling</code>             | xml.Node (read-only)     | Client and server scripts | The previous node in a node list or null if the current node is the first node.                                          |
| PROPERTY    | <code>Node.textContent</code>                 | string                   | Client and server scripts | The textual content of a node and its descendants.                                                                       |

## Parser Object Members

The following members are called on the xml.Parser object.

Return Type / Value Type

Supported Script Types

Parser.fromString(options)

Client and server scripts

Parses a string into a W3C XML document object.

| MEMBER TYPE | NAME                                    | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                 |
|-------------|-----------------------------------------|--------------------------|---------------------------|-------------------------------------------------------------|
| METHOD      | <code>Parser.fromString(options)</code> | xml.Document             | Client and server scripts | Parses a string into a W3C XML document object.             |
| METHOD      | <code>Parser.toString(options)</code>   | string                   | Client and server scripts | Converts (serializes) an xml.Document object into a string. |

XPath Object Members

The following members are called on the xml.XPath object.

Return Type / Value Type

Supported Script Types

XPath.select(options)

Client and server scripts

Selects an array of nodes from an XML document using an XPath expression.

| MEMBER TYPE | NAME                  | RETURN TYPE / VALUE TYPE | SUPPORTED SCRIPT TYPES    | DESCRIPTION                                                               |
|-------------|-----------------------|--------------------------|---------------------------|---------------------------------------------------------------------------|
| METHOD      | XPath.select(options) | xml.Node[]               | Client and server scripts | Selects an array of nodes from an XML document using an XPath expression. |

► Official NetSuite Script Samples (1 Sample(s))

Sample 1: N/xml Module Script Samples

```
<bookstore xmlns:b="http://www.qualifiednamespace.com/book">
  <b:book category="cooking">
    <b:title lang="en">Everyday Italian</b:title>
    <b:author>Giada De Laurentiis</b:author>
    <b:year>2005</b:year>
    <b:price>30.00</b:price>
  </b:book>
  <b:book category="children">
    <b:title lang="en">Harry Potter</b:title>
    <b:author>J K. Rowling</b:author>
    <b:year>2005</b:year>
    <b:price>29.99</b:price>
  </b:book>
  <b:book category="web">
    <b:title lang="en">XQuery Kick Start</b:title>
    <b:author>James McGovern</b:author>
    <b:author>Per Bothner</b:author>
    <b:author>Kurt Cagle</b:author>
    <b:author>James Linn</b:author>
    <b:author>Vaidyanathan Nagarajan</b:author>
    <b:year>2003</b:year>
    <b:price>49.99</b:price>
  </b:book>
  <b:book category="web" cover="paperback">
    <b:title lang="en">Learning XML</b:title>
    <b:author>Erik T. Ray</b:author>
    <b:year>2003</b:year>
    <b:price>39.95</b:price>
  </b:book>
</bookstore>
```